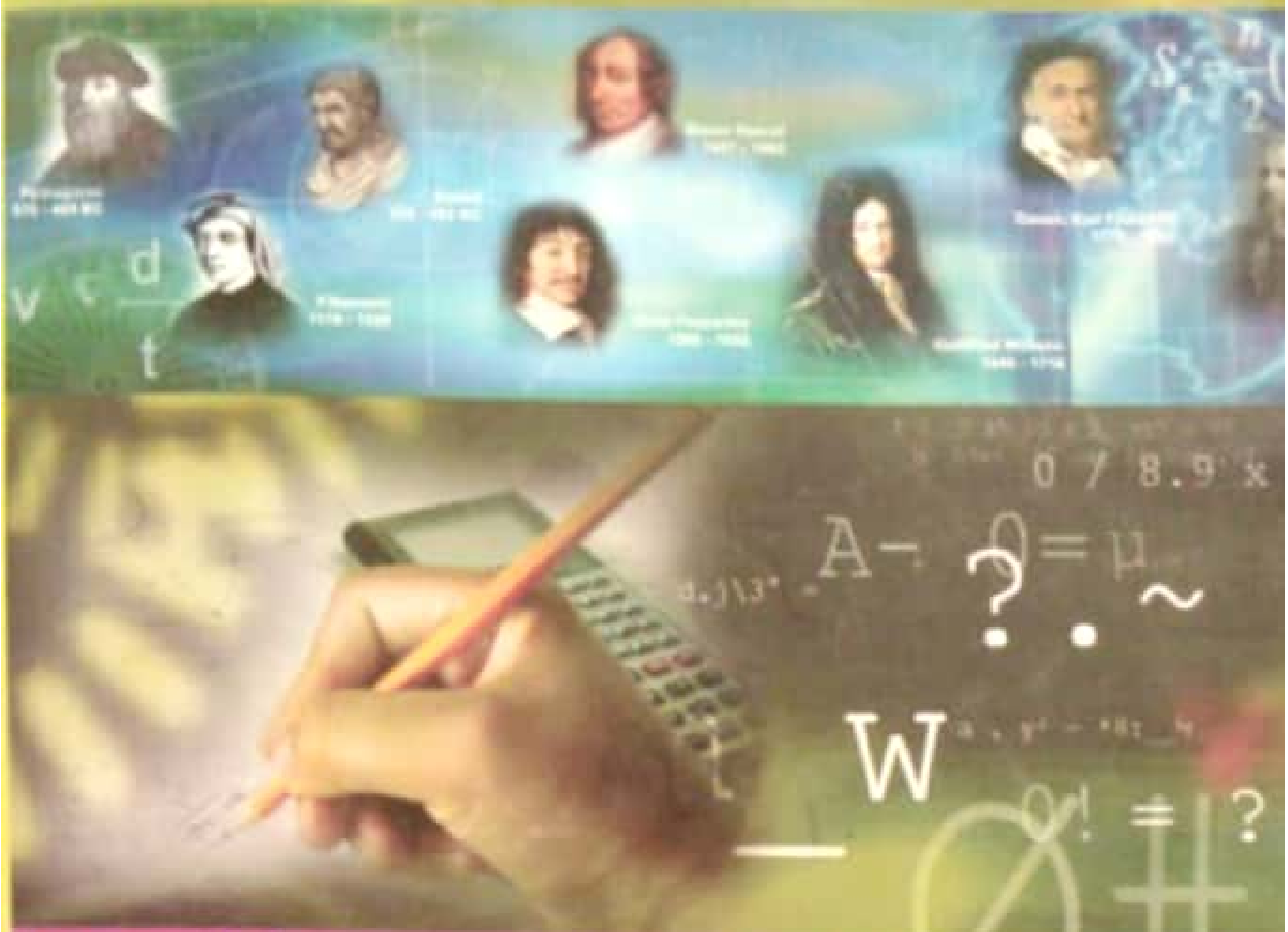

Caravan

According to FPSC/PPSC & NTS Latest Pattern

Mathematics MCQs

For:

Subject Specialist,
Senior Subject Specialist
& Lectureship

Prof. Zia-ul-Haq

Caravan Book House - Lahore

CONTENTS

NUMBER SYSTEM

➢ **Tick (✓) the correct one answer only.**

1. Real number system consists of
   (a) Natural numbers
   (b) Even and odd numbers
   (c) Rational and irrational numbers
   (d) None of these

2. Natural numbers are
   (a) Whole number
   (b) Rational numbers
   (c) Even and odd numbers
   (d) Irrational numbers

3. For "P" and "q" integers rational numbers are defined as
   (a) P/q where $q = 0$
   (b) P/q where $q \neq 0$
   (c) Pq where $q = 1$
   (d) $P - q$ where $p, q \neq 0$

4. Law of trichotomy for a, b ∈ (Field) is
   (a) $a > b, b > a$
   (b) $a = b$
   (c) $a > b, b > a, a = b$
   (d) None of these

5. Transitivity is defined ∀ a, b, c ∈ (Field)
   (a) $a > b \wedge b > c \Rightarrow a > c$
   (b) $a > b \wedge b > c \Rightarrow a \ngtr c$
   (c) $a < b \wedge b < c \Rightarrow a < c$
   (d) $a > b \wedge b < c \Rightarrow a = b = c$

6. Set N (natural number) and I (integers) are
   (a) Real field (b) Not orderd field
   (c) None (d) Orderd field

7. For a and b ∈ Rational commutative property w.r.t. addition is defined as
   (a) $a + b = b + a$ (b) $a - b = b - a$
   (c) $ab = ba$ (d) $\frac{a}{b} = \frac{b}{a}$

8. In the following set of numbers which ordered field
   (a) Set Q (rational numbers)
   (b) Set R (real numbers)
   (c) Both (a) and (b)
   (d) None of these

9. The following numbers which has not multiplicative inverse
   (a) Zero (b) One
   (c) Minus one (d) Two

10. Square root of 2 has no
    (a) Irrational number
    (b) Rational number
    (c) Even number
    (d) Odd number

11. For x, y Q(rational) of $x < y$ then true is
    (a) $-x < -y$ (b) $-x > -y$
    (c) $-x > \frac{1}{y}$ (d) $x < \frac{-1}{y}$

12. For x, y ∈ R(real). If $\frac{1}{x} > \frac{1}{y}$ then true is
    (a) $\frac{1}{x} < -\frac{1}{y}$ (b) $-x > -y$
    (c) $-\frac{1}{x} < -\frac{1}{y}$ (d) Both b and c

13. Q(set of rational) is ordered set. If for Q_1, Q_2, EQ

(a) $Q_1 < Q_2$ and $Q_2 - Q_1$

(b) $Q_2 < Q_1$

(c) $Q_1 = Q_2$

(d) None of these

14. For $E \subseteq S$ and $\alpha \in E$, then β is an upper bound of E if.

(a) $\alpha = \beta$ (b) $\alpha \le \beta \; \forall \; \alpha \in E$

(c) $\alpha \ge \beta \; \forall \in E$ (d) None of these

15. For $E \subseteq S$ and $\alpha \in E$, $\beta \in S$, then β is lower bound of E. If

(a) $B \le \alpha \; \forall \; \alpha \in E$ (b) $\alpha < \beta$

(c) $\alpha \ne \beta$ (d) $\alpha = \beta$

16. If n +ve integer has no perfect square then

(a) $\sqrt{n}$ perfect number

(b) Not rational

(c) Rational

(d) None of these

17. If r is a rational number and x is irrational number then

(a) $r + x$ is irrational (b) rx is irrational

(c) $r + x$ is rational (d) Both (a) and (b)

18. For every real number "$x > 0$" and for every integer "$n > 0$" there is only one real number "$y>0$" such that

(a) $y^n = x$ (b) $y^n \ne x$

(c) $y^n < x$ (d) $y^n > x$

19. For y and $x \in R$ there is only one integer $n > 0$ such that

(a) $y < x^{1/n}$ (b) $y = x^{1/n}$

(c) $x = y$ (d) None of these

20. If $|x| > 0$ the new can write

(a) Max(x, –x) (b) $x \ne -x$

(c) Min(x, –x) (d) Both (a) + (b)

21. For any real number x, there is an integer $n > 0$. Such that

(a) $x = n$ (b) $x < n$

(c) $x \ge n$ (d) $x \ne x$

22. For ever real number $\frac{y}{x}$ there is an integer n such that

(a) $\frac{y}{x} < n$ (b) $\frac{y}{x} > x$

(c) $\frac{y}{x} = n$ (d) $\frac{y}{x} \ne n$

23. If x and y be any real number then there exist Q rational. Such that

(a) $x < Q > y$ (b) $x < Q < y$

(c) $x < y < Q$ (d) $x = Q = y$

24. Set of Rational "Q"

(a) Not complete ordered field

(b) Field

(c) Complete ordered field

(d) None of these

25. If E be bounded set of real numbers and $\lambda = \text{Sup } E$ then for every $\in > 0$ and $\forall \; x_1 \in$

(a) $\lambda - \in x_1$ (b) $\lambda - \in < x_1$

(c) $\lambda - \in = x_1$ (d) $\lambda + \in \ge x_1$

26. For every real number x there is a set E of rationals numbers such that

(a) $x \ne \text{Sup } E$ (b) $x < \text{Sup } E$

(c) $x = \text{Sup } E$ (d) $x = E$

27. If A is a bounded set of reals and b real number then

(a) $\text{Sup }(b + A) \ne b + \text{Sup } A$

(b) $\text{Sup }(b + A) = \text{Sup } A$

(c) $\text{Sup }(b + A) = b + \text{Sup } A$

(d) None of these

28. For every real number y, there is set E of rationals. Such that

(a) $y = \text{Inf } E$ (b) $y < \text{Inf } E$

(c) $y = \text{Inf } E$ (d) $y = E$

29. If A is a bounded set of reals and b real number then

(a) $\text{Inf }(b + A) \ne b + \text{Inf }(A)$

(b) $\text{Inf }(b + A) = \text{Inf } A$

(c) $\text{Inf }(b + A) = \text{Sup } A$

(d) $\text{Inf }(b + A) = b + \text{Inf } A$

30. If $A \subseteq B$ belongs to real number, then
(a) $\text{Inf } B \le \text{Inf } A \le \text{Sup } A \le \text{Sup } B$
(b) $\text{Inf } B \ge \text{Sup } A$
(c) $\text{Sup } A \ge \text{Sup } B \le \text{Sup } (A + B)$
(d) None of these

31. If $-E \subseteq F$ (complete ordered field) and $-E$ bounded below then
(a) $\text{Inf } (E) = \text{Sup } (E)$
(b) $\text{Inf } (-E) = -\text{Sup } (E)$
(c) $\text{Inf } (-E) = \text{Sup } (-E)$
(d) $\text{Inf } (-E) \ne -\text{Sup } (E)$

32. If E is bounded set and $\lambda > 0$ then
(a) $\text{Sup } (E) = \lambda$
(b) $\text{Inf } (\lambda E) = \lambda$
(c) $\text{Inf } (\lambda E) = \text{Sup}$
(d) $\text{Sup } (\lambda E) = \lambda \text{ Sup } E$

33. If E bounded set of complete to ordered field then,
(a) $\text{Sup } (-E) \ne -\text{Inf } E$
(b) $\text{Sup}(-E) = -\text{Inf } E$
(c) $\text{Inf } (-E) = -\text{Inf } E$
(d) None of these

34. If E_1^1 is bounded below and L the set of lower bounds of E then for $\lambda \in L$
(a) $\lambda = \text{Inf } E$
(b) $\lambda > \text{Inf } E$
(c) $\lambda \ne \text{Inf } E$
(d) $\lambda \ne E$

35. If $P \ne 1$ is prime number then $x^2 = P$ has solution of numbers
(a) Rational
(b) Odd
(c) Natural
(d) Rational

36. Between two rational numbers there are
(a) Infinite rationals
(b) No rational
(c) Odd numbers
(d) Even numbers

37. For all real nos "x" and "y"
(a) $|x+y| = |x| + |y|$
(b) $|x+y| \le |x| + |y|$
(c) $|x-y| = |x| - |y|$
(d) $|x+y| < |x| - |y|$

38. For all real numbers "x" and "y"
(a) $||x| - |y|| = |x-y|$
(b) $||x| - |y|| < |x+y|$
(c) $||x| - |y| \ne |x-y|$
(d) $||x| - |y|| \le |x-y|$

39. For $x, y \in R^n$ then
(a) $|x \cdot y| \le ||x|| \, ||y||$
(b) $|x \cdot y| = ||x|| \, ||y||$
(c) $|x \cdot y| \ge ||x|| \, ||y||$
(d) $|x \cdot y| < ||xy||$

40. For $x, y \in R^n$ then
(a) $||x+y|| = ||x|| + ||y||$
(b) $||x+y|| \le ||x|| + ||y||$
(c) $||x+y|| < |x| + |y|$
(d) None of these

41. For $x, y \in R^n$ there could exist as
(a) $||x \cdot y|| \le ||x|| \cdot ||y||$
(b) $||x \cdot y|| = ||x|| \cdot ||y||$
(c) $||x \cdot y|| > ||x|| \cdot ||y||$
(d) Both (a) and (b)

42. If and only if $(x \cdot y) = 0$ then
(a) $||x+y||^2 \le ||x||^2 + ||y||^2$
(b) $||x+y||^2 = ||x||^2 + ||y||^2$
(c) $||x+y||^2 \ge ||x||^2$
(d) $||x+y||^2 \le ||y||^2$

43. If and only if $(x, y) = 0$ then
(a) $||x-y||^2 \ne ||x||^2 + ||y||^2$
(b) $||x-y||^2 \ge ||x||^2 + ||y||^2$
(c) $||x-y||^2 = |x+y|^2$
(d) $||x-y||^2 = ||x||^2 + ||y||^2$

44. For $x, y \in R$(real) $||$ram law is
(a) $||x+y||^2 + ||x-y||^2 = (||x||^2 + ||y||^2)$
(b) $||x+y||^2 + ||x-y||^2 = 0$
(c) $||x+y||^2 + ||x-y||^2 < ||x||^2 + ||y||^2$
(d) $||x+y||^2 + ||x-y||^2 = (2||x||^2 + ||y||^2)$

45. For x, y ∈ R^n we can write as
(a) $||x-y|| \geq ||x|| - ||y||$
(b) $||x-y|| = ||x|| - ||y||$
(c) $||x-y|| \leq ||x|| - ||y||$
(d) None of these

46. For x, y ∈ R^n there exist inequality
(a) $(x \cdot y) = \frac{1}{4}[||x+y||^2 - ||x-y||^2$
(b) $(x \cdot y) \leq \frac{1}{4}(||x+y||^2)$
(c) $(x \cdot y) = [||x+y||^2 - ||x-y||^2]$
(d) $(x \cdot y) \leq \frac{1}{4}[||x+y||^2 - ||x-y||^2$

47. For x, y ∈ r and $(x-y) \cdot (x+y) = 0$ then
(a) $||x|| \neq ||y||$ (b) $||x|| = ||y||$
(c) $||x|| < ||y||$ (d) Both (b) and (c)

48. A sequence is a function whose domain is the set of
(a) Integers (b) Rationals
(c) Irrationals (d) Primes

49. A sequence is usually written as
(a) $\{S_n\}_{n=0}^{1}$ (b) $\{S_n\}_{n=i}^{\alpha}$
(c) $\{S_n\}$ (d) None of these

50. A sequence is said to be decreasing if
(a) $S_{n+1} \leq S_n$ for all $n \geq 1$
(b) $S_{n+1} \geq S_n \; \forall \; n \geq 1$
(c) $S_{n+1} = S_n$ for all $n = 0$
(d) $S_{n+1} \leq S_{n-1} \; \forall \; n \geq 1$

51. A sequence $\{S_n\}$ is said to converge to limit S as $n \to \alpha$ if for n_0 and $\epsilon > 0$ we have
(a) $S_n - S = 0$ (b) $|S_n - X| > \epsilon$
(c) $|S_n - S| < \epsilon$ (d) $S_n = S$

52. A sequence $\{x_n\}$ is Cauchy sequence if for $\epsilon > 0$ and m, n > n_0 there is
(a) $|x_n - x_m| < \epsilon$ (b) $|x_m - x_n| = 0$
(c) $|x_n - x_m| > \epsilon$ (d) None of these

53. A Cauchy sequence of real numbers is
(a) Unbounded (b) Increasing
(c) Decreasing (d) Bounded

54. A sequence is said to be divergent if it is
(a) Unbounded (b) Bounded
(c) Increasing (d) Decreasing

55. $\{S_n\}$ a bounded monotomic sequence wil be increasing if it converge to its
(a) Infimum (b) Supremum
(c) Least bound (d) Upper bound

56. $\{S_n\}$ a bounded monotomic sequence wil be decreases if it converges to its
(a) Lower bound (b) Upper bound
(c) Infimum (d) Supermum

57. Every bound sequence has a subsequence which
(a) Divergent (b) Increasing
(c) Decreasing (d) Convergent

58. A metric space (X, d) is complete if every Cauchy sequence in X is
(a) Converges in X (b) Diverges
(c) Not exist (d) None of these

59. Every infinite sequence in a compact matric space X has a subsequence which
(a) Diverges in X (b) Converges in X
(c) Bounded (d) Not bound

60. Sum up to infinity of Geometric series is given by First term a and ratio r as
(a) $S_\alpha = \frac{a}{1+r}$ (b) $\frac{a}{1-r^2}$
(c) $\frac{1}{1-r}$ (d) $\frac{a}{1-r}$

61. If f is differentiable at x ∈ [a, b] then f at x is
(a) Continuous (b) Discontinuous
(c) Bounded (d) Infinite

62. A function f(x) is differentiable at point x if the following limit exist
(a) $\lim\limits_{t \to x} \frac{f(x)-f(t)}{t-x}$ (b) $\lim\limits_{t \to x} \frac{f(t)-f(x)}{t-x}$
(c) $\lim\limits_{t \to x} \frac{f(t)+f(x)}{t+x}$ (d) None of these

63. If "f" and "g" are differentiable at x then

(a) $(fg)' x = f'(x) \cdot g'(x) - f(x) g(x)$

(b) $(f'g)x = f'(x) + g'(x)$

(c) $(f'g)x = f'(x) - g'(x)$

(d) $(fg)'(x) = f'(x) g(x) + g'(x) f(x)$

64. If f(x) has a local maximum at point $x \in (a, b)$ then

(a) $f'(x) = 0$ (b) $f'(x) \neq 0$

(c) $f'(x) < 0$ (d) $f(x) < 0$

65. If f and g are differentiable in (a, b) then there is point $x \in (a, b)$ such that

(a) $[f(b) - f(a)]g'(x) = [g(b) - g(a)]f(x)$

(b) $[f(b) + f(a)] g'(x) = [g(b) + g(a)] f'(x)$

(c) $[f(b) - f(a)] g'(x) = [g(b) - g(a)] f'(x)$

(d) $[f(b) - f(a)] g(x) = [g(b) - g(a)] f'(x)$

66. f be function differentiable in (a, b) then there is point $x \in (a, b)$. Such that

(a) $f(b) - f(a) = (b - a) f'(x)$

(b) $f(b) - f(a) = (b - a) f(x)$

(c) $f(b) + f(a) = (b + a) f(x)$

(d) $f(b) f(a) = ba f(x)$

67. If "f" is differentiable in [a, b] then it is monotonically increasing if

(a) $f'(x) = 0$ (b) $f'(x) \leq 0$

(c) $f'(x) \geq 0$ (d) $f(x) = 0$

68. If f is differentiable in [a, b] then is monotonically decreasing if

(a) $f'(x) = 0$ (b) $f(x) = 0$

(c) $f'(x) \geq 0$ (d) $f'(x) \leq 0$

69. If f'(x) exist then it is constant function for all $x \in (a, b)$ if

(a) $f'(x) = 0$ (b) $f'(x) \geq 0$

(c) $f(x) = 0$ (d) None of these

70. If f'(x) exist on [a, b] and $f'(a) < \lambda < f'(b)$ then there is

(a) $f'(x) < \lambda$ (b) $f'(x) > \lambda$

(c) $f'(x) = \lambda$ (d) $f(x) = \lambda$

71. If f'(x) exist on [a, b] and $f'(a) > \lambda > f'(b)$ then there is

(a) $f(x) = \lambda$ (b) $f'(x) < \lambda$

(c) $f'(x) > \lambda$ (d) $f'(x) = \lambda$

72. Let $f : G \to y$ if $C \in G$ and $\lambda \in Y$ then there exist

(a) $(f^{-1})' \lambda = \frac{1}{f'(c)}$ (b) $f'(\lambda) \frac{1}{f(c)}$

(c) $f'(c) = \frac{1}{f'(c)}$ (d) $f^{-1}(\lambda) = f'(c)$

72. Associative property for real w.r.t. (+) is defined as

(a) $a + (b + c) = a \cdot (b + c)$

(b) $a + (b + c) = (a + b) + c$

(c) $a + (bc) = a + (b)$

(d) None of these

73. Let $f: R^n \to R$ if f is differentiable at $a \in R^n$ then it is

(a) Continuous at a

(b) Discontinuous at a

(c) Bounded

(d) None

74. Let $f : v \to R$ and $V \subseteq R^2$ then f(x, y) is differentiable at (a, b) If there are function ϕ_1 and ϕ_2 such that

(a) $f(x, y) + f(a, b) = \phi_1(x, y)(x + 1) + \phi_2(x, y)(y + b)$

(b) $f(x, y) - f(a, b) = \phi_1(x, y)(x - a) + \phi_2(x, y)(y - b)$

(c) $f(x, y) - f(a, b) = \phi_1(x, y)(x - a) - \phi_2(x, y)$

(d) $f(x, y) - f(a, b) = \phi_1(x, y) - \phi_2(x, y)(y - b)$

75. $\partial f/\partial x$ at (a, b) is defined as

(a) $\lim\limits_{h \to 0} \frac{f(a + h, b) - f(a, b)}{h}$

(b) $\lim\limits_{k \to 0} \frac{f(a, b + k) - f(a, b)}{k}$

(c) $\frac{f(a) - f(b)}{b - a}$

(d) $\lim\limits_{h \to 0} \frac{f(a - h) - f(b - h)}{h}$

76. If $z = f(x, y)$ has first order derivative then total differential dz is defined as

(a) $dz = \frac{\partial z}{\partial x} dx + \frac{\partial z}{\partial y} dy$

(b) $dz = \frac{\partial z}{\partial x}dx - \frac{\partial z}{\partial y}dy$

(c) $dz = \frac{\partial z}{\partial x} + \frac{\partial z}{\partial y}$

(d) $dz = \frac{\partial z}{\partial x} - \frac{\partial z}{\partial y}$

77. If $z = f(x, y)$ and $x = g(t)$, $y = h(t)$ then

(a) $\frac{dz}{dt} = \frac{\partial z}{\partial x} \cdot \frac{dx}{dt} - \frac{\partial z}{\partial y}\frac{dy}{dt}$

(b) $\frac{dz}{dt} = \frac{\partial z}{\partial x} \cdot \frac{dx}{dt} + \frac{\partial z}{\partial y}\frac{dt}{dt}$

(c) $\frac{dz}{dt} = \frac{\partial z}{\partial x} + \frac{\partial z}{\partial y}$ (d) $\frac{dz}{dt} = \frac{\partial z}{\partial x} - \frac{\partial z}{\partial y}$

78. If $f : V \to R$ where VCR^2 and f_x, f_y, f_{xy} exist in (a, b), then

(a) $f_{yx}(a, b) = f_{xy}(a, b)$

(b) $f_x(a, b) = f_y(a, b)$

(c) $f_{xy}(a, b) + f_y(a, b) = 0$

(d) None of these

79. ∇^2 is defined as

(a) $\frac{\partial^2 x}{\partial y^2} + \frac{\partial y^2}{\partial z^2} + \frac{\partial z^2}{\partial x^2}$ (b) $\frac{\partial x}{\partial y} - \frac{\partial y}{\partial z} - \frac{\partial z}{\partial x}$

(c) $\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$ (d) $\frac{\partial^2}{\partial x^2} - \frac{\partial^2}{\partial y^2} - \frac{\partial^2}{\partial z^2}$

80. Associative property w.r.t. "X" is defined as

(a) a(bc) = (ab)c (b) a + (bc) = a(bc)

(c) a(bc) = abc (d) a(bc) = a/bc

81. If $w = f(x, y)$ and $x = f(u, v)$, $y = f(u, v)$ then

(a) $\frac{\partial w}{\partial u} = \frac{\partial w}{\partial x} \cdot \frac{\partial x}{\partial u} + \frac{\partial w}{\partial y}$

(b) $\frac{\partial w}{\partial u} = \frac{\partial w}{\partial x} \cdot \frac{\partial x}{\partial u} - \frac{\partial w}{\partial y} \cdot \frac{\partial y}{\partial u}$

(c) $\frac{\partial w}{\partial u} = \frac{\partial w}{\partial x} \cdot \frac{\partial x}{\partial u} + \frac{\partial w}{\partial y} \cdot \frac{\partial y}{\partial u}$

(d) None of these

82. If $z = f(ax + by)$ then

(a) $b\frac{\partial z}{0x} + a\frac{\partial z}{0y} = 0$ (b) $b\frac{\partial z}{0x} = 0$

(c) $ab\frac{\partial z}{0y} = 0$ (d) $b\frac{\partial z}{\partial x} - a\frac{\partial z}{\partial y} = 0$

83. If $x = r\cos\theta$ and $y = r\sin\theta$ then

(a) $\left(\frac{\partial x}{\partial r}\right)_\theta = \cos\theta$ (b) $\left(\frac{\partial x}{\partial r}\right)_\theta = \sin\theta$

(c) $\left(\frac{\partial x}{\partial r}\right)_\theta = \frac{1}{\sec\theta}$ (d) Both (a) and (c)

84. If $F(x, y, z) = 0$ then

(a) $\left(\frac{\partial z}{0x}\right)_y \left(\frac{\partial x}{0y}\right)_y \left(\frac{\partial y}{\partial z}\right)_y = 0$

(b) $\left(\frac{\partial z}{\partial x}\right)_y \cdot \left(\frac{\partial x}{\partial y}\right)_z \cdot \left(\frac{\partial z}{\partial z}\right)_x = -1$

(c) $\frac{\partial z}{0x} + \frac{0z}{0x} + \frac{\partial z}{\partial z} = -1$

(d) None of these

85. $f : X \to y$ is said to be bounded above If the range f(x) in y is

(a) Bounded (b) Bounded below

(c) Bounded above (d) Zero

86. Let $f : X \to y$ is said to be bounded below if the range f(x) in y is

(a) Bounded below (b) Unbounded

(c) Bounded above (d) None of these

87. Let $f : X \to y$, $g : X \to y$ is $x, y \subseteq R$ and $f \geq 0$, $g \geq 0$ then

(a) Sup $(f + g) \leq$ Sup f + Sup g

(b) Sup (f + g) = Sup f + Sup g

(c) Sup $(f + g) \geq$ Sup f + Sup g

(d) Sup (f + g) = Sup (g) + 0

88. Let $f : X \to y$; X, Y $\subseteq$ R, and f is bounded on X then

(a) Sup $|f|$ – Inf$|f| \leq$ Sup f + Inf f

(b) Sup $|f|$ – Inf $|f| \leq$ Sup f – Inf f

(c) Sup f – Inf f = Sup $|f|$ – Inf $|f|$

(d) Sup f + Inf f = Sup $|f|$ + Inf $|f|$

89. $\lim_{x \to c} f(x)$, exist if and only if

(a) $\lim_{x \to \bar{c}} f(x) \neq \lim_{x \to c} f(x)$ (b) $\lim_{x \to \bar{c}} f - \lim_{x \to c} fx$

(c) $\lim_{x \to \bar{c}} f(x) = \lim_{x \to c} f(x)$ (d) None of these

90. $\lim_{x \to 0} \sin\frac{1}{x}$

(a) Does not exist (b) Exist

(c) 0 (d) 1

91. A function f: X → y x, y ⊆ R is continuous at point C ∈ X and ∈ > 0 and exist δ > 0 such that

(a) $|f(x) + f(c)| < \epsilon$ whenever $0 < |x - c| < \delta$

(b) $|f(x) - f(c)| > \epsilon$ whenever $|x - c| > \delta$

(c) $|f(x) - f(c)| < \epsilon$ whenever $0 < |x - c| < \delta$

(d) $|f(x) - f(c)| = 0$

92. $f(x) = \sqrt{x}$ on [0, α[is

(a) Continuous (b) Discontinuous

(c) Zero (d) −1

93. Suppose f ∈[a, b] and f(a) ≠ f(b) then for a < c < b there is number λ between f(a) and f(b) such as

(a) f(c) > λ (b) f(c) < λ

(c) f(c) = 0 (d) f(c) = λ

94. If f is continuous an strictly monotonic on an interval I then f^{-1} on f(I) is

(a) Continuous (b) Discontinuous

(c) $f^{-1} = 0$ (d) Not exist

95. Function f : A → B is (one-one) if

(a) Dom f ≠ A (b) Range f ≠ B

(c) Dom f = A (d) None

96. Function f : A → B is (In-to) if

(a) Range f ⊂ B (b) Range f = B

(c) Range f = A (d) Range f = 0

97. Function f : A → B is (on-to) if

(a) Range f ≠ A (b) Range f = A

(c) Range f ≠ B (d) Range f = B

98. Function "f" is differentiable "D" at C ∈ R iff

(a) $D_+ f(c) D_- f(c) = Df(c)$

(b) $D_+ f(c) \neq D_- f(c)$

(c) $D_+ f(c) D_- f(c)$

(d) $D_+ f(c) = D_- f(c) = 0$

99. If "f" is differentiable at c and "g" is differentiable at ∈ = f(c) then composite function ψ = gof at c is defined as

(a) $\psi'(c) = g(f(c)\, f'(c)$

(b) $\psi'(c) = g'(f(c)\, f'(c)$

(c) $\psi'(c) = g(f'(c)\, f(c))$

(d) None of these

100. If f is continuous on [a, b] and differentiable at]a, b[then there exist a point c a <<< b such that

(a) $f'(c)(b - a) = f(b) - f(a)$

(b) $f'(c)(b + a) = f(b) + f(a)$

(c) $f'(c)(b - a) = f(b) + f(a)$

(d) $f'(c)(b - a) = f(b)\, f(a)$

101. If $P = \{x_0, x_1, x_2, \ldots, x_n\} \in P(a, b)$ where "P" partition and "f" set of real valued functions bounded on [a, b] then "M_k" is defined as in kth interval

(a) $M_k = \{f(x)/ x_{k-1} < x < x_k\}$

(b) $M_k = \text{Sup } \{f(x)/a \le x \le b\}$

(c) $M_k = \text{Sup } \{f(x)/a \le x \le b\}$

(d) None of these

102. For the above statement "m_k" is defined

(a) $m_k = \text{Inf } \{f(x)/a \le x \le b\}$

(b) $m_k = \text{Inf } \{a \le x \le b\}$

(c) $m_k = \{f(x)/a \le x \le b\}$

(d) $m_k = \text{Inf } \{a \le b\}$

103. For the statement (101) "M" is defined as

(a) $M = \text{Inf } \{f(x)/a \le x \le b\}$

(b) $M = \{f(x)/x \le x \le b\}$

(c) $M = \text{Sup } \{f(x)/a \le x \le b\}$

(d) Both a and b

104. For the statement (101) "m" is defined as

(a) $m = \{f(x)/a \le x \le b\}$

(b) $m = \text{Inf } \{f(x)/a \le x \le b\}$

(c) $m = \{a \le x \le b\}$

(d) $m = \text{Inf } \{a \le x \le b\}$

105. For the statement (101) we can write as

(a) $M_k = M$ (b) $M_k \ge M$

(c) $M_k = m_k$ (d) $M_k \le M$

106. For the statement (101) we can write as
(a) $m \le m_k$ (b) $m = m_k$
(c) $m \ge m_k$ (d) None of these

107. For the statement (101) we can write as
(a) $m \le m_k \le M_k \le M$
(b) $m \le m_k \ge M_k \ge M$
(c) $m \ge m_k \ge M_k \le M$
(d) $m = m_k = M_k = M$

108. Upper Reimann Sum for $P \subseteq P$ (a, b) and function "f" is
(a) $U(f, P) = M_k \Delta x_k$
(b) $U(f, P) = \sum_{k=1}^{n} M_k \Delta x_k$
(c) $U(f, P) = M \Delta x$
(d) $U(f, P) = \sum_{k=1}^{n} M \Delta x$

109. Lower Reiman Sum for $P \subseteq P(a, b)$ an function f
(a) $L(f, P) = m\Delta x$
(b) $L(f, P) = \sum_{k=1}^{n} m\Delta x$
(c) $L(c, P) = m_k \Delta x_k$
(d) $L(f, P) = \sum_{k=1}^{n} m_x \Delta x_k$

110. If "f" and "α" be function defined in partition (a, b) then Reimann Stieltgs integral defined for lower sum
(a) $L(f, P, \alpha) = \sum_{k=1}^{n} m_k \alpha_k$
(b) $L(f, P, \alpha) = \sum_{k=1}^{n} m_k \Delta \alpha_k$
(c) $L(f, P, \alpha) = \sum_{k=1}^{n} m \Delta\alpha_k$
(d) $L(f, P, \alpha) = m\alpha(b - a)$

111. For statement (110) upper sum for Reimann Stlgs integral is defined as
(a) $U(f, P, \alpha) = \sum_{k=1}^{n} M_k \Delta\alpha_k$
(b) $U(f, P, \alpha) = \sum_{k=1}^{n} M \Delta\alpha_x$
(c) $U(f, P, \alpha) = \sum_{k=1}^{n} M_k$
(d) None of these

112. If "f" and "α" functions defined i partition P_1 and P_2 of [a, b] with $P_1 \subset P$ then we can write as
(a) $U(f, P_1, \alpha) \le U(f, P_2, \alpha)$
(b) $U(f, P_1, \alpha) = U(f, P_2, \alpha)$
(c) $U(f, P_1, \alpha) < L(f, P_2, \alpha)$
(d) $U(f, P_1, \alpha) \ge U(f, P_2, \alpha)$

113. For the statement (112) we can write f lower sum as
(a) $L(P_2, f, \alpha) \ge L(P_1, f, \alpha)$
(b) $L(P_2, f, \alpha) = L(P_1, f, \alpha)$
(c) $L(P_1, f, \alpha) > U(P_2, f, \alpha)$
(d) None of these

114. If f is bounded and integrable on [a, then there exist a number λ between t bounds of f such that
(a) $\int_a^b fdn = \lambda$
(b) $\int_a^b fdn = \int_a^b (b + a)$
(c) $\int_a^b fdn = \lambda (b - a)$
(d) $\int_a^b fdn < \lambda$

115. If f is continuous and integrable on [a, then there $c \in [a, b]$ such that
(a) $\int_a^b fdx = (b - a) f(c)$
(b) $\int_a^b fdx = (b + a) f(c)$
(c) $\int_a^b fdx = - fdx$
(d) $\int_a^b fdx = 0$

116. Function "f" is said to be Reimann Stieltjes integral if

(a) $\int_a^b fdx \leq \int_a^b fdx$ (b) $\int_a^b fdx \geq \int_a^b fdx$

(c) $\int_a^b fdx = \int_{\bar{a}}^b fdx$ (d) Both (a) and (b)

117. For Reiman integrable $\int_{\bar{a}}^b fdx =$

(a) Sup L(P, f, α) (b) Inf U(P, f, α)
(c) Σ L(P, f, α) (d) Σ U(P, f, α)

118. $\int_a^b$ fdx for Reiman Stieltjes integral

(a) U(P, f, α) (b) Inf U(P, f, α)
(c) L(P, f, α) (d) Sup L(P, f, α)

119. A function "f" in Cauchy criterion is Remiann integrable if

(a) $\int_a^b fdx = \int_{\bar{a}}^b fdx$

(b) $\int_a^b fdx \geq \int_a^b fdx$

(c) $\int_a^b fdx = \int_a^b fdx = \int_{\bar{a}}^b fdx$

(d) None of these

120. Cauchy criterion for Reimann Stieltjes integral is

(a) U(P, f, α) + L(P, f, α) < ϵ
(b) U(p, f, α) – L(P, f, α) > ϵ
(c) U(P, f, α) = P(P, f, α)
(d) U(P, f, α) – L(P, f, α) < ϵ

121. Let B $\in$ R (real) and f $\in$ RS_α on [a, b] and Bf belongs to Reiman Stieltjes integral then we can write as

(a) $\int_a^b \beta f(x)\, d\alpha(x) = \beta \int_a^b f(x)\, d\alpha(x)$

(b) $\int_a^b \beta f(x)\, d\alpha x \neq \int_a^b f(x)\, d\alpha(x)$

(c) $\int_{\bar{a}}^b f(x)\, d\alpha(x) - \int_a^b f(x)\, d\alpha(x) = 0$

(d) None of these

122. Let f be Reimann integrable on [a, b] f $\geq$ 0 then

(a) $\int_a^b f d\alpha = 0$ (b) $\int_a^b f d\alpha \geq 0$

(c) $\int_a^b f d\alpha \leq 0$ (d) $\int_a^b f d\alpha \neq 0$

123. If f $\in$ RS_α (Reimann Stieltjes) on [a, b] and |f| $\in$ RS_α

(a) $\left| \int_a^b f(x)\, dx \right| \leq \int_a^b \left| f(x) \right| dx$

(b) $\left| \int_a^b fx\, d\alpha(x) \right| = \int_a^b |fx|\, d\alpha(x)$

(c) $\left| \int_a^b f(x)\, d\alpha(x) \right| \int_a^b f(x)\, d\alpha(x)$

(d) None of these

124. When "f" and "g" are Reimann's Stieltjes integrable

(a) $fg = \frac{1}{4}[(f+g)^2 - (f-g)^2]$

(b) $fg = \frac{1}{4}[(f+g)^2 + (f-g)^2]$

(c) $fg = \frac{1}{4}[(f+g)^2]$

(d) $fg = \frac{1}{4}[(g-g^2)^2]$

125. Let f $\in$ RS_α on [a, b] and C any point such that a < c < b and f $\in$ RS_α on [a, b] and [c, b] then

(a) $\int_a^b f d\alpha = \int_a^c f d\alpha + \int_c^b f d\alpha$

(b) $\int_a^b f d\alpha = \int_a^c f d\alpha - \int_c^b f d\alpha$

(c) $\int_a^b f d\alpha = \int_a^c f\, d\alpha$

(d) None of these

126. If "f" is Monotomic on [a, b] then in [a, b] it is

(a) Differentiable (b) Non integrable
(c) Integrable (d) Both (a) and (c)

127. If "f" is Reimann integrable on [a, b] and $F(x) \int_a^x f(t)\,dt$ then in [a, b]

(a) f is continuous (b) Discontinuous
(c) Bounded (d) None of these

128. If f is Reiman integrable continuous and $F(x) \int_0^x f(t)\,dt$ then

(a) $F(x) = f(x)$ (b) $F'(x) = f(x)$
(c) $F(x) \neq f(x)$ (d) Zero

129. Suppose G be a primitive of a continuous function f on [a, b] then function F also primitive for constant c if

(a) $F(x) = G'(x) + c$ (b) $F'(x) = G(x)$
(c) $F(x) = G(x) + c$ (d) None of these

130. If "f" is Reimann integrable and "F" is primitive then

(a) $f'(x) = \int_a^b F(x)\,dx$

(b) $f(x) = \int_a^b F'(x)\,dx$

(c) $f'(x) = F'(x)$

(d) $\int_a^b f(x)\,dx = F(b) - F(a)$

131. Suppose that "u" has a continuous derivative as [c, d] and f continuous on Range u. Let a = u(c), b = u(d) then

(a) $\int_a^b f(x)\,dx = \int_c^d f(u)\,tu'(t)\,dt$

(b) $f(x)\,dx = f(u)\,tu'(t)dt$

(c) $\int_a^b f(x)\,dx = \int_c^d u'(t)\,dt$

(d) None of these

132. If $f \in R(\alpha)$ on [a, b] and $|f(x)| \leq M$ on [a, b] then

(a) $\int_a^b f d\alpha = M(\alpha(b) - \alpha(a))$

(b) $\int f d\alpha = M(\alpha(b) - \alpha(a)$

(c) $\left| \int_a^b f d\alpha \right| \leq M[\alpha(b) - \alpha(a)]$

(d) Both (b) and (c)

133. If $f \in IR(\alpha)$ and C is +ve constant and $f \in IR\,(c\alpha)$ then

(a) $\int_a^b f d(c\alpha) = c\int_a^b f d\alpha$

(b) $\int_a^b f d\alpha(c\alpha) = c\int f d\alpha$

(c) $\int_a^b f d(c\alpha) \neq c\int_a^b f d\alpha$

(d) $\int_a^b f d(c\alpha) = 0$

134. If $f \in IR(\alpha_1)$ and $f \in IR(\alpha_2)$ then

(a) $f \in IR(\alpha_1\alpha_2)$ (b) $f \in IR(\alpha_1 + \alpha_2)$

(c) $f \in IR\dfrac{\alpha_1}{\alpha_2}$ (d) None of these

135. For above statement (134) we can write as

(a) $\int_a^b f d(\alpha_1 + \alpha_2) = \int_a^b f d\alpha_1 + \int_a^b f d\alpha_2$

(b) $\int_a^b f d(\alpha_1 + \alpha_2) = \int f d\alpha_1\alpha_2$

(c) $\int_a^b f d(\alpha_1 + \alpha_2) + \int_a^b f d\alpha_1 + \int_a^b f d\alpha_2$

(d) Both (a) and (b)

136. If $\int_a^\alpha f(x)\,dx = \lim_{\lambda \to \alpha} \int_a^\lambda f\,dx$ exist then

(a) Divergent (b) Bounded
(c) Convergent (d) None of these

137. If $\int_a^\alpha f(x)\,dx = \lim_{\lambda \to \alpha} \int_a^\lambda f(x)\,dx$ does not exist then

(a) Convergent (b) Bounded
(c) None of these (d) Divergent

138. If f is bounded on [a, λ] for all λ the $\int_a^\alpha f(x)$ exist. If for a < k there exist as $\lambda'' > \lambda' > k > a$ as

(a) $\left| \int_{\lambda'}^{\lambda''} f(x)\,dx \right| < \epsilon$

(b) 0

(c) $\left| \int_{\lambda'}^{\lambda''} f(x)\,dx \right| \geq 0$

(d) Both (a) and (c)

139. Suppose "f" and "g" are Reimann integrable and $\int_a^\alpha g\,dx$ Conv, so does $\int_a^\alpha f\,dx$ Conv, then

(a) $0 \geq f(x) \geq g(x)$ (b) $0 < f(x) \leq g(x)$

(c) None of these (d) Both (a) and (b)

140. If "f" and "g" are Reimann integrable and $0 < c\, f(x) = 8$ and $\int_a^\alpha f(x)$. Diverges then

(a) $\int_a^\alpha g(x)$ convergent

(b) $\int_a^\alpha g(x) = 0$

(c) $\int_a^\alpha g(x)$ divergent

(d) None of these

141. Let $f \geq 0$ on $[a, \alpha[$ and $a > 0$ and $\lim_{\lambda \to \alpha} x^\alpha f(x) = l$ and $\int_a^\infty f(x)\,dx$ converges if

(a) $\alpha = 0$ (b) $\alpha < 0$

(c) $\alpha = 1$ (d) $\alpha \geq 1$

142. Given $\int_a^\alpha f(x)\,dx$ is unbounded then

(a) Diverges (b) Converges

(c) Zero (d) None of these

143. Let "g" bounded on $[a, \alpha[$ such that $\lim_{x \to \alpha} g(x) = 0$ and $\int_a^\lambda f(x)\,dx$ is bounded $\forall \lambda > a$, then $\int_a^\alpha f(x)\,g(x)$ is

(a) Divergent (b) Convergent

(c) Zero (d) None of these

144. $\int_a^\alpha f(x)\,dx$ is said to be absolutely convergent, if

(a) $\int_a^\alpha |f(x)|\,dx$ divergent

(b) $\int_a^\alpha |f(x)|\,dx = 0$

(c) $\int_a^\alpha f(x)\,dx$ convergent

(d) Both b and c

145. f is discontinuous at "a" and in Reimann integrable over $[a + \epsilon, b]\ \forall\ \epsilon > 0$ if $\lim_{x \to a} (x - a)^\alpha f(x) = l$, then $\int_a^b f(x)\,dx$ is convergent if

(a) $0 < \alpha < 1$ (b) $\alpha = 1$

(c) $0 > \alpha > 1$ (d) None of these

146. For above statement (145) $\int_a^b f(x)\,dx$ divergent

(a) $0 < \alpha < 1$ (b) $\alpha \geq 1$

(c) $\alpha = 0$ (d) Both b and c

147. $\int_0^a x^{\alpha-1} e^{-x}\,dx$ converges if

(a) $\alpha < 0$ (b) $\alpha = 0$

(c) $\alpha > 0$ (d) $\alpha = -1$

148. $V_a^b(f)$ be total variation of "f" on [a, b] and is finite then "f" is a function of

(a) Bounded variation

(b) Not bounded variation

(c) total bounded

(d) Infinite variations

149. If "f" is real valued and monotonic on [a, b] then f

(a) Bounded variation

(b) Not bounded variation

(c) Zero

(d) Discontinuous

150. If $f(x) = \sin x$ is of bounded variation then it is monotonic on

(a) $\left[\frac{\pi}{2}, \pi\right]$ (b) $[\pi, -\pi]$

(c) $\left[0, \frac{\pi}{2}\right]$ (d) Both (a) and (c)

151. If "f" is function of bounded variation in [a, b] then

(a) $\sum_{k=1}^{n} |Df| \neq f(b) - f(a)$

(b) $\sum_{k=1}^{n} |Df| = f(b)$

(c) $\sum_{k=1}^{n} |Df| = f(a)$

(d) $\sum_{k=1}^{n} |Df| = f(b) - f(a)$

152. If "f" is of bounded variation on [a, b] then on [a, b]

(a) Not bounded (b) Bounded

(c) Zero (d) Discontinuous

153. $\int_0^1 x^{m-1}(1-x)^{n-1}\,dx$ converges for "m > 0" and "n > 0"

(a) $-m > 0$ and $n > 0$

(b) $m > 0,\ n \geq 1$

(c) $m > 0$ and $n > 0$

(d) $m = 1,\ n = -1$

154. $\int_0^1 x^{m-1}(1-x)^{n-1}\,dx$ denoted by $\beta(m, n)$ known as

(a) Beta function (b) Gamma function

(c) Conservative (d) Both (a) and (c)

155. $\int_0^{\infty} e^{-x^2/2}\,dx =$

(a) $\sqrt{\frac{\pi}{2}}$ (b) $\frac{\pi}{2}$

(c) $= 0$ (d) π

156. If "f" and "g" are of bounded variation on [a, b] then

(a) $V_a^b(f+g) \geq V_a^b(f) + V_a^b(g)$

(b) $V_a^b(f-g) \leq V_a^b(f) - V_a^b(g)$

(c) $V_a^b(f+g) = 0$

(d) $V_a^b(f+g) \leq V_a^b(f) + V_a^b(g)$

157. If "f" and "g" are of bounded variation on [a, b] then for $\lambda \in R$ we can write as

(a) $V_a^b(\lambda f) \leq |\lambda|\ V_a^b f$

(b) $V_a^b(\lambda f) = 0$

(c) $V_a^b(\lambda f) \neq 0$

(d) $V_a^b(\lambda f) \geq |\lambda|\ V_a^b f$

158. If "f" and "g" are of bounded variation on [a, b] and $M_1 = \text{Sup}\{f(x): a \leq x \leq b\}$ and $M_2 = \text{Sup}\{g(x): a \leq x \leq b\}$

(a) $V_a^b(fg) \leq M_1 V_a^b(f)$

(b) $V_a^b f \cdot g \leq M_1 V_a^b(f) + M_2 V_a^b(f)$

(c) $V_a^b(fg) = M_1 V_a^b(f) + M_2 V_a^b(g)$

(d) $V_a^b(fg)\ V_a^b(g)$

159. Let "f" continuous [a, b] and of bounded variation on then [a, b]

(a) $f' = 0$ (b) $f' \leq 0$

(c) $f' \geq 0$ (d) f' exist

160. Let "f" and "g" be of bounded variation [a, b] then f/g is of bounded variation on [a, b] then

(a) f is continuous (b) g is continuous

(c) Both (a) and (b) (d) None of these

161. let a < c < b then "f" be of bounded variation and a < c < b then

(a) "f" is of bounded variation on [a, c]

(b) f is of bound variation on [c, b]

(c) Both (a) and (b)

(d) None of these

162. If "f" is of bounded variation on [a, c] and on [c, b] then

(a) $V_a^b(f) = V_a^c(f) + V_c^b(f)$

(b) $V_a^b(f) < V_a^c(f) + V_c^b(f)$

(c) $V_a^b(f) = V_a^c(f)$

(d) $V_a^b(f) \geq V_a^c(f) + V_c^b(f)$

163. A function "f" is of bounded variation on [a, b] if and only if "f" can be expressed as difference of two

(a) Increasing function

(b) Decreasing function

(c) None of these

(d) Both (a) and (b)

164. The function "f" is known as the pointwise limit function for sequence $\{f_n\}$ on X. We write

(a) $f(x) = \lim_{n\to\infty} f_n(x)$ (b) $f(x) = \lim_{n\to 0} f_n(x)$

(c) $f(x) \neq f_n(x)$ (d) None of these

165. Let "f" be a real valued function defined on set X, we write

(a) $||f|| = \{f(x): x \in X\}$

(b) $||f|| = \{|f(x)|: x \in X\}$

(c) $||f|| = \text{Sup}\{|f(x)|: x \in X\}$

(d) $||f|| \leq f(x): x \in X$

166. Sequence of real value function $\{f_n\}$ said to be converges uniformly to a function "f" on X. If

(a) $\lim_{n\to\infty} ||f_n - f|| = 0$

(b) $\lim_{n\to 0} ||f_n - f|| = 0$

(c) $f_n = f$

(d) $||f_n|| \to 0$ as $n \to \infty$

167. The series ΣU_n of continuous real valued functions converges uniformly on set X to "f" then on X

(a) f is discontinuous

(b) Constant

(c) Continuous

(d) $f = 0$

168. A sequence $\{f_n\}$ real valued function defined on X said to satisfy Cauchy criterion on "X" If for every positive real number ϵ and n_0, there is $m, n > n_0$

(a) $||f_m - f_n|| = \epsilon$ (b) $||f_m - f_n|| > 0$

(c) $||f_m - f_n|| = 1$ (d) $||f_m - f_n|| < \epsilon$

169. If a sequence $\{f_n\}$ real valued satisfied the Cauchy criterion uniformly on "X" it is

(a) Convergent (b) Divergent

(c) Bounded (d) Continuous

170. Series ΣU_n of real valued converges uniformly if for $\epsilon > 0$ and $m, n > n_0$ there is

(a) $\left|\left| \sum_{i=m+1}^{n} a_i \right|\right|$ (b) $\left|\left| \sum_{i=m+1}^{n} a_i \right|\right| < \epsilon$

(c) $\left|\left| \sum_{i=m+1}^{n} a_n \right|\right| = 0$ (d) $\sum_{i=m+1}^{n} a_i = 0$

171. Let series $\Sigma(x\, e^{-x})^k$ on [0, 1] is

(a) Diverges

(b) Converges

(c) Uniformly converges

(d) None of these

172. Let $\{U_n\}$ be a sequence of real valued function can defined on X. Such that there is real number $\lambda > 0$ and $m >> 0$

(a) $\left|\left| \sum_{i=m}^{n} U_i \right|\right| < 0$

(b) $\left|\left| \sum_{i=m}^{n} U_i \right|\right| < \lambda$

(c) $\sum_{i=m}^{n} U_i = 0$

(d) $\left|\left| \sum_{i=m}^{n} U_i \right|\right| \geq 0$

173. $\{U_n\}$ does not converge uniformly to U on [a, b], if

(a) U not integrable (b) U integrable

(c) $U = 0$ (d) U does not exist

174. Let $\sum_{n=1}^{\infty} U_n$ be series of function and U_n' exists on [a, b] then

(a) U_n' is continuous (b) U_n continuous

(c) $U_n = 0$ (d) $U_n' = 0$

175. Let $\sum_{n=1}^{\infty} U_n$ be series of functions and U_n' is continuous on [a, b] then

(a) $\Sigma U_n'$ diverges uniformly

(b) $\Sigma U_n'$ diverges

(c) $\Sigma U_n' = 0$

(d) $\Sigma U_n'$ converges uniformly on [a, b]

176. If $\{f_n\}$ be a sequence of real valued functions differentiable on [a, b] and $f_n'(x)$ converges uniformly then converges uniformly to f such that

(a) $f'(x) = \lim_{n\to\infty} f_n'(x)$

(b) $\underset{n\to\alpha}{\text{lt}}\ f_n'(x) = 0$

(c) $\underset{n\to\alpha}{\text{lt}}\ f_n'(x) = f(x)$

(d) $f'(x) = f_{(n)}'(x)$

177. $\int_0^\alpha e^{-xt}\frac{\sin t}{t}\,dt$ converges uniformly for all

(a) $x = 0$
(b) $x < 0$
(c) $x < 0$
(d) $x = +1$

178. Let "f" be Reimann integrable on $[a, \lambda]$, $a \le x < \alpha$ and $F_\lambda(x) = \int_a^\lambda f(x, t)\,dt$ then $\int_a^\lambda f(x, t)\,dt$ converges uniformly

(a) F_λ diverges
(b) F_λ converges uniformly
(c) $F_\lambda = 0$
(d) None of these

ANSWERS

1. c	2. b	3. b	4. c	5. a
6. b	7. a	8. c	9. a	10. a
11. a	12. d	13. a	14. b	15. a
16. b	17. d	18. a	19. b	20. a
21. b	22. a	23. b	24. a	25. b
26. c	27. c	28. a	29. d	30. a
31. b	32. d	33. b	34. a	35. d
36. a	37. b	38. d	39. a	40. b
41. a	42. b	43. d	44. d	45. a
46. d	47. b	48. a	49. b	50. a
51. c	52. a	53. d	54. a	55. b
56. c	57. d	58. a	59. b	60. d
61. a	62. b	63. d	64. a	65. c
66. a	67. c	68. d	69. a	70. c
71. d	72. a	72. b	73. a	74. b
75. a	76. a	77. b	78. a	79. c
80. a	81. c	82. d	83. d	84. b
85. c	86. a	87. a	88. b	89. c
90. a	91. c	92. a	93. d	94. a
95. c	96. a	97. d	98. a	99. b
100. a	101. b	102. a	103. c	104. b
105. d	106. a	107. a	108. b	109. d
110. b	111. a	112. d	113. a	114. c
115. a	116. c	117. a	118. b	119. c
120. d	121. a	122. b	123. c	124. a
125. a	126. d	127. a	128. b	129. c
130. d	131. a	132. c	133. a	134. b
135. a	136. c	137. d	138. a	139. b
140. c	141. d	142. a	143. b	144. c
145. a	146. b	147. c	148. a	149. a
150. c	151. b	152. b	153. c	154. a
155. a	156. d	157. a	158. b	159. d
160. c	161. c	162. a	163. a	164. a
165. c	166. a	167. c	168. d	169. a
170. b	171. c	172. b	173. a	174. a
175. d	176. a	177. c	178. b	

ABSTRACT ALGEBRA

➤ Tick (✓) the correct one answer only.

1. If A and B two sets then for "$a \in A$" and "$b \in B$" Binary relation $(A \times B)$ is defined as for real R
   (a) $(a, b) \in R$ (b) $(a, b) \notin R$
   (c) $a = b$ (d) Both (b) and (c)

2. If $A \times B = \{(1, 2), (1, 4), (3, 2), (3, 4)\}$ then set "A" and B
   (a) $\{1, 2\} = A$, $B = \{2, 4\}$
   (b) $A = \{0, 1\}$, $B = \{2, 4\}$
   (c) $\{1, 3\} = A$, $B = \{2, 4\}$
   (d) $A = \{1, 2\}$ $B = \{3, 4\}$

3. For all "$a \in A$" R is reflexive relation if and only if
   (a) $(a, a) \in R$ (b) $a = a$
   (c) $(a, a) \notin R$ (d) $a \in R$

4. A relation R on A is symmetric if "$a, b \in A$" and $(a, b) \in R$
   (a) $a = b$ (b) $a \neq b$
   (c) $A = ?$ (d) $(b, a) \in R$

5. A relation on "R" said to transitive if $(a, b) \in R$, $(b, c) R$
   (a) $(a, c) \in R$ (b) $a = c$
   (c) $(a, c) \notin R$ (d) $a \in R$

6. A relation R is antisymetric if for $(a, b) \in R$ and $(b, a) \in R$ implies as
   (a) $a \neq b$ (b) $a < b$
   (c) $a = b$ (d) $a = b^{-1}$

7. If relation R is reflexive symmetric and transitive then relation is called
   (a) Equivalence (b) Antisymmetric
   (c) $R = R^{-1}$ (d) None of these

Two function ϕ: A B and Ψ: $A \to B$ said to be equal if
   (a) $\phi(a) \neq \Psi(a)$ (b) $\phi(a) = \Psi(a)$
   (c) $\Psi(a) = 0$ (d) $\Psi(a) = 0$

9. A mapping $I_A : A \to A$ is the identity mapping function if
   (a) $I_A = \{(a, a); a \in A\}$
   (b) $I_A = \{(a, b); a, b \in A\}$
   (c) $I_A = \{a \in A\}$
   (d) $I_A = \{(a, b); a \in A\}$

10. For fixed $C \in R$, $\phi_c = \{(x, c): x \in R\}$ called the
   (a) Identity function
   (b) 1 – function
   (c) Constant function
   (d) None of these

11. For some $a \in A$, $b \in B$ if $(a, b) \in \Psi$ and $b = \Psi(a)$ then
   (a) Ψ is 1 – 1 (b) Ψ is on to
   (c) Ψ is constant (d) None of these

12. A function $f: X \to Y$ called surjective if
   (a) $R_f = X$ (b) R_f = constant
   (c) $R_f = R$ (d) $R_f = Y$

13. Let $\Psi \to A \to B$ be a function and for $a \in A$, $b \in B$ $\Psi(a) \neq \Psi(b)$ for $a \neq b$ function is called
   (a) One-one (b) On-to
   (c) Identity (d) Constant

14. Function $f: X \to Y$ called injective if for $x_1, x_2 \in X$ and $x_1 = x_2$ there is
   (a) $f(x_1) = x_2$ (b) $f(x_1) = 0$
   (c) $f(x_1) = f(x_2)$ (d) Both (a) and (c

15. A function Ψ: $A \to B$ which is injective well as surjective is called function
   (a) Bijective (b) Constant
   (c) Identity (d) None of these

51. Any two cyclic group of same order are
(a) Isomorphic (b) Monomorphic
(c) Homomorphic (d) None of these

52. Every subgroup of a cyclic group is
(a) Non cyclic (b) Abelian
(c) Cyclic (d) Both b and c

53. A homomorphic image of cyclic group is
(a) Abelian (b) Isomorphic
(c) Non cyclic (d) Cyclic

54. Let $C = < a, a^n = e >$ be a cyclic group then a^k is generator of C iff k and n are relative
(a) Prime (b) Even
(c) Odd (d) Both a and c

55. For complexes (i.e. set) X, Y of a group "G" X and Y are permutable if
(a) $X = Y$ (b) $XY \neq YX$
(c) $XY = YX$ (d) $X \neq Y$

56. If H is subgroup of G and $a \in G$ the complex $aH = \{ah : h \in H\}$ called
(a) Right coset (b) Coset
(c) Left coset (d) None of these

57. G be a finite group of order "n" and H a subgroup of G of order m then
(a) m/n (b) $m = n$
(c) $m \neq n$ (d) $m = n = 0$

58. G be a finite group of order "n" and "k" be the index of sub group H of G then
(a) $K = n$ (b) k/n
(c) $k \neq n$ (d) Both (a) and (b)

9. Every group whose order is a prime number is necessary
(a) Abelian (b) Non-abelian
(c) Non cyclic (d) Cyclic

Let H and k be subgroups of G then HK is a subgroup of G iff
(a) $HK = KH$ (b) $H = K$
(c) H/K (d) $HK \neq KH$

Let X be a subset of a group G, then the set $\{a \in G : aX = Xa\}$ is said to be
(a) Cyclic (b) Noncyclic
(c) Normalizer (d) Equal

62. For any subset X of G then normalizer subgroup of G is represented by
(a) N(X) (b) $N_G(X)$
(c) $N = X$ (d) $N = G = X$

63. Let H and G be group and "$H \subseteq G$" then
(a) $N_G(X) \supseteq H$ (b) $hH = H = Hh$
(c) $N_G(X) = H$ (d) None of these

64. Let "X" subset of "G" then centralizer denoted by
(a) $N_G(X)$ (b) $G(X) = X$
(c) $C_G(X)$ (d) $C_G(x)$

65. For any subset "X" of group G then $\forall x \in X$ and $a \in G$
(a) $C(X) = \{a \in G : a_x = xa\}$
(b) $C_{(G)}X = \{a \in G : ax = xa\ \forall x \in X\}$
(c) $C(X) = C_G(x)$
(d) $C(X) \neq C_G(X)$

66. For any subset "X" of a group G $C_G(X) = \{a \in G : ax = xa, \forall x \in X$ then $C_G(X)$ is
(a) Power set of G (b) Superset of G
(c) Subgroup of G (d) Both (a) and (b)

67. The set of those elements of "G" which commute with every element of G is called
(a) Norm of G (b) Home of G
(c) Both a and b (d) Center of G

68. G be a group and "$a, \in G$" then the element $x\ a\ x^{-1}$ is called the
(a) Conjugate element of G
(b) Norm of G
(c) Center of G
(d) None of these

69. For a, b $\in$ G and for some "$x \in G$" i called conjugate of a if
(a) $b = ax$ (b) $b = a^{-1}x$
(c) $b = x\,ax^{-1}$ (d) $b = ax^{-1}$

70. Let "G" be group and "$a \in G$" then th number of elements in conjugacy cla "C_a" and the index of normalizer of "a" G are

90. Every subgroup of abelian group is group
(a) Normal (b) Center
(c) Equivalence (d) Both (a) and (c)

91. Let Z be the group of integers under addition H = <0, ± n, ± 2n, ... > then "H" in "Z" is
(a) Equivalent (b) Normal
(c) Center (d) Conjugate

92. Every group G has at least number of normal subgroups
(a) 1 (b) 3
(c) 2 (d) 4

93. Every group of order P(Prime) is
(a) Mixed group (b) Normal group
(c) Abelian group (d) Simple group

94. If $\phi: G \to G$ be Homomorphism and let $K = \{K \in G: \phi(K) = e', e' \in G'\}$ is called
(a) Kernal of ϕ (b) Normal of ϕ
(c) Center (d) None of these

95. $\phi: G \to G'$ be a surjective Homomorphism then normal subgroup of G is
(a) Factor group (b) Kernal ϕ
(c) Normal (d) Both (b) and (c)

96. Let $\phi: G \to G'$ be surjective Homomorphism then isomorphic to G' is
(a) Ker ϕ (b) Normal
(c) Factor group G/K
(d) None of these

97. Let $\phi: G \to G'$ and Ker $\phi = \{g \in G: \phi(g) = e\}$ is
(a) Normalin G (b) Abelian
(c) Conjugate (d) Center in G

98. Let $\phi: G \to G'$ with Ker $\phi = \{g \in G, \phi(\delta) = e\}$ is normal in G then
(a) G/Kr ϕ = G (b) $\frac{G}{\text{Ker}\,\phi} \cong G'$
(c) $\frac{G}{\text{Ker}\,\phi} = G'$ (d) None of these

99. Let A and B subgroups of a group G. Such that A is normal G, then normal subgroup of B is
(a) $A \cup B$ (b) $A - B$
(c) $A \cap B$ (d) $A' \cup B'$

100. If A, B be the subgroups of group G, such that A is normal in G then
(a) $A \cong B/A \cap B$ (b) $B \cong \frac{A}{A \cap B}$
(c) $\frac{B}{A} \cong \frac{AB}{B}$ (d) $\frac{AB}{A} \cong \frac{B}{A \cap B}$

101. Let A and B be normal subgroup of group such that $A \subseteq B$ then
(a) ${}^{G/A}/_{B/A} \cong G/B$ (b) ${}^{G/A}/_{B/A} = G/B$
(c) ${}^{G/A}/_{B/A} \cong \frac{B}{A}$ (d) $\frac{A}{B} \cong \frac{B}{A}$

102. A group G is abelian if and only iff G/c(G) is
(a) Normal (b) Cyclic
(c) None cyclic (d) None of these

103. Every group of order P^2 where P is prime number
(a) Cyclic (b) Normal
(c) Abelian (d) Both (a) and (c)

104. Every group of order "m" where "m" $m \leq 5$ is
(a) Abelian (b) Normal
(c) Conjugate (d) Cyclic

105. Every group in which each non-identity element is of order 2 is
(a) Normal (b) Cyclic
(c) Abelian (d) Conjugate

106. The set A(G) of all automorphism of group G is
(a) Group (b) Not group
(c) Normal group (d) None of these

107. Let (G, +) be the group of integers under addition for each $k \in z$, the mapping $\phi_k: z \to z$ defined by $\phi_k(n) = k(n)\ \forall\ n \in z$ is
(a) Auto morphism
(b) Sendomorphism
(c) Homomorphism
(d) None of these

108. For $a \in$ G(group) the mapping $I_a: G \to G\ \forall\ x \in G$ an automorphism is defined as

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(a) $I_a(x) = a x^{-1}$ (b) $I_a(x) = x a x^{-1}$

(c) $I_a(x) = a x a^{-1}$ (d) None of these

109. For each $x \in \gamma(G)$ is groups we can write

(a) $I_x = I_e$ (b) $I_x \neq I_e$

(c) $I_x I_y = I_{xy}$ (d) Both (a) and (c)

110. If $a, b \in G$ (group) then commutative of a and b is denoted by

(a) (a, b) (b) $[a, b]$

(c) $\left[\frac{a}{b}\right]$ (d) None of these

111. If $a, b \in G$ (group) then the element of commutator of a and b is given by

(a) $ab\,b^{-1}$ (b) $a^{-1}bb$

(c) $ab\,a^{-1}b^{-1}$ (d) $ab \subset e$

112. If I(G) be the group of inner automorphism of G and let $\xi(G)$ be the center of G, then

(a) $G \cong I(G)$ (b) $I(G) \cong \xi(G)$

(c) $I(G) \cong G$ (d) $G/\xi(G) \cong I(G)$

113. Sub group G generated by all commutators $[a, b]$, $a, b \in G$ is called the

(a) Commutator subgroup

(b) Normal subgroup

(c) Abelian (d) Both a and b

114. Commutator subgroup in G is

(a) Conjugate (b) Normal

(c) Abelian (d) $\xi(G)$

115. If G'(commutator subgroup) of G then G/G'

(a) Normal (b) Conjugate

(c) $\xi(G)$ (d) Abelian

116. If K is normal subgroup of G such that G/K is abelian that

(a) $K \supseteq G$ (b) $K \subseteq G'$

(c) $K \supseteq G'$ (d) $K = G$

117. For group $H \subseteq G$ is characteristic if for each automorphism α of G

(a) $\alpha(H) = H$ (b) $\alpha(H) = e$

(c) $\alpha(H) = H$ (d) None of these

118. For any G the center $\xi(G)$ of G is subgroup as

(a) Abelian (b) Characteristic

(c) Normal (d) Conjugacy

119. If F be subgroup of 'G' and F is fully invariant endomorphism ϕ to G, $\phi: G \to G$ then

(a) $\phi(F) \subseteq F$ (b) $\phi(F) = F$

(c) $\frac{\phi(F)}{F} = \phi$ (d) None of these

120. Let G be group and $a, b \in G$ such that ab = ba and $(O(a) \cdot O(b)) = 1$ then order O will be as

(a) $O(ab) \neq O(a)\,O(b)$

(b) $O(ab) = O(a) \cdot O(b)$

(c) $O(ab) = O(a)/O(b)$

(d) None of these

121. Non-empty subset H of a group G is normal subgroup of G if $\forall x, y \in H$ and $g \in G$, we have

(a) $(gx)(gy) \in H$ (b) $g(x)(gy)^{-1} \notin H$

(c) $g(x), g(y) \in H$ (d) $g(x)(gy)^{-1} \in H$

122. If N is normal subgroup of G, then order 0

(a) $0\left(\frac{G}{N}\right) = \frac{0(G)}{0(N)}$ (b) $0\left(\frac{G}{N}\right) \neq \frac{0(G)}{0(N)}$

(c) $0(G) = 0(N)$ (d) None of these

123. Let <z, +> and <E, +> be groups of integers and even integers with mapping $f: z \to E$ S.t $f(x) = 2x \ \forall x \in z$ then f is

(a) Monomorphism (b) Epimorphism

(c) Homomorphism (d) None of these

124. If H and K normal subgroup of G and $H \subseteq K$ then $\frac{K}{H}$ and $\frac{G}{H}$ are

(a) Normal (b) Abelian

(c) Commute (d) Centralize

125. For any group G and identity e we can write as

(a) $\frac{G}{\{e\}} \cong G$ (b) $\frac{G}{G} \cong e$

(c) Both (a) and (b)

(d) $G \cong \{e\}$

126. If $f: C \to C$ (Complex) such that $f(z) = \bar{z}$ then f is
(a) Automorphism (b) Epimorphism
(c) Homorphism (d) None of these

127. $f: G \to G$ s.t. $f(x) = x^{-1}$ is automorphism iff G is
(a) Normal (b) Conjugate
(c) Abelian (d) Centralizer

128. If $G_1 \cong G_2$ then
(a) $I(G_1) \cong I(G_2)$
(b) $Auto(G_1) \cong Auto(G_2)$
(c) Both (a) and (b)
(d) None of these

129. Automorphism group of finite group is
(a) Finite (b) Infinite
(c) Normal (d) Abelian

130. Automorphism and inner automorphism of G is
(a) Abelian (b) Normal
(c) Conjugate (d) None of these

131. Every group order P^6 where P is prime number is ———
(a) Cylcic (b) Ibelian
(c) Normal (d) Congugate

132. Let P be prime and a, b, $\in$ G and P divides $0(G)$ then
(a) $(ab)^P \neq a^P b^P$ (b) $(ab)^P = ab$
(c) $(ab)P = a^P b^P$ (d) $ab = a^P b$

133. Group obtained by the direct product of sy lows p-group is
(a) Normal (b) Abelian
(c) Center (d) Commutator

34. Let $0(G) = Pq$, where P, q are distinct primes $p < q$, $P \times q - 1$ then G is
(a) Abelian (b) Normal
(c) Cyclic (d) None of these

5. G be finite group and prime "P" such that $P/0(G)$ then there exist $x \in G$
(a) $0(x) = P$ (b) $0(x) \neq P$
(c) $0(x) = e$ (d) $0(x) = 0(G)$

136. Let "G" be any group and $G^n = \langle x^n : x \in G \rangle$ then G^n is
(a) Fully invariant (b) Conjugate
(c) Normal (d) Abelian

137. Every group has at least characteristics subgroups
(a) 2 (b) 1
(c) 3 (d) 4

138. Every fully invariant subgroup is
(a) Abelian (b) Normal
(c) Characteristics (d) Conjugate

139. Every characteristics subgroup of a group is
(a) Abelian (b) Conjugate
(c) Characteristic (d) Normal

140. Every fully invariant subgroup of a group is
(a) Normal (b) Conjugate
(c) Abelian (d) None of these

141. H be a characteristic subgroup of a normal group K of group G then H in G is
(a) Conjugate (b) Abelian
(c) Normal (d) Center

142. Property of the internal direct product of the subgroups H and K of G is
(a) G is generated by H and K
(b) H and K are normal
(c) H is normal to K
(d) Both (a) and (b)

143. A property of the internal direct product of the subgroups H and K of G is
(a) $H \cap K = \{e\}$ (b) $H \cup K = \phi$
(c) $H \mid K = H \cap K$ (d) None of these

144. A property of the internal direct product of two subgroup H and K in G is
(a) H & K are not generator of G
(b) H & K are normal
(c) $H \cup K = \{e\}$ (d) $H \cap K \neq \{e\}$

145. $Q = \{\pm I, \pm c^e, \pm j, \pm k\}$ can not be expressed as the direct product of its subgroups
(a) Yes (b) No
(c) May or may not be
(d) None of these

146. Let $G = \langle a, b: a^2 = b^2 = (ab)^2 = 1 \rangle = \{1, a, b, ab\}$ and $H = \{1, a: a^2 = 1\}$ and $K = \{1, b: b^2 = 1\}$ be normal subgroups of G such that $H \cap K = \{e\}$ then
(a) G is not direct product
(b) G is direct product
(c) $H \cup K = K$ (d) $H \cap K = H$

147. For H and K subgroups of G, if every element of H is permutable with every element of K then
(a) G is direct product of its H and K
(b) $H \cap K \neq \{e\}$
(c) G is not direct product of H and K
(d) None of these

148. For statement (147) every element of G has
(a) Not unique expression
(b) Unique expression
(c) $H \cup K = \{e\}$
(d) $H \cap K = K$

149. Let $G = (H \times K)$ product and H_1 be normal subgroup of H then H_1 in G is
(a) Abelian (b) Commutator
(c) Normal (d) Conjugate

150. Let H and K be subgroups of G and $G = H \times K$ then for their center ξ
(a) $\xi(G) = \xi(H) \times \xi(K)$
(b) $\xi(G) \neq \xi(H) \times \xi(K)$
(c) $\xi(G) = \xi(H)/\xi(K)$
(d) $\xi(G) = \xi(K) \cdot H$

151. If H and K are subgroups and $G = H \times K$ then H and K are called
(a) Normal (b) Direct factor
(c) Conjugate (d) Center

152. Every group of order "P" P a prime number is
(a) Decomposable
(b) Indecomposeable
(c) Normal (d) None of these

153. The group $G = \langle a, b: a^3 = b^2 = (ab)^2 = 1 \rangle$ is
(a) Normal (b) Decomposable
(c) Indecmposeable (d) Both a and c

154. Decomposable cyclic group has order of number which is
(a) Composite (b) Even
(c) Odd (d) Prime

155. A non empty set R is called Ring in which
(a) (R, +) is abelian group
(b) (R, +) is non abelian
(c) (R, +) semi group
(d) Both (b) and (c)

156. A non empty set R is called Ring in which (·) multiply
(a) (R, ·) is abelian group
(b) (R, +) non abelian group
(c) (R, ·) is semi group
(d) (R, +) is semi group

157. If in a non empty set "R" left and right distribution law hold in a then R is called
(a) Group (b) Ring
(c) Semi group (d) Field

158. The set z of integers under usual addition + and multiplication (·) is
(a) Group (b) Ring
(c) Field (d) None of these

159. The set C (Complex) under simple addition and multiplication is
(a) Group (b) Semigroup
(c) Not ring (d) Ring

160. The set C(Complex) under complex addition and multiplication is
(a) Group (b) Semigroup
(c) Not ring (d) Ring

161. A Ring (R, +, ·) said to be commutative ring if $\forall$ a, b $\in$ R there exist
(a) $a \cdot b = b \cdot a$ (b) $a + b = b + a$
(c) $\frac{a}{b} = \frac{b}{a}$ (d) $ab/a = \frac{ba}{a}$

162. A ring (R, +, ·) is said to be a ring with identity if there is an element "e" in R such that $\forall x \in R$
(a) $xe = ex$ (b) $xe = ex = x$
(c) $x = e$ (d) $x \neq e$

163. A ring (R, +, ·) commutative with identity then R is called
(a) Group (b) Ring
(c) Field (d) Semigroup

164. Rings Q(+, ·), (R, +, ·) and (C, +, ·) where Q, R and C are rational, real and complex are
(a) Field (b) Semigroup
(c) Group (d) None of these

165. A ring (R, +, ·) with identity is said to be
(a) Ring (b) Field
(c) Division Ring (d) Normal

166. A ring R(+, ·) said to be ring with zero division if it has
(a) Nonzero division (b) Identity
(c) Zero division (d) Both a and b

167. A commutative ring with identity and non zero divisor is called as
(a) Integral domain (b) Divisor
(c) Abelian group (d) Both a and b

168. Every field is
(a) Group (b) Semigroup
(c) Divisor (d) Integral domain

169. Zp where P(Prime) is
(a) Conjugate group (b) Group
(c) Divisor (d) Integral domain

70. A commutative ring R with identity and left or right commutation law holds is
(a) Integral domain (b) Semigroup
(c) Divisor (d) Group

71. Zp, where P is prime number is an
(a) Group (b) Semigroup
(c) Division ring (d) Integral domain

2. A commutative ring R with identity and left cancellation law is
(a) Integral domain (b) Division ring
(c) Group (d) None of these

. If R and S be two rings then S is a subring of R if
(a) $S = R$ (b) $S \subseteq R$
(c) $S \cap R = R$ (d) $S \cup R = R$

174. Ring "R" is said to have characteristic of $n \neq 0$ if "n" is smallest +ve integer n such that $\forall a \in R$
(a) $na = 0$ (b) $na = a$
(c) $na \neq 0$ (d) Both a and b

175. Let D be an integral domain with characteristic "n" if $n \neq 0$, then "n" is a
(a) Even number (b) Odd
(c) Composite (d) Prime

176. The characteristic of (Z, +, ·), (Q, +, ·), (R, +, ·) is
(a) 1 (b) 2
(c) 3 (d) Zero

177. A ring "R" is Cacled Bolean Ring if for $x \in R$
(a) $x = e$ (b) $x^2 = x$
(c) $x^2 = e$ (d) $x^2 = \frac{x}{e}$

178. One property of ring Homomorphism of
(a) $\phi(a + b) = \phi(a) + \phi(b)$
(b) $\phi(a - b) = \phi(a) - \phi(b)$
(c) Both (a) and (b)
(d) None of these

179. A bijective Homomorphism $\phi: R \to R'$ called is known as
(a) Monomorphism (b) Homomophism
(c) Automorphism (d) None of these

180. An injective Homomorphism $\phi: R \to R'$ is called
(a) Monomorphism (b) Epimorphism
(c) Endomorphism (d) None of these

181. A surjective Homomorphism $\phi: R \to R'$ is called an
(a) None of these (b) Epimorphism
(c) Endomorphism (d) Automorphism

182. An injective homomorphism $\phi: R \to R'$ is called an
(a) Epimorphism (b) Embedding
(c) Epimorphism (d) Isomorphism

183. Subset "U" of "R" said to be left ideal in $R \ \forall a \in U, R \in U$
(a) $a + b \in U$ (b) $ab \in u$
(c) $a + b \notin u$ (d) $a - b \in u$

184. For ring R both ideals of R are
(a) {0}, {R}
(b) {0}
(c) R
(d) None of these

185. Trivial ideals of R are
(a) {0}
(b) R
(c) {0}, R
(d) Both (a) + (b)

186. Non trivial ideals of R are
(a) {0}
(b) R
(c) {0}, R
(d) Other than {0} and R

187. Let "Z" be set of integers and E be the set of all even integers then E is
(a) Proper ideal of Z
(b) Improper ideal of Z
(c) None of these

188. If a ring R with identity has no proper left (or right) ideal then R is
(a) Integral domain
(b) Zero division
(c) Division ring
(d) None of these

189. A field has ideal
(a) Proper
(b) No proper
(c) Field itself
(d) Both (a) and (c)

190. Let $\phi: R \to R'$ be a ring Homomorphism and set $\{x \in R: \phi(x) = 0\}$ be ideal of R, then it is denoted by
(a) Division ring
(b) Ker ϕ
(c) Zero division
(d) Integral domain

191. Quotient ring is denoted by
(a) R – u
(b) R + u
(c) Ru
(d) R/u

192. One of the operation of quotient ring $\frac{R}{u}$ with ideal u and $x \in R$ is defined by
(a) $x + u + y + u = x + y + u$
(b) $x + u + y + u = x + y$
(c) $x + u + y + u = x + y + 2u$
(d) $x + u + y + u = 2u$

193. For statement (192) one operation for $\frac{R}{u}$ is
(a) $(x + u)(y + u) = xy + u^2$
(b) $xy + u = (x + u)(y + u)$
(c) $(x + u)(y + u) = x^2y^2 + u$
(d) $(x + u)(y + u) = x + y + u$

194. Let "Z" be the ring of integers the set of all even integers is
(a) Non ideal in z
(b) Division ring
(c) Ideal in z
(d) Integral domain

195. Factor ring is denoted by where u
(a) R + u
(b) R – u
(c) Ru
(d) R/u

196. Let $\phi: R \to R'$ be a surjective ring homorphism then there is a Kernal k of ϕ which his an ideal of R then
(a) $\frac{R}{K} \cong R'$
(b) $\frac{R}{K} = R'$
(c) $\frac{R}{K} = K$
(d) $RK = K$

197. For any ideal K of a ring R
(a) R/K is not ring
(b) Rk is ring
(c) $\frac{R}{K}$ is ring
(d) R + K is ring

198. Let $\phi: R \to R'$ is surjective ring Homomorphism then ϕ is an isomorphism iff
(a) Ker ϕ = {0}
(b) Ker ϕ = e
(c) Ker ϕ = R
(d) Ker ϕ = {0}

199. A Homomorphism $\phi: R \to R'$ Ker ϕ = {0} then ϕ is
(a) Injective
(b) Surjective
(c) Ker ϕ = e
(d) None of these

200. Let R be ring. An ideal M of R is said to be maximal if for any ideal K of R such
(a) M = K = R
(b) $M \neq K \neq R$
(c) $M \subseteq K \subseteq R$
(d) M = R

201. If R be a commutative ring with identity I and M be an ideal in R, then M is maximal ideal iff R/M is
(a) Field
(b) Ideal
(c) Division ring
(d) Integral domain

202. An ideal "P" of a ring "R" said to be prime ideal if for a, b ∈ R
(a) ab ∉ P
(b) ab = P
(c) ab ∈ P
(d) ab = e

vector space

203. An ideal P of a ring "R" is prime ideal if for a, b ∈ R, ab ∈ P is
(a) a ∈ P (b) b ∈ P
(c) a ∈ P or b ∈ P (d) None of these

204. Let "R" be a commutative ring with identity and "P" an ideal of R then P is prime ideal if and only if R/P is
(a) Integral domain (b) Division ring
(c) Zero Divisor (d) None of these

205. If R be ring and $\frac{\text{"R"}}{\text{P}}$ is an integral domain then P is
(a) Ideal (b) Prime ideal
(c) Division ring (d) Integral domain

206. Every maximal ideal of commutative ring with identity is a
(a) Ideal (b) Integral domain
(c) Division ring (d) Prime ideal

207. The Ring "Q" of quarterian is a
(a) Division ring (b) Ideal
(c) Integral domain (d) None of these

208. Let R be ring then defined as {Z ∈ R: zx = xz, ∀ x ∈ R} is known as
(a) Ideal in z (b) Center of ring
(c) Integral domain (d) None of these

209. The center of ring is a
(a) Subring of R
(b) Not subring
(c) Integral domain
(d) Subring of normal of R

210. Every integral domain can be embedded in field of
(a) Ideals (b) Quotients
(c) Prime ideals (d) None of these

211. A field P is said to be a prime field if it has
(a) Improper field
(b) Field
(c) Proper field
(d) No proper subfield

12. For each prime P
(a) Zp is prime field (b) Zp is field
(c) Zp is ideal (d) None of these

213. One of the property of vector space is abelian group under binary operation "+" and if for every β, α ∈ F v ∈ V
(a) $\alpha(v + w) = \alpha v + \alpha w$
(b) $\alpha(\beta v) \neq \alpha\beta v$
(c) $1 \cdot v \neq v$
(d) None of these

214. For statement (213) we can have
(a) $(\alpha + \beta)v = \alpha v$
(b) $(\alpha + \beta)v = \alpha v + \beta v$
(c) $1 \cdot v \neq v$
(d) $\alpha(\beta v) = \alpha v$

215. For statement (213) we can have
(a) $\alpha(\beta v) \neq (\alpha\beta)v$ (b) $\alpha(\beta v) = \alpha\beta(v)$
(c) $1 \cdot v \neq v$ (d) $\alpha v = v$

216. For statement (213) we can have
(a) $\alpha\beta(v) = \alpha v$
(b) $(\alpha - \beta)v = (\alpha + \beta)v$
(c) $vv = v$
(d) $1(v) = v$

217. If for a non-empty subset w of vector space "V" (i) v, v ∈ w ⇒ v + w ∈ w (ii) w ∈ w, α ∈ F ⇒ αw ∈ w then w is
(a) Subspace (b) Not subspace
(c) Super space (d) None of these

218. If 'w' be a subset of "V" and for, v, ∈ w and α, β ∈ F we have, αv + βw ∈ w then w is
(a) Group (b) Subspace
(c) Ring (d) Subring

219. A non empty subset 'w" of vector space V if for v, w ∈ w and α ∈ F satisfies the condition (i) v + w ∈ w and (ii) α w ∈ w called
(a) Subring (b) Ring
(c) Group (d) Subspace

220. If "u" and "w" are subspaces of vector space V over F then if u + w = {u + w, u ∈ u and w ∈ w} then
(a) Group (b) Subgroup
(c) Subspace (d) None of these

221. Intersection of two subspaces is a
(a) Subspace (b) Group
(c) Superspace (d) Ring

222. If "u" and "v" are vector spaces over F, and the mapping T u into v is defined by for $u_1, u_2 \in u, \alpha \in u$ $(T\, u_1 + u_2) = T(u_1) + T(u_2)$ and $T(\alpha u_1) = \frac{\alpha}{1}(Tu_1)$ then T is
(a) Monomorphism (b) Epimorphism
(c) Homomorphism (d) None of these

223. The Homomorphism mapping "T": $u \to v$ defined as $\{u \in u/T(u) = 0\}$ called
(a) Ker T (b) Monomorphism
(c) Isomorphism (d) None of these

224. A one to one Homomorphism is called as
(a) Kernal (b) Monomorphism
(c) Isomorphism (d) None of these

225. Kernal of a Homomorphism is a
(a) Subspace (b) Isomorphic
(c) Conjugate (d) Ring

226. If v and w are subspace then quotient space v/w has one operations for $v_1, v_2 \in v$ and $\alpha \in R$.
(a) $(v_1 + w) + (v_2 + w) = v_1v_2 + w$
(b) $(v_1 + w) + (v_2 + w) = (v_1 + v_2)w$
(c) $(v_1 + w) + (v_2 + w) = (v_1 + v_2) + w$
(d) None of these

227. For statement (226) one operation is
(a) $\alpha(v_1 + w) = \alpha v_1$
(b) $\alpha(v_1 + w) = w$
(c) $\alpha(v_1 + w) \neq w$
(d) $\alpha(v_1 + w) = \alpha v + w$

28. If T is a Homomorphism of "u" into "v" with Kernal w then "v" is isomorphic to
(a) u/w (b) u + w
(c) u − w (d) uw

29. If "u_i" $i = 1, 2, 3, \ldots, n$ be subspace of a vector space v "v" is said to be the internal direct sum of "u_i" if every element "$v \in V$} can be written as
(a) $V = u_1, u_2, u_3, \ldots, u_n$
(b) $V = u_1 + u_2 + u_3 + u_4 \ldots u_n$
(c) None of these
(d) Both (a) and(b)

230. External direct sum $v_1, v_2, v_3, v_4 \ldots v_n$ is written as
(a) $v_1 + v_2 + v_3 + \ldots + v_n$
(b) $v_1v_2 + v_2v_3 + v_3v_4 + \ldots v_n$
(c) $v_1 \oplus v_2 \oplus v_3 \oplus v_4 \oplus \ldots \oplus v_n$
(d) $v_1v_2v_3v_4 \ldots v_n$

231. If V be vector space and $v_1, v_2, v_3, \ldots, v_n \in V$ and $\alpha_i \in F$ then linear combination is defined as
(a) $v_1 + v_2 + v_3 + \ldots + v_n$
(b) $\alpha_1 + \alpha_2 + \alpha_3 + \ldots + \alpha_n$
(c) $\alpha_1v_1 + \alpha_2v_2 + \alpha_3v_3 + \ldots \alpha_nv_n$
(d) Both a and b

232. The set of all linear combinations of finite set of elements of S is called
(a) Linear space of S
(b) Spanning set
(c) Direct sum
(d) None of these

233. Let S be a non-empty subset of a vector space v, then linear span L(S) is a
(a) Vector space (b) Spanning set
(c) Direct sum (d) Linear span

234. Linear span L(S) is the smallest subspace of "V" containing
(a) S (b) V
(c) Both a and b (d) None of these

235. A vector space "V" is said to be finite dimensional if there is a finite subset S in V such
(a) $V \neq L(S)$ (b) $V = \Sigma V_i$
(c) $V = L(S)$ (d) None of these

236. Let vectors $v_1, v_2, v_3, \ldots, v_n \in v$ and their exist $\alpha_1, \alpha_2, \alpha_3, \ldots, \alpha_n \in F$ not all zero such that $\alpha_1v_1 + \alpha_2v_2 + \alpha_3v_3 + \ldots \alpha_nv_n = 0$ then vectors $v_1, v_2, v_3, \ldots, v_n$ are linearly
(a) Dependent (b) Independent
(c) Span (d) Non zero

237. If in the above statement $(\alpha_1v_1 + \alpha_2v_2 + \alpha_3v_3 + \ldots, \alpha_nv_n \neq 0)$ called
(a) Independent (b) Independent
(c) Span (d) Non zero

238. The vector (1, 1, 1), (1, 0, 1) and (0, 1, 1) are linearly
(a) Independent (b) Dependent
(c) Non zero (d) None of above

239. A set which contains a linearly dependent subset is itself linearly
(a) Independent (b) Dependent
(c) Non zero (d) None of these

240. If for vectors $v_1, v_2, v_3, \ldots, v_n \in v$, for $\lambda_i \in F$ (i = 1, 2, 3, ..., n) we have $\lambda_1 v_1 + \lambda_2 v_2 + \lambda_3 v_3 + \ldots \lambda_n v_n$ has unique representation then is linearly
(a) Independent (b) Dependent
(c) Zero (d) Both (b) and (c)

241. A set of non zero vectors $v_1, v_2, \ldots, v_n \in V$, if one of them is a linear combination of proceeding vectors known as linearly
(a) Independent (b) Dependent
(c) Zero (d) Both a and c

242. If there are vectors v1, v2, ∈ v and "$v_1 + v_2$", "$v_1 - v_2$" are linearly independent then $v_1, v_2, \in v$ are
(a) Linearly dependent
(b) Base (c) Independent
(d) None of these

243. One of the property for a subset "S" of a vector space V is called a basis of V, if
(a) S is linearly independent
(b) S is a base
(c) S is only spanning set
(d) Both a and b

14. For statement (243)
(a) S is linearly dependent
(b) S is a base
(c) S is only spanning set
(d) V is linear span of S

5. If V be vector-space of dimension n, then every set of n linearly independent vectors of v forms a
(a) Basis of v
(b) Spanning set of v
(c) Both a and b
(d) None of these

246. If V is a finite dimension and w is a subspace of v, then
(a) Dim W = dim V (b) Dim W > dim V
(c) Dim V = 0 (d) Dim W ≤ dim V

247. IF V is a finite to dimension and W is a subspace of V, then
(a) Dim V/W = dim V – dim W
(b) Dim V = dim W
(c) Dim (VW) = 0
(d) Dim V/W = 0

248. If A and B are finite dimensional vector spaces of vector space V, then for A + B
(a) Dim (A + B) = dim A + dim B
(b) Dim (A + B) = dim AB
(c) Dim(A + B) = dim A + dim B – dim(A ∩ B)
(d) None of these

249. If T maps a basis of v_1 and basis of v_2 then T is an
(a) Isomorphism (b) Monomorphism
(c) Homomorphism (d) Epimorphism

250. Two finite dimensional vector spaces are isomorphic iff they are of the
(a) Different dimensions
(b) Same dimensions
(c) Basis of each other
(d) Both b and c

251. Dual space for vector spaces "V" an "W" over a field F is denoted by
(a) (V,W) (b) Hom$\left(\frac{V}{W}\right)$
(c) Hom(V, W) (d) Both a and c

252. For vector spaces V and W, the set of all vector spaces Homomorphism of "V" and "W" is denoted by
(a) Hom(V, W) (b) Hom$\left(\frac{V}{W}\right)$
(c) (V, W) (d) $\left(\frac{V}{W}\right)$

253. If V and F be two vector spaces the Hom(V, W) is a
(a) Linear space (b) Vector space
(c) Metric space (d) None of these

254. If "V" and "W" are of dimensions "m" and "n" over F, then Hom(V, W) is of dimensions
(a) $m + n$ (b) $\frac{m}{n}$
(c) $m - n$ (d) mn

255. If $\text{Dim}_F V = m$, then $\dim_F(V, V)$
(a) m^2 (b) m
(c) $m + 1$ (d) $m - 1$

256. If $\dim_F V = m$ then $\dim_F(V, F)$
(a) m^2 (b) m
(c) $m - 1$ (d) $m + 1$

257. If V is finite dimensional over F then V (dual space) is
(a) Homomorphism to V
(b) Epimorphism to V
(c) Isomorphic to V
(d) Both (a) and (b)

258. If W is a subspace of V then
$A(W) = \{f \in V \mid f(W) = 0, \forall W \in W\}$ called
(a) Annihilator of W
(b) Subspace of W
(c) Vector space
(d) None of these

259. Annihilator A(W) is subspace of
(a) V(Vector space) (b) L(V)
(c) V (dual space) (d) None of these

260. If "U" and "W" are subspaces of V, such that $U \subset W$ then for annihilator A(U) and A(W)
(a) $A(u) \supset A(W)$ (b) $A(u) = A(w)$
(c) $u = W$ (d) $A(u) \subsetneq A(W)$

261. If "W" is subspace of vector space V, then W is isomorphic to
(a) A(W) (b) V
(c) V A(W) (d) V/A(W)

262. For statement (261) we can write as
(a) Dim A(W) = dim V – dim W
(b) Dim A(W) = dim W
(c) Dim(A) = dim V + dim A(W)
(d) None of these

263. T: $V_1(F) \to V_2(F)$ be Homomorphism then kernel of T is called
(a) Basis
(b) Null space of T
(c) Dimensions of v_1/v_2
(d) Both (a) and (c)

264. The dimensions of N(T) called
(a) Basis of T
(b) Dimensions of N(T)
(c) Nullity of T
(d) Both (a) and (b)

265. Let T: $V_1(F) \to V_2(F)$ be vector space Homomorphism then
(a) Dim $v_1(F)$ = dim $v_2(F)$
(b) Dim $v_1(F)$ = Rank T
(c) Dim $v_1(F) \neq$ dim $v_2(F)$
(d) Dim $v_1(F)$ = Nullity T + Rank T

266. Let T: $V_1 \to V_2$ be Homomorphism with v_1 and v_2 same dimensions then T is on-to-one iff
(a) T is one-to-one (b) T is on to
(c) Both (a) and (b) (d) None of these

267. If V is of dimensions n over F, then V
(a) Isomorphic to F^n $V \cong F^n$
(b) Homomorphism to n
(c) Monomorphism to n
(d) None of these

268. An associative ring "A" is called algebra over F, If "A" is a vector-space over "F" such that all a, b ∈ A, α ∈ F
(a) $\alpha(a, b) = (\alpha a)b = a(b\alpha)$
(b) $\alpha(ab) \neq a(b\alpha)$
(c) $\alpha(ab) = a(b\alpha)$
(d) None of these

269. Hom(V, V) is an algebra over F then V is
(a) Vector space
(b) Not vector-space
(c) Metric-space
(d) Both a and c

1. Complex number has two parts namely
(a) Real (b) Imaginary
(c) None of these (d) Both a and b

2. In complex number "i^{2}" is equal to
(a) -1 (b) 1
(c) 0 (d) -2

3. A number of the form P/q where P and q are integers and $q \neq 0$ called number
(a) Irrational (b) Rational
(c) Even (d) Odd

4. A number which can be represented infinite non-recurring decimals called
(a) Even (b) Odd
(c) Irrational (d) Rational

5. An ordered pair denoted by $z = (x, y) = x + iy$ where "x" and "y" are real and imaginary parts respectively called number
(a) Real (b) Imaginary
(c) Odd (d) Complex

6. If $z_1 = x_1 + iy_1$ and $z_2 = x_2 + iy_2$ be two complex numbers then $(z_1 + z_2)$ is equal to
(a) $(x_1 + x_2) + i(y_1 + y_2)$
(b) $x_1x_2 + iy_1y_2$
(c) $x_1x_2 + i\ y_1y_2$
(d) $x_1 + i(y_1 + y_2)$

7. In the above statement $(z_1 - z_2)$ is equal to
(a) $(x_1 - x_2) + (y_1 - y_2)$
(b) $(x_1 - x_2) + i(y_1 - y_2)$
(c) $x_1y_1 - x_2y_2$
(d) $(x_1y_1 - i(x_2y_2)$

8. If $z_1 = (x_1 + iy_1)$ and $z_2 = (x_2 + iy_2)$ then $(z_1 \times z_2)$ is
(a) $(x_1x_2 - y_1y_2) - i(x_1y_2 + x_2y_1)$
(b) $(x_1x_2 - y_1y_2) + i(x_1y_2 + x_2y_1)$
(c) $x_1x_2 + iy_1y_2$
(d) None of these

9. The polar form of complex number is
(a) $z = r(\cos\theta + \sin\theta)$
(b) $z = r\cos\theta i\sin\theta$
(c) $z = r\cos\theta$
(d) $z = ir\sin\theta + \cos\theta$

10. For complex number $z = r(\cos\theta\ \ i\sin\theta)$
(a) $r = x + y$ (b) $r = x^2 + y^2$
(c) $r^2 = x^2 + y^2$ (d) $r^2 = x + y$

11. For complex number $z = r(\cos\theta + i\sin\theta)$
(a) $\theta = \tan^{-1} y$ (b) $\theta = \tan^{-1}\frac{y}{x}$
(c) $\theta = \tan\frac{y}{x}$ (d) $\theta = \tan^{-1} yx$

12. $|Z|$ for a complex number is equal to
(a) $|z| = r^2$
(b) $|z| = r = \sqrt{x^2 + y^2}$
(c) $|z| = z$
(d) None of these

13. $|Z| = 0$ if and only if x for $z = x + iy$
(a) $x = 0$ (b) $y = 0$
(c) both $x = 0, y = 0$
(d) None of these

14. For $z = x + iy$, $|z|$ is equal to
(a) $x^2 + y^2$ (b) $x^2 - y^2$
(c) x^2y^2 (d) x^2/y^2

15. Conjugate of $z = x + iy$ is given by
(a) $z = x - iy$ (b) $z = xy$

16. For complex number $z = x + iy$, $|z|^2$ is given by

(a) $z \cdot \bar{z}$ (b) $z - \bar{z}$

(c) $z + \bar{z}$ (d) None of these

17. Re(z) in complex number $x + iy$ is

(a) y (b) x

(c) $x - iy$ (d) (x) (iy)

18. $I_{mg}(z)$ in complex number $x + iy$ is

(a) y (b) x

(c) $x - iy$ (d) $x(iy)$

19. For complex number $z = x + iy$ when $y = 0$ then

(a) $|z|^2 = |I_{mg}(z)|^2$

(b) $|z|^2 = I_{mg}z$

(c) $|z|^2 \geq |I_{mg}z|^2$

(d) $z = |I_{mg}\ z|^2$

20. In two complex numbers $z_1 = x_1 + iy_1$ and $z_2 = x_2 + iy_2$

(a) $|z_1z_2| = |z| \cdot |z_2|$

(b) $|z_1z_2| = |z_1|$

(c) $|z_1z_2| = |z_2|$

(d) None of these

21. For the above statement (20)

(a) $\left|\frac{z_1}{z_2}\right| \neq \left|\frac{z_1}{z_2}\right|$ (b) $\left|\frac{z_1}{z_2}\right| \neq ?$

(c) $\left|\frac{z_1}{z_2}\right| = \left|\frac{z_1}{z_2}\right|$ (d) $\left|\frac{z_1}{z_2}\right| = \frac{z_1}{z_2}$

22. $|z_1 + z_2|^2$ for complex numbers z_1 and z_2 is

(a) $|z1 + z2|2 = z_1^2 + z_2^2$

(b) $|z_1 + z_2|^2 = (z_1 + z_2)\overline{(z_1 + z_2)}$

(c) $|z_1 + z_2|^2 = z_1^2 - z_2^2$

(d) $|z_1 + z_2|\overline{(z_1 + z_2)}$

23. For example numbers "z_1" and "z_2" we can have

(a) $|z_1 - z_2| \geq |z_1| - |z_2|$

(b) $|z_1 - z_2| > |z_1| - |z_2|$

(c) $|z_1 - z_2| = |z_1| - |z_2|$

(d) None of these

24. For any complex number $z_1 = x_1 + iy_1$ the $\text{Arg}(z_1)$ is

(a) $\theta = \tan\frac{y}{x}$ (b) $\theta = \tan x\,c$

(c) $\theta = \tan^{-1}\frac{y}{x}$ (d) $\theta = \tan^{-1} y$

25. For any two complex numbers z_1 and z_2 $\text{Arg}(z_1z_2)$

(a) $\theta_1 + \theta_2$ (b) $\theta_1 - \theta_2$

(c) θ_1/θ_2 (d) $\theta_1\theta_2$

26. For any two complex numbers z_1 and z_2 $\text{Arg}\left(\frac{z_1}{z_2}\right)$ is

(a) $\theta_1 + \theta_2$ (b) $\frac{\theta_1}{\theta_2}$

(c) $\theta_1\theta_2$ (d) $\theta_1 - \theta_2$

27. $\left|\frac{az + b}{a\bar{z} + a}\right| = 1$ for

(a) $|z| = 0$ (b) $|z| = -1$

(c) $|z| = 1$ (d) $|z|^2 = 0$

28. $\left|\frac{a + b}{1 + \bar{a}b}\right| \leq 1$ for

(a) $|a| < 1$ (b) $|b| < 1$

(c) Both a and b (d) $|a| < 0$

29. $|z_1 + z_2|^2 + |z_1 - z_2|^2 =$

(a) $2[|z_1|^2 + |z_2|^2]$ (b) $z_1^2z_2^2$

(c) $z_1^2 - z_2^2$ (d) $|z_1^2 - z_2^2|^2$

30. If $w = f(z)$ where $z = x + iy$ and $w = u + i$ then for $w = z^2$ we can write

(a) $u = x^2 - y^2$ (b) $v = 2xy$

(c) Both a and b (d) None of these

31. For above statement $w = f(z)$ is function o

(a) Single value (b) Double value

(c) n-values (d) Zero

32. $W = (z)^{1/n}$ is an function of value
(a) Single (b) Double
(c) n-value (d) Zero

33. If $\lim_{z \to z_0} f(z) = w_0$ then we can write as
(a) $|f(z) - w_0| < \epsilon$ for which $0 < |z - z_0| < \delta$
(b) $|f(z) - w_0| > \epsilon$ for which $0 < |z - z_0| < \delta$
(c) $|f(z - w_0)| = 0$
(d) None of these

34. $\lim_{z \to 1} \left(\frac{z^n - 1}{z - 1}\right)$, when n is +ve integer
(a) 1 (b) 0
(c) n (d) -1

35. The limit of function w = f(z) is
(a) Unique (b) Not unique
(c) Zero (d) 1

36. If f(z) is a constant function then Cauchy equation
(a) Does not hold (b) Hold
(c) Satisfied not (d) Zero

37. The necessary and sufficient condition for function w = f(z) to be analytic function is
(a) $\partial u/\partial x = \frac{\partial v}{\partial y}$ and $\frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$
(b) $\frac{\partial u}{\partial x} = \frac{\partial u}{\partial y}$
(c) $\frac{\partial v}{\partial x} = \frac{\partial v}{\partial y}$
(d) $\frac{\partial u}{\partial y} = \frac{\partial v}{\partial x}$

38. Cauchy Reimann equation for a function to be analytic are
(a) $\frac{\partial u}{\partial x} = -\frac{\partial v}{\partial y}$
(b) $\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}$ and $\frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$
(c) $\frac{\partial v}{\partial x} = -\frac{\partial v}{\partial y}$
(d) $\frac{\partial u}{\partial y} = \frac{\partial v}{\partial x}$

39. If a function f(z) is analytic equations necessarily hold
(a) Cauchy Reimann
(b) $\frac{\partial u}{\partial x} = \frac{\partial v}{0k}$ (c) $\frac{\partial u}{\partial x} = 0$
(d) None of these

40. If Cauchy Reimann equation are satisfied then the function
(a) Continuous not
(b) Chauchy Reimann equation hold
(c) Cauchy Reimann equation may or may not be hold
(d) None of these

41. The essential characteristic for a function to be analytic is that it does not involve
(a) $\bar{z}$ (b) z
(c) z and $\frac{1}{z}$ (d) Both b and c

42. The necessary condition for function f(z) to be analytic is that
(a) $\frac{\partial f}{\partial \bar{z}} = 0$ (b) $\frac{\partial f}{\partial z} = 0$
(c) $\frac{\partial f}{\partial z \partial \bar{z}} = 0$ (d) $\frac{\partial f}{\partial x} = 0$

43. A real valued function u(x, y) or v(x, y) is said to be harmonic if
(a) $\nabla^2 u \neq 0$ (b) $\nabla^2 u = 0$
(c) $\nabla u = 0$ (d) $\nabla^2 u = 1$

44. Harmonic function satisfies the formal differential equation
(a) $\frac{\partial^2 u}{\partial z^2} = 0$ (b) $\frac{\partial^2 u}{\partial \bar{z}^2} = 0$
(c) $\frac{\partial^2 u}{\partial x^2}$ (d) $\frac{\partial^2 u}{\partial z \partial \bar{z}} = 0$

45. When f(z) = u(r, θ) + iv(r, θ) polar form then Cauchy Reimann equation are given by
(a) $\frac{\partial u}{\partial r} = \frac{1}{r}\frac{\partial v}{\partial \theta}$ and $\frac{\partial v}{\partial r} = -\frac{1}{r}\frac{\partial u}{\partial \theta}$
(b) $\frac{\partial u}{\partial \theta} = r\frac{\partial v}{\partial \theta}$ (c) $\frac{\partial v}{\partial r} = r\frac{\partial u}{\partial \theta}$
(d) None of these

46. Laplace equation in polar form is

(a) $\frac{\partial^2 u}{\partial r^2}+\frac{1}{r}\frac{\partial u}{\partial r}+\frac{1}{r^2}\frac{\partial u}{\partial \theta^2}=0$

(b) $\frac{\partial u}{\partial r}+\frac{1}{r}\frac{\partial u}{\partial \theta}+\frac{1}{r}\frac{\partial^2 u}{\partial r^2}\neq 0$

(c) $\frac{\partial^2 \theta}{\partial r^2}+\frac{1}{\theta}\frac{\partial u}{\partial r}=0$

(d) None of these

47. A level curve of a real value function u(x, y) defined in domain D is given by

(a) u(x, y) = 0 (b) u(x, y) = 1

(c) u(x, y) = c (d) None of these

48. The condition for orthogonal system is

(a) $\frac{\partial u}{\partial x}\cdot\frac{\partial v}{\partial x}+\frac{\partial u}{\partial y}\cdot\frac{\partial v}{\partial y}=0+1$

(b) $\frac{\partial u}{\partial x}\cdot\frac{\partial v}{\partial x}=-\frac{\partial u}{\partial y}\cdot\frac{\partial v}{\partial y}$

(c) $\frac{\partial u}{\partial x}\cdot\frac{\partial v}{\partial y}-\frac{\partial u}{\partial y}-\frac{\partial v}{\partial x}=0$

(d) $\frac{\partial u}{\partial x}=-\frac{\partial v}{\partial y}$

... function with constant modulus

54. Sin z = (where z is complex variable)

(a) $\frac{e^{z}+e^{-z}}{2}$ (b) $\frac{e^{z}-e^{-z}}{2}$

(c) $\frac{e^{iz}+e^{-iz}}{2i}$ (d) $\frac{e^{iz}-e^{-iz}}{2i}$

55. Case z = (where z is complex variabl

(a) $\frac{e^{iz}+e^{-iz}}{2}$ (b) $\frac{e^{iz}-e^{iz}}{2}$

(c) $\frac{e^{iz}+e^{iz}}{2i}$ (d) None of th

56. Sin (iz) =

(a) Sin hz (b) i sin hz

(c) i sin z (d) i cas(z)

56. Cas (iz) =

(a) i cosec (b) cosec

(c) cosec (d) i sin hz

57. tan z = where z is complex number

(a) $\frac{e^{iz}-e^{-iz}}{e^{iz}+e^{-iz}}$ (b) $\frac{1}{i}\left(\frac{e^{iz}-e^{-iz}}{e^{iz}+e^{-iz}}\right.$

(c) $e^{iz}+e^{-iz}$ (d) None of the

... is complex constant

62. $|\sin z|^2$ = where $(z - x + iy)$

(a) $\sin^2 n + \cos^2 n$ (b) $\sin^2 x + \sin h^2 y$

(c) $\sin^2 x = \sin^2 hy$ (d) None of these

63. $|\operatorname{Cos} z|^2$ = (where $z = x + iy$)

(a) $\operatorname{Cos}^2 x \sin^2 hy$ (b) $\operatorname{Cos} x = \sin^2 hy$

(c) $\operatorname{Cos}^2 x - \sin^2 x$ (d) None of these

64. Sin hz = where (Σ complex number)

(a) $\frac{e^z - e^{-z}}{2i}$ (b) $\frac{e^z + e^{-z}}{2}$

(c) $\frac{e^z - e^{-z}}{2}$ (d) None of these

65. Sin h(iz) = (where z is complex)

(a) sin z (b) i sin(hz)

(c) sin z (d) i sin z

66. The fundamental period of "sin hz" and cos hz is

(a) $2\pi i$ (b) πi

(c) $\frac{\pi}{2} i$ (d) 0

67. If $e^w = z$ (where both z and w are complex)

(a) W = log e (b) W = log z

(c) $\frac{w}{z}$ (d) None of these

68. f'(z) (where z is complex)

(a) z (b) z^2

(c) $\frac{1}{z}$ (d) $\frac{1}{z^2}$

69. log f complex number, we can have

(a) log z = log |z|

(b) log z = log |z| + iθ

(c) log z = |z|

(d) log z = iθ

70. For complex numbers z_1 and z_2 we can have

(a) $\log(z_1 z_2) = z_1 z_2$

(b) $\log(z_1 z_2) = z_1$

(c) $\log(z_1 z_2) = 0$

(d) $\log(z_1 z_2) = \log z_1 + \log z_2$

71. e^{iy} = (where y is any integer)

(a) cos y + i sin y (b) cos y − i sin y

(c) cos y + sin y (d) None of these

72. $e^{2k\pi i}$ where (k = 0, 1, 2, 3, ...) value is

(a) 0 (b) −1

(c) c (d) 1

73. $e^{k\pi i}$ = where (k = 0, ±1, ±2, ±3, ...)

(a) 0 (b) 1

(c) −1 (d) None of these

74. If $z \neq 0$ and w_1, w_2 are complex nos.

(a) $z^{w_1} \cdot z^{w_2} = z^{w_1 + w_2}$

(b) $z^{w_1} \cdot z^{w_2} = 0$ (c) $z^{w_1} w_2 = w_2 z_2$

(d) $z^{w_1} z^{w_2} = 0$

75. $e^{i\pi/2}$ =

(a) 1 (b) −1

(c) 0 (d) i

76. For complex "z" $\frac{d}{dz}(\sin^{-1} z)$ =

(a) $\frac{1}{1+z^2}$ (b) $\frac{-1}{\sqrt{1+z^2}}$

(c) $\frac{-1}{\sqrt{1-z^2}}$ (d) $\frac{1}{\sqrt{1-z^2}}$

77. For complex z $\frac{d}{dz}(\cos^{-1} z)$ =

(a) $\frac{-1}{\sqrt{1-z^2}}$ (b) $\frac{1}{\sqrt{1-z^2}}$

(c) $\frac{1}{\sqrt{1+z^2}}$ (d) $\frac{-1}{\sqrt{1+z}}$

78. For complex number z and closed centre c

(a) $\int_c z^n \, dz \neq 0$ (b) $\int_c z^n \, dz = 1$

(c) $\int_c z^n \, dz = 0$ (d) $\int_c z^n \, dz = -1$

79. For complex nos. "z" and "z_0"

$\int_c (z - z_0)^n \, dz$ = (for $n \neq -1$)

(a) −1 (b) 1

(c) α (d) 0

80. $\int_c (z - z_0)^n\, dz =$ (where $n = -1$) is equal to

(a) $2\pi i$ (b) 2π
(c) 0 (d) -1

81. If C is continuous curve and $|f(z)| =$ "M" and "L" length of contour c then

(a) $\int_c f(z)\, dz = M$ (b) $\int_c f(z)\, dz = L$

(c) $\int_c f(z)\, dz = ML$ (d) $\int_c f(z)\, dz \le ML$

82. $\left| \int_c \bar{z}\, dz \right| =$ (where c is a semi circle)

(a) $\le \pi$ (b) 0
(c) -1 (d) 1

83. $\int_c \frac{dz}{z} =$ (where c is a square)

(a) π (b) $2\pi i$
(c) 0 (d) πi

84. $\int_{-i}^{i} |z|\, dz =$ (where c is straight line)

(a) 0 (b) -1
(c) 1 (d) i

85. If c is a curve enclosing the area S with P and Q as function of x and y then

$\int_c Pdx + Qdy =$

(a) $\int_s \int \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dxdy$

(b) $\int_s \int \left(\frac{\partial Q}{\partial x} + \frac{\partial P}{\partial y} \right) dxdy$

(c) $\int_s \int \left(\frac{\partial Q}{\partial x} \cdot \frac{\partial P}{\partial y} \right) dxdy$

(d) $\int_s \int \left(\frac{\partial P}{\partial x} - \frac{\partial Q}{0y} \right) dxdy$

86. If f'(z) is analytic then for closed contour

c $\int_c f(z)\, dz =$

(a) -1 (b) 1
(c) 0 (d) None of these

87. $\int_c \frac{1}{z^2} =$ (where z is complex)

(a) 0 (b) 1
(c) -1 (d) π

88. If f(z) complex valued function consisting of contour c_1 and c_2 then

(a) $\int_{c_1} f(z)\, dz \ne \int_{c_2} f(z)\, dz$

(b) $\int_{c_1} f(z)\, dz = \int_{c_2} f(z)\, dz$

(c) $\int_{c_1} f(z)\, dz = \int_{c_2} f(z)\, dz = 0$

(d) None of these

89. $\int_c \frac{dz}{(z-a)^n}$ for all $n > 1$ is equal to

(a) 1 (b) -1
(c) α (d) 0

90. $\int_c \frac{dz}{(z-a)^n}$ for $n = 1$ is equal to

(a) $2\pi i$ (b) πi
(c) 1 (d) -1

91. $\int_c \frac{1}{z}\, dz$ where c semicircular are $|z| = 1$ from -1 to

(a) 0 (b) $-\pi i$
(c) 1 (d) -1

92. If f(z) b analytic function in simply connected domain enclosed by closed contour C containing $z = a$ then

(a) $f(a) = 0$

(b) $\frac{1}{2\pi i} \int_c \frac{f(-z)}{z-a}\, da = 0$

(c) $f(a) = \frac{1}{2\pi i} \int_c \frac{f(z)}{z-a}$

(d) None of these

93. f(z) be analytic with in the boundary of C and let "a" be any point within C, then in general we can write as

(a) $f^n(a) = \frac{nlo}{2\pi i} \int_c \frac{f(z)}{(z-a)^{n+1}}\, dz$

(b) $f(a) = 0$

(c) $f^n(a) \int_c \frac{(z-a)^n}{f(a)}$

(d) None of these

94. Let $f(z)$ be analytic on closed contour $C\ |z-a| = r$ and $|f(z)| \le M$ then

(a) $f^n(a) = 0$ (b) $|f^n(a)| \le \frac{n!o}{r^n} M$

(c) $f^n(0) = 1$ (d) $f^n(a) = M$

95. Let "f(z)" be analytic and continuous in closed contour C then

(a) $\int_c f(z)\,dz = 1$ (b) $\int_c f'(z)\,dz = 0$

(c) $\int_c f(z)\,dz = 0$ (d) None of these

96. Let $f(z)$ be analytic and $F(z)$ indefinite integral of $f(z)$ then

(a) $\int_a^b f(z)\,dz = F(b) - F(a)$

(b) $\int_a^b f(z)\,dz = 0$

(c) $\int_a^b f(z)\,dz = F(b)$

(d) None of these

97. Sum of n terms of infinite series for first term "a" and common ratio r is given by

(a) $S_\alpha = \frac{a}{1+r}$ (b) $S_\alpha = \frac{a}{r}$

(c) $S_\alpha = \frac{1}{1-r}$ (d) $S_\alpha = \frac{a}{1-r}$

98. A series $\Sigma\, U_n$ is said to be absolutely conv if

(a) $\Sigma |U_n|$ convergent

(b) ΣU_n convergent

(c) $\Sigma |U_n| = 0$

(d) None of these

99. A series ΣU_n is conditionally convergent if

(a) $\Sigma |U_n|$ divergent

(b) $\Sigma |U_n|$ convergent

(c) $\Sigma U_n = 0$ (d) None of these

100. The ratio test for the power series of the form $\sum_{n=0}^{\alpha} a_n z^n$ is given by

(a) $\underset{n\to\alpha}{\text{let}} \left| \frac{a_{n+1}}{a_n} \right| = 1$

(b) $|z| \ge 1$ (c) $|z| \underset{n\to\alpha}{\text{lt}} \left|\frac{a_{n+1}}{a_n}\right| < 1$

(d) None of these

101. The geometric series $\sum_{n=0}^{\alpha} z_n$ converges if

(a) $|z| = 1$ (b) $|z| \ge 1$

(c) $|z| = 0$ (d) $|z| < 1$

102. The radius of convergence "R" of a power series is given by

(a) $R = \frac{1}{|a_n|^{1/n}} \underset{n\to\alpha}{\text{lt}}$ (b) $R \underset{n\to\alpha}{\text{lt}} \left| \frac{a_n}{a_{n+1}} \right|$

(c) Both a and b (d) None of these

103. $\underset{n\to\alpha}{\text{let}}\ n^{1/n} =$ (where n natural number)

(a) 1 (b) 0

(c) −1 (d) Both a and c

104. Every power series inside its circle of convergence represents

(a) Non analytic function

(b) Analytic function

(c) Constant function

(d) None of these

105. Expansion of analytic function in a power series is

(a) Unique (b) None of these

(c) Zero (d) Not unique

106. The geometric series $\sum_{n=0}^{\alpha} z_n$ converges if

(a) $|z| = 1$ (b) $|z| > 1$

(c) $|z| < 1$ (d) $|z| = 0$

107. For the series $\sum_{n=0}^{\alpha} a_n z^n$, radius of conv. is given

(a) $\left|\frac{a_{n+1}}{a_n}\right| \underset{n\to\alpha}{\text{lt}}$ (b) $\underset{n\to\alpha}{\text{lt}} \left|\frac{a_n}{a_{n+1}}\right|$

(c) $|a_{n+1}| \underset{n\to\alpha}{\text{lt}}$ (d) $\underset{n\to\alpha}{\text{lt}} |a_n|$

108. Taylor's series for f(z) with circle c at center z = a is given by

(a) $f(z) = \sum_{n=0}^{\alpha} \frac{(z-a)^n}{n!} f^n(a)$

(b) $f(z) = \sum_{n=0}^{\alpha} \frac{z-a}{n} f^n(a)$

(c) $f(z) = \sum_{n=0}^{\alpha} \frac{z+a}{n} f^n(a)$

(d) $\sum_{n=0}^{\alpha} \frac{z+a}{n!}$

109. The Maclaurins series for f(z) when a = 0

(a) $f(z) = \frac{z^n}{n!} f^n(0)$ (b) $\sum_{n=0}^{\alpha} \frac{z^n}{1} f^n(0)$ n_0

(c) f(z) = 0 (d) None of these

110. Let f(z) analytic bounded by two concentric circles $c_1 + c_2$ with center at z = a radius R and r "R > r" is given by

(a) $f(a+h) = a_0 + a_1h + a_2h^2 + \dots a_nh_n + \frac{b_1}{h} + \frac{b_2}{h} + \dots \frac{b^n}{h^n}$

(b) $f(a+h) = a_0 + a_1h + a_2h^2 + \dots a_nh_n + \frac{b_1}{h} + \frac{b_2}{h^2} + \dots \frac{b^n}{h^n}$

(c) None of these

111. For the above statement (110) we can trace a_n as

(a) $\frac{1}{2\pi i}\int_{c_1} \frac{f(z)\,dz}{(z-a)^{n+1}}$ (b) $\frac{1}{2\pi i}\int_{c_1} \frac{f(z)\,dz}{(z-a)^n}$

(c) $\frac{1}{2\pi i}\int_{c_1} \frac{f(z)\,dz}{z^n}$ (d) $\frac{1}{2\pi}\int_{c} \frac{f(z)}{(z-a)}dz$

112. For the above statement (110) we can trace "b_n"

(a) $b_n = \frac{1}{2\pi i}\int_{c_2} f(z)(z-a)\,dz$

(b) $b_n = \int_{c_2} f(z)(z-a)^n$

(c) $b_n = \frac{1}{2\pi i}\int_{c_2} f(z)(z-a)^{n-1}\,dz$

(d) None of these

113. Let f(z) be a complex valued function and if z = a is zero of f(z) then

(a) f(a) = 0 (b) f(a) ≠ 0

(c) f(a) = 2 (d) f(a) = a

114. If f(a) = 0 and f(a) = 0 but f''(a) ≠ 0 then zero is called of order

(a) 1 (b) 3

(c) 0 (d) 2

115. If f(z) Ceases to be analytic at z = a then point z = a is called

(a) Singularities (b) Poles

(c) Zeroes (d) None of these

116. In the laurants expansion $a_0 + a_1z + a_2z^2 + \dots a_nz^n$ is called the part

(a) Analytic (b) Principal

(c) Both a and b (d) None of these

117. In the laurants expansion $\frac{b_1}{z} + \frac{b^2}{z^2} + \dots + \frac{b^n}{z^n} + \dots$ is called the part

(a) Analytic (b) Principal

(c) Both (d) None of these

118. If the principle part of f(z) at $z = z_0$ consists of no term then z_0 called a

(a) Singularly

(b) Pole

(c) Removable singularly

(d) Both a and b

119. If the principal part of f(z) at $z = z_0$ contains and infinite number of terms $z = z_0$ then it is called singularity

(a) Artificial (b) Essential

(c) Pole (d) Zero of f

120. If the principal part of a function f(z) at $z = z_0$ consist of finite number of terms say m, then f(z) has a pole of order

(a) m − 1 (b) m + 1

(c) m (d) 0

121. A singularly of finite order is called a

(a) Pole

(b) Zero of f(z)

(c) Artificial singularly

(d) None of these

122. Singularity of infinite order called singularly
(a) Removable (b) Infinite order
(c) Essential (d) None of these

123. A function which has poles as its only singularities in the finite part of the plane called function
(a) Mesomorphic (b) Analytic
(c) Entire (d) Non-integral

124. A function f(z) which has no singularity in the finite part of the plane or at infinity is
(a) Variable (b) On-to
(c) One-one (d) Constant

125. Let z_0 isolated singularity of f(z) and if $|f(z)|$ is bounded in some of z_0 then z_0 is
(a) Pole
(b) Removable Singularity
(c) Residue
(d) None of these

126. The function $f(z) = e^{1/z}$ has essential singularity at
(a) $z = 1$ (b) $z = -1$
(c) $z = 0$ (d) $z = \alpha$

127. Cauchy-integral formula at z = a is
(a) $f^n(a) = \frac{n_0^1}{2\pi i} \int_c \frac{f(z)\,dz}{(z-a)^{n+1}}$
(b) $f^n(a) = \int_c \frac{f(z)\,dz}{(z-a)^{n+1}}$
(c) $f^n(a) = \int \frac{f(z)\,dz}{(z-a)^{n+1}}$
(d) None of these

128. The residue of f(z) = tank (z) is
(a) 0 (b) −1
(c) π (d) 1

129. If f(z) is analytic (Regular) except at a finite N_0 of poles with in a closed contour c and continuous then $\int_c f(z)\,dz = 0$
(a) $2\pi i \sum_{i=1}^{n} R_i$ (b) $2\pi i \sum_{i=1}^{n-1} R_i$
(c) $\sum_{i=1}^{n} R_i$ (d) None of these

130. If there is complex number $z = e^{i\theta}$ then
(a) $\cos\theta = z + \frac{1}{z}$
(b) $\frac{1}{2}\left(z - \frac{1}{z}\right) = \cos\theta$
(c) $\cos\theta = \frac{1}{2}\left(z + \frac{1}{z}\right)$
(d) $\frac{1}{2i}\left(z + \frac{1}{z}\right)$

131. If there is complex number $z = e^{i\theta}$ then $\sin\theta = 0$
(a) $\frac{1}{2i}\left(z - \frac{1}{z}\right)$ (b) $\frac{1}{2}\left(z - \frac{1}{z}\right)$
(c) $\left(z + \frac{1}{z}\right)$ (d) $\frac{1}{2i}\left(z + \frac{1}{z}\right)$

132. For complex number $z = e^{i\theta}$ we can have
(a) $d\theta = \frac{1}{2i}\frac{dz}{z}$ (b) $d\theta = \frac{dz}{iz}$
(c) $d\theta = i\,dz$ (d) $d\theta = 0$

133. If ΣR be the sum of residues at poles then $\int_c f(z)\,dz$ is given by
(a) $2\pi \Sigma R$ (b) $2\pi R$
(c) $2\pi i\, \Sigma R$ (d) None of these

134. If ΣR^+ denotes sum of residue lying in the upper half plane then
(a) $\int_{-\alpha}^{\alpha} f(x)\,dx = 2\pi i\, \Sigma R^+$
(b) $\int_{-\alpha}^{\alpha} f(x)\,dx = 2\pi\, \Sigma R$
(c) $\int_{-\alpha}^{\alpha} f(x)\,dx = 2\pi i$
(d) $\int_{-\alpha}^{\alpha} f(x)\,dx = \Sigma R$

135. If ΣR denote the sum of residue at the chosen poles in the upper half plane then
(a) $\int_{-\alpha}^{\alpha} f(x) \sin m x\,dx = 2\pi i\, \Sigma R$

(b) $\int_{-a}^{a} f(x) \sin dx = 0$

(c) $\int_{-a}^{a} f(x) \sin mn\, dx = I_m\,(2\pi i\Sigma R)$

(d) None of these

136. For the above statement (135)

(a) $\int_{0}^{a} f(x) \cos mn\, dx = Re\,(\pi i\Sigma R)$

(b) $\int_{0}^{a} f(x) \cos mn\, dx = 0$

(c) $\int_{0}^{a} f(x) \cos mn\, dx = \pi i\Sigma R$

(d) $\int_{0}^{a} f(x) \cos mn\, dx = \pi i$

137. Let f(z) be analytic function of z in Domain D of z-plane then transformation affected w = f(z) is conformal at all points of D if

(a) $f'(z) = 0$ (b) $f'(z) \neq 0$
(c) $f'(z) = 1$ (d) None of these

138. The orthogonal curves in one-plane give rise to orthogonal curves in other plane under

(a) Mapping
(b) Conformal mapping
(c) Both a and b
(d) None of these

139. Transformation of the form W = AZ + B is called linear if

(a) $A = 0$ (b) $B - 0$
(c) $A \neq 0$ (d) $A = 1$

140. Linear transformation is a

(a) Conformal mapping
(b) Mapping
(c) Both a and b
(d) None of these

141. The transformation T defined by $W = T(z) \frac{az+b}{cz+d}$ where a, b, c and d are complex constant called bilinear if

(a) $ad - bc \neq 0$ (b) $ab - cd \neq 0$
(c) $ab + cd = 0$ (d) $ac = 0$

142. If $W = T(z) = \frac{az+b}{cz+d}$ then inverse transformation T^{-1} is defined as

(a) $Z = \tan^{-1} W = \frac{dw+b}{cw-a}$

(b) $Z = \tan^{-1} W = \frac{-dw+b}{cw+a}$

(c) $Z = \tan^{-1} W = \frac{dw-a}{cw+b}$

(d) None of these

143. In the above statement No 142 the determinant of transformation is

(a) $ad + bc$ (b) $ab - cd$
(c) $a + cd$ (d) $ad - bc$

144. The transformation defined in statement (142) called normalized if ad - bd =

(a) 0 (b) -1
(c) 1 (d) Both a and b

145. In the Bilinear transformation if $ad - bc \neq 0$ then Bilinear transformation is

(a) 1 - 1 (b) On-to
(c) 1-to-1 (d) None of these

146. Product of any finite number of bilinear transformation is also a transformation

(a) Linear (b) Bilinear
(c) Both a and b (d) None of these

147. Set of all bilinear transformations under the product of transformation forms

(a) Group (b) Abelian group
(c) Identity group (d) Ker of group

148. A bilinear transformation maps circle into a

(a) Straight line (b) Parabola
(c) Circle (d) Square

149. A bilinear transformation maps straight line into

(a) Circle (b) Straight line
(c) Circle (d) None of these

150. The general equation of a circle or a straight line in z-plane may be written as

(a) $AZ\bar{Z} + BZ + \bar{B}Z + D = 1$

(b) $AZ\bar{Z} + BZ + \bar{B}Z + D = 2$

151. The general equation of circle or straight line $AZ\bar{Z} + B\bar{Z} + \bar{B}Z + D = 0$ is circle if

(a) $A \neq 0$ and $B\bar{B} - AD > 0$

(b) $A = 0$ and $B\bar{B} - AD < 0$

(c) $B\bar{B} - AD < 0$

(d) $A \neq 0$ and $B\bar{B} - AD < 0$

152. The points which coincide with their transforms under a linear transformation are called the

(a) Variable point (b) Fixed point

(c) Both a and b (d) None of these

153. The normal form of every bilinear transformation with the fixed point α, β an be take as

(a) $\frac{w-\alpha}{w-\beta} = k\left(\frac{z-\alpha}{z-\beta}\right)$

(b) $\frac{w+\alpha}{w+\beta} = k\left(\frac{z+\alpha}{z+\beta}\right)$

(c) $\frac{w-\alpha}{w-\beta} = k\left(\frac{z+\alpha}{z+\beta}\right)$

(d) $\frac{w+\alpha}{w+\beta} = k\left(\frac{z-\alpha}{z-\beta}\right)$

154. Every bilinear transformation which has only one fixed point say α, we can write as

(a) $\frac{1}{w+\alpha} = \frac{1}{w+\alpha} + k$

(b) $\frac{1}{w-\alpha} = \frac{1}{w+\alpha} + k$

(c) $\frac{1}{w-\alpha} = \frac{1}{z-\alpha}$

(d) $\frac{1}{w-\alpha} = \frac{1}{z-\alpha}$

155. The angle of rotation by the transformation $W = \frac{1}{z}$ at the points $z = 1$ is

(a) 0 (b) $-\pi$

(c) π (d) -1

156. Angle of rotation produced by the transformation $W = \frac{1}{z}$ at point $z = i$ is

(a) 0 (b) 1

(c) -1 (d) π

157. $e^{\pi/2\, i} =$

(a) -1 (b) i

(c) 0 (d) 2

158. Equation of tangent line of a point on a curve is given by

(a) $\underline{R} = \underline{r} - u\underline{t}$ (b) $\underline{R} = R + u$

(c) $\underline{R} = \underline{r} + u\underline{t}$ (d) $\frac{\underline{R}}{\underline{r}} = 0$

159. Equation of normal plane is given by

(a) $(\vec{R} - \vec{r}) \cdot \vec{t} = 0$ (b) $(\vec{R} - \vec{r}) \cdot \vec{b} = 0$

(c) $(\vec{R} - \vec{r}) \cdot \vec{N} = 0$ (d) 0

160. The radius of curvature "f" when k is curvature is given by

(a) $f = \frac{1}{2k}$ (b) $f = k$

(c) $f = 1$ (d) $f = \frac{1}{k}$

161. If $\vec{R}$ is any point and $\vec{r}$ position vector in the plane $\vec{t}$ and $\vec{n}$ be tangent and normal then equation of oscillating plane is given by

(a) $[\vec{R} - \vec{r}, \vec{t}, \vec{n}] = 0$

(b) $[\vec{R} - \vec{r}, t, r] = 0$

(c) None of these

162. If T, b and n are tangent, binormal and normal respectively then

(a) $\underline{t} \cdot \underline{t} = \underline{n} \cdot \underline{n} = \underline{b} \cdot \underline{b} = 1$

(b) $t \cdot t = n \cdot n = -1$

(c) $n \cdot n = b \cdot b = 0$

(d) None of these

163. For the statement (162) we can write as

(a) $t \times t = n \times n = 0$

(b) $\underline{t} \times \underline{t} = \underline{b} \times \underline{b} = 0$

(c) Both a and b

(d) None of these

164. Equation of the binomial is given by

(a) $\underline{R} = \underline{r} + u\underline{b}$ (b) $\underline{R} = \underline{r} - u\underline{b}$

(c) $\underline{R} = \underline{r}$ (d) $\frac{\underline{R}}{\underline{r}} = 0$

165. Radius of Torsion σ is given by which "τ" is Torsion is given by

(a) $\sigma = \tau$ (b) $\sigma = \frac{1}{\tau}$

(c) $\tau = \frac{1}{\sigma}$ (d) Both b and c

166. One of the Serret Frenet formula are

(a) $t' = kn$ (b) $t' = \tau n$

(c) $t' = \tau b$ (d) $t' = n'$

167. One of the Serret Frenet formulas are

(a) $b' = \tau k$ (b) $b' = -\tau n$

(c) $b' = \tau b$ (d) $b' = tn$

168. One of the Serret Frent formula are

(a) $n' = \tau b + kt$ (b) $n' = \tau b - t$

(c) $n' = \tau b - kt$ (d) None of these

169. If t-tangent, n-normal and b-binormal than moving trihedron is

(a) (t, b) (b) (t, n, b)

(c) (t, n) (d) (n, b)

170. For position vector $\vec{r}$, t-tangent, n-normal + b-binormal

(a) $r' = \tau b$ (b) $r' = t$

(c) $r' = t$ (d) $r' = \tau$

171. If "τ" is Torsion, "k" curvature and "r" position vector then τ is given by

(a) $\tau = \frac{1}{k^2}[r', r'', r''']$ (b) $\tau = \frac{1}{k^2}[r' \cdot r'']$

(c) $\frac{1}{k^2}[r', r'']$ (d) None of these

172. If "r" be position vector, "n" normal, "t" tangent then for curvature "k"

(a) $r'' = kt$ (b) $r'' = rn$

(c) $r'' = kn$ (d) $r'' = k$

173. If position vector r is function of arc length then the formula for curvature k is

(a) $k = r'' = \sqrt{(x'')^2 + (y'')^2 + (z'')^2}$

(b) $k = r'' = \sqrt{(x'')^2 + (y'')^2}$

(c) $k = 0$

(d) $k = \sqrt{x''^2 + y''^2}$

174. For the statement (173) if S be are length and k curvature with position vector $\vec{r}$

(a) $K = \frac{\vec{r}' \times \vec{r}''}{|r'|^3}$ (b) $\frac{\vec{r}' \times \vec{r}''}{|S'|^3}$

(c) $\frac{\vec{r}' \times \vec{r}''}{|S'|^3}$ (d) $\frac{\vec{r}' \times \vec{r}''}{|r'|^2}$

175. If K(Curvature) is zero at all the points then curve is a

(a) Plane (b) Circle

(c) Straight line (d) Zero

176. If τ (Torsion) is zero at all the points then curve is a

(a) Plane (b) Circle

(c) Parabola (d) Straight line

177. Necessary and sufficient condition for curve to be plane is

(a) $[r' r'' r'''] = 1$ (b) $[r', r'', r'''] = 0$

(c) $[r, r'] = 0$ (d) None of these

178. If tangent and binormal at a point of a curve make angles θ, ϕ respect with fixed direction, then

(a) $\frac{\sin\theta\, d\theta}{\sin\phi\, d\phi} = -\frac{k}{\tau}$ (b) $\frac{\sin\theta\, d\theta}{\sin\phi\, d\phi} = -\frac{k}{\tau}$

(c) $\frac{\sin\theta\, d\theta}{\sin\phi\, d\phi} = 0$ (d) None of these

179. If plane of curvature at every point of a curve passes through fixed point then curve is a

(a) Circle (b) Straight line

(c) Plane (d) None of these

180. If "n" is normal and "S" the arc length then scew curvature "n'" is given by

(a) $n' = \sqrt{\tau^2 + k^2}$ (b) $n' = \tau^2 - k^2$

(c) $n' = k'$ (d) $n' = n$

181. Equation of enter of curvature "C" with "f" radius of curvature, $\vec{r}$ position vector t normal n is given by

(a) $c + rn = 0$ (b) $c = r + \rho n$

(c) $c = r - \rho n$ (d) Zero

182. If radius of curvature "r" is constant for given curve C then K_1 curvature of C_1 and K curvature of C

(a) $k_1 k = 0$
(b) $\frac{k_1}{k}$
(c) $k_1 \neq k$
(d) $k_1 = k$

183. Torsion of the locus of center of curvature and Torsion of the given curve has relation

(a) $\tau_1 \alpha \frac{1}{\tau}$
(b) $\tau_1 \tau = -1$
(c) $\tau_1 \tau = 0$
(d) $\tau_1 = \tau$

184. If $\vec{r}$ be the position vector and S are length then

(a) $r'' \cdot r''' = 0$
(b) $r', r'' = 0$
(c) $r \cdot r'' = 1$
(d) $r' \cdot r''' = 0$

185. If $\vec{r}$ be position vector and S are length then

(a) $r' \cdot r'' = k^2$
(b) $r' \cdot r'' = -k^2$
(c) $r \cdot r \cdot r'' = k^2$
(d) $r' \cdot r = 0$

186. The principal normal at consecutive points do not intersect under Torsion (τ) is

(a) $\tau = 1$
(b) $\tau = \tau'$
(c) $\tau = 0$
(d) $\tau \cdot (f) = 0$

187. Plane of curvature always passes through

(a) One point
(b) Four point
(c) Three point
(d) Two parts

188. Circle of curvature always passes through

(a) Three points
(b) Two points
(c) One point
(d) None of these

189. Oscillating sphere passes through points

(a) 1
(b) 3
(c) 4
(d) 2

190. For the curvature k and k_1 of c an c_1 and Torsion τ and τ_1 of c and c_1 we can write as

(a) $kk_1 = \tau\tau_1$
(b) $k = k_1$
(c) $\tau = \tau_1$
(d) $s = \tau$

191. Necessary and sufficient conditions s for a curve to be a spherical curve is that

(a) $\frac{\rho}{\sigma} + \frac{d}{ds}\left(\frac{\rho'}{\tau}\right) = 1$
(b) $\frac{\rho}{\sigma} + \frac{d}{ds}\left(\frac{\rho'}{\tau}\right) = 0$
(c) $\frac{\rho}{\sigma} + \frac{d}{ds}\left(\frac{1}{\tau}\right) = 1$
(d) $\frac{\rho}{\sigma} + \frac{d}{ds}(\rho') = 1$

192. For intrinsic equation of the curve curvature and torsion is function of

(a) x and y
(b) S
(c) r
(d) Both a and c

193. Necessary and sufficient conditions for a curve to be a helix is that the ratio of

(a) $\frac{k}{\tau} = 1$
(b) $\frac{k}{\tau} = 0$
(c) $\frac{k}{\tau} = -1$
(d) $\frac{k}{\tau} = \text{constant}$

194. The curve is straight line if

(a) $\frac{k}{\tau} = 0$
(b) $\frac{k}{\tau} = 1$
(c) $\frac{k}{\tau} = 2$
(d) $\frac{k}{\tau} = -1$

195. Spherical indicatrix lies on the surface of a

(a) Unit plane
(b) Unit sphere
(c) None of these

196. For the space curve c, number of family of evaluates are

(a) Finite
(b) Infinite
(c) 2
(d) One

197. Locus of the center of curvature is an evaluate only when the curve is a

(a) Circle
(b) Straight line
(c) Plane
(d) Sphere

198. An equation of the form z = f(x, y) is called

(a) Monge's form
(b) Analytic form
(c) Implicit
(d) Explicit

199. Equation of normal plane at a point "P" t the surface is

(a) F(x, yz) = 1
(b) F(x, y, z) = 0
(c) F(x, y, z) = −1
(d) F(y) = x

200. Sum of squares of the intercepts made by the tangent plane to the surface $x^{2/3} + y^{2/3} + z^{2/3}$ is
(a) 1
(b) −1
(c) 0
(d) Constant

201. Equation F(x, y, z, a) = 0 represents the family of
(a) Circles
(b) Planes
(c) St-lines
(d) Spheres

202. The equation for the characteristic of the surface of parametric value "a" is
(a) $\frac{\partial F(a)}{\partial(a)} = 0$
(b) $\frac{\partial F(a)}{\partial a}$
(c) $\frac{\partial F(a)}{\partial a} = -1$
(d) $\frac{\partial F(a)}{\partial a} = a$

203. The locus of the characteristic of F(a) = 0 as "a" varies is called the
(a) Evaluate
(b) Envelops
(c) Involute
(d) Circle

204. Each characteristic touches the
(a) Edge of regression
(b) Circle
(c) Sphere
(d) St-lines

205. The envelope of the oscillating plane is called oscillating
(a) Sphere
(b) Plane
(c) Developable
(d) None of these

206. The envelope of the normal plane of a twisted curve is called the polar
(a) Sphere
(b) Developable
(c) Circle
(d) St-line

207. The envelope of the rectifying plane of a curve is called the developable
(a) Oscillating
(b) Polar
(c) Rectifying
(d) Sphere

208. Any relation f(u, v) = 0 on the surface represents a
(a) Curve
(b) Sphere
(c) Circle
(d) St-line

209. The curve on the surface, along which one of the parameters remain constant are called curves
(a) Parametric
(b) Polar
(c) Nonpolar
(d) Circle

210. The normal section of the surface having the maximum or minimum curvature are called sections
(a) Normal
(b) Principal
(c) Tangents
(d) Circle

211. The tangent lines along the principal sections at a point are called principal
(a) Principal sections
(b) Directions
(c) None of these

212. In minimal surface, the first curvature at all the points is
(a) 1
(b) −1
(c) 0
(d) None of these

213. The two principal directions at any point of a surface are
(a) Parallel
(b) Straight
(c) Orthogonal
(d) Circle

214. A surface is developable iff its specific curvature at all points is
(a) One
(b) Zero
(c) −1
(d) None of these

215. A surface generated pay the rotation of the plane curve about an axis in its plane is called a
(a) Circle of revolution
(b) Sphere of revolution
(c) Surface of revolution
(d) Straight line

216. A surface whose oscillating plane at each point contains the normal to the surface at the point called
(a) Geodesics
(b) Circles
(c) Spheres
(d) Lines

ANSWERS

1. d	2. a	3. b	4. c	5. d
6. a	7. b	8. b	9. a	10. c
11. b	12. b	13. c	14. a	15. d
16. a	17. b	18. a	19. c	20. a
21. c	22. b	23. a	24. c	25. a
26. d	27. c	28. c	29. a	30. c
31. a	32. c	33. a	34. c	35. a
36. b	37. a	38. b	39. a	40. c
41. a	42. b	43. b	44. d	45. a
46. a	47. c	48. b	49. b	50. c
51. a	52. b	53. c	54. d	55. a
56. b	56. c	57. b	58. a	59. d
60. a	61. c	62. b	63. a	64. c
65. d	66. a	67. b	68. c	69. b
70. d	71. a	72. d	73. c	74. a
75. d	76. d	77. a	78. c	79. d
80. a	81. d	82. a	83. b	84. d
85. a	86. c	87. a	88. b	89. d
90. a	91. b	92. c	93. a	94. b
95. c	96. a	97. d	98. a	99. a
100. c	101. d	102. c	103. a	104. b
105. a	106. c	107. b	108. a	109. b
110. b	111. a	112. c	113. a	114. d
115. a	116. b	117. b	118. c	119. b
120. c	121. a	122. c	123. a	124. d
125. b	126. c	127. a	128. d	129. a
130. c	131. a	132. b	133. c	134. a
135. c	136. a	137. b	138. b	139. c
140. c	141. a	142. b	143. d	144. c
145. a	146. b	147. b	148. c	149. b
150. c	151. a	152. b	153. a	154. c
155. c	156. a	157. b	158. c	159. a
160. d	161. a	162. a	163. c	164. a
165. d	166. a	167. b	168. c	169. b
170. b	171. a	172. c	173. a	174. b
175. c	176. a	177. b	178. a	179. c
180. a	181. b	182. d	183. a	184. b
185. b	186. c	187. d	188. a	189. c
190. a	191. b	192. b	193. d	194. a
195. b	196. b	197. c	198. a	199. b
200. d	201. d	202. a	203. b	204. a
205. c	206. b	207. c	208. a	209. a
210. b	211. b	212. c	213. c	214. b
215. c	216. a			

VECTOR – TENSOR – MECHANICS

➢ Tick (✓) the correct one answer only.

1. Vector is a physical quantity which has
   (a) Magnitude (b) Direction
   (c) Both a and b (d) None of these

2. For a vector, negative vector has same magnitude
   (a) Opposite direction
   (b) Same direction
   (c) None of these

3. If a vector is multiplied by (–2) then
   (a) Direction same magnitude 2 times
   (b) Direction reverse magnitude 2 times
   (c) Magnitude two times
   (d) Direction reverse only

For a negative of a vector, change in
   (a) Magnitude (b) Direction
   (c) Value (d) None of these

A linear expansion $\lambda a_1 + u a_2 + \ldots v a_n$ where $\vec{a}_1, \vec{a}_2, \vec{a}_3 \ldots \vec{a}_n$ are vectors and λ, u, v, ... are scalars called
   (a) Combination (b) Multiplication
   (c) Linear combination

It is the above statement $\lambda \vec{a}_1 + u\vec{a}_2 + \ldots v a_n = 0$ then this linear combination is called
   (a) Independent (b) Dependent
   (c) Addition (d) None of these

The necessary and the sufficient condition for two vectors to be parallel or antiparallel is that the vectors are linearly
   (a) Independent (b) Dependent
   (c) Zero (d) One

8. If the vectors are at right angles to each other than the basis is said to be
   (a) Normal (b) Parallel
   (c) Orthogonal (d) Tangent

9. Vector $\vec{A}$ in mathematical form is writ
   (a) $\vec{A} = |A|$ (b) $\vec{A}\ |A|\ \hat{A}$
   (c) $|A| = \hat{A}$ (d) $\vec{A} = \hat{A}$

10. Vector $\vec{A}$ in rectangular Cartesian co-ordinate system is represented by
   (a) $\vec{A} = x\hat{i} + y\hat{y} + z\hat{k}$
   (b) $\vec{A} = x\hat{i} - y\hat{\partial} - z\hat{k}$ (c) None of these

11. The magnitude of $\vec{A} = x\hat{i} + y\hat{z} + z\hat{k}$ is given by
   (a) $\sqrt{x^2 - y^2 - z^2}$ (b) $x^2 + y^2 + z^2$
   (c) $x^2 - y^2 - z^2$ (d) $\sqrt{x^2 + y^2 + z^2}$

12. When two vectors are multiplied we can give
   (a) Scalar (b) Vector
   (c) Both a and b (d) None of these

13. When two vectors dot(·) multiplied we get a
   (a) Vector (b) Scalar
   (c) Straight line (d) Circle

14. Vectors are added by a special rule know as
   (a) Addition (b) Head rule
   (c) Head to tail rule (d) None of these

15. When the vectors cross (X) multiplied w get
   (a) Vector (b) Scalar
   (c) Both a and b (d) None of these

16. If $\vec{a}$ and $\vec{b}$ two vectors making angle "Q" each other than
(a) $\vec{a} \cdot \vec{b} = \cos\theta$ (b) $\vec{a} \cdot \vec{a} = a\,b\cos\theta$
(c) $\vec{a} \cdot \vec{b} = ab$ (d) None of these

17. For unit vectors $\hat{i}$, $\hat{j}$ and $\hat{k}$ along X, Y and Z-axis respectively then
(a) $i \cdot i = j \cdot j = 1$ (b) $\hat{i} \cdot \hat{j} = \hat{j} \cdot k = 0$
(c) Both a and b (d) None of these

18. If $\vec{a}$ and $\vec{b}$ two vectors making angle θ with each other than
(a) $\vec{a} \times \vec{b} = ab\sin\theta\,\hat{e}$
(b) $\vec{a} \times \vec{b} = ab\sin\theta$
(c) $\vec{a} \times \vec{b} = 0$

19. If $\vec{a} \times \vec{b} = 0$ then angle "θ" between them is
(a) $\theta = 90°$ (b) $\theta = 0°$
(c) $\theta = 270°$ (d) $\theta = 45°$

20. If $\vec{a} \cdot \vec{b} = 0$ then angle "θ" between the vectors is
(a) $\theta = 0°$ (b) $\theta = 45°$
(c) $\theta = 180°$ (d) $\theta = 90°$

21. For unit vectors $\hat{i}$, $\hat{j}$ and $\hat{k}$ along X, Y and Z-axis is respectively then
(a) $\hat{i} \times \hat{j} = k$ (b) $j \times k = i$
(c) $i \times i = 0$ (d) All a, b and c

22. In cross product (X) of two vectors resultant on the plane formed by two is
(a) Perpendicular (b) Parallel
(c) Antiparallel (d) All of them

3. Scalar triple product of three vectors $\vec{a}, \vec{b} + \vec{c}$ is given by
(a) $\vec{a}, \vec{b} \times \vec{c}$ (b) $\vec{a} \times \vec{b} \times \vec{c}$
(c) $\vec{a} \times \vec{b}$ (d) $c \times b \times a$

For scalar triple product of vectors $\vec{a}$, $\vec{b}$ and $\vec{c}$
(a) $\vec{a} \cdot \vec{b} \times \vec{c} = \vec{a} \times \vec{b} \cdot \vec{c}$
(b) $\vec{a} \cdot \vec{b} \times \vec{c} = \vec{b} \cdot \vec{c} \times \vec{a}$
(c) $\vec{a} \cdot \vec{b} \times \vec{c} = \vec{a} \cdot \vec{b} \cdot \vec{c}$
(d) $\vec{a} \cdot \vec{b} \times \vec{c} = \vec{a} \times \vec{b} \times \vec{c}$

25. For two vectors $\vec{a}$ and $\vec{b}$ we can write as
(a) $\vec{a} \times \vec{b} = -\vec{b} \times a$ (b) $\vec{a} \cdot \vec{b} = \vec{b} \cdot \vec{a}$
(c) Both a and b (d) None of these

26. For vector product of three vectors $\vec{a}$, $\vec{b}$ and $\vec{c}$ we can get
(a) $\vec{a} \times (\vec{b} \times \vec{c}) = a \cdot (b - a \cdot bc)$
(b) $a \times (b \times c) = a \cdot b(+a \cdot a)$
(c) $\vec{a} \times (\vec{b} \times \vec{c}) = 0$
(d) $\vec{a} \times (\vec{b} \times \vec{c}) = \vec{a} \cdot \vec{c}\vec{b} + \vec{a} \cdot b \cdot \vec{c}$

27. For three vectors $\vec{a}$, $\vec{b}$ and $\vec{c}$ we can write
(a) $(a \times b) \times c \neq a \times (b \times c)$
(b) $(a \times b) \times c = (b \times c) \times a$
(c) $a \times b \times c = a \times (b \times c)$
(d) Both a and b

28. Scalar product of $\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$ and $\vec{b} = b_1\hat{i} + b_2\hat{j} + b_3\hat{k}$ then $\vec{a} \times \vec{b}$ is equal to
(a) $a_1b_1 + a_2b_2 + a_3b_3$
(b) $a_1b_1 - a_2b_2 - a_3b_3$
(c) $a_1b_1 + a_3b_3 = 0$
(d) None of these

29. $(a + b)\,(b + c) \times (c + a)$
(a) [a b c] (b) 2[a b c]
(c) Zero (d) $(a \times b)$

30. $(a \times b) \times (c \times d) =$
(a) [a c b]d – [b c d]b
(b) [a b c]d + [d c a]b
(c) [a c d]b – [b c d]a
(d) [b, cd]b + [b c d]a

31. $[a \times b \;\; b \times c \;\; c \times a] =$
(a) $[a\,b\,c]^2$ (b) [a b c]
(c) $[a\,b\,c]^{-1}$ (d) [a b d]

32. The necessary condition for three points to be collinear is that for vectors $\underline{a}$, $\underline{b} \times \underline{c}$
(a) $p\underline{a} + q\underline{b} - r\underline{c} = 0$ (b) $p\underline{a} + q\underline{b} + r\underline{c} = 0$
(c) $p\underline{a} + q\underline{b} = 0$ (d) None of these

33. The altitude of a triangle are
(a) Concurrent (b) Go to vertices
(c) Parallel (d) Perpendicular

34. The line segment joining the vertex of an iso celle triangle to the mid point of base is perpendicular to the
(a) Hypotenuse (b) Base
(c) Altitude (d) None of these

35. The medians of triangle are concurrent and their point of intersection divides each median in the ratio
(a) 2: 3 (b) 3: 1
(c) 1: 1 (d) 2: 1

36. There is bisection of diagonals in
(a) Triangle (b) Parallelogram
(c) Square (d) Both b and c

37. The line segments joining the mid point of the adjacent sides of a quadrilateral form a
(a) Parallelogram (b) Square
(c) Triangle (d) None of these

38. If two pairs of opposite edges of a tetrahedron are orthogonal, then the remaining pair of opposite edges are also
(a) Parallel (b) Orthogonal
(c) Linear (d) Triangle

39. The line segment joining the mid points of opposite edges of tetrahedron are concurrent and bisect each other at the points
(a) Different (b) Common
(c) One (d) Two

40. Vector equation of straight line with vector $\vec{r}$ and vectors $\vec{a} \times \vec{b}$ with parameter t is
(a) $\vec{r} = \vec{a} + t\vec{b}$ (b) $\vec{r} = \vec{a} + \vec{b}$
(c) $\vec{r} = \vec{a} - \vec{b}$ (d) $\vec{r} = \vec{a} - t\vec{b}$

41. If Ψ is scalar point function, then the relation $\Psi(x, y, z) = c$ (constant) represent
(a) Circle (b) Straight line
(c) Surface (d) Vector

42. The formation $\frac{\partial \Psi}{\partial x}\hat{i} + \frac{\partial \Psi}{\partial y}\hat{j} + \frac{\partial \Psi}{\partial z}\hat{k}$ called the
(a) Grade Ψ (b) Div. Ψ
(c) Curve Ψ (d) None of these

43. Grade of Ψ is a field of
(a) Scalar (b) Vector
(c) Space (d) None of these

44. Grade of Ψ is represented by
(a) $\nabla \cdot \Psi$ (b) $\nabla \times \Psi$
(c) $\nabla\Psi$ (d) None of these

45. For vector $\vec{A}\ a_1\hat{i}\ a_2\hat{j}\ a_3\hat{k}$ the relation $a_1\frac{\partial}{\partial x} + a_2\frac{\partial}{\partial y} + a_3\frac{\partial}{\partial z}$ is denoted by
(a) ∇A (b) $\nabla\ \vec{A}$
(c) $\nabla \times \vec{A}$ (d) Both a and b

46. Divergence of Ψ is a field of
(a) Scalars (b) Vectors
(c) Constants (d) None of these

47. Divergence of Ψ is represented by
(a) $\nabla \times \Psi$ (b) $\nabla\Psi$
(c) $\nabla \cdot \Psi$ (d) ∇

48. Operational relation $\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$ is denoted by
(a) ∇^2 (b) ∇
(c) $\nabla \times \nabla$ (d) $\nabla^2 - \nabla$

49. ∇^2 is a quantity
(a) Vector (b) Scalar
(c) Both a and b (d) None of these

50. Divergence grade Ψ
(a) $\nabla\Psi$ (b) ∇
(c) $\nabla^2\Psi$ (d) Ψ

51. Curl $(\nabla\Psi)$ =
(a) 0 (b) 1
(c) -1 (d) None of these

52. Div(Curl) $\vec{A}$ =
(a) 1 (b) -1
(c) None (d) 0

53. Curl curl $\vec{A}$ =
(a) Grad div $\vec{A} - \nabla^2 A$
(b) Grad div $\vec{A} + \nabla^2 A$

(c) Grad div A

(d) Grad div A – Curl A

54. For ϕ and Ψ scalars grad $(\phi\Psi)$ =

(a) Ψ grad ϕ + ϕ grad Ψ

(b) Ψ grad ϕ – ϕ grad Ψ

(c) Ψ grad ϕ

(d) ϕ grad Ψ

55. For vectors $\vec{A}$ and $\vec{B}$ grad $(\vec{A}\cdot\vec{B})$ =

(a) $\vec{B}\times$ Curl A + $\vec{A}\times$ Curl B

(b) $\vec{B}\times$ cure A – $\vec{A}\times$ Curl B

(c) B × curl A + A × curl B + B · ∇A + A · ∇B

(d) None of these

56. For scalar Ψ and vector $\vec{A}$ div$(\Psi\vec{A})$

(a) Ψ div A + $\nabla\Psi\cdot$A

(b) Ψ div A – $\nabla\Psi\cdot$A

(c) Ψ^2 div A – A^2 div Ψ

(d) $\nabla^2\Psi - \nabla A$

57. For vector $\vec{A}$ and $\vec{B}$ div$(\vec{A}\times\vec{B})$ =

(a) $\vec{B}$ curl $\vec{A}$ + $\vec{A}\cdot$ curl $\vec{B}$

(b) $\vec{B}\cdot$ curl $\vec{A}$ – $\vec{A}$ curl $\vec{B}$

(c) A curl B + ∇ · B

(d) A · ∇B + B · ∇A

58. For scalar Ψ and vector A curl (ΨA) =

(a) Ψ Curl $\vec{A}$ + grad $\Psi\times$A

(b) Ψ curl A – grad $\Psi\times$A

(c) Ψ + grad $\Psi\times$A

(d) Grad $\Psi\times$A + A · $\nabla\Psi$

59. For vectors $\vec{A}$ and $\vec{B}$ curl $(\vec{A}\times\vec{B})$ =

(a) A div B – B div A

(b) B · ∇A – A · ∇B

(c) A div B – B div A + B · ∇A – A · ∇B

(d) None of these

60. Unit tangential T of the curve $r(t) = \cos t\,\hat{i} + \sin t\,\hat{y} + t\hat{k}$

(a) $\frac{1}{2}(i\times k)$ (b) $-\frac{1}{2}(i+j)$

(c) $\frac{1}{2}(-j+k)$ (d) $\hat{i}+\hat{j}+\hat{k}$

61. Grad f(r) = where f'(r) represent derivative

(a) $f'(r)\,(r^{-1})\,\vec{r}$ (b) $f(r)\,\vec{r}$

(c) $f'(r)$ (d) $f'(r)\,r$

62. If $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$ then

(a) Div $\vec{r}$ = 3 (b) Div $\vec{r}$ = 0

(c) Div $\vec{r}$ = 1 (d) Div $\vec{r}$ = 0

63. If $r = x\hat{i} + y\hat{j} + z\hat{k}$ then curl $\vec{r}$ = 0

(a) 1 (b) –1

(c) 3 (d) 0

64. $(dr\cdot\nabla)$ = where $\vec{r}$ is vector then

(a) $d\vec{r}$ (b) $d\vec{A}$

(c) 0 (d) dr × A

65. Curl $(\vec{a}\times\vec{r})$ = (where $\vec{a}$ is constant vectors)

(a) $\vec{a}$ (b) 0

(c) $2\vec{a}$ (d) 1

66. Grad $(\vec{a}\cdot\vec{r})$ = (where $\vec{a}$ is constant vector)

(a) $\vec{a}$ (b) 0

(c) $2\vec{a}$ (d) –1

67. If $\vec{A}$ = A(t) and t = t(4) then $\frac{d\vec{A}}{dt}$ =

(a) 0 (b) –1

(c) 1 (d) None of thes

68. For vector function A(t) we can write

(a) $A\times\frac{dA}{dt} = 1$ (b) $A\times\frac{dA}{dt} = -1$

(c) $A\times\frac{dA}{dt}$ (d) None of the

69. $\frac{d}{dt}(A\times A)'$ =

(a) A × A' (b) A × A''

(c) A × A' + A'' × A (d) 2A''

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(c) $\int_S ds \times \vec{A} = \int_V \nabla Adv$

(d) None of these

102. Divergence theorem in grad form is given by

(a) $\int_S ds \cdot A = \int_V \nabla \cdot Adv$

(b) $\int_S ds \times A = \int_V \nabla \times Adv$

(c) $\int_S ds \cdot \Psi = \int_V \nabla \Psi Adv$

(d) None of these

103. Stokes theorem is given by

(a) $\int_C (\nabla \times \vec{A}) \cdot \hat{n} ds = \Phi \vec{A} \cdot \vec{dr}$

(b) $\int_C (\nabla \times \vec{A}) \cdot \hat{n} ds = \int_S \nabla \cdot va$

(c) $\int_S (\nabla \times \vec{A}) dx = \Phi \vec{A} \cdot \vec{V} dv$

(d) $\int_C (\nabla \times \vec{A}) ds = 0$

104. Stokes theorem in cross product form is

(a) $\int_C \vec{A} \times \vec{dr} = \int_S (ds \times \nabla) \times \vec{A}$

(b) $\int_C \vec{A} \times \vec{dr} = \int (\vec{ds} \times \nabla) \times \vec{A}$

(c) $\int_C A \times dr = \int_S ds \times A$

(d) None of these

105. Stokes theorem grad form is given by

(a) $\int_C dr = \int_S (dx \times \nabla)\psi$

(b) $\int_C dr\, \psi = -\int ds\, \psi$

(c) $\int_C dr\, \psi = -\int_S (ds \times \nabla)\psi$

(d) None of these

106. Standard form of Green's theorem is given by

(a) $\int_C (Pdx + Qdy) = \int_S \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) dx\, dy$

(b) $\int_C (Pdx + Qdy) = \int_S \frac{\partial Q}{\partial x} \frac{\partial P}{\partial y}$

(c) $\int_C \frac{\partial Q}{\partial x} \frac{\partial P}{\partial y} dy = 0$

(d) None of these

107. In tensor Analysis kronucker delta symbol

(a) $\partial_{ij} = 1 (i, j = 1, 2, 3)$

(b) $\partial_{ij} = 0 (i \neq j)$

(c) Both a and b

(d) None of these

108. In tensor analysis $A_i =$

(a) $\sum_j A_j \partial_{ij}$ (b) $\sum_j \partial_{ij}$

(c) ∂_{ii} (d) ∂_{ij}

109. In tensor analysis $e_i =$

(a) $\sum_j \partial_{ij}$ (b) $\sum_j A_i \partial_{jj}$

(c) $\sum_j e_j \partial_{ij}$ (d) None of these

110. In tensor analysis $\partial_{ik} = 0$

(a) $a_{ij} b_{jk}$ (b) $\sum_j a_{ij} b_{jk}$

(c) $\sum_j a_{ij}$ (d) $\sum_j b_{jk}$

111. In tensor analysis ∂_{jk}

(a) $\sum_i a_{ij} a_{ik}$ (b) $\sum_j a_{ij} a_{jj}$

(c) $\sum_i a_{ii}$ (d) $\sum_j a_{ij} b_{ij}$

112. Transformation equation for coordinates in tensor formation are

(a) $x_i' = \sum_j a_{ij} x_j$ (b) $x_i = \sum_j a_{ij} x_j$

(c) $x_j = \sum_i a_{ij} b_i$ (d) None of these

113. In tensor the transformation matrix $R = (a_{ij})$ is known as matrix of

(a) Base (b) Dependent

(c) Orthogonal (d) None of these

114. For matrix $R = (a_{ij})$ the transformation is proper if det R =
(a) −1 (b) 0
(c) +1 (d) Both a and b

115. For matrix $R = (a_{ij})$ the transformation is improper if det R
(a) 0 (b) 1
(c) α (d) −1

116. For matrix $R = (a_{ij})$ the transformation is called identity if det R
(a) I(identity matrix)
(b) 1 (c) −1
(d) 0

117. For matrix $R = (a_{ij})$ matrix of coordinates and T_2T_1 product of transformation is defined then this product is
(a) Vector space (b) Group
(c) Ring (d) Metric space

118. The transformation equation of scalars in tensor notation is
(a) $A_i = \sum_j a_{ji} A'_j$ (b) $A_i = \sum_j a_{ji}$
(c) $A_i = \sum_j a_{ij}$ (d) None of these

119. $A'_i = a_{ij} A_j$ in tensor notation in a Cartesian tensor of ran
(a) 4 (b) 3
(c) 2 (d) 1

120. $A'_{ij} = a_{ik} a_{jl} A_{kl}$ in tensor notation is a Cartesian tensor of rank
(a) 2 (b) 0
(c) −1 (d) 1

121. In tensor notation, if $A = (A_{ij})$, $A' = (A'_{kl})$ and $R = (a_{ij})$ the new can write as
(a) $A = R^t A R$ (b) $A' = R A R^t$
(c) $A = A'$ (d) $AA' = I$

122. If $A_{i1i2...ir}$ and $B_{i1i2...ir}$ are tensors of rank r then $(A_{i1i2...ir} \pm B_{i1i2...ir})$ also a tensor of rank
(a) $r - 1$ (b) $r + 1$
(c) r (d) 0

123. If $A_{i1i2...ir}$ and $B_{j1j2...js}$ are given tensors of rank r and s then the product $A_{i1i2...ir} B_{j1j2...js}$ is a tensor of rank
(a) rs (b) $r - s$
(c) r/s (d) $r + s$

124. A single contraction of a tensor $A_{i1i2...ir}$ will yield a tensor of rank
(a) $r - 2$ (b) $r + 2$
(c) $\frac{r}{2}$ (d) 0

125. The multi-indexed symbol in tensor is checked by the
(a) Quotient theorem
(b) Componedo
(c) Dividnedo
(d) None of these

126. The property of symmetry (Antisymmetry) of tensor is coordinate system is
(a) Dependent (b) Independent
(c) None of these

127. The tensor expressed as a sum of symmetric and antisymmetric tensors is a tensor of rank =
(a) 1 (b) 0
(c) −1 (d) 2

128. Tensor equation under coordinat transformation is
(a) Invariant (b) Not same
(c) Zero (d) Both b and c

129. If $A_i B_i$ is scalar and A_i arbitrary vect then B_i is a
(a) Scalar (b) Zero
(c) Vector (d) 1

130. Aij is symmetric tensor of rank 2 and B antisymmetric tensor of rank 2 then $A_{ij}B$ =
(a) 1 (b) −1
(c) 0 (d) +2

131. If Ψ is scalar and A_{ij} be second ra tensor then "ΨA_{ij}" tensor ha rank
(a) 2 (b) 1
(c) 0 (d) −1

132. If $A_{ij}B_j$ is a vector and A_{ij} arbitrary tensor of rank 2 then B_j is a
(a) Scalar (b) Vector
(c) Both a and b (d) Identity

133. The antisymmetric tensor A_{ij} in N- dimension has components
(a) N (b) N + 1
(c) N − 1 (d) $\frac{N(N-1)}{2}$

134. Alternating or levi-civita tensor "ϵ_{ijk}" has indexed symbols
(a) 1 (b) 2
(c) 3 (d) Zero

35. In levi-civita tensor when i, j and k are in cyclic permutation of 1, 2 and 3 then $\epsilon_{ijk} =$
(a) 0 (b) 1
(c) −1 (d) 2

36. In alternating tensor when i, j and k are in anticyclic permutation then
(a) 0 (b) 1
(c) −1 (d) 2

37. Alternating tensor ϵ_{ijk} has neither cyclic nor anticyclic permutation then $\epsilon_{ijk} =$
(a) 1 (b) −1
(c) 2 (d) 0

38. Transformation equation for levi-civita tensor $E'_{ijk} =$
(a) $a_{il}\, a_{jm}\, a_{kn}\, \epsilon_{lmn}$ (b) $a_{il}\, a_{jm}$
(c) $a_{il}a_{jm}$ (d) $a_{il}a_{jn}$

9. For levi-civita tensor we can write as
(a) $\epsilon_{ijk} \neq {}^{-}\epsilon_{jik}$ (b) $\epsilon_{ijk} = -\epsilon_{jik}$
(c) ϵ_{ijk} (d) ϵ_{kji}

0. Product of levi-civita tensor in the form of kronecker tensor is given by $\epsilon_{ijk}\epsilon_{lmk} =$
(a) $\delta_{il}\,\delta_{jm} - \delta_{im}\,\delta_{jl}$
(b) $\delta_{im}\,\delta_{jn} + \delta_{jm}\,\delta_{il}$
(c) $\delta_{il}\,\delta_{jm} + \delta_{im}\,\delta_{jl}$
(d) None of these

141. If $\vec{A}_i$ and $\vec{B}i$ (i = 1, 2, 3, ...) be given two vectors then $\vec{A} \cdot \vec{B}$ is given by
(a) $\sum_i A_iA_j$ (b) $\sum_k A_iA_k$
(c) $\sum_i A_iB_i$ (d) None of these

142. If there is vector $\vec{A} = \sum_i A_i I_i$ and $\vec{B} \sum_j B_j I_j$ then $(A \times B)_i =$
(a) $\epsilon_{ijk}A_jB_k$ (b) ϵ_{ijk}
(c) $-\epsilon_{ijk}A_jB_k$ (d) $\epsilon_{ijk}A_gB_k$

143. In vector notation $\vec{A} \times \vec{B} = 0$
(a) AB (b) $-\vec{B} \times \vec{A}$
(c) $-\vec{A} \times \vec{B}$ (d) −AB

144. If $\vec{A}$, $\vec{B}$ and $\vec{C}$ are vectors in proper tensor notations then scalar triple product $\vec{A} \times \vec{B} \cdot \vec{C}$ is given by
(a) $\epsilon_{ijk}A_iB_jC_k$ (b) $\epsilon_{ijk}A_jB_kC_i$
(c) $\epsilon_{ijk}A_jB_k$ (d) ϵ_{ijk}

145. In scalar triple product $\vec{A} \times \vec{B} \cdot \vec{C} =$
(a) $\vec{B} \cdot \vec{A} \times \vec{C}$ (b) $\vec{B} \times \vec{C} \cdot \vec{A}$
(c) $-\vec{C} \cdot \vec{A} \times \vec{B}$ (d) ABC

146. If $\vec{A}$, $\vec{B}$ and $\vec{C}$ are vectors in proper tensor notations then in form of levi-civita tensor we can write vector triple product $[\vec{A} \times (\vec{B} \times \vec{e})]_i$ as
(a) $\epsilon_{ijk}\epsilon_{klm}A_jB_lC_m$ (b) $\epsilon_{ijk}\epsilon_{klm}$
(c) $\epsilon_{gik}A_jB_lC_m$ (d) None of these

147. If $A_{i1,i2...ir}$ is Cartesian tensor of rank r then $\frac{d}{dx_j}(A_{i1,i2...ir})$ is a tensor of rank
(a) r (b) r − 1
(c) r + 1 (d) Zero

148. In tensor notation $(\text{grad}\ \Psi)_i =$
(a) $\frac{\partial \Psi}{\partial x_i}$ (b) $\frac{\partial \Psi}{\partial x}$
(c) $\frac{\partial \Psi}{\partial y}$ (d) $\frac{\partial^2 \Psi}{\partial \Psi^2}$

149. In tensor notation Div(A) is

(a) $\frac{\partial^2 A_i}{\partial x_i}$ (b) $\frac{\partial A_i}{\partial y_i}$

(c) $\frac{\partial A_j}{\partial y_i}$ (d) $\frac{\partial A_i}{\partial x_i}$

150. Curl $\vec{A}$ in tensor notation $(\text{Curl } A)_k$ =

(a) $\epsilon_{ijk} \frac{\partial A_j}{\partial x_i}$ (b) $\epsilon_{ijk} \frac{\partial A_j}{\partial x_j}$

(c) $\epsilon_{ijk} \frac{\partial A_j}{\partial x_k}$ (d) $\frac{\partial A_j}{\partial x_i}$

151. Mass per unit volume is known as
(a) Acceleration (b) Viscosity
(c) Density (d) Velocity

152. The force associated with mass m and acceleration due to gravity "g" is
(a) Weight (b) Moment
(c) Velocity (d) Acceleration

153. A body subject to no force is either at rest or moves uniformly is first law of
(a) Motion (b) Rest
(c) Gravity (d) None of these

154. If two forces are represented as adjacent sides of parallelogram then there resultant is given by the parallelogram
(a) Side (b) Diagonal
(c) None of these

155. Moment of momentum is called
(a) Impulse (b) Equilibrium
(c) Angular moment (d) None of these

156. Moment of inertia I_{xx} (about x-axis) of a body of mass m is given by

(a) $\Sigma m_i(r_i^2 - x_i^2)$ (b) $\Sigma m_i(r_i^2 x_i^2)$

(c) $\Sigma m_i(r_i^2)$ (d) None of these

157. Product of inertia "I_{xy}" of mass "m" is given

(a) $-\Sigma m_i y_i z_i$ (b) $+ \epsilon m_i x_i z_i$

(c) $-\Sigma m_i x_i y_i$ (d) Zero

158. Moment of inertia I_{yy} (about y-axis) is

(a) $\Sigma m_i (r_i^2 - z_i^2)$ (b) $\Sigma m_i(r_i^2 - y_i^2)$

(c) $\Sigma m_i(r_i^2 z_i^2)$ (d) None of these

159. The moment of inertia of thin uniform rod of length 2a mass M about its center is

(a) Ma^2 (b) $\frac{Ma}{2}$

(c) $\frac{Ma^2}{2}$ (d) $\frac{Ma^2}{3}$

160. The Moment of inertia of thin uniform rod of mass "M" and length "l" about its center is

(a) $\frac{1}{12} Ml^2$ (b) $\frac{1}{3} Ml$

(c) $\frac{1}{4} Ml^2$ (d) $\frac{1}{2} M^2 l^2$

161. The Moment of inertia of thin uniform rod of mass "M" of length "2a" about one of its end

(a) $\frac{1}{3} Ma^2$ (b) $\frac{4}{3} Ma^2$

(c) $\frac{1}{4} Ma^2$ (d) $\frac{1}{2} Ma^2$

162. The moment of inertia of thin uniform and of length "l" and mass "M" about one of its end is

(a) $\frac{1}{4} Ml^2$ (b) $\frac{1}{2} Ml^2$

(c) $\frac{1}{3} Ml^2$ (d) Zero

163. If A and B are moment of inertia of lamina about two ⊥ axis in its plane then the moment of inertia about the line through point of intersection and ⊥ to the lamina "C" is

(a) $C = a + b$ (b) $C = a - b$

(c) $C = ab$ (d) $C = \frac{a}{b}$

164. The moment of inertia of uniform rectangular laminas of mass M sides length 2a + 2b

(a) $I_{xx} = \frac{1}{3} b^2 M$ (b) $\frac{1}{3} a^2 M$

(c) Both a and b (d) None of these

165. The product of inertia for above statement no. (164) is

(a) $I_{xy} = 0$ (b) $I_{xy} = 1$

(c) $I_{xy} = -1$ (d) $I_{xy} = 0$

66. The moment of inertia of a hemisphere about its axis of symmetry is with mass "m"

(a) $\frac{2}{3}r^2 m$ (b) $\frac{2}{5}r^2 m$

(c) $\frac{2}{5}rm$ (d) $\frac{3}{2}r^2 m$

67. The moment of inertia of mass "m" hemisphere about axis perp. to axis of symmetry is

(a) $\frac{1}{2\pi}\frac{m}{r^3}$ (b) $\frac{3}{2\pi}\frac{m}{r^3}$

(c) $\frac{3}{2}\frac{m}{r^3}$ (d) Zero

68. The moment of inertia of hemisphere about axis of symmetry is

(a) Different (b) Double

(c) Same (d) None of these

69. The moment of inertia of a uniform circular cylinder about its axis where a be radius of base

(a) $I_{zz} = \frac{am}{2}$ (b) $I_{zz} = a^3 m$

(c) $I_{zz} = \frac{a^2 m}{2}$ (d) $I_{zz} = \frac{am^2}{2}$

0. The moment of inertia $\perp$ to the axis of cylinder that is I_{xx} or I_{yy} is (with height "h" and base radius a)

(a) $\frac{ma^2}{4} + \frac{mh^2}{12}$ (b) $\frac{ma^2}{4} + \frac{mh^2}{3}$

(c) $\frac{ma^2}{4} + \frac{mh^2}{12}$ (d) None of these

1. The moment of inertia for a rectangular parallelepiped about its edges and corners with side lengths a, b and c along X, Y and Z axis

(a) $I_{xx} = \frac{m}{3}(b^2 + c^2)$ (b) $I_{yy} = \frac{m}{3}(a^2 + c^2)$

(c) Both a and b (d) None of these

172. The moment of inertia sides "a" of a cube of mass "M" about X-axis and Y-axis is

(a) $\frac{2}{3}Ma^2$ (b) $\frac{1}{3}Ma^2$

(c) $\frac{1}{2}Ma$ (d) $\frac{2}{3}Ma^2$

173. The product of inertia of a cube of sides a is

(a) $-\frac{1}{4}Ma^2$ (b) $\frac{1}{2}Ma^2$

(c) Ma^2 (d) Zero

174. The volume of rectangular parallelepiped of sides a, b and c is v =

(a) ab^2 (b) abc

(c) $a^2b^2c^2$ (d) $\frac{ab}{c}$

175. Volume of circular cylinder cylinder of radius a and height h is

(a) πah (b) $\pi a^2 h$

(c) $4\pi a^2$ (d) $4\pi a$

176. Volume of sphere of radius a is given by

(a) $4\pi a^2$ (b) $\frac{1}{2}\pi a^3$

(c) $\frac{4}{3}\pi a^3$ (d) πa^2

177. The moment of inertia of uniform circular wire of radius "a" about x-axis

(a) $\frac{1}{2}a^2 m$ (b) $\frac{1}{2}am$

(c) $\frac{1}{2}a^2 m^2$ (d) $\frac{1}{4}a^2 m$

178. The circumference of a wire of radius a given by

(a) $2\pi a^2$ (b) $2\pi a$

(c) $2\pi a^3$ (d) $4\pi a^2$

179. The moment of inertia of uniform circular disc of radius "a" is

(a) $\frac{1}{4}ma^2$ (b) ma^2

(c) $\frac{1}{3}ma^3$ (d) $\frac{1}{6}ma^2$

80. Moment of inertia of uniform circular disc of radius of radius a about X-axis and Y-axis
(a) Different (b) 1
(c) Same (d) Zero

81. The moment of inertia of uniform circular disc of radius a through center and perpendicular to disc is
(a) $\frac{1}{2}ma^2$ (b) ma^2
(c) Zero (d) $\frac{1}{3}ma^3$

82. The circular area of uniform disc of radius a is
(a) πa^3 (b) $\frac{\pi}{2}a^2$
(c) πa^2 (d) $\pi^2 a^2$

3. The moment of inertia of an elliptic disc of axis is 2a and 2b about major axis
(a) $\frac{1}{4}Mb^2$ (b) $\frac{1}{4}M\pi^2$
(c) $\frac{1}{4}\pi m$ (d) $\frac{1}{4}M^2$

4. The moment of inertia of an elliptic disc of axis 2a and 2b about minor axis
(a) $\frac{1}{2}Mb$ (b) $\frac{1}{4}Ma^2$
(c) $\frac{1}{2}Ma^2$ (d) $\frac{1}{3}Ma$

5. The moment of inertia of hollow sphere about diameter of radius a is
(a) $\frac{2}{3}Ma^2$ (b) $\frac{2}{3}M^2$
(c) $\frac{2}{3}M^2a^2$ (d) $\frac{1}{3}mb^2$

6. The moment of inertia of solid sphere about diameter is
(a) $\frac{2}{3}Ma^2$ (b) $\frac{2}{3}Ma$
(c) $\frac{2}{5}Ma^2$ (d) $\frac{2}{5}Ma$

7. The moment of inertia of uniform solid sphere of radius "a" about X-axis, Y-axis and Z-axis
(a) Different (b) Same
(c) Zero (d) 1

188. Volume of a uniform solid sphere of radius a is
(a) $\frac{4}{3}\pi a^2$ (b) $\frac{4}{3}\pi a$
(c) $\frac{4}{3}\pi a^2$ (d) $\frac{4}{3}\pi a^4$

189. The moment of inertia for ellipsoid about X-axis is given by
(a) $\frac{8}{3}m(b^2+c^2)$ (b) $\frac{8}{3}mb^2$
(c) $\frac{8}{3}ma^2$ (d) Zero

190. Volume of ellipsoid with a, b and c intercepts on X-axis, Y-axis and Z-axis is
(a) a^2bc (b) ab^2c
(c) abc^2 (d) abc

191. Product of inertia for ellipsoid is
(a) $-2mab$ (b) $-2mbc$
(c) $-2mac$ (d) All(a, b and c)

192. The axis relative to which the product of inertia vanishes (zero) called
(a) Principal axis (b) Axis
(c) Circles (d) Parallel linear

193. The principal axis to each other are
(a) Parallel (b) Orthogonal
(c) Zero (d) None of these

194. Rotational K.E. of a body with angula[r] velocity W and radius vector $\vec{r}$ is given b[y]
(a) $K.E. = r^2w$ (b) $K.E. = rw^2$
(c) $v = \frac{1}{2}rw$ (d) $K.E. = \frac{1}{2}\underline{W} \times L$

195. Equation of momental ellipsoid with A, [B] and C moment of inertia and F, G, [H] product of inertia is given by
(a) $Ax^2 + By^2 + CZ^2 + 2FYZ + 2GZx + 2Hxy = 1$
(b) $Ax^2 + By^2 + CZ^2 - 2FYZ + 2GZx - 2Hxy = 1$
(c) $Ax^2 - By^2 - CZ^2 - 2FYZ - 2GZx - 2H[xy] = 0$
(d) None of these

196. Two distribution of matters are said to be equimomental system if they have same mass and same
(a) Dimensions
(b) Velocity
(c) Moment of inertia
(d) None of these

197. If there is a rod having masses $\frac{M}{6}$ at both ends and $\frac{2}{3}$M at center then it will be
(a) Equimomental (b) Stable
(c) Motion (d) None of these

198. When A', B' and H' are moment of inertial about z = 0 in XY plane then equation of momental ellipsoid is given by
(a) $A'x^2 - B'y^2 + 2H'xy = 0$
(b) $A'x^2 + B'y^2 + 2H'xy = 1$
(c) Zero
(d) None of these

199. A uniform rectangular lamina sides 2a + 2b then direction of principal axis with angle "α" is
(a) $\tan 2\alpha = \frac{3ab}{2(a^2 + b^2)}$
(b) $\tan \alpha = \frac{3a^2b^2}{2(a^2 - b^2)}$
(c) $\tan \alpha = \frac{3ab}{2(a^2 - b^2)}$
(d) None of these

200. The triangle ABC will have equimomental system if the particles placed at the mid points of the sides have masses each
(a) m (b) $\frac{m}{2}$
(c) $\frac{m}{4}$ (d) $\frac{m}{3}$

201. The equations of motion of rigid body when we have fixed axis
(a) $L_x^{\cdot} = W^{\cdot}I_{xz} - WI_{y}$
(b) $L_y^{\cdot} = W^{\cdot}I_{yz} + WI_{x}$
(c) $L_z^{\cdot} = W^{\cdot}I_{zz}$
(d) All of a b and c

202. General Euler's dynamic equation with moment of inertia "I" angular velocity W and angular momentum L is given by
(a) $L^{\cdot} = I\,\underline{W}^{\cdot} + \underline{W} \times L$
(b) $L^{\cdot} = I\,\underline{W}^{\cdot} - \underline{W} \times L$
(c) $L^{\cdot} = I\,\underline{W}^{\cdot}$
(d) $L^{\cdot} = \underline{W} \times L$

203. If $L = (L_1, L_2, L_3)$ is the angular momentum and A, B and C are principal moment of inertia then one of Euler's equation of motion
(a) $L_1 = AW_1^{\cdot} + (B - C)W_2W_3$
(b) $L_1 = AW_1 + (B + C)W_2W_3$
(c) $L_1 = AW_1^{\cdot} - (B - C)W_2W_3$
(d) $L_1 = AW_1^{\cdot} + (B + C)W_2^{\cdot}W_3^{\cdot}$

ANSWERS

1. c	2. a	3. b	4. b	5. c
6. a	7. b	8. c	9. b	10. a
11. d	12. c	13. b	14. c	15. a
16. b	17. c	18. a	19. b	20. d
21. d	22. a	23. a	24. b	25. c
26. a	27. d	28. a	29. b	30. c
31. a	32. b	33. a	34. b	35. d
36. d	37. a	38. b	39. b	40. a
41. c	42. a	43. b	44. c	45. b
46. a	47. c	48. a	49. b	50. c
51. a	52. d	53. a	54. a	55. c
56. a	57. b	58. a	59. c	60. c
61. d	62. a	63. d	64. b	65. c
66. a	67. a	68. c	69. b	70. c
71. a	72. b	73. d	74. a	75. b
76. c	77. a	78. d	79. a	80. b
81. c	82. a	83. a	84. c	85. a
86. b	87. c	88. a	89. a	90. a
91. c	92. b	93. a	94. c	95. a
96. c	97. a	98. b	99. c	100. a
101. b	102. c	103. a	104. b	105. c

106. a	107. c	108. a	109. c	110. b
111. a	112. a	113. c	114. c	115. d
116. a	117. b	118. a	119. d	120. a
121. b	122. c	123. d	124. a	125. a
126. b	127. d	128. a	129. c	130. c
131. a	132. b	133. d	134. c	135. b
136. c	137. d	138. a	139. b	140. a
141. c	142. d	143. b	144. a	145. b
146. a	147. c	148. a	149. d	150. a
151. c	152. a	153. a	154. b	155. c

156. a	157. c	158. b	159. d	160. a
161. b	162. c	163. a	164. c	165. a
166. b	167. b	168. c	169. c	170. a
171. c	172. d	173. a	174. b	175. b
176. c	177. a	178. b	179. a	180. c
181. a	182. c	183. a	184. b	185. a
186. c	187. b	188. d	189. a	190. d
191. d	192. a	193. b	194. d	195. a
196. c	197. a	198. b	199. c	200. d
201. d	202. a	203. c		

TOPOLOGY & FUNCTIONAL ANALYSIS

➤ **Tick (✓) the correct one answer only.**

1. Members of T(topology) called
   (a) Open sets (b) Compliment set
   (c) Empty set (d) Subset

2. If any number of sets have their union and intersection open with ϕ and X open then it is called
   (a) Real space
   (b) Complete space
   (c) Topological space
   (d) None of these

3. The ordered pair (X, T) is called
   (a) Topological space
   (b) Cartesian product
   (c) Metric space
   (d) None of these

4. If T_1 and T_2 between two topology on "X" and $T_1 \subset T_2$ then T_1 is called
   (a) Stronger
   (b) Finer
   (c) Coarser
   (d) Weaker and finer

Let $X \neq \phi$, and P(X) is power set of X, then P(X) is
   (a) Base (b) Metric on X
   (c) Complete space (d) Topological

For X, P(X) is called topology
   (a) Discrete (b) Indiscrete
   (c) Complete (d) Base

$\{X, \phi\}$ is a topology on X and is called topology
   (a) Indiscrete (b) Discrete
   (c) Complete (d) None of these

8. An open ball in metric topology is denoted by
   (a) B(X, r) (b) B′(x, r)
   (c) (X, r) (d) (x′, r)

9. Topology
   $T = \{u|a, b[, a, b \in R\ b > a\}$ is called
   (a) Usual topology
   (b) Metric topology
   (c) Discrete topology
   (d) None of these

10. If compliment of every openset is finite then it is
   (a) Usual topology
   (b) Finite topology
   (c) Co-finite topology
   (d) Metric topology

11. Let (X, T_x) be topological space and $y \subset X$ and $Y \neq \phi$ then $T_y = \{y \cap u : u \in T_x\}$ defined is called
   (a) Relative topology
   (b) Usual topology
   (c) co-finite topology
   (d) None of these

12. If there is topology $T_y = \{Y \cap u; u \in T_X\}$ in "$Y \cap u$" in "T_y"
   (a) Closed (b) Empty
   (c) Open (d) Finite

13. If (Y, T_y) is a subspace of (X, T_X) then (X, T_X) called
   (a) Parent space
   (b) Usual topology
   (c) Co-finite topology
   (d) None of these

14. If X is finite then each topology defined on X is

(a) Infinite (b) Finite
(c) Closed (d) Empty

15. Let (X, T) be a topological space and A be nonempty subset of X, then X/A be open then A is

(a) Infinite (b) Empty
(c) Open (d) Closed

16. Let(X, T) be topological space, union of finite number of closed set is

(a) Open (b) Finite
(c) Closed (d) Infinite

17. If (X, T) be topological space, intersection of finite number of closed set is

(a) Closed (b) Open
(c) Finite (d) Infinite

18. For (X, T) topological space ϕ and X are always

(a) Open (b) Closed
(c) Finite (d) Both a and b

19. If (X, T) topological closure of a subset "A" of X is denoted by

(a) $\bar{A}$ (b) A'
(c) A° (d) None of these

20. Intersection of all the closed supersets of A is called set

(a) Closed (b) Closure
(c) Open (d) Infinite

21. Let (X, T) be topological space an $A \subseteq X$ then A is closed iff

(a) $\bar{A} = A$ (b) $A = A'$
(c) A is empty (d) A is open

22. If $\bar{A}$ be closure of A then

(a) $\bar{A} \neq A$ (b) $\bar{A} = \phi$
(c) $\bar{\bar{A}} = \bar{A}$ (d) $\bar{A} = A'$

23. For any subset A an B of (X, T)

(a) $A \subseteq B \Rightarrow \bar{A} \subseteq \bar{B}$
(b) $\overline{A \cap B} = \bar{A} \cup \bar{B}$
(c) Both (a) and (b)
(d) None of these

24. For any subset "A" and "B" of (X, T)

(a) $\overline{A \cup B} \neq \bar{A} \cup \bar{B}$
(b) $\overline{A \cap B} \subseteq \bar{A} \cap \bar{B}$
(c) $A' \cap B' \neq \phi$
(d) None of these

25. In (X, T) topological space interior of A is denoted by

(a) $\bar{A}$ (b) A'
(c) A^0 (d) None of these

26. If (X, T) be topological space then interior of A is defined as

(a) $A^0 = \{UB, B \in I, B \subseteq A\}$
(b) $A^0 = \{UA\}$
(c) $A^0 = UA'$
(d) None of these

27. In t(A) or A^0 is an

(a) Closed set (b) Super set
(c) Empty set (d) Open set

28. If (X, T) be topological space them the $A \subseteq X$ and largest open set contained in A is

(a) A^0 (b) $\bar{A}$
(c) A' (d) None of these

29. Let (X, T) be topological space and $A \subseteq X$ then

(a) $A^0 \subseteq A$ (b) $A = A^0$
(c) $A \neq A^0$ (d) $A^0 = A'$

30. Let (X, T) be topology T $A \subseteq X$ then A is open iff

(a) $A^0 \neq A$ (b) $A = A^0$
(c) $A - A^0 = \phi$ (d) None of these

31. For an interior of a set A^0

(a) $(A^0)^0 = A^0$ (b) $A^0 = \bar{A}$
(c) $A^0 = A'$ (d) $A^0 = \phi$

32. For any two subsets A & B of 'X' If $A \subseteq B$

(a) $A^0 \supseteq B^0$ (b) $A^0 \neq B^0$
(c) $A^0 \subseteq B^0$ (d) $A \cap B = \phi$

33. In topological space (X, T) and A & B be subset of X $(A \cap B)^0$ =

(a) $A^0 \cup B^0$ (b) $A^0 - B^0$

(c) $A^0 \cap B$ (d) $A^0 \cap B^0$

34. For (X, T) topology space and A & B be subset of X

(a) $(A \cup B)^0 = A^0 \cup B^0$

(b) $(A \cup B)^0 \neq A^0 \cup B^0$

(c) $(A \cup B)^0 \supseteq A^0 \cup B^0$

(d) Both (b) and c

35. For (X, T) topological space $A \subseteq X$ then the set A = {UB, B ∈ T, B ⊆ X | A} is called

(a) Exterior of A (b) A^0

(c) A' (d) None of these

36. In topological space (X, T), $A \subseteq X$ then

(a) Ext(A) = Int(X)

(b) Ext(A) = Int(X − A)

(c) Ext A ≠ Int X − A

(d) Ext(A) = ϕ

37. Let (X, T) be topological space and $A \subseteq X$ then

(a) (Int A) ∪ (Ext A) = ϕ

(b) (Int A) − (Ext A) = ϕ

(c) (Int A) ∩ (Ext A) = ϕ

(d) Int A = Ext A

38. In(X, T) topological space Ext(A) is an

(a) Open set (b) Closed set

(c) Empty set (d) Subset

39. For (X, T) topological space the largest open set contained in (X − A) is

(a) Closed set (b) Open set

(c) Exterior of set (d) None of these

40. For any two subset A & B of (X, T) Ext(A ∪ B)

(a) Ext(A) ∩ Ext(B) (b) Ext A − Ext B

(c) Ext(A ∩ B) (d) In (A ∪ B)

41. For any two subsets A & B of (X, T)

(a) Ext(A ∩ B) = Ext(A) ∪ Ext(B)

(b) Ext(A ∩ B) = Int A + Int B

(c) Ext(A ∩ B) ⊇ Ext(A) ∪ Ext.(B)

(d) None of these

42. Let (X, T) topological space and $A \subseteq X$ then Int(X − A) =

(a) Int(X − A') (b) $X - \bar{A}$

(c) X − A (d) X − A'

43. Let (X, T) topology space and $A \subseteq X$ then

$\overline{X - A}$ =

(a) X − Int A' (b) X − A'

(c) $X - A^0$ (d) X − Int A

44. If (X, T) be topological space and $A \subseteq X$ then if there is a point "x ∈ X" which is neither an interior point and nor an exterior point called

(a) Frontier point (b) Boundary of A

(c) Both a and b (d) None of these

45. Frontier of A (Frt(A)) =

(a) X − {Ext(A) ∪ Int(A)}

(b) X − Ext(A)

(c) X-Int A

(d) None of these

46. Frontier of A Fr(A) =

(a) A ∩ X/A (b) $\bar{A} \cap \overline{X/A}$

(c) A ∪ X/A (d) A − (X − A)

47. If (X, T) be topological space and

$A \subseteq X$ then $\bar{A}$ =

(a) A − Fr(A) (b) A ∩ Fr(A)

(c) A' − Fr(A) (d) A ∪ Fr(A)

48. Let (X, T) be topological space and $A \subseteq X$ then A^0 =

(a) A + Fr(A) (b) A − Fr(A)

(c) A' − A (d) A' − Fr(A)

49. If Frontier of A is subset of any set then that set is

(a) Closed set (b) Open set

(c) Closure of set (d) Super set

50. If A be subset of (X, T), there is A is both open and closed then A has empty

(a) Frontier (b) Closed set

(c) Open set (d) Sub set

51. In topological space (X, T), $x_0 \in X$ and N $\subseteq$ X and "u" open set, then N nbhd of x_0 if

(a) $x_0 \in N \subseteq u$ (b) $x_0 \in u \subseteq N$

(c) $x_0 \in N$ (d) $x_0 \in u \subseteq X$

52. For nbhd system N_{x_0} of topology (X, T) and $A \subseteq X$ containing a member of N_{x_0} then

(a) $A \notin N_{x_0}$ (b) $A = N_{x_0}$

(c) $A \in N_{x_0}$ (d) None of these

53. For (X, T) topological space $A \subseteq X$ and U open set containing x is limit point of A if

(a) $U \cap (A/x) = \phi$ (b) $U \cap (A/x) \neq \phi$

(c) $UAX = \phi$ (d) $U \cap A' = A$

54. For topological space X and A subset of X A is dense if

(a) $A = X$ (b) $A' = X$

(c) $\bar{A} = X$ (d) $\bar{A} = \phi$

55. For topological space X and $A \subseteq X$, A is somewhere dense if

(a) $\bar{A} = \phi$ (b) $A^0 \neq \phi$

(c) $\bar{A} \neq \phi$ (d) $(\bar{A})^0 \neq \phi$

56. For topological space X and $A \subseteq X$. A is nowhere dense if

(a) $(\bar{A})^0 = \phi$ (b) $\bar{A} = A$

(c) $\bar{A} = \phi$ (d) None of these

57. Let A be subset of (X, T) then $\bar{A}$ =

(a) $A \cap A^d$ (b) $A - A^d$

(c) $A \cup A^d$ (d) A − {open set}

58. A is closed if and only if

(a) $A^d = A$ (b) $A^d \subseteq A$

(c) $A^d = \phi$ (d) $A = \phi$

59. Subset A of X is dense if there is open set "U" such that

(a) $A \cap U = \phi$ (b) $A \cap U = A'$

(c) $A \cap U \neq \phi$ (d) $A' \cap U = \phi$

60. In (R, T) topological space, Q set of rationals is dense if where R real Nos.

(a) $\bar{Q} = R$ (b) $\bar{Q} \neq R$

(c) $Q \subseteq R$ (d) $Q/R = \phi$

61. A space (X, T) is seprable if there is A subset of X which is countable and

(a) $A = X$ (b) $\bar{A} \neq X$

(c) $\bar{A} = X$ (d) $\bar{A} = \phi$

62. If U open set of (X, T) and $B \subseteq T$, then "B" is called basis for T if for each open set U

(a) $B = \{B\alpha : \alpha \in \Omega \neq \phi, U \in T,$
$U = UB\alpha, \Omega' \in \Omega\}$

(b) $B = \{B_\alpha \neq \phi\}$

(c) $B = \{B_\alpha' = \phi,\}$

(d) None of these

63. Let T = {U]a, b[where a, b ∈ R usual topology on R and
B = {]a, b[, a b ∈ R} then B is a

(a) Base (b) Closed

(c) Both a and b (d) None of these

64. A family B of subset of (X, T) is a base for T if and only if for $\{B_\alpha = \alpha \in \Omega\} = B$

(a) $X = UB_\alpha$

(b) For $B_1, B_2, B_3 \in B, x \in B_3 \subseteq B_1 \cap B_2$

(c) Both (a) and (b)

65. A topological space (X, T) is called first countable space if every local base of every point of X is

(a) Countable (b) Uncountable

(c) Infinite (d) Finite

66. A topological space (X, T) is second countable, if T have a countable

(a) Subset (b) Sub base

(c) Base (d) None of these

67. If (X, T) topological space and $C = \{U_\alpha : \alpha \in \Omega, U_\alpha \in t\}$ then C is a cover for X if

(a) $X = \cap U_\alpha \; \alpha \in \Omega$

(b) $X = U_\alpha$

(c) $X = \phi$

(d) $X = UU_\alpha \; \alpha \in \Omega$

68. Closed subspaces of lindelöf space is

(a) Separable (b) Lindelöf
(c) Base (d) Subspace

69. Every second countable space is

(a) Lindelöf (b) Base
(c) Separable (d) Ist countable

70. Every second countable space is

(a) Separable (b) Lindelöf
(c) Base (d) Both (a) and (b)

71. Let (X, T) and (Y, T^*) b two topological spaces and f: x → Y is constant mapping then

(a) f continuous (b) Discontinuous
(c) One-one (d) On-to

72. Let(X, Y) be topological space A f: x → Y is continuous if and only if in X

(a) $f^{-1}(v)$ is open (b) $f^{-1}(v)$ is closed
(c) Both a, b (d) None of these

73. A function f: x → y is continuous iff for every subset A of X

(a) $f\bar{A} \neq \overline{f(A)}$ (b) $f(\bar{A}) \subseteq \overline{f(A)}$
(c) $\bar{A} \subseteq \overline{f(A)}$ (d) $A = \bar{A}$

74. If f: x → y and g: y → z be continuous functions then composite function go f : y → z is

(a) Continuous (b) Discontinuous
(c) On-to (d) One-one

75. If in X and Y topology f: x → y is bijective, f^{-1} is continuous and f is continuous them "f" is

(a) Monomorphism (b) Epimorphism
(c) Homomorphism (d) None of these

76. Let f: x → y be bijective function and continuous then f^{-1} is continuous if and only if

(a) f is open (b) f is closed
(c) f is isomorphic (d) None of these

77. If (X, T) topology space is compact iff every open cover of x has subcover

(a) Infinite (b) Finite
(c) Odd (d) Even

78. Every closed subspace of compact is

(a) Seprable (b) Finite
(c) Compact (d) Infinite

79. Let X be infinite set with co-finite topology "T" then (X, T) is space

(a) Compact (b) Seprable
(c) Complete (d) Lindelöf

80. In a topological space X if every open cover of X has finite sub cover then it is called

(a) Seprable (b) Complete
(c) Compact (d) Lindelöf

81. Every closed and bounded interval in R is

(a) Compact (b) Seprable
(c) Open (d) Closed

82. In topology space (X, T) if every sequence in X has convergent subsequence then X is

(a) Compact
(b) Sequentially compact
(c) Seprable
(d) None of these

83. Every finite subset of topological space is

(a) Complete
(b) Compact
(c) Sequentially compact
(d) None of these

84. Le "X" topological space then if X is compact then

(a) Countable compact
(b) Seprable
(c) Closed
(d) Base

85. In topological space X very sequentially compact space is

(a) Seprable
(b) Closed (c) Open
(d) Countable compact

86. A space X is said to satisfy Bolzaan Weirstrass property iff every infinit subset of X has

(a) Limit point in X (b) Closed in X
(c) Open in X (d) None of these

87. Let (X, T) be topological space then it is To-space if for any x, y ∈ X
(a) $x \neq y$ (b) $x = y$
(c) $x = y = \phi$ (d) None of these

88. A space X is a To-space iff for any a, b ∈ X and a ≠ b implies that
(a) $\{\bar{a}\} \neq \{\bar{b}\}$ (b) $\{\bar{a}\} = \{\bar{b}\}$
(c) $\{\bar{a}\} = \phi$ (d) $\{\bar{b}\} = \phi$

89. A topological space X is T_1-space if for a, b, ∈ X a ≠ b there exist two open sets "U" and v such that
(a) $a \in u$, $b \in v$ and $b \notin u$ and $a \notin v$
(b) $u \cap v = \phi$
(c) $a \in v$ and $b \notin v$
(d) None of these

90. Every To-space is
(a) Not T_1 space (b) T_1-space
(c) Closed space (d) None of these

91. A topological space (X, T) is a T_1-space iff every singleton subset of X is
(a) Open (b) Closed
(c) Base (d) Sub-base

92. If X is T_1-space then each subset "A" of X is inter section of its open
(a) Subsets (b) Base
(c) Supersets (d) Closed

93. A space X is T_1 iff every subset finite of X is
(a) Open (b) Subset
(c) Closed (d) None of these

94. Every finite T_1-space is
(a) Discrete (b) Indiscrete
(c) Lindelöf (d) Seprable

95. If a T_1-space no finite subset has limit point
(a) 2 (b) 3
(c) 4 (d) One

96. A topological space (X, T) is T_2(Housdroff) if for x, y ∈ X x ≠ y there are u and v open sets such that x ∈ u and y ∈ v there is
(a) $u \cap v \neq \phi$ (b) $u \cap v = \phi$
(c) $u = v$ (d) $u \cup v = u$

97. Subspaces of Housdroff-spaces are
(a) T_0 spaces (b) T_1 spaces
(c) T_2 spaces (d) T_3 spaces

98. If X is T_2 (Housdroff space) the dignal $D = \{(x, x) : x \in x\}$ is closed in
(a) $X \times X$ (b) X
(c) D (d) None of these

99. Every compact subset of Housdroff-space is
(a) Open (b) Closed
(c) Base (d) Both a and c

100. If X and Y are topological spaces and A is closed subset of X then closed subset of Y is
(a) X' (b) $\bar{A}$
(c) f(A) (d) A'

101. A bijective continuous function from compact space to a Housdroff space is
(a) Homomorphism (b) Epimorphism
(c) Monomorphism (d) None of these

102. Every countably compact metric space is
(a) Bounded (b) Totally bounded
(c) Subspace (d) None of these

103. In topology space X, for open set u and x ∈ u there exist open sets u + v such that $A \subseteq U$ and $U \cap V = \phi$ called space
(a) T_1 space (b) T_2 space
(c) Regular (d) To

104. Every subspace of a regular space is
(a) Regular space (b) T_1 space
(c) T_2 space (d) T_0 space

105. A regular T_1 space is called as
(a) T_2 space (b) T_0 space
(c) T_3 space (d) None of these

106. Every Regular (T_3 space) is a
(a) T_0-space (b) T_1-space
(c) T_3-space (d) T_2 -space

107. If for any open set "u" in X and $x \in u$, there is open set v containing x such that $x \in \bar{v} \subseteq u$ then space is called
(a) T_2-space (b) T_1-space
(c) T_0-space (d) None of these

108. If each element of "X" has a nbhd base consisting of closed sets then space is
(a) T_0-space (b) T_1-space
(c) T_2-space (d) T_3-space

109. Every subspace of a completely regular space is
(a) Completely regular
(b) T_0-space (c) T_1-space
(d) T_2-space

110. Every subspace of Tychnoff space is
(a) T_0 space (b) T_1 space
(c) T_2 space (d) Tychonoff

111. Every Tychonoff space is
(a) Housroff (b) Complete
(c) Normal (d) T_0-space

112. In topological X, if for any disjoint pair of closed sets A and B there exist open sets U and V such that $A \subseteq U$ and $B \subseteq V$ and $U \cap V = \phi$ then
(a) Normal space (b) T_0-space
(c) T_1-space (d) T_2-space

113. Every metric-space is
(a) T_0-space (b) Normal space
(c) T_1-space (d) T_2-space

114. Normal space may not be a
(a) T_1-space (b) T_3-space
(c) T_2-space (d) T_0-space

115. Every closed subspace of a normal space is
(a) T_0-space (b) Normal
(c) T_1-space (d) T_2-space

116. A closed continuous image of normal space is
(a) T_1-space (b) T_2-space
(c) Normal (d) T_0-space

117. A normal T_1-space is
(a) T_4-space (b) T_2-space
(c) T_3-space (d) T_0

118. If in topological space X, there are non-empty open sets U and V such that $A \cap B = \phi$ and $A \cup B = X$ called
(a) T_1 space
(b) Disconnected space
(c) T_2-space
(d) T_3-space

119. A topological space represented by union of two disjoint non-empty closed sets called
(a) Connected (b) T_0-space
(c) T_1-space (d) T_2-space

120. Every indiscrete space is
(a) T_1-space (b) T_0-space
(c) Connected (d) Normal

121. An infinite set with co-finite topology is
(a) T_2 space
(b) Connected space
(c) T_1-space (d) T_0

122. R(real nos.) is a
(a) T_0-space
(b) T_1-space
(c) Connected space
(d) T_2 space

123. Any continuous image of connected space is
(a) Connected (b) T_0-space
(c) T_1-space (d) T_2-space

124. The range of a continuous real function on a connected space is an
(a) T_0-space (b) T_1-space
(c) Interval (d) T_2-space

125. In topological space "X" if it contains non empty open set called space
(a) Connected (b) Disconnected
(c) T_0-space (d) T_1 -pace

126. Let $X = \bigcup_{\alpha \in I} X_\alpha$ where each X_α is connected and $\cap_{\alpha \in I} X_\alpha \neq \phi$ then X is

(a) T_0-space (b) T_1-space
(c) Connected (d) T_2-space

127. The union of connected subsets with non empty intersection is

(a) T_0-space (b) Connected
(c) T_1-space (d) T_2

128. Let C by a connected subset of X and for some subset A of X $C \subset A \subseteq \bar{C}$ then

(a) A is connected
(b) $\bar{C}$ is connected
(c) Both a and b connected
(d) None of these

129. A topological space X, in which there exist a continuous mapping of X on to the two point discrete space {0, 1} is called

(a) Disconnected (b) T_0-space
(c) T_1-space (d) T_2-space

130. A topological space "X" in which there does not exist a continuous mapping of X onto the two points discrete space {0, 1} is called

(a) T_0-space (b) T_1-space
(c) Connected (d) T_2-space

131. If in a topological X every continuous mapping "f" from X into the two point discrete space {0, 1} is constant called space

(a) Connected (b) Disconnected
(c) T_0 (d) T_1-space

132. The components of a connected space X is the

(a) Whole space X (b) T_0-space
(c) T_1-space (d) T_2-space

133. Let X-be topological space, the for each $x \in X$ there is exactly one component of X containing

(a) x (b) Empty set
(c) X' (d) None of these

134. Let X be a topological space, then a connected Clopen subspace of X is a

(a) Component of X (b) T_0-space
(c) T_1-space (d) T_2-space

135. Let X be topological space then every component in X is

(a) Open (b) Closed
(c) T_2 space (d) T_1 space

136. Every discrete space is

(a) T_0-space
(b) T_1-space
(c) Totally disconnected
(d) None of these

137. Every totally disconnected space is a

(a) Housedroff space
(b) T_2-space
(c) T_1-space
(d) T_0-space

138. If a Housdorff space X has an open base whose sets are also closed then X is

(a) T_0-space
(b) T_1-space
(c) Totally disconnected
(d) None of these

139. If there is a selection $\{A_\alpha : \alpha \in I\}$ of subset of X such that $X = \bigcup_{\alpha \in I} A_\alpha$ then it is said

(a) Cover of X (b) Open set of X
(c) Closed set of X (d) Frontier of X

140. In a topological space if every cover has a finite subcover then it is called

(a) T_0-space (b) T_1-space
(c) T_2-space (d) Compact space

141. Every finite topological space is

(a) Normal space (b) Compact space
(c) T_0-space (d) T_1-space

142. Let (X, T) be a topological space, where "T" consists of finite number of elements then X is

(a) T_0-space (b) T_1-space
(c) Compact space (d) T_2 space

143. An infinite set with co-finite topology is
(a) Compact space (b) Normal space
(c) T_2-space (d) None of these

144. The real line R is
(a) Compact (b) Not compact
(c) Normal (d) T_0

145. The homeomoprhic image of compact space is
(a) Compact (b) Normal
(c) T_0-space (d) T_1-space

146. Any continuous bijective function from a compact space X to a Housdroff space Y is a
(a) Base (b) Closed set
(c) Homeomorphism (d) None of these

147. If there is a sequence $\{x_n\}$ such that $\underset{n \to \alpha}{Lt}\, x_n = x$ then sequence is called
(a) Convergent (b) Divergent
(c) Zero (d) One

148. A sequence in a metric space (X, d) converge to only
(a) Two points (b) One point
(c) No point (d) None of these

149. A sequence $\{x_n\}$ for which $\epsilon > 0$ there exist natural number $n_0(\epsilon)$ such that $d(x_n, x_m) < \epsilon \ \forall\ n_1\ m > n_0$ called sequence
(a) Decreasing (b) Increasing
(c) Nested (d) Cauchy

150. In a metric space (X, d) every convergent sequence is sequence named
(a) Cauchy (b) Decreasing
(c) Increasing (d) None of these

151. A Cauchy sequence is convergent in X if, it has a subsequence
(a) Cauchy (b) Divergent
(c) Convergent (d) Increase

152. If a sequence $\{x_n\}$ converge to point x then every subsequence $\{x_{nk}\}$ will converge to point
(a) 1 (b) Zero
(c) x (d) Other then x

153. In a metric space (X, d), if every Cauchy sequence X is convergent in X, called
(a) Complete space (b) Normal space
(c) Banach space (d) None of these

154. A subspace "Y" of a complete metric space is complete iff Y is
(a) Open (b) Closed
(c) Normal (d) Sequence

155. If in a sequence $A_{n+1} \subseteq A_n$ and diameter $\delta(A_n)$ tends to zero as $n \to \alpha$ then $\{A_n\}$ is sequence
(a) Sub (b) Normal
(c) Nested (d) Complete

156. If in complete metric space $\{F_n\}$ be decreasing sequence of closed subsets of X such that $\delta(F_n) \to 0$ as $n \to \alpha$ then $\bigcap_{n=1}^{\alpha}$ contains exactly points
(a) Two (b) One
(c) Zero (d) Three

157. The space C of complex numbers with a usual metric is space
(a) Complete (b) Normal
(c) Banach (d) Dual

158. In metric space (X, d) If A & B be two non-empty subsets of X, then A is said to be dense with respect to B if
(a) $B \subseteq A$ (b) $B = A$
(c) $B \subseteq \bar{A}$ (d) $A = \phi$

159. If A be subset of (X, d) and 4 open set then
(a) $U \cap A \neq \phi$ (b) $U \cap A = \phi$
(c) $U \subseteq A$ (d) $U = A$

160. A subset A of X is said no where dense in X if $\bar{A}$ do not contain any
(a) Closed ball (b) Open ball
(c) Subset (d) None of these

161. If in a metric space (X, d) if A is no where dense in X then $X/\bar{A}$ is
(a) Dense in X (b) Subset in X
(c) Union in X (d) None of these

162. If a subset A of (X, T) expressed as countable union of its nowhere dense subsets then it is
(a) 2nd Category (b) Ist category
(c) Dense (d) Closed

163. In usual metric space R set Q(rationals) is of
(a) Ist category (b) 2nd category
(c) Dense (d) Closed

164. A complete metric space is of
(a) Dense (b) Closed
(c) Second category (d) None of these

165. A point $x \in X$ is called isolated point if {x}
(a) Open
(b) Closed
(c) Both open + closed
(d) None of these

166. A complete metric space x which no isolated point is
(a) Uncountable (b) Countable
(c) Dense (d) Complete

167. For natural number "N" a function
$\| \; \| N \to R$ defined
$\| \cdot \| x = \| x \|$ called
(a) Normal of x (b) Countable
(c) Dense (d) Uncountable

168. For all x, y, z in a normal space N
(a) $d(x + y + z) = d(x, z)$
(b) $d(x + z, y + z) = d(x, y)$
(c) $d(x + z, y + z) = d(x, y)$
(d) None of these

169. Th space R^n (Real) n-types is.
(a) Separable space (b) Banach space
(c) Complete space (d) Normal space

170. The following space are normal spaces
(a) R^n (Real) (b) C^n complex
(c) Both a and b (d) None of these

171. The space C_0 of all sequences which converge to zero is a
(a) Normal space (b) Complete space
(c) Linear space (d) None of these

172. Complete normal linear space is called
(a) Banach space
(b) Convergent space
(c) Super space
(d) None of these

173. The space R^n (Real) is space
(a) Complete (b) Banach
(c) To-space (d) T_1-space

174. The space C[a, b] of all continuous functions from [a, b] to R and C is space
(a) Banach (b) Complete
(c) T_1-space (d) T_0-space

175. A normal space is called a Banach space if every Cauchy is
(a) Divergent in it (b) Convergent in it
(c) T_0-space (d) T_1

176. Show that the space "l^{α}" of all bounded sequence of real numbers is a space
(a) Complete (b) T_1-space
(c) T_0-space (d) Banach space

177. An infinite series $\sum_{n=1}^{\alpha} x_n$ is convergent i[f] its sequence of partial sums is
(a) Divergent (b) Convergent
(c) Complete (d) Banach

178. N be linear space and C be subset of N, is said to be convex if for any $x, y \in C$ a[nd] α, β [0, 1] iff
(a) $\alpha n + (1 - \alpha) \in C$
(b) $\alpha(n + (1 + \alpha)y \in C$
(c) $\alpha x + \alpha y = 0$
(d) None of these

179. A subspace of a linear space is
(a) Banach (b) Convex set
(c) Normal (d) None of these

180. Let C be convex set, then for any scal[ar] α, α_c is
(a) Convex (b) Convergent
(c) Divergent (d) T_1-space

181. The closure of a convex subset of a normal space is a

(a) T_0-space (b) T_1-space

(c) Convex set (d) Normal space

182. If the norms $||\cdot||$ and $||\cdot||_0$ on N_1 are equivalent then there exist non-zero positive real numbers "a" and "b" such that

(a) $a||x||_0 \leq ||x|| \leq b||x||_0$

(b) $a||x||_0 \leq b$

(c) $a||x||_0 = b||x||$

(d) None of these

183. Any two norms on a finite dimensional linear space are

(a) T_0 space (b) T_1 space

(c) Equivalent (d) Countable

184. The closed unit Ball

$\bar{B}(0, 1) = \{x \in X, ||x|| \leq 1\}$ is

(a) Convex (b) Open

(c) Equivalent (d) Countable

185. Any two norms on a finite dimensional linear space are

(a) Closed (b) Countable

(c) Equivalent (d) None of these

186. The pair $(V, <\cdot, \cdot>)$ is called an

(a) Inner product space

(b) Banach space

(c) Hilbert space

(d) None of these

187. In Hilbert space for $x, y \in V$(linear space)

(a) $<y_1, x> = -<x, y>$

(b) $<y, x> = \overline{<x, y>}$

(c) $<x, y> = <0, x>$

(d) $<x, y> = -\overline{<x, y>}$

188. For any two element "x", "y" in an inner product space v

(a) $|<x, y>| \leq ||x|| \cdot ||y||$

(b) $|<x, y>| = ||x|| \cdot ||y||$

(c) $|<x, y>| \leq -||x|| \cdot ||y||$

(d) $|<x, y>| \geq ||x|| \, ||y||$

189. For Hilbert spaces $x, y \in v$(linear space)

(a) $||x+y||^2 > <x+y>$

(b) $||x+y||^2 = <x+y, x+y>$

(c) $||x+y||^2 = <x, y>$

(d) $||x+y||^2 = -<y, x>$

190. Every inner product space is a

(a) T_0-space (b) T_1-space

(c) Metric space (d) Both a and d

191. In an inner product space V and x, y in v if $x + y$ then

(a) $||x+y||^2 = ||x||^2 + ||y||^2$

(b) $||x+||^2 = ||x||^2$

(c) $||x+y||^2 \leq ||x||^2 + ||y||^2$

(d) $||x+y||^2 = -|x| + |y|$

192. A sequence $\{x_k\}$ of nonzero vectors in an inner product space "V" is said to be orthognal if for all $i \neq j$

(a) $<x_i, x_j> = 1$ (b) $<x_i, x_j> = 0$

(c) $<x_i> = <x_j>$ (d) $<x_i, x_j> = -1$

193. Let H be Hilbert $A \subseteq H$. An element $x \in H$ is said to orthognal to A $\forall y \in A$ if

(a) $<x_1, y> = 1$ (b) $<x, y> = 0$

(c) $<x> = <y>$ (d) 1

194. The set of those elements of H which are orthognal to A(Annihilators of A) represented by

(a) A' (b) $\bar{A}$

(c) $A^{\perp}$ (d) A^0

195. Let A & B be subsets of Hilbert space "H" then

(a) $A \subseteq A^{\perp\perp}$ (b) $A \supseteq A^{\perp\perp}$

(c) $A \neq A^{\perp\perp}$ (d) $A = A^{\perp}$

196. For A & B be subsets of Hilbert space H and $A \subseteq B$ then for $A^{\perp}$ and $B^{\perp}$ we have

(a) $A^{\perp} \subseteq B^{\perp}$ (b) $A^{\perp} = B^{\perp}$

(c) $B^{\perp} \subseteq A^{\perp}$ (d) $A = B^{\perp}$

181. The closure of a convex subset of a normal space is a

(a) T_0-space (b) T_1-space
(c) Convex set (d) Normal space

182. If the norms $||\cdot||$ and $||\cdot||_0$ on N_1 are equivalent then there exist non-zero positive real numbers "a" and "b" such that

(a) $a||x||_0 \le ||x|| \le b||x||_0$
(b) $a||x||_0 \le b$
(c) $a||x||_0 = b||x||$
(d) None of these

183. Any two norms on a finite dimensional linear space are

(a) T_0 space (b) T_1 space
(c) Equivalent (d) Countable

184. The closed unit Ball

$\bar{B}(0, 1) = \{x \in X, ||x|| \le 1\}$ is

(a) Convex (b) Open
(c) Equivalent (d) Countable

185. Any two norms on a finite dimensional linear space are

(a) Closed (b) Countable
(c) Equivalent (d) None of these

186. The pair $(V, <\cdot, \cdot>)$ is called an

(a) Inner product space
(b) Banach space
(c) Hilbert space
(d) None of these

187. In Hilbert space for $x, y \in V$(linear space)

(a) $<y_1, x> = -<x, y>$
(b) $<y, x> = <\overline{x, y}>$
(c) $<x, y> = <0, x>$
(d) $<x, y> = -<\overline{x, y}>$

88. For any two element "x", "y" in an inner product space v

(a) $|<x, y>| \le ||x|| \cdot ||y||$
(b) $|<x, y>| = ||x|| \cdot ||y||$
(c) $|<x, y>| \le -||x|| \cdot ||y||$
(d) $|<x, y>| \ge ||x|| \, ||y||$

189. For Hilbert spaces $x, y \in v$(linear space)

(a) $||x + y||^2 > <x + y>$
(b) $||x + y||^2 = <x + y, x + y>$
(c) $||x + y||^2 = <x, y>$
(d) $||x + y||^2 = -<y, x>$

190. Every inner product space is a

(a) T_0-space (b) T_1-space
(c) Metric space (d) Both a and d

191. In an inner product space V and x, y in v if $x + y$ then

(a) $||x + y||^2 = ||x||^2 + ||y||^2$
(b) $||x + ||^2 = ||x||^2$
(c) $||x + y||^2 \le ||x||^2 + ||y||^2$
(d) $||x + y||^2 = -|x| + |y|$

192. A sequence $\{x_k\}$ of nonzero vectors in an inner product space "V" is said to be orthognal if for all $i \ne j$

(a) $<x_i, x_j> = 1$ (b) $<x_i, x_j> = 0$
(c) $<x_i> = <x_j>$ (d) $<x_i, x_j> = -1$

193. Let H be Hilbert $A \subseteq H$. An element $x \in H$ is said to orthognal to A $\forall\, y \in A$ if

(a) $<x_1, y> = 1$ (b) $<x, y> = 0$
(c) $<x> = <y>$ (d) 1

194. The set of those elements of H which are orthognal to A(Annihilators of A) represented by

(a) A' (b) $\bar{A}$
(c) $A^{\perp}$ (d) A^0

195. Let A & B be subsets of Hilbert space "H" then

(a) $A \subseteq A^{\perp\perp}$ (b) $A \supseteq A^{\perp\perp}$
(c) $A \ne A^{\perp\perp}$ (d) $A = A^{\perp}$

196. For A & B be subsets of Hilbert space H and $A \subseteq B$ then for $A^{\perp}$ and $B^{\perp}$ we have

(a) $A^{\perp} \subseteq B^{\perp}$ (b) $A^{\perp} = B^{\perp}$
(c) $B^{\perp} \subseteq A^{\perp}$ (d) $A = B^{\perp}$

197. For A & B subset of Hilbert space H when $A \subseteq B$ then

(a) $(A \cup B)^{\perp} \neq A^{\perp} \cap B^{\perp}$

(b) $(A \cup B)^{\perp} = A^{\perp} \cap B^{\perp}$

(c) $(A - B)^{\perp} = A^{\perp}$

(d) $(A - B)^{\perp} = B^{\perp}$

198. If A be subset of Hilbert space H then

(a) $A^{+} = B^{++}$ (b) $A^{++} = A^{+++}$

(c) $A^{+} = A^{+++}$ (d) $A^{+} = A$

199. If A be a subset of Hilbert space H then

(a) $A^{++} = A^{++}$ (b) $A \cap A^{+} \subseteq \{0\}$

(c) $A \cap A^{+} \subseteq A$ (d) None of these

200. The subsets $A^{\perp}$ of Hilbert space H is

(a) Open (b) Closed

(c) Complete (d) Normal

201. Direct sum of $v_1, v_2 \in v$ represented by

(a) $v_1 + v_2$ (b) $v_1 \oplus v_2$

(c) v_1/v_2 (d) None of these

202. If v_1 and v_2 be subspaces of v then v_1 and v_2 is called direct sum of

(a) $v_1 \cap v_2 = v_1$ (b) $v_1 \cap v_2 = v_2$

(c) $v_1 \cap v_2 = \{0\}$ (d) ϕ

203. Let A be a complete subspaces of inner product space X, and $A^{\perp}$ is orthognal component of A then

(a) $X = A \oplus A^{\perp}$ (b) $X = A + A^{\perp}$

(c) $X = A/A^{\perp}$ (d) $X = A^{\perp}$

204. If A be a closed subspace of a Hilbert space H then

(a) $H = A + A^{+}$ (b) $H = A \oplus A^{+}$

(c) $H = A - A^{+}$ (d) None of these

205. If A be a closed subspace of a Hilbert space H and if π is orthognal projection of H into A then for all $x_1, x_2 \in H$

(a) $<\pi(x_1), x_2> = <x_1, \pi(x_2)>$

(b) $\pi(x_1), x_2> = <\pi(x_1>$

(c) $<x, \pi(x_2)> = 0$

(d) Both b and c

206. The null space of π(orthognal projection) if a subset A of H Hilbert space is

(a) A (b) $A^{\perp}$

(c) $\bar{A}$ (d) '

207. For all $x \in H$ (Hilbert space) and π(orthognal projection) we can write as

(a) $<\pi x, x> = \pi$

(b) $<\pi x, x> \neq ||\pi x||^2$

(c) $<\pi(x), x> = ||\pi(x)||^2$

(d) None of these

208. If A and B be closed subspaces of a Hilbert space H and π & π be projection of H onto A and B then $A \perp B$ iff

(a) $\pi\pi = 0$ (b) $\pi\pi = 1$

(c) $\pi\pi = -1$ (d) $\pi = \pi$

ANSWERS

1. a	2. c	3. a	4. c	5. d
6. a	7. a	8. a	9. a	10. b
11. a	12. c	13. a	14. b	15. d
16. c	17. a	18. d	19. a	20. b
21. a	22. c	23. c	24. b	25. c
26. a	27. d	28. a	29. b	30. b
31. a	32. c	33. d	34. d	35. a
36. b	37. c	38. a	39. c	40. a
41. c	42. b	43. d	44. c	45. a
46. b	47. d	48. b	49. a	50. a
51. b	52. c	53. b	54. c	55. d
56. a	57. c	58. b	59. c	60. a
61. c	62. b	63. a	64. c	65. a
66. c	67. d	68. b	69. a	70. d
71. a	72. a	73. b	74. a	75. c
76. b	77. b	78. c	79. a	80. c
81. a	82. b	83. c	84. a	85. d
86. a	87. b	88. a	89. a	90. a
91. b	92. c	93. c	94. a	95. d
96. b	97. c	98. a	99. b	100. c
101. a	102. b	103. c	104. a	105. c
106. d	107. a	108. d	109. a	110. d
111. a	112. a	113. b	114. c	115. b
116. c	117. a	118. b	119. a	120. c
121. b	122. c	123. a	124. c	125. b
126. c	127. b	128. c	129. a	130. c
131. a	132. a	133. a	134. a	135. b
136. c	137. a	138. c	139. a	140. d

141. b	142. c	143. a	144. b	145. a	176. d	177. a	178. a	179. b	180. a
146. c	147. a	148. b	149. d	150. a	181. c	182. a	183. c	184. a	185. c
151. c	152. c	153. a	154. b	155. c	186. a	187. b	188. a	189. b	190. c
156. b	157. a	158. c	159. a	160. b	191. a	192. b	193. b	194. c	195. a
161. a	162. b	163. a	164. c	165. c	196. c	197. c	198. c	199. b	200. b
166. a	167. a	168. b	169. d	170. c	201. b	202. c	203. a	204. b	205. a
171. a	172. a	173. b	174. a	175. b	206. b	207. c	208. a		

➢ **Tick (✓) the correct one answer only.**

1. If there is bijective mapping from set A to set B, then the sets are said to be
   (a) Equal (b) Equivalent
   (c) Subset (d) Superset

2. In sets "~" is an relation
   (a) Equivalence (b) Equal
   (c) Superset (d) Subset

3. A set is said to be infinite if it is equivalent to its
   (a) Subset (b) Superset
   (c) Proper Set (d) Equal

4. A set is said to be denumerable, if it is equivalent to set
   (a) Natural number
   (b) Real Numbers
   (c) Complex numbers
   (d) Infinite

A set is said to be countable if it is
(a) Finite (b) Denumerable
(c) Infinite (d) Both (a) and (b)

A set is said to be uncountable if it is neither or nor
(a) Finite (b) Infinite
(c) Denumerable (d) Both a and c

Every infinite sequence of distinct elements is
(a) Denumerable (b) Numerable
(c) Finite (d) None of these

For N (natural number) the set $N \times N$ is
(a) Denumerable (b) Numerable
(c) Power set (d) Empty

Every infinite set contains a denumerable
(a) Super set (b) Subset
(c) Finite (d) Proper set

10. Every subset of denumerable set is
    (a) Countable (b) Finite
    (c) Infinite (d) Super set

11. Every subset of denumerable set is
    (a) Finite (b) Infinite
    (c) Countable (d) Uncountable

12. Every subset of countable set is
    (a) Countable (b) Uncountable
    (c) Finite (d) Infinite

13. Set of real number IR is
    (a) Finite (b) Infinite
    (c) Countable (d) Uncountable

14. Union of denumerable family of pairwis disjoint denumerable set is
    (a) Numerable (b) Denumerable
    (c) Countable (d) Uncountable

15. Set of Rational numbers 'Q' is
    (a) Denumerable (b) Countable
    (c) Uncountable (d) Empty

16. Unit interval [0, 1] is not
    (a) Numerable (b) Denumerable
    (c) Finite (d) Infinite

17. Any interval [a, b] , $a < b$, $a, n \in R$ is nor
    (a) Denumerable (b) Numerable
    (c) Countable (d) Uncountable

18. Let A be any set and α denotes collection of sets which are equivalent t then α called
    (a) Rational Numbers
    (b) Even
    (c) Cardinal number
    (d) None of these

19. For any cardinals α, β and γ we can have
(a) $\alpha + (\beta + \gamma) = (\alpha + \beta) + \gamma$
(b) $\alpha(\beta + \gamma) = \alpha\beta + \alpha\gamma$
(c) $\alpha(\beta\gamma) = (\alpha\beta)\gamma$
(d) a, b & c

20. Let α be the Cardinality of 'A' and β be the Cardinality of B i.e. $\alpha = \# A$ & $\beta = \# B$ then
(a) $\beta^{\alpha} = \# (B^{A})$ (b) $\beta^{\alpha} = \alpha^{\beta}$
(c) $\alpha = \beta$ (d) $A = B$

21. If α, β and γ be the Cardinality of sets A, B and C then in real number sense we can write
(a) $\alpha^{\beta} \cdot \alpha^{\gamma} = \alpha^{\beta+\gamma}$ (b) $(\alpha^{\beta})^{\gamma} = \alpha^{\beta\gamma}$
(c) Both a and b (d) None of these

22. Let S be the set of points on Cartesian plane, with rational components then show that S is
(a) Numerable (b) Denumerable
(c) Countable (d) Uncountable

23. Let 'X' bet and C(X) be collection of characteristic functions defined on X then
(a) $C(X) \cong 2^{X}$ (b) $C(X) - 2^{X}$
(c) $C^{X} = 2^{X}$ (d) None of these

24. If "n" is any finite Cardinal and $a = \# (IN)$ the
(a) $a + 1 = a + a$ (b) $a + 2 = a^{2}$
(c) $a + b = a$ (d) $a - n = a$

25. If $X \supseteq X \supseteq X_1$ and $X - X_1$ then
(a) $X - Y$ (b) $X = Y$
(c) $X^{Y} = 1$ (d) $X^{1} = 0$

26. If $A \cong B$ & $B \cong A$ then $A - B$ hence for Cardinl number α & β then $\alpha \le \beta$ & $\beta \le \alpha$ then
(a) $\alpha \neq \beta$ (b) $\alpha = \beta$
(c) $\alpha/\beta = 0$ (d) $\alpha = \alpha - \beta$

27. If # R = = C then
(a) $C \le 2^{\alpha}$ (b) $C = 2^{\alpha-1}$
(c) $C = 2$ (d) $2^{\alpha+1} = C$

28. If $C = \# [0, 1]$ then $C \cdot C = C^{2} =$
(a) $C + 1$ (b) C
(c) $C - 1$ (d) $C - 2$

29. If α and β are Cardinals then show that
(a) $\alpha^{\beta} \cdot \alpha^{\gamma} = \alpha^{\beta+\gamma}$ (b) $\alpha^{\beta} \cdot \alpha^{\gamma} = \alpha^{\beta}$
(c) $\alpha^{\beta} \cdot \alpha^{\gamma} = \alpha^{\beta-\gamma}$ (d) $\alpha^{\beta} \cdot \alpha^{1} = \alpha^{\beta-1}$

30. If $A \le B$ & $B \le C$ then $A \le C$ and if $\# A = \alpha$, $\#(B) = \beta$ and $\#(C) = \gamma$ and $\alpha \le \beta$, $\beta \le \gamma$ then
(a) $\alpha = \gamma$ (b) $\alpha \le \gamma$
(c) $\alpha \le \gamma + 1$ (d) $\alpha = \alpha - 1$

31. For any set A, if $\#(A) = \alpha$ then
(a) $\alpha < 2^{\alpha}$ (b) $\alpha = 2^{\alpha}$
(c) $\alpha \neq 2^{\alpha}$ (d) $\alpha = 2$

32. If β is any infinite Cardinal and $a = \#(IN)$ then
(a) $a + \beta = -\beta$ (b) $a + \beta = \beta$
(c) $\alpha = \beta$ (d) $a \neq \beta$

33. Prove that $[0, 1] \sim IR =]-\alpha, {}^{+}\alpha[$ and α, β, γ are Cardinality of sets A, B & C respectively then
(a) $(\alpha\beta)^{\gamma} = \alpha^{\gamma}\beta^{\gamma}$ (b) $(\alpha\beta)^{1} = \alpha\beta$
(c) $(\alpha\beta)^{\gamma} = \alpha^{\gamma}$ (d) None of these

34. If $X \le Y$ and $Y \le X$ then
(a) $X = Y$ (b) $X \neq Y$
(c) $X - Y$ (d) $XY = 1$

35. One of the property of non-empty set A with R relation defined in A is
(a) R is reflexive (b) $xRx \; \forall \; x \in A$
(c) $xRx = 1$ (d) Both a and b

36. Let $A = \{2, 3, 4, 5, 6\}$ and R be defined as xRy if
(a) x divides y
(b) x does not divide by
(c) $x = y$
(d) $x = y = 0$

37. If R is partial order is said to be total order if for $a, b \in A$
(a) $a \le b$ (b) $b \le a$
(c) $a \neq b$ (d) Both a and b

8. The natural order defined in IN is
(a) Total order (b) Cardinal order
(c) Both a and b (d) None of these

9. The order defined by x divides y in set $A = \{1, 2, 3, 4, 5, 6\}$ is a
(a) Total order (b) Partial order
(c) Cardinal order (d) None of these

0. Let A & B be totally ordered set, then the Cartesian product $(A \times B)$ is
(a) Partial order (b) Cardinal order
(c) Total order (d) None of these

1. Let A be an ordered set, an element $a \in A$ is said to be the first element of A if $\forall x \in a$
(a) $a = x$ (b) $a \leq x$
(c) $a \neq x$ (d) $a = x + 1$

2. Let $S = \{x : x \in Q, 2 \leq x^2 \leq 3\}$ then has
(a) No first element (b) Last element
(c) No last element (d) Both a and c

3. $(W \times W)$ where "W" is set of whole number is
(a) Denumerable (b) Numerable
(c) Countable (d) Uncountable

4. $(N \times N)$ where "N" is set of natural numbers is
(a) Number (b) Denumerable
(c) Finite (d) Both a and c

5. If $A = \{1, 3, 5, 7, ...\}$ and $B = \{2, 4, 6, 8, ...\}$ then "$A \times B$" is
(a) Countable (b) Uncountable
(c) Number (d) Denumerable

6. A sequence of distinct elements is
(a) Denumerable (b) Numerable
(c) Countable (d) Infinite

7. If $A_1, A_2, A_3, ..., A_n$ are pairwise disjoint denumerable sets then $\cup_{i=1}^{n} A_i$ is
(a) Numerable (b) Denumerable
(c) Finite (d) Infinite

18. Let "A" be any set and P(A) be its power set, then
(a) $|P(A)| = |A|$ (b) $|P(A)| = 3|A|$
(c) $|P(A)| = 2^{|A|}$ (d) $|P(A)| = 0$

40. If there are two sets A & B both have Cardinality C, then we can write as
(a) $A | B$ is non-empty finite
(b) $A | B$ is empty finite
(c) $A | B$ is zero
(d) None of these

50. If there are two sets A & B both have Cardinality C, then $A | B$ is
(a) Uncountable (b) Denumerable
(c) Countable (d) Zero

51. Let $\alpha \leq \beta$, then for any Cardinal number γ
(a) $\alpha + \gamma \leq \beta + \gamma$ (b) $\alpha\gamma \leq \beta\gamma$
(c) $\alpha\gamma = \beta\gamma$ (d) Both a and b

52. For any set A and power set of A P(A) we can write as
(a) $|A| < P(A)$ (b) $|A| = P(A)$
(c) $|A| < P(A)$ (d) None of these

53. If X, Y and X_1 be the sets such that $x \supset y \supset Y_1$ the we can write as
(a) $X - X_1$ (b) $X - Y$
(c) $X_1 - Y$ (d)Both a and b

54. If a and c have their usual meaning of Cardinality then we can prove
(a) $2^a = c$ (b) $2^a < c$
(c) $a^2 > c$ (d) $a^2 = c$

55. One of the property of collection R of subsets of X being a ring of X is that for A, $B \in R$
(a) $A \cap B \in R$ (b) $A \cup B \in R$
(c) $A' - B' \in R$ (d) $A'/B' = 9$

56. If A, $B \in R$ then one of the property of collection R of subsets of X being a Ring over R is that
(a) $A | B \in R$ (b) $A' | B' \in R$
(c) $A'/B \in R$ (d) $A' \cap B' \in R$

57. If R is a Ring then for any A, $B \in R$ we have
(a) $\phi \in R$ (b) $A \Delta B \in R$
(c) Both a and b (d) None of these

58. If R is a Ring then for any A, $B \in R$ can have
(a) $A \cap B \in R$ (b) $A \Delta B \in R$
(c) $\phi \in R$ (d) All a, b and c

59. If A, B ∈ R then A∖B ∈ R and $U_{n=1}^{n} A_n \in R$ and $R \subseteq X$, then X is called ring
(a) Sigma (b) Boolean
(c) Commutative (d) Associative

60. Show that every ring R is not necessary a
(a) Identity ring
(b) σ-Ring
(c) Identity ring
(d) Commutative ring

61. For A, B ⊆ X, Boolean algebra A of subsets of non empty X then
(a) $A \cup B \in \mathcal{A}$ (b) $A^c \in \mathcal{A}$
(c) Both a and b (d) None of these

62. If for all $A \in \mathcal{A}$ and $U_{n=1}^{\infty} A_n \in R \in \mathcal{A}$ then
(a) σ-Algebra (b) Boolean Algebra
(c) σ-Ring (d) Identity ring

63. For any set X, {ϕ, X} is a Algebra
(a) Boolean (b) σ-(sigma)
(c) Identity (d) Algebra

64. Countable union of elements of $\mathcal{A}$ (collection of subset X)
(a) Elements of $\mathcal{A}$ (b) $\mathcal{A}'$
(c) $\mathcal{A} \cap \mathcal{A}$ (d) $\mathcal{A} \cup \mathcal{A}'$

65. Let 'X' be any set then 2^X is
(a) σ-Algebra (b) Boolean algebra
(c) Both a and b (d) None of these

66. Every algebra is an
(a) Group (b) Ring
(c) Subring (d) Metric

67. A ring (or σ-ring) is an algebra (or σ-Algebra iff
(a) X ∈ R
(b) X ∈ W
(c) X ∈ Q (rationales)
(d) None of these

68. The intersection of any number of σ-Algebra on a set X is a
(a) σ-Algebra on X (b) Ring
(c) Subring (d) Group

69. Union of two σ-Algebra on a set may not be
(a) Ring (b) Group
(c) Subring (d) σ-Algebra

70. Let G be a family of subsets of a set X, then there is a smallest σ-Algebra containing
(a) R(Real) (b) G
(c) A group (d) A ring

71. Let $\mathcal{A}$ be a σ-Algebra of subsets of X and $\mu : \mathcal{A} \in R$ and $\{A_i\}$ be a sequence of subsets in $\mathcal{A}$, then μ is said to be finitely subadditive if
(a) $\mu\left(\bigcup_{i=1}^{n} E_i\right) \geq \sum_{i=1}^{n} u(E_i)$
(b) $u\left(\bigcup_{i=1}^{n} E_i\right) \leq \sum_{i=1}^{n} u(E_i)$
(c) $u\left(\bigcup_{i=1}^{n} E_i\right) < \bigcup_{i=1}^{n} (E_i)$
(d) None of these

72. For the statement No (71) is countably subadditive if
(a) $u\left(\bigcup_{i=1}^{n} E_i\right) \leq \sum_{i=1}^{n} u(E_i)$
(b) $u\left(\bigcup_{i=1}^{n} E_i\right) = \bigcup_{i=1}^{n} E_i$
(c) $u\left(\bigcup_{i=1}^{n} E_i\right) \geq \sum_{i=1}^{n} u(E_i)$
(d) $u\left(\bigcup_{i=1}^{n} E_i\right) = \bigcup_{i=1}^{n} E_i$

73. For the statement (71) is finitely additive if
(a) $u\left(\bigcup_{i=1}^{n} E_i\right) \leq \sum_{i=1}^{n} (E_i)$
(b) $u\left(\bigcup_{i=1}^{n} E_i\right) = \sum_{i=1}^{n} u(E_i)$
(c) $u\left(\bigcup_{i=1}^{n} E_i\right) \geq \sum_{i=1}^{n} u(E_i)$
(d) $u\left(\bigcup_{i=1}^{n} E_i\right) = \sum_{i=1}^{n} E_i$

74. For the statement No.(71) is σ-additive if
(a) α is commutative
(b) σ is associative

(c) $u\left(\bigcup_{i=1}^{n} E_i\right) \leq \sum_{i=1}^{n} u(E_i)$

(d) None of these

75. If $A \in \mathcal{A}$ and $A \subseteq B$ and $u(A) \leq \mu(B)$ then μ is

(a) Monotone (b) Commutative

(c) Additive (d) Both a and c

76. One of the property to be measure for if $\mathcal{A}$ be a σ-Algebra a function $\mu: \mathcal{A} \to [0, \alpha]$ is said measure if

(a) $u(\phi) = 0$ (b) $u(\phi) = \phi$

(c) $u(1) = 1$ (d) $u(\sigma) = 0$

77. For the statement No.76 one of the property to measure

(a) $u(A) \leq 0 \ \forall A \in \mathcal{A}$

(b) $u(A) \geq 0 \ \forall A \in \mathcal{A}$

(c) $u(A) = 0$

(d) None of these

78. $\mathcal{A}$ be sigma algebra and A function $u : \mathcal{A} \to [0, \infty]$ then one of the property for u be measure for a sequence (A_n) of pairwise disjoint is

(a) σ-additivity of μ

(b) Finity addivity

(c) Monotone

(d) Countably subadditivity

79. A measure function is Monotone if for A, B $\in \mathcal{A}$ and $B \subset A$, for $x : \mathcal{A} \to [0, \infty]$ we have

(a) for $B \subset A$, $\mu(B) \leq u(A)$

(b) $u(B) = u(A)$

(c) $u(B) > u(A)$

(d) $u(B) = u(A)$

80. The one of the general property of measure of $u : \mathcal{A} \to [0, \infty]$ for A, B $\in \mathcal{A}$ with $B \subset A$ and $u(B) < \alpha$

(a) $u(A + B) = u(A) + u(B)$

(b) $u(A) = u(B)$

(c) $u(AB) = u(A) \cdot u(B)$

(d) $u(A - B) = u(A) - u(B)$

81. Let $\mathcal{A}$ be a α-algebra of subsets of X and u be a measure defined on $\mathcal{A}$ then the triplet $(X, \mathcal{A}, u)$ is called

(a) Measure space (b) Borel space

(c) Metrit space (d) Mull space

82. If $(X, \mathcal{A}, u)$ be a measure space and $u(E \Delta F) = 0$th

(a) $u(E) = u(F) \neq (EuF) \neq u(E \cap F)$

(b) $u(E) = u(F) = u(EuF) = u(E)$

(c) $u(E) \neq u(E \cap F)$

(d) $u(F) \neq u(E \cap F)\, u(E)$

83. In a measure space $(X, \mathcal{A}, u)$ finite, if $u(X) < \infty$ then u is called

(a) Infinite (b) Finite measure

(c) Total measure (d) None of these

84. A measure space $(X, \mathcal{A}, u)$ for $\{A_n\}$ sequence of X in $\mathcal{A}$ for $X = u \sum_{n=1}^{\infty} A_n$ and $u(A_n) < \infty$ called

(a) Finite measure

(b) Infinite measure

(c) σ-finite measure space

(d) σ-space

85. If there is a mapping $u: 2^N \to [0, \infty]$ by $u(A) = \infty$ for infinite and u(A) number of points in A then θ is

(a) Counting measure

(b) Finite measure

(c) Infinite (d) Zero

86. Let $X \neq \phi$ be a set a mapping $u : 2^X \to [0, \infty]$ by $u(E) = 0$ if $P \notin E$ & $u(E) = 1$ if $P \in E$ then u is

(a) Counting measure

(b) Infinite measure

(c) Finite measure

(d) None of these

87. Let $X \neq \phi$ be a set. A mapping $u : 2^X \to [0, \infty]$ by $u(E) = 0$ if $E = \phi$ and $u(E) = \infty$ if $E \neq \phi$ then u is

(a) Note-finite (b) not σ-finite

(c) α-finite (d) both a and b

88. Sum and scalar multiple $(a > 0)$ of a measure is a

(a) Measure (b) Infinite

(c) Finite (d) None of these

89. One of the property of a function $u : 2x \to [0, \alpha]$ is called outer measure is

(a) $u(\phi) = 0$ (b) $u(\phi) \neq 0$

(c) $u(\phi) = 1$ (d) $u(\phi) = 0$

6. For the statement # 105 E is measureable iff

(a) E^c is measureable

(b) E^c not measureable

(c) $E^c = E$

(d) None of these

7. For the statement # 106 E is measureable if

(a) ϕ' & IR are measureable

(b) ϕ & IR are measureable

(c) E is subadditive

(d) E is additive

8. The union of two measureable

(a) Measureable (b) Additive

(c) Subadditive (d) None of these

9. The intersection of two measureable set is

(a) Additive (b) Subadditive

(c) Measureable (d) None of these

0. Let A be any set of real numbers and $\{E_1, E_2, \ldots, E_n\}$ be a family of pairwise disjoint measureable sets then

(a) $m\left[A\cap\left(\bigcup_{i=1}^{n} E_i\right)\right] = \sum_{i=1}^{n} m(A\cap E_i)$

(b) $\sum_{i=1}^{n} m(A\cap E_i) = 0$

(c) $m\left[A\cap\left(\bigcup_{i=1}^{n} E_i\right)\right] = 0$

(d) None of these

1. Let A be any set of real numbers and $\{E_i\}$ be a sequence of pairwise disjoint measureable sets then

(a) $m\left[A\cap\left(\bigcup_{i=1}^{n} E_i\right)\right] = 0$

(b) $\sum_{i=1}^{n} m(A\cap E_i) = 0$

(c) None of these

(d) $m\left[A\cap\left(\bigcup_{i=1}^{n} E_i\right)\right] = \sum_{i=1}^{n} m(A\cap E_i)$

2. m is countably additive (α-additive) on the sequence of

(a) Measureable sets

(b) Additive

(c) Subadditive

(d) None measureable

113. If F is measureable set and m $(A\Delta G) = 0$ G is

(a) Infinite set

(b) Finite

(c) Measureable set

(d) None of these

114. The interval (a, ∞) is measureable set for each

(a) $a \in W$ (b) $a \in O$

(c) $a \in P$ (d) $a \in IR$

115. For any a, b in IR with $a<b$ the interval (a, b) is

(a) Measureable (b) Additive

(c) Subadditive (d) Closed

116. Any closed set in IR is

(a) Additive (b) Measureable

(c) Subadditive (d) Zero

117. A subset E of IR is measureable if and only if for a given $\epsilon>0$ there is an openset $0 \supseteq E$ with

(a) $m(0|E)<\epsilon$ (b) $(0/E) = 0$

(c) $m(0/E)>\epsilon$ (d) None of these

118. For a subset E of IR & $\epsilon>0$ there is openset "0" $\supseteq E$ with

(a) $F \supseteq E$ (b) $F/E = 0$

(c) $F \subseteq E$ (d) $F \cap E = \phi$

119. If a set G countable intersection of opensets then it is said to be

(a) G set (b) G' set

(c) None of these (d) G_δ set

120. If $a, b \in IR$ then $[a, b] = \bigcap_{i=1}^{\infty}\left(a-\frac{1}{n}, b+\frac{1}{n}\right)$ is a

(a) G_δ set (b) Closed set

(c) Open set (d) Borel set

121. Every G_δ set is

(a) Closed (b) Open

(c) Measureable (d) Both (a) and (c)

139. Let $\{E_n\}$ be a decreasing sequence of measureable sets such that $m(E_K) < \infty$ for some $K \in N$, then

(a) $m\left(\bigcap_{n=1}^{\infty} E_n\right) = m\left(\bigcap_{n=1}^{\infty} E_n\right)$

(b) $m\left(\bigcap_{n=1}^{\infty} E_n\right) \neq m\, E_n$

(c) $m\left(\bigcap_{n=1}^{\infty} E_n\right) = mm\left(\bigcup_{n=1}^{\infty} E_n\right)$

(d) None of these

140. Smallest α-Algebra B which contains all of the open sets of IR is called

(a) Algebra (b) Borel algebra
(c) δ-Algebra (d) σ-Algebra

141. For the statement (140) each of the element called

(a) σ-set (b) δ-set
(c) Boyal set (d) None of these

142. Every Borel set is

(a) Measureable (b) Measure zero
(c) Lebesgue (d) σ-Algebra

143. For every Borel set B we can write as

(a) B = M (b) B ⊇ M
(c) B ≠ M (d) B ⊆ M

144. Borel Measure space for m & IR is represented by

(a) (R, B, m) (b) (IR, B)
(c) (R, O, B) (D) R, B

145. Any singleton, finite and countable set is a Borel set with Borel measure

(a) 1 (b) –1
(c) Zero (d) Infinite

146. For x, y ∈ A = [0, 1] sum modulo 1 of x & y is denoted by

(a) $x + y = \begin{cases} x + y & \text{if } x + y < 1 \\ x + y - 1 & \text{if } x + y \geq 1 \end{cases}$

(b) $x + y = \begin{cases} x - y & \text{if } x + y < 1 \\ x - y - 1 & \text{if } x - y = 0 \end{cases}$

(c) $x + y = x - y$ if $x + y < 1$

(d) $x + y = \begin{cases} x + y = -1 \\ x - y = 1 \end{cases}$

147. Let $E \subseteq A = [0, 1)$ & $y \in IR$ then we define the translation modulo 1' to be the set

(a) $E + y = \{x + y; x \in E\}$
(b) $E = \{x + y; x \in E\}$
(c) $E + y = \{x + y\}$
(d) $E + y = \{X + x + y\}$

148. Let $E \subseteq A = [0, 1)$ be a measureable set, then for each $y \in A$ the set "E + y" is

(a) Variant (b) Measureable
(c) Zero (d) Borel

149. Let $E \subseteq A = [0, 1)$ be measureable set, then for each $y \in A$ & the set $E + y$ we can write

(a) $m(E) = m(E + y)$ (b) $m(E) = m(y)$
(c) $m(E) = m(y)$ (d) None of these

150. Let E be measureable set, then a function $f : E \to IR$ is said to be legesgue measureable function if and only if

(a) $\{x \in E: f(x) > \alpha\}$ (b) $\{x \in E: f(x) < \alpha\}$
(c) $f(x) < \alpha(d)$ (d) $\{x \in E: f(x) = \alpha\}$

151. If f be an extended real valued functions defined on a measureable set E then one of the following is

(a) f is measureable
(b) f is Boolean
(c) f is non zero
(d) f is non-measureable

152. One of the property for statement (151) is given by

(a) $\{x \in E: f(x) \leq \alpha, \forall \alpha \in IR\}$
(b) $\{x \in E: f(x) = \alpha\}$
(c) $\{x \in E: f(x) > \alpha \ \forall \alpha \in R\}$
(d) None of these

153. Let "f" be an extended real valued function defined on a measureable set E then $\{x \in E: f(x) = \alpha\}$ is

(a) Borel measure (b) Measureable
(c) σ-additive (d) None of these

154. If "f" be extended real valued function defined on set A ∀ x ∈ A we can write as

(a) $f^{+}(x) = \begin{cases} f\cdot(x) & \text{if } f(x) \geq 0 \\ 0 & \text{if } f(x) < 0 \end{cases}$

$= \text{Max}(F(x_1), 0) = (fv_0)$

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(b) $f^-(x) = \begin{cases} 0 & \text{if } f(x) \geq 0 \\ -f(x) & \text{if } f(x) < 0 \end{cases}$

$= \text{Max}(-f(x_1), 0) = (-f)v_0)$

(c) Both (a) and (b)

(d) None of these

155. For a measureable function "f" we can have "f^-"

(a) $\frac{|f| - f}{2}$ (b) $\frac{|f| + f}{2}$

(c) $|f| < 0$ (d) $|f| > 0$

156. For a measureable function "f" we can have "f^-"

(a) $f^- = \frac{|f| - f}{2}$ (b) $\frac{|f| + f}{2}$

(c) $\frac{|f|}{f}$ (d) $\frac{-|f|}{f}$

157. If "f" measureable function then we can have

(a) $|f| = f^+ - f^-$ (b) $|f| = \frac{f^+}{f^-}$

(c) $f = f^+ - f^-$ (d) $f^+ = f^-$

158. Let "f" be extended real valued function then we can have

(a) $f = f^+ - f^-$ (b) $f^+ + f^-$

(c) $f^- = f^+$ (d) $|f| = f^-$

159. If "f" & "g" be extended real valued function then Max(fg)

(a) $\frac{(f+g) - (f-g)}{2}$ (b) $\frac{|f+g| + (f-g)}{2}$

(c) $\frac{(f+g) + |f-g|}{2}$ (d) $|f+g| - f(g)$

160. If f & g be extended real value function then Min(f,g)

(a) $\frac{(f+g) + (f-g)}{2}$ (b) $|f+g| - f(g)$

(c) $\frac{(f+g) + |f-g|}{2}$ (d) $\frac{(f+g) - |f-g|}{2}$

161. Let $X \neq \phi$ and $A \subseteq X$ then the characteristics function $\chi_A : X \to \mathbb{R}$ is defined by

(a) $\chi_A(x) = \begin{cases} 1 & \text{if } x \in A \\ 0 & \text{if } x \notin A \end{cases}$

(b) $\chi_A = \begin{cases} 1 & \text{if } x \notin A \\ 0 & \text{if } x \in A \end{cases}$

(c) $\chi_x = \{0 \text{ if } x \in A\}$

(d) $\chi_A = \{0 \text{ if } x \in A\}$

162. If A & B are the subset of a set of X the for characteristics function χ, the following holds

(a) $\chi_\phi = 1, \chi_x = 0$ (b) $\chi_\phi = 0, \chi_x = 1$

(c) $\chi_x = 0$ (d) $\chi_\phi = 1$

163. If $A \subseteq B$ then for characteristics function we can write

(a) $\chi_A = \chi_B$ (b) $\chi_A < \chi_B$

(c) $\chi_A > \chi_B$ (d) Both a and b

164. For statement (163) it can be written as

(a) $\chi_{A \cap B} = \chi_A \cdot \chi_B$ (b) $\chi_{A \cap B} = \chi_A - \chi_B$

(c) $\chi_{A \cup B} = \chi_A \cdot \chi_B$ (d) $\chi_{A/B} = A + B$

165. For A & B be subsets of X then for characteristics function χ it can be written as

(a) $\chi_{A \cup B} = \chi_A + \chi_B$

(b) $\chi_{A \cup B} = \chi_A + \chi_B - \chi_{A \cap B}$

(c) $\chi_{A \cup B} = 0$

(d) $\chi_{A \cup B} = \chi_A - \chi_B$

166. For the statement No.(165) it can be written as

(a) $\chi_{A-B} = \chi_A + \chi_B$ (b) $\chi_{A-B} = \chi_A - \chi_B$

(c) $\chi_{A-B} = \chi_A - \chi_{A \cap B}$ (d) $\chi_{A-B} = 0$

167. For the statement No.(165) characteristic function $\chi_{A \times B}$ can be written as

(a) $\chi_A \cdot \chi_B$ (b) $\chi_A - \chi_B$

(c) $\chi_A + \chi_B$ (d) χ_A / χ_B

168. If $A = \bigcup_{n=1}^{\infty} A_n$ and $\{A_n\}$ is a sequence of pairwise disjoint subset of X then ?

(a) $\sum_{n=1}^{\infty} \chi_{A_n}$ (b) $\sum_{n=1}^{\infty}$ [illegible]

(c) $\sum_{n=1}^{\infty} \chi_A$ (d) χ_{A_n}

169. χ_A characteristics function is measureable if A is

(a) σ-Algebra (b) ρ-Algebra

(c) Measureable (d) Zero

170. If f & g be two real valued measureable functions defined in same domain "D" and $C \in IR$ then the function "f + C" is

(a) Measureable (b) Boolean

(c) Zero (d) None of these

171. For the statement (170) (fg) and (f/g) are function

(a) Unjective (b) Measureable

(c) Constant (d) Zero

172. For the statement (170) & "f" + "g" be measureable then measureable function is

(a) $f \pm g$ (b) $f^2 - g^2$

(c) f^2/g (d) $\frac{g^2}{f}$

173. For the statement (170) f^2 and "fvg" are

(a) Monatomic (b) Zero

(c) Measureable (d) on-to

174. For the statement (170) | f | is

(a) Measureable (b) Monotonic

(c) on-to (d) Zero

175. Let $f: X \to Y$ be a function and $Z \subset X$ then the function $g: Z \to Y$ is called the restriction of f on z if $\forall x \in z$

(a) $f(x) \neq f(x)$ (b) $\frac{g(x)}{f(x)} = 0$

(c) $\frac{(g)(x)}{f(x)} = 1$ (d) $g(x) = f(x)$

176. If f be extended real-valued functions in D & A be measureable subset of D then restriction of f to A is also

(a) Measureable (b) Monotonic

(c) One-one (d) on-to

177. Let D & E are measureable sets & f be function with domain $D \cup E$ then f is measureable iff its restriction D & E are

(a) Measureable (b) Monotonic

(c) Constt (d) Zero

178. Let "f" be measureable function in D, then f is measureable if and only if the function

$$g(x) = \begin{cases} f(x) & \text{if} \quad x \in D \\ 0 & \text{if} \quad x \notin D \end{cases} \quad \text{is}$$

(a) Monatomic (b) Constt

(c) Measureable (d) Constt

179. An extended real-valued function defined on a set of measure zero is

(a) Monotonic (b) Measureable

(c) Constt (d) Zero

180. Let $\{f_n\}$ be a sequence of extended real-valued measureable function with same domain D then $\max_{1 \le k \le n} f_k \ \forall n \in N$ is

(a) Measureable (b) Monotonic

(c) Constant (d) Zero

181. For the statement (180) $\min_{1 \le k \le n} f_k \ \forall n \in N$ is

(a) Constant (b) Zero

(c) Monotonic (d) Measureable

182. If "f" be sequence of intended real-valued measureable function with same domain D then $\sup_{n \in N} f_n$ is

(a) Monotonic (b) Measureable

(c) One-one (d) On-to

183. If "f_n" be sequence of extended real valued measureable function with domain D then "$\inf_{n \in N} f_n$" is

(a) Monotonic (b) Measureable

(c) One-one (d) On-to

184. If in the statement 182 "f" is measureable if

(a) $\lim_{n \to \infty} = 0$ (b) $\lim_{n \to \infty} f_n = \infty$

(c) $\lim_{n \to \infty} f_n = f$ (d) None of these

185. A continuous function defined on a measureable set is
(a) Measureable (b) σ-Algebra
(c) One-one (d) None of these

186. Let 'f' be continuous function and G is an open set, then {x:f(x) ∈ G} is a
(a) Closed set (b) One-one
(c) On-to (d) Measureable

187. Let "f" be a real-valued function defined on a measureable domain D and G is open set in IR then f is measureable if and only if measureable is
(a) f^{-1} (b) f(G)
(c) f/G (d) f + G

188. If F is a closed subset in IR f^{-1} (F) is measureable if and only if measureable is
(a) f (b) f^{-1}
(c) f' (d) Both a and c

189. Let "f" & "g" be measureable function defined on the same measureable domain D then the following sets are measurvable x ∈ D
(a) $\frac{f(x)}{g(x)}$ (b) f'(x)
(c) f(x) > g(x) (d) None of these

190. For the statement No.189 the following sets
(a) g(x) > f(x) (b) f(x) = g(x)
(c) f'(x) = 0 (d) Both a and b

191. For a function "f" defined in an interval [a, b] is step function, if for each sub-interval (x_{i-1}, x_i) f(x) has value where (i = 1, 2, 3, 4, ...n)
(a) Zero (b) 1
(c) -1 (d) Constant

192. Step function & characteristics function are examples of
(a) Composite function
(b) Simple function
(c) Constant function
(d) None of these

193. Every constable set is measureable with measure
(a) 1 (b) -1
(c) Zero (d) Finite

194. The differential equation of the form $x^2\frac{d^2y}{dx^2} + x\frac{dy}{dx} + (x^2 - n^2) = y$ is known as
(a) Bessel function
(b) Constant function
(c) One-one function
(d) Bessel equation

195. The solution of the equation (194) is known as
(a) Bessel function
(b) Bessel equation
(c) Quadratic equation
(d) None of these

196. One of the Recurrence Relation for $J_n(x)$ Bessel function
(a) $J_n'(x) = nJ_n(x)\ xJ_{n+1}(x)$
(b) $x J_n'(x) = nJ_n(x) - xJ_{n+1}(x)$
(c) $J_n'(x) = nJ_n(x)$
(d) $x J_n'(x) = nJ_n(x) + xJ_{n+1}(x)$

197. $x J_n'(x) = nJ_n(x) + xJ_{n+1}(x)$ is one of the recurrence relation for function
(a) Composite (b) Bessel's
(c) One-one (d) On-to

198. One of the recurrence relation for $J_n(x)$ Bessel function
(a) $2J_n'(x) = J_{n-1}(x) - J_{n+1}(x)$
(b) $2J_n'(x) = J_{n-1}(x) - J_{n+1}(x)$
(c) $2J_n'(x) = J_{n+1}(x) + J_{n-1}(x)$
(d) $2J_n'(x) = 0$

199. With Bessel function "J_n" the relation is one of the $\frac{d}{dx}[x^{-n} J_n(x)] = -x^{-n}J_{n+1}(x)$ is relation
(a) Recurrence (b) Bessel
(c) Function (d) Measureable

200. One of the recurrence relation for Bessel function
(a) $\frac{d}{dx}[J_n(x)] = x^n J_{n-1}(x)$

(b) $x J_n$ ...

(c) $J_n'(x) = nJ_n(x)$

(d) $x J_n'(x) = nJ_n(x) + xJ_{n+1}(x)$

197. $x J_n'(x) = nJ_n(x) + xJ_{n+1}(x)$ is one of the recurrence relation for function

(a) Composite (b) Bessel's

(c) One-one (d) On-to

198. One-of the recurrence relation for $J_n(x)$ Bessel function

(a) $2J_n'(x) = J_{n-1}(x) - J_{n+1}(x)$

(b) $2J_n'(x) = J_{n-1}(x) - J_{n+1}(x)$

(c) $2J_n'(x) = J_{n+1}(x) + J_{n-1}(x)$

(d) $2J_n'(x) = 0$

199. With Bessel function "J_n" the relation is one of the $\frac{d}{dx}[x^{-n} J_n(x)] = -x^{-n}J_{n+1}(x)$ is relation

(a) Recurrence (b) Bessel

(c) Function (d) Measureable

200. One of the recurrence relation for Bessel function

(a) $\frac{d}{dx}[J_n(x)] = x^n J_{n-1}(x)$

(b) $\frac{d}{dx}[xJ^n(x)] = x^n J_{n-1}(x)$

(c) $J_n x = x^n J_{n-1}(x)$

(d) $J_n' x = 0$

201. Generating function for Bessel function $J_n(x)$ is

(a) $\text{Exp}\left\{\frac{x}{2}\left(t+\frac{1}{t}\right)\right\}$

(b) $\text{Exp}\left\{\frac{x}{2}t^2 - t^{1/2}\right\}$

(c) $\text{Exp}\left\{\frac{x}{2}\left(t-\frac{1}{t}\right)\right\}$

(d) None of these

202. For Bessel function generating function $\text{Exp}\left\{\frac{x}{2}\left(t-\frac{1}{t}\right)\right\}$ is equal to

(a) $\sum_{n=-\infty}^{\infty} t^n J_n(x)$ (b) $\sum_{n=1}^{\infty} t^n J_n(x)$

(c) $\sum_{n=0}^{1} t^n J_n(x)$ (d) $\sum_{n=-\infty}^{\infty} t^n J_{n+1}(x)$

203. For Bessel function $J_n(x)$ when n +ve integer we have $J_n(x)$

(a) $(-1)^{n+1}J^{n-1}(x)$ (b) $(-1)^n J_{n+1}(x)$

(c) $(-1)^n J_n x$ (d) None of these

204. For Bessel functions $J_n(x)$ when n is +ve we can have for $J_n(-x) =$

(a) $(-1)^n J_n(x)$ (b) $(-1)^{n+1}J_{n+1}(x)$

(c) $(1)^{n-1}J^{n-1}(x)$ (d) Zero

205. For Bessel function we can have as for $J_{-1/2}(x) =$

(a) $\sqrt{\frac{2}{\pi}}\cos x$ (b) $\sqrt{\frac{2}{\pi x}}\cos x$

(c) $\sqrt{\frac{2}{\pi x}}$ (d) $\sqrt{\frac{2}{\pi}}$

206. For Bessel function we can have as for $J_{1/2}(x) =$

(a) $\sqrt{\frac{2}{\pi x}}$ (b) $\sqrt{2\pi x}$

(c) $\sqrt{\frac{2}{\pi x}}\sin x$ (d) Zero

207. In Bessel function $J(x)\ (J_{1/2}(x))^2 + J_{-1/2}(x))^2 =$

(a) $\frac{2}{\pi x}$ (b) $\frac{1}{\pi x}$

(c) $\frac{\pi}{x}$ (d) None of these

208. For Bessel function $J(x)$ we can have $J_{3/2}(x) =$

(a) $\sqrt{\frac{2}{\pi x}}\left(\frac{\sin x}{x} - \cos x\right)$

(b) $\sqrt{\frac{2}{\pi x}}\left(\frac{\sin x}{x} + \cos x\right)$

(c) $\sqrt{\frac{2}{\pi x}}$

(d) $\frac{2}{\pi x}$

209. For Bessel function $J(x)$ we can have $J_{-3/2}(x) =$

(a) $\sqrt{\frac{2}{\pi x}}\left(\frac{\cos x}{x} - \sin x\right)$

(b) $-\sqrt{\frac{2}{\pi x}}\left(\frac{\cos x}{x} - \sin x\right)$

(c) $\sqrt{\frac{2}{\pi x}}$ (d) Zero

210. The differential equation of the form equation $(1-x^2)\frac{d^2y}{dx^2} - 2x\frac{dy}{dx} + x(x+1)y = 0$ is called

(a) Legendre's (b) Bessels

(c) Gamma (d) Linear

211. The solution set of the differential eq(210) called function

(a) Besseles (b) Legender

(c) Linear (d) Gamma

212. Legendre's polynomial of degree n of the Ist kind is denoted by

(a) $P_n(x)$ (b) $P(x)$

(c) $P_n(x)$ (d) None of these

213. Generating function for $P_n(x)$ then solution set of $(1 - 2xt + t^2)^{-1/2}$ is given by

(a) $\sum_{n=0}^{n} t^n P_n(X)$ (b) 1

(c) 0 (d) $\sum_{n=0}^{n} t^n P_n(X)$

214. For Legender generating function $P_n(X)$ we have $P_n(X) =$

(a) $(-1)^{n+1} P_n(X)$ (b) $(-1)^n P_n(X)$

(c) $P_n(X)(-1)^{n-1}$ (d) None of these

215. For Legender generating function $P_n(X)$ we have $P_n(1) =$

(a) 0 (b) -1

(c) 1 (d) None of these

216. One of the recurrence relation for $P_n(X)$ is $(2n + 1)(X) P_n(X) =$

(a) $(n + 1) P_{n+1}(X) + nP_{n-1}(X)$

(b) $(n + 1) P_{n+1}(X)$

(c) $nP_{n-1}(X)$

(d) Zero

217. One of the Recurrence relation $P_n(X)$ is $nP_n(X) =$

(a) $(2n - 1)P_{n-1}(X)$ (b) $(2n - 1)P_{n-1}(X)$

(c) Zero (d) None of these

218. One of the recurrence relation for $P_n(X)$ is $nP_n(X) =$

(a) $XP_n'(X)$ (b) $P_{n-1}'(X)$

(c) $XP_n'(X) - P_{n-1}'(X)$ (d) None of these

219. One of the recurrence relation for $P_n(X)$ is $(n + 1) P_n(X) =$

(a) $P_{n+1}'(X) - XP_n'(X)$

(b) $P_{n+1}'(X)$

(c) $P_n'(X)$

(d) Zero

220. Rodrigue's Formula for $P_n(x) =$

(a) $\frac{1}{2^n n!} \frac{d^n}{dx^n}(x - 1)^n$

(b) $\frac{1}{2^n n!} \frac{d^n}{dx^2}(x^2 - 1)$

(c) $\frac{1}{2^n n!} \frac{d^n}{dx^n}(x^2 - 1)^n$

(d) None of these

221. Orthogonal property of Legendre's polynomials

(a) $\int_{-1}^{1} P_m(X) P_n(X)\,dx = \frac{2}{2n+1}\delta_{mn}$ where $\delta_{mn} = 0$ $m \neq n$, $\delta_{mn} = 1$ $m = n$

(b) $\int_{-1}^{1} P_m(X) P_n(X)\,dx\,\delta_{mn}$

(c) $\int_{-1}^{1} P_m(X) P_n(X)\,dx = 0$

(d) None of these

222. The Pochhammer symbol $(a)_n$ is defined by

(a) $n = \begin{cases} 1 & \text{if } n = 0 \\ a^2(a^2+1)\ldots a^2+n-1 & \text{if } n \geq 1 \end{cases}$

(b) $(a)_n = \begin{cases} 1 & \text{if } n = 0 \\ a(a+1)\ldots(a+n-1) & \text{if } n \geq 1 \end{cases}$

(c) $(a)_n = \{a^2(a^2-1)\ldots a^n-n-1 \text{ if } n \leq 1\}$

(d) None of these

223. One of the Pochhammer symbol property is

(a) $a_{n+1} = a(a + 1)_n$

(b) $a_{n+1} = (a + n)(n + 1)$

(c) $a_{n+1} = (a + n)(a - n)$

(d) $a_{n+1} = a_n(a + 1)$

224. One of the Pchhammer symbol $(a)_n$ is

(a) $(n)_{n+1} = x(a)_n$

(b) $(a)_{n+1} = (a + x)(a)_n$

(c) $(a)_{n+1} = (a + n)(a)_{n+1}$

(d) $(a)_{n+1} = (a + n)(a - 1)_{n-1}$

225. The properties of the Pochhamer symbol is (a_n)

(a) $(1)_{n+1} = n+1!$

(b) $(1)_n = n!$ (c) $(-n)_n = (-1)^n\, n!$

(d) both (b) and (c)

226. One of the properties of Pochhammer symbol is

(a) $(a-1)a_{n-1} = (a-1)_{n-1}(a+n-2)$

(b) $a(a_{n-1}) = (a-n)_{n-1}(a+n-2)$

(c) $(a+1)a_{n+1} = (a+1)_{n+1}(a+n)$

(d) None of these

227. One of the property of Pochhammer symbol is

(a) $(a-k)_{n-k} = (-1)^{n-k}(1-a-n)_{n-k}$

(b) $(a+k)_{n+k} = (-1)^{n-k}(1+a+n)$

(c) $(ak)_{n+k} = (1+a+n)_{n+k}$

(d) $(a+k)_{n+k} = (1)^{n+k}(1+a+n)$

228. The one of the property for Pochhammer symbol is

(a) $(a)_{n-k} = (a-1)_n\,(a+n)_k$

(b) $(a)_{n+k} = (a_n)\,(a+n)_k$

(c) $a_{n+k} = (a_n)\,(a+n)_{k+1}$

(d) $(a_{n+k}) = a_n(a+n)_{k-1}$

229. For Polhhammer symbol $(a)_{-n} =$

(a) $\dfrac{(-1)^{n+1}}{(1+a)_n}$ (b) $\dfrac{(-1)^{n-1}}{(1+a)_{n-1}}$

(c) $\dfrac{(-1)^{n+1}}{(1+a)(n+1)}$ (d) $\dfrac{(-1)^n}{(1-a)_n}$

230. For Pochhamer symbol it can be shown $(-n)_k =$

(a) $\dfrac{(-1)^k\, n!}{(n-k)!}$ (b) $\dfrac{(-1)^{k+1}\, n+1!}{(n+k)!}$

(c) $\dfrac{(n+1)}{(n+k)!}$ (d) None of these

231. For Pochhamar symbols it can be shown that $(1+n)_n =$

(a) $4^n\left(\frac{1}{3}\right)_n$ (b) $2^n\left(\frac{1}{2}\right)_n$

(c) $4^n\left(\frac{1}{2}\right)_n$ (d) $4^n\left(\frac{3}{2}\right)_n$

232. $\binom{n}{k} =$

(a) $\dfrac{(-1)^k(-n)_k}{k!}$ (b) $\dfrac{(1)^k\,(n)^k}{k+1!}$

(c) $\dfrac{(-1)^k(-n)^{k+1}}{k!}$ (d) None of these

233. Gamma function for improper integral is defined

(a) $\int_0^{\infty} x^{n+1}e^{x}dx$ (b) $\int_0^{\infty} x^{n-1}e^{x}dx$

(c) $\int_{+0}^{\infty} x^{n-1}e^{-x}dx$ (d) None of these

234. Hypergeometric function $F(\alpha, \beta, \gamma; x) =$

(a) $\sum_{n=0}^{\infty}\dfrac{(\alpha)_n(\beta)_n}{n!(\gamma)_n}x^n$ (b) $\sum_{n=0}^{\infty}\dfrac{(\alpha)_{n-1}(\beta)_n}{n!(\gamma)_{n-1}}$

(c) $\dfrac{\alpha_{n-1}\beta_n}{n!(\gamma)_{n-1}}$ (d) None of these

235. Gauss's theorem is given by $F(\alpha, \beta, \gamma; 1) =$

(a) $\dfrac{\Gamma(\gamma-\beta-\alpha)}{\Gamma(\gamma-\alpha)\Gamma(\gamma-\beta)}$ (b) $\dfrac{\Gamma(\alpha)\Gamma(\gamma-\beta-\alpha)}{\Gamma(\gamma-\alpha)\Gamma(\gamma-\beta)}$

(c) $\dfrac{\Gamma(\gamma)}{\Gamma(\beta)\Gamma(\alpha)}$ (d) None of these

236. Vandermonde's theorem is given by as $F(-n, \beta, \gamma; 1) =$

(a) $\dfrac{(\gamma+\beta)_n}{(\gamma)_n}$ (b) $\dfrac{(\gamma-\beta)_n}{(\gamma)_n}$

(c) $\dfrac{(\gamma+\beta)_n}{(\gamma)_n}$ (d) $\dfrac{(\gamma+\beta)_n}{\gamma^n}$

237. For Vandermonde's theorem $x\,F\left(\frac{1}{2}, \frac{1}{2}, \frac{3}{2}; x^2\right)$

(a) $\sin^{-1}x$ (b) $\tan^{-1}x$

(c) $\cos^{-1}x$ (d) $\log(1+x)$

242. For Vandermonds theorem $x\,F\left(\frac{1}{2}, 1, \frac{3}{2}; -x^2\right) =$

(a) $\cos^{-1}x$ (b) $\sin^{-1}x$

(c) $\log(1+x)$ (d) $\tan^{-1}x$

243. For Vandermonde's formula $F(1, 1, 1; x) =$
(a) $(1+x)^{-1}$ (b) $(1-x)^{-1}$
(c) $(1-x^2)^{-1}$ (d) $(1+x^2)^{-1}$

244. The differential equation
$x\frac{d^2y}{dx^2} + (\beta - x)\frac{dy}{dx} - \alpha y = 0$ where α & β constant, is known as
(a) Confluent hypergeometric equation
(b) Heat equation
(c) Wave equation
(d) Single Harmonic equation

245. For hypergeometric $\frac{d}{dx}F(\alpha+1, \beta+1; x) =$
(a) $\alpha(F)(\alpha+\beta+1)$
(b) $\frac{\alpha}{\beta}F(\alpha+1, \beta+1; x)$
(c) Zero
(d) $\frac{\alpha}{\beta}(\alpha+1; x)$

246. Kummer's relation $F(\alpha, \beta; x) =$
(a) $e^x F(\beta-\alpha, \beta; -x)$ (b) $e^x F(\beta-\alpha, -x)$
(c) $e^x F(\beta, \alpha)$ (d) $e^x F(\beta-\alpha, \beta; x)$

ANSWERS

1. b	2. a	3. c	4. a	5. a
6. d	7. a	8. a	9. b	10. a
11. c	12. a	13. b	14. b	15. a
16. b	17. a	18. c	19. d	20. a
21. c	22. b	23. b	24. c	25. a
26. b	27. a	28. b	29. a	30. b
31. a	32. b	33. a	34. c	35. d
36. a	37.	38. a	39. b	40. a
41. c	42. d	43. a	44. d	45. a
46. a	47. b	48. c	49. a	50. c
51. d	52. a	53. d	54. a	55. b
56. a	57. c	58. d	59. c	60. b
61. c	62. a	63. b	64. a	65. a
66. b	67. a	68. a	69. d	70. b
71. b	72. a	73. b	74. c	75. a
76. a	77. b	78. a	79. a	80. d
81. a	82. b	83. b	84. c	85. a
86. c	87. d	88. a	89. a	90. b
91. d	92. b	93. b	94. d	95. a
96. b	97. b	98. c	99. a	100. b
101. a	102. a	103. d	104. a	105. d
106. a	107. b	108. a	109. c	110. a
111. d	112. a	113. c	114. d	115. a
116. b	117. a	118. c	119. d	120. a
121. d	122. a	123. c	124. a	125. c
126. a	127. c	128. b	129. c	130. b
131. c	132. a	133. c	134. a	135. c
136. a	137. b	138. c	139. a	140. b
141. c	142. a	143. c	144. a	145. a
146. a	147. a	148. b	149. a	150. a
151. a	152. a	153. b	154. c	155. b
156. a	157. c	158. b	159. c	160. d
161. a	162. b	163. d	164. a	165. b
166. c	167. a	168. a	169. c	170. a
171. b	172. a	173. c	174. a	175. d
176. a	177. a	178. c	179. b	180. a
181. d	182. b	183. b	184. c	185. a
186. d	187. a	188. a	189. c	190. d
191. d	192. b	193. c	194. d	195. a
196. b	197. b	198. b	199. a	200. b
201. c	202. a	203. c	204. a	205. b
206. c	207. a	208. a	209. b	210. a
211. b	212. c	213. a	214. b	215. [illegible]
216. a	217. b	218. c	219. a	220. [illegible]
221. a	222. b	223. a	224. b	225. [illegible]
226. a	227. a	228. b	229. d	230. [illegible]
231. c	232. a	233. c	234. a	235. [illegible]
236. b	237. c	238. a	239. b	240. [illegible]
241. a	242. d	243. b	244. a	245. [illegible]
246. a				

MATHEMATICAL PHYSICS

Tick (✓) the correct one answer only.

The differential equation which involves one dependent variable w.r.t. independent variable called
(a) Linear differential equation
(b) Ordinary differential equation
(c) Higher order equation
(d) None of these

The highest ordinary derivative that appears in the ordinary differential equation called differential equation
(a) Order (b) Degree
(c) Linearity (d) Homogeneity

Order of differential equation
$x^2 \frac{d^2u}{dx^2} + x\frac{du}{dx} + u = x$ is
(a) 1 (b) 0
(c) 2 (d) None of these

The degree of differential equation
$\frac{du}{dx} + \sqrt{u} = x$ is
(a) 0 (b) 1
(c) None of these (d) 2

The order of differential equation
$\frac{du}{dx} + 4\cos x = \sin x$ is
(a) 1 (b) 0
(c) 2 (d) −1

The degree of differential equation
$\frac{d^2u}{dx^2} + \lambda u = P(x)$
(a) 0 (b) 1
(c) 3 (d) 2

7. If differential equation has first degree in dependent variable it is called
(a) Linear order differential equation
(b) Homogeneous
(c) None homogeneous
(d) None of these

8. A differential equation is non linear if it has function
(a) Radical (b) Trigonometric
(c) Complex (d) Real

9. The differential equation
$\frac{du}{dx} + x\cos u = \sin x$ is
(a) Linear
(b) Homogeneous
(c) None linear
(d) None homogeneous

10. $u'' + 3u' + 2u = e^x$ is a
(a) None homogeneous
(b) Homogeneous
(c) Linear
(d) None linear

11. A differential equation which have one or more dependent variable called
(a) Ordinary differential equation
(b) Partial ordinary differential equation
(c) Linear
(d) None linear

12. A differential equation which have one or more independent variable called
(a) Ordinary differential equation
(b) Linear
(c) None linear
(d) Partial ordinary differential equation

29. Laplace equation in one dimension using electrostatic flux law is

(a) $E = -\frac{\partial \phi}{\partial n} n$ (b) $E = \frac{\partial^2 \phi}{\partial n}$

(c) $E = 0$ (d) $E = \frac{\partial \phi}{\partial n}$

30. Mathematically partial differential equation for classification is given by

(a) $Au_{xx} + Bu_{xy} + Cu_{yy} + Du_x + Eu_y + Fu + G = 0$

(b) $Au_{xx}\ Bu_{yy} + Cu_{xy} + Eu + Fu = 0$

(c) $Au_{yy} + Bu_{xx} + Cu_{xy} + C^2u + F^2u^2 = 0$

(d) None of these

31. For the statement 30(a) the partial differential equation is ellipse if

(a) $B^2\text{-}4AC = 0$ (b) $B^2\text{-}4AC < 0$

(c) $B^2 + 4AC = 0$ (d) $B^2 + 4AC > 0$

2 For the statement 30(a) the partial differential equation is parabolic if

(a) $B^2\text{-}4A^2C = 0$ (b) $B^2\text{-}4A^2C^2 = 0$

(c) $B^2 + 4AC = 0$ (d) $B^2\text{-}4AC = 0$

3. For the statement 30(a) the partial differential equation is hyperbolic if

(a) $B^2\text{-}4AC > 0$ (b) $B^2 + 4AC = 0$

(c) $B^2\text{-}4AC < 0$ (d) $B^2 + 4AC = 0$

. Wave equation in two dimensions is given by

(a) $U_{xx} + U_{yy} = \frac{1}{c^2}[U_{tt} - F(x,y,t)]$

(b) $U_{xx} + U_{yy} = \frac{1}{c^2} U_{tt}$

(c) $U_{xy} + U_{yx} = \frac{1}{c^2}$

(d) $U_{xx} + U_{yx} = 0$

For f mass per unit volume the wave equation in 3-D is for external force f per unit mass acting on surface ds

(a) $\nabla u = \frac{1}{c}[U_{tt} + f]$ (b) $\nabla^2 u = \frac{1}{c^2}[U_{tt} + f]$

(c) $\nabla^2 u = 0$ (d) $\nabla^2 u = \frac{1}{c^2}[U_{tt} + f]$

36. For statement (35) for non-homogeneous surface

(a) $\nabla u = \frac{\rho}{T} U_{tt}$ (b) $\nabla^2 u = \frac{\rho}{T} U_t$

(c) $\nabla^2 u = \frac{\rho}{T} U_{tt}$ (d) $\nabla^2 u = 0$

37. For statement (35) for homogenous surface is

(a) $\nabla^2 u = \frac{1}{c^2} U_{tt}$ (b) $\nabla^2 u = \frac{1}{c} U_{tt}$

(c) $\nabla^2 u = C\, U_{tt}$ (d) None of these

38. The heat equation in one-dimensional for k(x, t, u) coefficient of normal conductivity is given by

(a) $U_t = k\frac{\partial u}{\partial t}$ (b) $U_{xx} \frac{1}{k}\frac{\partial u}{\partial t}$

(c) $U_{xx} = \frac{1}{k^2}\frac{\partial u}{\partial t}$ (d) $U_{xy} = \frac{1}{k^2}$

39. The heat equation in 3-D for k(diffusitivity) is

(a) $\nabla u = \frac{1}{k}\frac{\partial u}{\partial t}$ (b) $\nabla u = \frac{\partial u}{\partial t}$

(c) $\nabla^2 u = \frac{1}{k}\frac{\partial u}{\partial t}$ (d) Zero

40. Simple harmonic motion differential equation is

(a) $\frac{d}{dx}\left[1 \cdot \frac{d}{dx}\right] + \ln x = 0$

(b) $\frac{d}{dt}\left[1 \cdot \frac{d}{dx}\right] + \ln x = 0$

(c) $\frac{d}{dt}\left[1 \cdot \frac{d}{dx}\right] = 0$

(d) $\frac{d}{dt}\left[1 \cdot \frac{d}{dx}\lambda\right] = 0$

41. Simple harmonic oscillator's differential equation is

(a) $\frac{d}{dk}\left[1 \cdot \frac{du}{dx}\right] + w^2u = 1$

(b) $\frac{d}{dx}\left[1 \cdot \frac{du}{dx}\right] + w^2u = 0$

(c) $\frac{d^2}{dx^2}\left[1 \cdot \frac{du}{dx}\right] + w^2u = 0$

(d) None of these

53. The function $\int_0^1 x^2\phi(x)\, dx = \phi(t)$ is

(a) Linear (b) Non-linear
(c) Homogeneous (d) None of these

54. $\bar{u}(s) = \int_0^\infty e^{-st}\, u(t)\, dt$ where
$S = x + iy$(complex) is transformation
(a) Laplace (b) Fourier
(c) Linear (d) Non-linear

55. Laplace $L\{1\} = \int_0^\infty e^{-st} \cdot 1\, dt =$ (for $S>0$)

(a) S (b) S^2
(c) $\frac{1}{S}$ (d) 0

56. Laplace $L\{t^n\} =$

(a) $\frac{n}{S^{n+1}}$ (b) $\frac{n!}{S^{n+1}}$
(c) $\frac{n}{S^{n-1}}$ (d) $\frac{n}{S}$

57. Laplace $L(c) = \int_0^\infty e^{-st} \cdot c =$ for $s>1$ is

(a) $\frac{1}{S}$ (b) CS
(c) $\frac{C}{S}$ (d) $=0$

8. Laplace transformation
$Lu(t) = \int_0^\infty e^{-st}\, u(t)\, dt$ for $u(t) = \sin kt$ and $s>0$ is given by

(a) $\frac{k}{k^2+s^2}$ (b) $\frac{k}{s^2-k^2}$
(c) $\frac{1}{S-a}$ (d) $\frac{k}{S+a}$

. $L[u(t)] = \int_0^\infty e^{-st}\, u(t)\, dt$ for $u(t) = \cos kt$ is given

(a) $\frac{k}{S^2+k^2}$ (b) $\frac{S}{S^2+k^2}$
(c) $\frac{S}{S^2-k^2}$ (d) None of these

60. Laplace $L\{u(t)\} = \int_0^\infty e^{-st}\, u(t)\, dt$ for $u(t) = t^n$ is

(a) $\frac{n}{S^{n+1}}$ (b) $\frac{2n!}{S^{n+1}}$
(c) $\frac{n^2}{S^{n+1}}$ (d) $\frac{n!}{S^{n+1}}$

61. Laplace $L\{u(t)\} = \int_0^\infty e^{-st}\, u(t)\, dt$ for $u(t) = \sin hkt$

(a) $\frac{k}{S^2-k^2}$ (b) $\frac{k}{S^2+k^2}$
(c) $\frac{k^2}{S^2+k^2}$ (d) $\frac{k^2}{S^2}$

62. Laplace $L\{u(t)\} = \int_0^\infty e^{-st}\, u(t)\, dt$ for $u(t) = \cos hkt$

(a) $\frac{k}{S^2-k^2}$ (b) $\frac{S}{S^2+k^2}$
(c) $\frac{S}{S^2-k^2}$ (d) $\frac{S^2}{S^2+k^2}$

63. Laplace $L\{u(t)\} = \int_0^\infty e^{-st}\, u(t)\, dt$ for $u(t) = e^{at}\sin kt$

(a) $\frac{k}{(S-a)^2+k^2}$ (b) $\frac{k}{(S+a)^2+k^2}$
(c) $\frac{k^2}{(S+a)^2}$ (d) None of these

64. Laplace $L\{u(t)\} = \int_0^\infty e^{-st}\, u(t)\, dt$ for $u(t) = e^t\cos kt$

(a) $\frac{k}{(S-a)^2}$ (b) $\frac{k^2}{(S+a)^2}$
(c) $\frac{k}{(S-a)^2+k^2}$ (d) $\frac{k^2}{S^2}$

65. Laplace L[u(t)] = $\bar{u}(S) = \int_0^\infty e^{-st} u(t)\, dt$ for $u(t) = e^{at} = e \sin hkt$ is

(a) $\frac{k}{(S-a)^2-k^2}$ (b) $\frac{k}{(S+a)^2-k^2}$

(c) $\frac{k}{(S-a)^2+k^2}$ (d) $\frac{k^2}{S^2}$

66. Laplace L[u(t)] = $\bar{u}(S) = \int_0^\infty e^{-st} u(t)\, dt$ = for $u(t) = e^{at} \cos hkt$ is

(a) $\frac{k}{(S-a)^2+k^2}$ (b) $\frac{S}{(S-a)^2+k^2}$

(c) $\frac{S-a}{(S-a)^2-k^2}$ (d) $\frac{S}{k}$

67. Inverse Laplace transformation L^{-1} [$\bar{u}(S)$] = for $\bar{u}(S) = \frac{1}{S}$ is

(a) 1 (b) 0

(c) −1 (d) Constant

68. Inverse Laplace transformation L^{-1} [$\bar{u}(S)$] = for $\bar{u}(S)$ = sin kt

(a) $\frac{k^2}{S^2-k^2}$ (b) $\frac{k}{S^2+k^2}$

(c) $\frac{S}{k+S}$ (d) $\frac{k}{S^2+k^2}$

69. Inverse Laplace transformation L^{-1} [$\bar{u}(s)$] = for $\bar{u}(s)$ = cos kt is

(a) $\frac{S}{S^2+k^2}$ (b) $\frac{1}{S-a}$

(c) $\frac{1}{S+a}$ (d) $\frac{1}{S^2-k^2}$

70. Inverse Laplace transformation L^{-1} [$\bar{u}(s)$] = for $\bar{u}(s) = e^{at}$ is

(a) $\frac{1}{S+a}$ (b) $\frac{k}{S^2-k^2}$

(c) $\frac{k}{S^2+k^2}$ (d) $\frac{1}{S-a}$

71. Inverse Laplace transformation L^{-1}[$\bar{u}(s)$] = where u(t) $\bar{u}(s) = \frac{n!}{S^{n+1}}$

(a) t^n (b) t^n

(c) n (d) 1

72. Inverse Laplace transformation L^{-1} [$\bar{u}(s)$] = for $u(s) = \frac{k}{(S-a)^2+k^2}$ is

(a) e^{at} cos kt (b) e^{at} sin kt

(c) sin h kt (d) cos h kt

73. Inverse Laplace transformation L^{-1} [$\bar{u}(s)$] for $\bar{u}(s) = \frac{S-a}{(S-a)^2+k^2}$ is

(a) e^{at} sin kt (b) e^{at} cos kt

(c) e^{at} (d) e^{-at}

74. Inverse Laplace transformation L^{-1} [$\bar{u}(s)$] = for $\bar{u}(s) = \frac{k}{S^2-k^2}$

(a) e^{at} (b) e^{-at}

(c) sin h kt (d) cos kt

75. Inverse Laplace L^{-1}[$\bar{u}(s)$] for $\bar{u}(s) = \frac{S}{S^2-k^2}$

(a) cos h kt (b) sin h kt

(c) 0 (d) e^{akt}

76. Inverse transformation L^{-1} [$\bar{u}(s)$] = for $\bar{u}(s) = \log_e\left(\frac{S+b}{S+a}\right)$

(a) $\frac{e^{-at}+e^{-bt}}{t}$ (b) $\frac{e^{-at}-e^{-bt}}{t}$

(c) $\frac{et+e^{-t}}{2}$ (d) $\frac{e^{2at}-e^{2bt}}{2}$

77. The function $f(z) = \frac{1}{z}$ then f(z) is analytic except at z =

(a) 1 (b) −1

(c) 0 (d) 2

78. The function $f(z) = \frac{1}{z}$ then f(z) has isolated singularity at

(a) z = −1 (b) z = 2

(c) z = 0 (d) z = e

79. The function $f(x) = \frac{1}{\sin\frac{\pi}{2}}$ has singularities

(a) Finite (b) Infinite
(c) 0 (d) −1

80. A function $u(t) = u(t + nT)$ for all t where nT is constant then having period

(a) 1 (b) 0
(c) T (d) −1

81. The inverse Laplace transforms of $\frac{1}{S(S^2+1)}$ is

(a) 1 − cost (b) 1 + cost
(c) sin t (d) tan t

82. The Fourier transforms operator and its inverse are

(a) Linear (b) Non-linear
(c) Constant (d) Zero

83. For Fourier property of translation of $\{u(x + a)\} =$

(a) $e^{i\xi a}\ \bar{u}(\epsilon)$ (b) $e^{i\xi a}\ \bar{u}(a)$
(c) $e^{i\xi a}\ \bar{u}(\epsilon)$ (d) Zero

84. The forueir shifting property translation is for if $\{u(x - a)\} =$

(a) $e^{\epsilon a}\ \bar{u}(\epsilon)$ (b) $e^{-i\xi a}\ \bar{u}(e)$
(c) Zero (d) $e^{-\xi a}\ \bar{u}(\epsilon)$

85. Multiplication by an exponential property in fourneir series if is $L\{e^{ax} \cdot u(x)\} =$

(a) $\{f\{u(x)\}_{x\to\xi+ia}$ (b) $\{f\{(x)\}_{x\to\xi+ia}$
(c) $\{f\{(x)\}_{x\to ia}$ (d) None of these

86. Multiplication by an exponential property is foureir series T is if $\{e^{-ax}\ u(x)\} =$

(a) $\{T\{u(x)\}_{x\to ia}$ (b) $\{T\{u(x)\}_{x\to\xi-ia}$
(c) Zero (d) None of these

87. For fourneir exponential property $T\ \{N^{-dx^2}\}$ =

(a) Zero (b) $e^{-\alpha x^2}$
(c) $\frac{Ne^{-\alpha^2}}{\sqrt{2\alpha}}$ (d) $\frac{Ne^{-\xi^2/4\alpha}}{\sqrt{2\alpha}}$

88. If f(x) is pairwise smooty and absolutely integrable then $\lim_{|k|\to\infty} f(x) =$

(a) $\lim_{|k|\to\infty} f(x) = 1$ (b) $\lim_{|k|\to\infty} f(x) = 0$
(c) $\lim_{|k|\to\infty} f(x) = -1$ (d) None of these

89. Convolution of two functions f(x) and g(x) over the infinite interval $-\alpha < x < \alpha$ is defined as

$-\alpha < x < \alpha \frac{1}{\sqrt{2\pi}} \int_{-\alpha}^{\infty} f(\eta)\ g(x - \eta) d\eta =$

(a) $f(x) | g(x)$ (b) $f(x) \cdot g(x)$
(c) $f(x) / g(x)$ (d) 0

90. Let $T\{f(x)\} = \bar{f}(\xi)$ and $T\{g(x)\} = \bar{g}(\xi)$ then $T\{\bar{f}(\xi) \cdot \bar{g}(\xi)\} =$

(a) $f(x) \cdot g(x)$ (b) $f(x)/g(x)$
(c) $f(x) - g(x)$ (d) $f^2(x)$

91. For the statement $T\{f(x) \cdot g(x)\} =$

(a) $\bar{f}(\xi) | \bar{g}(\xi)$ (b) $\bar{f}(\xi) \cdot \bar{g}(\xi)$
(c) $\bar{f}(\xi) - \bar{g}(\xi)$ (d) Zero

92. Fournier sine transformer $T_{(s)}$ over a function f(x) is $\sqrt{\frac{2}{\pi}} \int_0^{\alpha} f(x) \sin \xi(x)\ dx =$

(a) $\bar{f}_s(\xi)$ (b) $\bar{f}_s(0)$
(c) $\bar{f}_s(x)$ (d) $T_{(s)}(x)$

93. In Fourier transformation $\lim_{|k|\to\infty} T\{u(x)\} =$

(a) 1 (b) −1
(c) 2 (d) 0

94. The value of common integral $\int_{-\alpha}^{\alpha} \frac{\sin}{x} dx$ =

(a) π (b) Zero
(c) −1 (d) 1

95. The value of common integral $2 \int_0^{\alpha} \frac{\sin x}{x} dx$ =

(a) 0 (b) −1
(c) π (d) $-\frac{\pi}{2}$

96. The value of integral $\int_0^{\infty} \frac{\sin x}{x} dx =$

(a) π (b) $-\pi$

(c) $\frac{\pi}{4}$ (d) $\frac{\pi}{2}$

97. The subject variational methods is a subject of

(a) Maxima (b) Minima

(c) Integers (d) Both a and b

98. Necessary condition for a curve $y = f(x)$ for Maxima & minima is

(a) $f'(x) = 0$ (b) $f'(x) = 1$

(c) $f'(x) = -1$ (d) $f'(x) = c$

99. A critical point for "f" at any point C is that for which $f'(c) =$

(a) 0 (b) Does not exist

(c) Finite (d) Both a and b

100. The critical point for which $f'(x) = 0$ are called

(a) Stationary point (b) Constant point

(c) Variable point (d) None of these

01. $I[Y(X)] = \int_a^b f(x, y, y')\, dx$ is well defined number called

(a) Function (b) Functional

(c) Domain (d) Range

02. The necessary condition for $y = f(x)$ at $x = x_0 \in (a, b)$ then, for critical point

(a) $f'(x_0) = 0$

(b) $f'(x_0)$ does not exist

(c) Both a and b

(d) None of these

3. In the above statement No 102 sufficient condition for maximum at x_0 is

(a) $f''(x) < 0$ (b) $f''(x) = 0$

(c) $f''(x) > 0$ (d) None of these

. In the above statement No. 102 sufficient condition for min at x_0 is

(a) $f''(x) < 0$ (b) $f''(x) = 0$

(c) $f''(x) > 0$ (d) None of these

105. Necessary condition to exist extrema is that for (x_0, y_0) in an interior point of f, there exist

(a) $f_x(x_0, y_0) = 0$ (b) $f_y(x_0, y_0) = 0$

(c) $f_{xy}(x_0, y_0) = 0$ (d) Both a and b

106. Let $f_{xx}(x_0, y_0) = A$ and $f_{yy}(x_0, y_0) = B$, $f_{xy}(x_0, y_0) = C$ then function is maximum at (x_0, y_0) if

(a) $AB - C^2 > 0$ (b) $A < 0$

(c) $A > 0$ (d) Both a and b

107. Let $f_{xx}(x_0, y_0) = A$ & $f_{yy}(x_0, y_0) = B$ & $f_{xy}(x_0, y_0) = C$ function is min if at (x_0, y_0)

(a) $A < 0$ (b) $AB - C^2 > 0$

(c) $A > 0$ (d) Both b and c

108. For statement No.107 there is no information about maximum or minimum

(a) $AB - C^2 = 0$ (b) $AB - C^2 < 0$

(c) $AB - C^2 > 0$ (d) None of these

109. Length of curve $AB = \int_A^B ds$ is given by

(a) $\int_A^B \sqrt{1 - y'^2}\, dx$ (b) $\int_A^B \sqrt{1 + Y'^2}\, dx$

(c) $\int_A^B \sqrt{1 + Y^2}\, dx$ (d) Zero

110. Euler-Lagrange equation in one dimension is

(a) $\frac{\partial f}{\partial y} + \frac{d}{dx} = 0$ (b) $\frac{\partial f}{\partial y} + \frac{d}{dx}\frac{\partial f}{\partial y'} = 0$

(c) $\frac{\partial f}{\partial y} - \frac{d}{dx}\left(\frac{\partial f}{\partial y'}\right)$ (d) None of these

111. First order Euler Lagranges differential equation

(a) $\frac{\partial f}{\partial y'} = C$ (b) $\frac{\partial f}{\partial y} = 0$

(c) $\frac{\partial f}{\partial y'} = 9$ (d) $\frac{\partial f}{\partial y'} = \pm 1$

112. First integral of Euler's Lagrange differential equation is given by

(a) $f + y'\frac{\partial f}{\partial y} = 0$ (b) $f - y'\frac{\partial f}{\partial y} = 0$

(c) $f - y'\frac{\partial f}{\partial y'} = c$ (d) None of these

113. The curve $\int_0^{\pi/2} (y'^2 - y^2)dx; y(0) = 0$

$y\left(\frac{\pi}{2}\right) = 1$ is extremized for the functional

(a) $y = \sin x$ (b) $y = \cos x$

(c) $y = \tan x$ (d) $y = 0$

14. In Hamilton's principle for K.E.T. and P.E. V intimate (t_1, t_2) i.e. $\int_{t_1}^{t_2} (T - v)\, dt$ is

(a) Zero (b) Stationary

(c) 1 (d) -1

15. Euler-Langranges equation when integrand has one dependent and two independent variables i.e. for

$I[u(x, y)] = \iint_R f(x, y, u, u_x, u_y)\, dxdy$ is

(a) $\frac{\partial f}{\partial u} + \frac{\partial}{\partial y}\frac{\partial f}{\partial uy} = 0$

(b) $\frac{\partial}{\partial u} + \frac{\partial f}{\partial uy} = 1$

(c) $\frac{\partial f}{\partial u} - \frac{\partial}{\partial x}\frac{\partial f}{\partial ux} - \frac{\partial}{\partial y}\frac{\partial f}{\partial uy} = 0$

(d) None of these

The Euler-Lagranges differential equation having more than one independent variable is called differential equation

(a) Ostrogradisky (b) Cauchy

(c) Zeeman (d) None of these

The Euler's Lagranges equation when the integrand involves one dependent & three independent variable as

$\iiint_R f(x, y, z, u, u_x, u_y, u_z)\, dxdydz$ is given by

(a) $\frac{\partial f}{\partial u} - \frac{\partial}{\partial y}\frac{\partial f}{\partial uy} = 0$

(b) $\frac{\partial f}{\partial u} + \frac{\partial}{\partial x}\frac{\partial f}{\partial ux} + \frac{\partial}{\partial y}\frac{\partial f}{\partial uy} = 0$

(c) $\frac{\partial f}{\partial u} - \frac{\partial}{\partial x}\frac{\partial f}{\partial ux} - \frac{\partial}{\partial y}\frac{\partial f}{\partial uy} - \frac{\partial}{\partial z}\frac{\partial f}{\partial uz} = 0$

(d) Non of these

118. Euler' largrange equation when integrand has two dependent & one independent variable

$\int_{x_1}^{x_2} y(x, y_1, y_2, y_1', y_2')\, dx$ is given by for y_1 dependent

(a) $\frac{\partial f}{\partial y_1} - \frac{d}{dx}\frac{\partial f}{\partial y_1'} = 0$ (b) $\frac{\partial f}{\partial y_1} + \frac{d}{dx}\frac{\partial f}{\partial y_1'} = 1$

(c) $\frac{\partial f}{\partial y_2} + \frac{d}{dx}\frac{\partial f}{\partial y_2} = 0$ (d) None of these

119. Eulers Lagranges equation for above statement No 118 for y_2 dependent variable is

(a) $\frac{\partial f}{\partial y_2} + \frac{d}{dx}\frac{\partial f}{\partial y_2'} = 1$ (b) $\frac{\partial f}{\partial y_1} + \frac{d}{dx}\frac{\partial f}{\partial y_2'} = 0$

(c) $\frac{\partial f}{\partial y_2} - \frac{d}{dx}\frac{\partial f}{\partial y_2'} = 0$ (d) $\frac{\partial f}{\partial y_2} - \frac{d}{dx}\frac{\partial f}{\partial y_1'} = 0$

120. The arc length of the curve in polar coordinate

(a) $\int_{\theta_2}^{\theta_1} \sqrt{r^2 + r'^2}$ (b) $\int_{\theta_2}^{\theta_1} \sqrt{r'^2}$

(c) $\int_{\theta_2}^{\theta_2} \sqrt{r^2 + r_1^2}$ (d) None of these

121. The arc length from $A \to B$ in 3-dimensions curve is given by

(a) $\int_A^B \sqrt{(dx)^2 - (dy)^2 - (dz)^2}$

(b) $\int_A^B \sqrt{(dx)^2 + (dy)^2 + (dz)^2}$

(c) $\int_A^B \sqrt{dx^2 + dy^2}$

(d) None of these

122. The Geodesic curve for cylinder $x^2 + y^2 = a^2$ is

(a) $\frac{y}{x} = \tan\frac{z - \alpha_1}{\alpha_2}$ (b) $\frac{x}{y} = \tan\frac{z + \alpha_1}{\alpha_2}$

(c) $x = \tan\frac{\alpha_1}{\alpha_2}$ (d) None of these

123. The shortest curve $y = y(x)$ between the points of plane (x_0, y_0) and (x_1, y_1) is obtained by minimizing the functional of

(a) $\int_{x_1}^{x_2} (1 + y'^2)\, dx$ (b) $\int_{x_1}^{x_2} (1 - y'^2)^{1/2}\, dx$

(c) $\int_{x_1}^{x_2} (1 + y'^2)^{1/2}\, dx$ (d) None of these

124. The largrangian for particle for central field of force of equation of motion is given by

(a) $m\ddot{r} - mr\dot{\theta}^2 = f(r)$ (b) $m\ddot{r} + mr\dot{\theta}^2 = f(r)$

(c) $m\dot{r} + m\dot{r}^2 = f'(r)$ (d) None of these

125. The solid surface of revolution of given curve which for a given surface area has max value is

(a) Square (b) Rectangle

(c) Sphere (d) Triangle

ANSWERS

1. b	2. a	3. c	4. d	5. a
6. b	7. a	8. b	9. c	10. a
11. b	12. d	13. b	14. d	[illegible]
16. a	17. b	18. a	19. b	[illegible]
21. b	22. a	23. b	24. c	[illegible]
26. a	27. b	28. a	29. a	[illegible]
31. b	32. d	33. a	34. a	[illegible]
36. c	37. a	38. b	39. c	[illegible]
41. b	42. a	43. c	44. a	[illegible]
46. a	47. b	48. d	49. a	[illegible]
51. a	52. b	53. c	54. a	[illegible]
56. b	57. c	58. a	59. b	[illegible]
61. a	62. c	63. a	64. d	[illegible]
66. c	67. a	68. d	69. a	[illegible]
71. a	72. b	73. b	74. c	[illegible]
76. b	77. c	78. c	79. b	[illegible]
81. b	82. a	83. a	84. b	[illegible]
86. b	87. d	88. b	89. a	[illegible]
91. b	92. a	93. d	94. a	[illegible]
96. d	97. d	98. a	99. d	[illegible]
101. b	102. c	103. a	104. c	[illegible]
106. d	107. d	108. a	109. b	[illegible]
111. a	112. c	113. a	114. b	[illegible]
116. a	117. c	118. a	119. c	[illegible]
121. b	122. a	123. b	124. a	[illegible]

NUMERICAL ANALYSIS

Tick (✓) the correct one answer only.

1. Every equation f(x) = 0 of degree n has only
   (a) 1 roots (b) 2 root
   (c) 0 roots (d) n roots

2. If f(x) = 0 is divisible by x − a, then x = a is always a root
   (a) Real (b) Imaginary
   (c) Complex (d) Real

3. When integral line become parallel to f(x) at x = c then
   (a) $f'(c) = 1$ (b) $f'(c) = 0$
   (c) $f'(c) = -1$ (d) $f'(c) = c$

4. If in equation f(a) and f(b) have opposite signs then root of f(x) = 0 lies between
   (a) x = a, x = −b (b) x = −a, x = b
   (c) x = a, x = b (d) x = −a, x = −b

Every equation of an odd degree has real root
(a) One (b) Two
(c) Three (d) n

Every equation of an even degree with last term −ve sign has a pair of roots
(a) Imaginary (b) Real
(c) Complex (d) None of these

Every equation of even degree with last term with −ve sign has a pair of roots
(a) Real (b) Imaginary
(c) Complex (d) Zero

Every equation of an even degree with last term with −ve sign has a pair of roots
(a) Rational (b) Irrational
(c) Even (d) Odd

9. For quadratic equation $a_2x_2 + a_1x + a_0$ and x_1 and x_2 are roots of this then values of these are given
   (a) $\dfrac{-a_1 \pm \sqrt{a_1^2 - 4a_2a_0}}{2a_2}$
   (b) = Zero
   (c) ± 1
   (d) None of these

10. For the statement given in statement (9)
    (a) $x_1 + x_2 = -\dfrac{a_2}{a_1}$ (b) $x_1 + x_2 = -\dfrac{a_1}{a_2}$
    (c) $x_1 + x_2 = \dfrac{a_1}{a_0}$ (d) Zero

11. For the statement given in statement (9)
    (a) $x_1x_2 = -\dfrac{a_1}{a_2}$ (b) $x_1x_2 = -\dfrac{a_2}{a_1}$
    (c) $x_1x_2 = \dfrac{a_0}{a_2}$ (d) $x_1x_2 = \pm 1$

12. Considering bi-quadratic equation $a_4x_4 + a_3k_3 + a_2k_2 + a_1x + a_0 = 0$ having roots x_1, x_2, x_3 and x_4 then sum of roots
    (a) $-\dfrac{a_3}{a_4}$ (b) $\dfrac{a_2}{a_1}$
    (c) $\dfrac{a_4}{a_1}$ (d) $-\dfrac{a_3}{a_1}$

13. For the statement No.(12) product of roots is given
    (a) $-\dfrac{a_2}{a_1}$ (b) $\dfrac{a_3}{a_0}$
    (c) $\dfrac{a_0}{a_4}$ (d) $-\dfrac{a_3}{a_1}$

30. Find a real root of the equation $x^3 - 4x^2 + x - 10 = 0$
(a) 4.412 (b) 4.009
(c) 4.3069 (d) 3.996

31. The roots near $x = 3$ of equation $e^x - \log_e x = 20$ upto four decimal places is
(a) 3.109 (b) 3.96
(c) 3.746 (d) 3.500

32. The real root of the equation $x^3 + x^2 - 1 = 0$ by using iteration method is
(a) 0.7549 (b) 0.8100
(c) 0.7951 (d) None of these

33. The roots of the equation upto 4 decimal places $x = e^{-x}$ near $x = 0.5$ is
(a) 0.5671 (b) 0.5594
(c) 0.5814 (d) 0.491

34. The root of the equation $2x - \cos x - 3 = 0$ correct to 3dp taking $x_0 = \frac{\pi}{2}$ is
(a) 1.4913 (b) 1.532
(c) 1.523 (d) None of these

35. The regular falsi method to find real roots of an equation $f(x) = 0$ has close resemblance with
(a) Bisection method
(b) Tri-section
(c) Old method
(d) None of these

36. The real root of $x\log_{10}x - 1.2 = 0$ upto 4dp (decimal-places) of the equation
(a) 2.7906 (b) 2.7406
(c) 2.7102 (d) 2.7230

37. The real roots of $2x - 3\sin x - 5 =$ upto 5 decimal places is
(a) 1.3410 (b) 1.3682
(c) 1.3592 (d) None of these

38. The correct root upto 4dp between 0.4 & 0.6 of the equation $\sin x = 5x - 2$ is
(a) 0.4950 (b) 0.4960
(c) 0.399 (d) 0.4107

39. The real root of the equation $x^6 - x^4 - x^3 - 1 = 0$ between 1.4 & 1.5 upto 4d.p. is
(a) 1.4041 (b) 1.4036
(c) 1.4123 (d) 1.4009

40. The +ve root of $x^3 - x - 10 = 0$ is
(a) 2.0112 (b) 2.003
(c) 2.0946 (d) 2.1130

41. A +ve root of nonlinear equation $x_3 - 4x_2 + x - 10 = 0$ is
(a) 4.3069 (b) 4.2919
(c) 4.3110 (d) 4.0029

42. A +ve real root of non linear equation $e^x + x - 10 = 0$ is
(a) 2.0706 (b) 2.1340
(c) 2.0912 (d) 1.9963

43. The Newton's formula for any type of equation is
(a) $x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$ (b) $x_n = x_{n+1} - \frac{f(x_n)}{f'(x_n)}$
(c) Zero (d) ± 1

44. For values $(x, x_0, x_1, x_2, \ldots, x_n)$ & $(y, y_0, y, y_i - y_n)$ then the width of interval h is given
(a) $x_3 - x_1$ (b) $x_1 - x_0$
(c) $x_2 - x_1$ (d) Both b and c

45. The smallest eign value of a matrix A is the reciprocal of the determinant eign value of the matrix is
(a) A^{-1} (b) A^2
(c) $A - A = 0$ (d) None of these

46. The statement given in 44 is called a Newton
(a) Gregory formula (b) Simpson
(c) None of these (d) a and b

47. For $y = \sin\frac{\pi}{2}x$ as a polynomial coincides with it at $x = 0, 1, 2, 3$, is $f(x) =$
(a) $\frac{x^3}{3} + 2x^2 + \frac{8x}{3}$ (b) $\frac{x^3}{3} - 2x^2 + \frac{8x}{3}$

78. Highest suffix-lowest suffix will give the diff-eq
(a) Degree (b) Order
(c) Mode (d) None of these

79. A general solution of a difference equation of order n is solution which involves arbitrary constants
(a) n (b) n + 1
(c) n - 2 (d) Zero

80. Equation
$y_{n+k} + a_1 y_{n+k-1} + a_2 y_{n+k-2} + \ldots a_n y_n = 0$ is called a linear
(a) Diff. equation
(b) Non homogeneous
(c) Homogeneous
(d) Constant

81. $y_{n+k} + a_1 y_{n+k-1} + a_2 y_{n+k-2} + \ldots a_n y_n = f(x)$ is called
(a) Homogeneous
(b) Diff. equation
(c) Non homogeneous
(d) None of these

82. Law of superposition for u_n & v_n any two solutions of homogeneous difference equations for constants α_1 and α_2 is given by
(a) $\alpha_1 U_n \ \alpha_2 V_n$ (b) $\alpha_1 U_n - \alpha_2 V_n$
(c) $\alpha_1 U_n / \alpha_2 V_n$ (d) Both a and b

83. For difference eq. f(x) = constant, to find y particular solution we shall substitute
(a) $y_n = 1$ (b) $y_n = -1$
(c) $y_n = c$ (d) $y_n = 0$

84. For non-homogenous difference equation of the form $f = \alpha\ a^k$ then in order to find y we have to put
(a) $y_k = a^k$ (b) $y_k = Ca^k$
(c) $y_k = C^2 a^k$ (d) $y_k = ba^k$

85. For non-homogenous diff-equation of polynomial say $f_k = 1 + k^2, 1 + k + k^2 \ldots$ etc the in order to find y particular solution we put
(a) y_k = constant
(b) $y_k = \pm 1$
(c) $y_k = B_0 + B_1 k + B_2 k^2 + \ldots$
(d) None of these

86. For difference eq. of the form $f_k = \sin Ak$ or $\cos kB$ to find y we shall make substitution
(a) $y_k = \pm$
(b) $y_k = 0$
(c) $y_k = Ca^k$
(d) $y_k = C_1 \sin Ak + C_2 \cos Ak$

87. If diff-eq of the form $f_k = a^k \sin Ak$ or $a^k \cos Ak$ then to find y we make substitution
(a) $y_k = a^k (C_1 \sin Ak + C_2 \cos Ak)$
(b) $y_k = a^k (C_1 \sin Ak - C_2 \cos Ak)$
(c) $y_k = a^k C_1 \sin Ak$
(d) $y_k = ak\ C1 \cos ak$

88. The method for solving differential equation $y_{n+1} =$
$y_n + hf\left[xn + \frac{h}{2}, y_n + \frac{h}{2} f(x_n, y_n)\right]$ is called
(a) Modified Euler's method
(b) R.K. Method
(c) Both a and b
(d) None of these

89. For solving in general
$y_{n+1} = y_n + hy'_n + \frac{h^2}{2!} y'' + \frac{h}{3!} y''' + \ldots$ is a method
(a) Taylor's series (b) Euler's
(c) R.K. Method (d) None of these

90. $y_{n+1} = y_n + h f(x_n, y_n)$ is formula
(a) R.K. (b) Euler's
(c) Differential (d) None of these

91. Matrix norm is defined as $\|A\| =$
(a) $\max = A$, $\|A\| = 1$ (b) $\max \|Ax\|$, $\|x\| = 1$
(c) $\|Ax\|$ (d) None of these

107. For forwarded and backward operator $\Delta\nabla =$

(a) $(1+\Delta)\nabla$ (b) ∇^2
(c) $\Delta - \nabla$ (d) δ

108. For forwarded and backward operator $\delta^2 =$

(a) $\nabla\Delta$ (b) $\Delta - \nabla$
(c) ∇^2 (d) Both a and b

109. For shift operator E and forward Δ we can have as $(E^{1/2} + E^{-1/2})(1+\Delta)^{1/2} =$

(a) $2 + \Delta$ (b) Δ
(c) Δ^2 (d) $\Delta + 1$

110. For forward and shift operator $E\left(\frac{\Delta}{E}\right)^2 e^x =$

(a) Ee^x (b) e^x
(c) $e^x - 1$ (d) None of these

111. For forwarded operator Δ we can have as $\Delta^2 =$

(a) $(1+\Delta)\delta$ (b) $(1+\Delta)\delta^3$
(c) $(1+\Delta)\delta^2$ (d) δ^2

112. For forwarded and backward operator $(1+\Delta)(1-\nabla) =$

(a) 1 (b) -1
(c) 0 (d) 2

113. For u(Averaging operator), δ (central difference operator) and E(shift operator) $E^{1/2} =$

(a) $u\delta$ (b) $u - \delta$
(c) $u + \frac{1}{2}\delta$ (d) Zero

114. According to above statement $E^{-1/2} =$

(a) $u - \frac{1}{2}\delta$ (b) $u + \delta$
(c) u/δ (d) ± 1

115. For forward operator Δ we can have as $\Delta^2 =$

(a) $(1-\Delta)\delta^2$ (b) $(1+\Delta)\delta^2$
(c) $\Delta\delta^2$ (d) $\nabla\delta$

116. For central difference operator δ it can be written as $\Delta(\Delta+1)^{-1/2} =$

(a) δ (b) ∇
(c) ∇^2 (d) Δ

117. According to Trapezoidal rule for $n = 1$ we can write as $\int_{x_0}^{x_1} y\,dx =$

(a) $\frac{h}{2}(y_0 - y_1)$ (b) $\frac{h}{2}(y_0 + y_1)$
(c) $\frac{h}{3}(y_0 + y_1)$ (d) Zero

118. We can take forward operator Δ as

(a) $\frac{1}{h}\left(\Delta + \frac{\Delta^2}{2} + \frac{\Delta^3}{3} + \frac{\Delta^4}{4} \dots\right)$

(b) $\frac{1}{h}\left(\Delta - \frac{\Delta^2}{2} + \frac{\Delta^3}{3} - \frac{\Delta^4}{4} \dots\right)$

(c) $\frac{1}{h}$

(d) None of these

ANSWERS

1. d	2. d	3. b	4. c	5. a
6. b	7. c	8. b	9. a	10. b
11. c	12. a	13. c	14. b	15. d
16. a	17. b	18. c	19. c	20. a
21. b	22. b	23. a	24. c	25. a
26. b	27. b	28. a	29. b	30. c
31. d	32. a	33. a	34. c	35. a
36. b	37. c	38. a	39. b	40. c
41. a	42. a	43. a	44. d	45. a
46. a	47. b	48. c	49. d	50. b
51. a	52. b	53. c	54. d	55. a
56. b	57. b	58. a	59. c	60. b
61. c	62. a	63. b	64. b	65. c
66. a	67. b	68. a	69. b	70. c
71. a	72. a	73. c	74. a	75. b
76. c	77. a	78. b	79. a	80. c
81. c	82. a	83. c	84. b	85. c
86. d	87. a	88. c	89. a	90. b
91. b	92. c	93. a	94. c	95. a
96. c	97. a	98. c	99. a	100. b
101. c	102. a	103. a	104. d	105. a
106. b	107. c	108. d	109. a	110. b
111. c	112. b	113. c	114. a	115. b
116. a	117. b	118.		

PREVIOUS SOLVED PAPERS

APPLIED MATHS PAPERS

PAPER I (2001)

COMPULSORY QUESTION

Q.8. Write only the correct answer in the Answer Book. Don't reproduce the questions.

1. An equation of a plane determined by the points $P_1(2, -1, 1)$, $P_2(3, 2, -1)$ and $P_3(-1, 3, -2)$ is:
   (a) $11x + 5y' = 13x = 30$
   (b) $2x + y - z = 15$
   (c) $x - y + z = 10$
   (d) None of these

2. The projection p of a vector in the direction of a vector is
   (a) $\bar{a} \cdot \bar{b}$ (b) $\frac{\bar{a}\,\bar{b}}{|\bar{b}|}$
   (c) $\frac{\bar{a}\,\bar{b}}{|\bar{a}|}$ (d) None of these

3. Let a(x, y, z) be a differentiable function at a point P(x, y, z) in a certain region. The directional derivative of a in the direction of is
   (a) $V\delta\bar{a}$ (b) $\frac{E\delta\bar{a}}{|\bar{a}|}$
   (c) $\frac{E\delta\cdot\bar{a}}{|E\delta|}$ (d) None of these

4. Let $\vec{V}$ be a vector field, in a region R, whose partial derivatives are continuous at points of R. If curl $\vec{V} = 0$ at the points of R, then
   (a) $\vec{V}$ is said to be rotational in R
   (b) $\vec{V}$ is said to be irrational in R
   (c) $\vec{V}$ is said to be solenoidal in R
   (d) None of these

5. Let f be a differentiable function that represent a surface S. Then the gradient of (none zero vector) at a point P of S is
   (a) A normal vector on S at P
   (b) A vector in the tangent plane at P
   (c) An oblique vector on S at P
   (d) None of these

6. A potential function is scalar function which satisfies
   (a) Heat equation (b) Laplace equation
   (c) Wave equation (d) None of these

7. The spherical coordinates system is
   (a) An orthogonal system
   (b) Not an orthogonal system
   (c) Equivalent to Cartesian coordinates system
   (d) Neon of these

8. The range of a projectile is maximum if the angle of project ion is
   (a) $\frac{\pi}{3}$ (b) $\frac{\pi}{6}$
   (c) $\frac{\pi}{4}$ (d) None of these

9. The centre of mass of a set of particle is the point with respect to which
   (a) Linear momentum of the set of particles is zero
   (b) Sum of the vectors joining the set of particles is zero
   (c) The mass is zero
   (d) None

10. The center of mass of a right circular solid cone of height h is
   (a) $\frac{2}{3}h$ (b) $\frac{3}{4}h$

(c) $\frac{1}{1}$ h (d) None of these

11. Let A be a unit vector. The $\vec{A}$ and $\frac{d\vec{A}}{dt}$ will be
(a) Parallel to each other
(b) Anti-parallel to each other
(c) Perpendicular to each other
(d) None of these

12. The components of the force at any point, in a conservation field, are the negative of the components of the gradient of
(a) The potential energy at the point
(b) The kinetic energy at the point
(c) The Lagrangian at the point
(d) None of these

ANSWERS

1. d	2. c	3. b	4. a	5. b
6. a	7. a	8. c	9. b	10. c
11. a	12. a			

PAPER II

COMPULSORY QUESTION

Q 8. Write on the correct answer in the Answer Book. Don't reproduce the questions.

1. The differential equation $[1 + (y')^2]^{1/2} = y$ has the order and degree respectively
(a) 2, 1 (b) 2, 2
(c) 1, 2 (d) None of these

2. Elimination of constants a, b from the equation $y = ae^{3x} + be^{3x}$ gives a differential equation of order
(a) 1 (b) 2
(c) 3 (d) None of these

3. The differential equation $y + y = xy^3$ may be called as linear equation of
(a) Cauchy (b) Bessel
(c) Bernoulli's (d) None of these

4. The differential equation $x^2y + xy + (x^2 - k^2)y = 0$ is
(a) Gauss equation
(b) Bessel equation
(c) Legendre equation
(d) None of these

5. The differential equation $(1 - x^2)y - 2xy + n(n + 1)y = 0$ is
(a) Gauss equation
(b) Bessel equation
(c) Legendre equation
(d) None of these

6. The equation $y + 4y + y = 0$, $y(0) = 1$, $y(0) = 2$, define
(a) Initial value problem
(b) Boundary value problem
(c) None of these

7. The differential equation $(x + y - 10)\,dx - (y - x + 2)\,dy = 0$ is
(a) Linear (b) Exact
(c) Homogeneous (d) None of these

8. Particular integral of $(D^2 + 4y)\sin 3x$ is
(a) $-\frac{1}{5}\sin 3x$ (b) $-\frac{1}{5}\cos 3x$
(c) $-\frac{1}{5}\sin 3x$ (d) None of these

9. The equation $\frac{\delta^2 u}{\delta^2 u} + \frac{5\delta^2 u}{\delta y^2} = \frac{\delta u}{\delta z}$, where x, y, are variable, is a partial differential equation of order and degree.
(a) 2, 1 (b) 2, 2
(c) 1, 2 (d) None of these

10. The partial differential equation $lnp + q = z^2$ is
(a) Of order 1 and is linear
(b) Of order 1 and is not linear
(c) Not of order 1 and is linear
(d) None of these

11. A partial differential $A\frac{\delta^2 u}{\delta x^2} + B\frac{\delta^2 u}{\delta y^2} + C\frac{\delta^2 u}{\delta x\,\delta y} = \sin x + 2y$, where A, B, C are real constants is

(a) Linear
(b) Homogeneous
(c) Non-Homogeneous
(d) None of these

12. The solution of $xp + yq =$ is

(a) $f(x, y) = 0$ (b) $f\left(\frac{x}{y}, \frac{x}{z}\right) = 0$
(c) $f(xy, yz) = 0$ (d) $f(x^2, y^2) = 0$
(e) None of these

13. If E is the shift operator and Δ the forward difference operator then $E - \Delta =$

(a) -1 (b) 1
(c) 0 (d) None of these

14. An equation of the form $AU_{xx} + 2BU_{xy}CU_{yy} = F(x, y, u, u_x, u_y)$ is said to be elliptic if

(a) $AC - B^2 > 0$ (b) $AC - B^2 = 0$
(c) $AC - B^2 < 0$ (d) None of these

15. Heat equation $U_t = C^2U_{xx}$ is

(a) Parabolic (b) Elliptic
(c) Hyperbolic (d) None of these

16. If a is given as first approximation to a root of the equation $f(x) = 0$ and $f(a)$ is very small, then the Newton Rapshon method is

(a) Applicable (b) Not-applicable
(e) Both a, b (d) None of these

17. Lagranges interpolation formula is used for

(a) qual-interval (b) Un-equal interval
(c) Half-interval (d) None of these

18. If A_λ, B_μ are components of a contra variant and covariant tensor of rank one, then $C^\lambda_\mu = A^\lambda B_\mu$ ae the components of a mixed tensor of rank

(a) One (b) Two
(c) Three (d) None of these

19. $\delta_{ik}\, \epsilon_{ikm}$ has the value

(a) 0 (b) -1
(c) 3 (d) None of these

20. The smallest +ve root of $x^3 - 5x + 3 = 0$ lies between

(a) 1 and 2 (b) 0 and 1
(c) 2 and 3 (d) None of these

ANSWERS

1. a	2. b	3. a	4. c	5. c
6. a	7. b	8. b	9. a	10. b
11. c	12. e	13. b	14. c	15. c
16. b	17. b	18. a	19. d	10. b

PURE MATH PAPER – I

COMPULSORY QUESTIONS

Q.8. Write only the correct answer in the Answer Book. Don't reproduce the questions.

1. The set {i, –1, 1, –1} is

(a) Semi group under addition
(b) Group under addition
(c) Group under multiplication
(d) None of these

2. Number of subgroup of order one of an infinite group G is

(a) Zero (b) 1
(c) 2 (d) Infinite
(e) None of these

3. A cyclic group of order n is generate by

(a) n element (b) $(n - 1)$ elements
(c) Two elements (d) One element
(e) None of these

4. Let H be a subgroup of order m of a group of order n, the number of right cossets of H in G is

(a) n (b) $n - m$
(c) $m - n$ (d) $\frac{n}{m}$
(e) None of these

5. The dimension of a vector space V is the number of

(a) Linearly independent vectors in V
(b) Linearly dependent vector in V
(c) Linearly independent vectors spanning V
(d) None of these

6. The characteristics of an integral domain is

(a) Zero (b) A prime
(c) Zero or a prime (d) None of these

7. The eigenvalue is related to the corresponding eigenvector (for a matrix A) as
(a) $|A - \lambda I|\, x = 0$ (b) $[A - \lambda I]\, x = b$
(c) $Ax = \lambda x$ (d) None of these

8. For two vectors $\vec{A}$ and $\vec{B} \cdot \vec{A}$ gives
(a) Cos of angle between $\vec{A}$ and $\vec{B}$
(b) Area of parallelogram with $\vec{A}$ and $\vec{B}$ as its adjacent sides
(c) Vector perpendicular to $\vec{A}$ and $\vec{B}$
(d) None of these

9. If θ is angle between two vectors $\vec{A}$ and $\vec{B}$, then $\dfrac{\vec{A} \times \vec{B}}{|\vec{A}|\,|\vec{B}|}$ gives
(a) $\tan\theta$ (b) $\cos\theta$
(c) $\sin\theta$ (d) $\sec\theta$

10. For the curve $y = \tan x$, the number of asymptotes is
(a) 1 (b) 2
(c) Infinite (d) None of these

11. If $lx + my + nz + d = 0$ is equation of the plane, then l, m and n are
(a) Direction cosines of a line in plane
(b) Direction ratios of a line in plane
(c) Direction cosines of normal to plane
(d) Direction ratios of a line making an angel θ with plane

12. The line $\dfrac{x - x_0}{a} = \dfrac{y - y_0}{b} = \dfrac{z - z_0}{c}$ passes through
(a) $P(x_0, y_0, z_0)$
(b) $P(x_0, y_0, z_0)$ and Q (a, b, c)
(c) Q(a, b, c)
(d) None of these

13. The centre of the sphere $x^2 + y^2 + z^2 + 2x + 4y + 9z + 13 = 0$ is the point
(a) $\left(1, 2, \frac{9}{2}\right)$ (b) $\left(-1, -2, \frac{9}{2}\right)$
(c) (1, 2, 3) (d) $-1, 2, -3)$
(e) None of these

14. $\dfrac{x^2}{a^2} + \dfrac{y^2}{b^2} + \dfrac{z^2}{c^2} = 1$ is equation of
(a) Sphere (b) Hyperbolic
(c) Parabolic (d) Ellipsoid

15. The perpendicular distance of points (3, 2, 1) form the plane $2x + 3y - z = 5$ is
(a) 2 (b) 4
(c) 6 (d) 8
(e) None of these

$\sqrt{6+6-1}$

16. A commutative ring with identity and no divisors of zero is called
(a) Field (b) Ideal
(c) Integral domain (d) None of these

17. An $m \times n$ matrix A and a $p \times q$ matrix B are conformable for multiplication AB
(a) Always (b) when $p = m$
(c) when $p = n$ (d) when $q = n$
(e) None of these

18. Ker T where T: U(F) → V (F) is linear transformation is
(a) $\{x \mid x \in U, T(x) \neq 0\}$
(b) V(F)
(c) $\{y \mid y \in V, y = T(0)\}$
(d) $\{x \mid x \in U, T(x) \neq 0\}$
(e) None of these

19. Isomorphism is
(a) A bisection
(b) An onto homomorphism
(c) Bijctive homomorphism
(d) None of these

20. Angle between lines $\dfrac{x-2}{3} = \dfrac{2y+3}{-1} = \dfrac{z-1}{5}$ and $\dfrac{x+1}{2} = \dfrac{y-2}{1} = \dfrac{z-2}{-1}$ is
(a) Zero (b) $\frac{\pi}{4}$
(c) $\frac{\pi}{2}$ (d) π

ANSWERS

1. c	2. d	3. d	4. d	5. c
6. c	7. a	8. a	9. c	10. b
11. a	12. a	13. a	14. d	15. e
16. c	17. c	18. c	19. b	20. c

PAPER – II

COMPULSORY QUESTION

Q.8. Write only the correct answer in the Answer Book. Don't reproduce the questions.

1. The function $f(x)\frac{x^2-a^2}{x-a}$ is discontinuous at
   (a) $x=1$ (b) $x=a$
   (c) $x=0$ (d) $x=\sqrt{a}$
   (e) None of these

2. $f(x)=\cos x$ has maximum value at
   (a) $x=0$ (b) $x=1$
   (c) $x=\frac{\pi}{2}$ (d) $x=\frac{3\pi}{2}$
   (e) None of these

3. $\lim \frac{\sin x}{x}$ is
   (a) Zero (b) 1
   (c) Undefined (d) –1
   (e) None of these

4. Derivative of the function $f(x)$ $\tan x$ at $x=\frac{\pi}{4}$ is
   (a) 2 (b) $\frac{1}{2}$
   (c) 1 (d) Zero
   (e) None of these

5. For an increasing function f.
   let $x_1 < x_2$ then
   (a) $f(x_1) > f(x_2)$ (b) $f(x_1) < f(x_2)$
   (c) $f(x_1) = f(x_2)$ (d) None of these

6. Area under the curve $f(x) = ex + 2$ bounded by $x = 0$, $x = 2$ and x-axis given by
   (a) 3 (b) $e^3 + 2$
   (c) $e^2 + 1$ (d) $e^2 + 3$
   (e) None of these

7. Normal to the parabola $y^2 = 12x$ at $(3, -6)$ is
   (a) $y = x = 3$ (b) $y = x - 9$
   (c) $y + x + 3 = 0$ (d) None of these

8. Equation of tangent to the circle $x^2 + y^2 = a^2$ at (x_1, y_{10}) is given by
   (a) $x_2 + y_2 + 2gx + 2fy + c = 0$
   (b) $x_2 + y_2 + 2gx_1 + 2fy_2 + c = 0$
   (c) $xx_1 + yy_1 + 2gx_1 + 2fy_1 - \varepsilon = 0$
   (d) $xx_1 + yy_1 + g(x + x_1) + f(y + y_1) + c = 0$
   (e) None of these

9. In a complete metric space
   (a) Every sequence is bounded
   (b) Every sequence converges
   (c) Every Cauchy sequence converges
   (d) There is no convergent sequence
   (e) None of these

10. The open ball of radius 1 and center at zero in R is given by
    (a) (0, 1) (b) [0, 1]
    (c) (–1, 1) (d) {0}

11. For the positive term series and if $\sum_1^\infty a_n \sum_1^\infty b_n$ if $a_n \le b_n \; \forall_n = 1, 2, \ldots$, if $\sum_1^\infty b_n$ is convergent then:
    (a) $\sum_1^\infty a_n$ diverges
    (b) $\sum_1^\infty a_n$ converges
    (c) $\sum_1^\infty a_n$ converges absolutely
    (d) None of these

12. Polar form for the complex number $z = 3 - 4i$ is
    (a) $5e^{i\theta}$ (b) $5e^{-i\theta}$
    (c) $5e^{2i\theta}$ (d) $e^{i\theta}$

13. Log $(x + iy)$ is given by
    $\left(|z| = \sqrt{x^2+y^2}, \theta = \tan^{-1}\frac{y}{x} \right)$
    (a) $\log |z| = i\theta$ (b) $\text{long } |z| + i\lambda\theta$
    (c) $\log (|z| + i\lambda\theta)$ (d) $\log (|z| + i\theta)$
    (e) None of these

14. A curve $Z = f(t)$ is smooth if for $t \in [a, b]$
    (a) $f'(t) = 1$ (b) $f'(t) = 0$
    (c) $f'(t) \ne 0$ (d) $f(a) = f(b)$
    (e) None of these

11. For a practice moving in a central force the angular momentum is
(a) Conserved (b) Zero
(c) Variable (d) None of these

12. The centre of mass of semicircular lamina $x^2 + y^2 = a^2$ in the upper half lies on
(a) The origin (b) x-axis
(c) y-axis (d) None of these

13. If the amplitude of oscillation of a particle performing simple harmonic is doubled, then its time period is
(a) Doubled (b) Halved
(c) Unchanged (d) None of these

14. The ratio of coefficient of static friction and coefficient of kinetic friction is
(a) Greater than 1 (b) Less than 1
(c) Equal to 1 (d) None of these

15. The simple harmonic motion, the acceleration at distance x is
(a) $-\lambda x$ (b) λx
(c) λx^2 (d) $-\lambda x^2$

16. The range of the projectile is maximum, if the angle of projection is
(a) $\frac{\pi}{3}$ (b) $\frac{\pi}{4}$
(c) $\frac{\pi}{6}$ (d) $\frac{2\pi}{3}$

17. The transverse component of acceleration is
(a) r (b) $r\theta$
(c) $2r\theta$ (d) $r - r(\theta)^2$

18. The orbit of plane is
(a) Circle (b) Eclipse
(c) Hyperbola (d) Parabola

19. The angular of speed of the earth about its axis is
(a) 7.29×10^{-5} rad/sec
(b) 7.5×10^{-5} rad/sec
(c) 6.89 rad/sec
(d) None of these

20. The moment of inertia of hollow sphere of radius a and mass M about a diameter is
(a) $\frac{1}{2}Ma^2$ (b) $\frac{1}{3}Ma^2$
(c) Ma^2 (d) $2Ma^2$

ANSWERS

1. d	2. a	3. a	4. a	5. b
6. a	7. c	8. c	9. b	10. c
11. b	12. b	13. b	14. b	15. a
16. b	17. d	18. b	19. a	20. b

PAPER II

COMPULSORY QUESTION

Q.8. Write only the correct answer in the Answer Book. Don't reproduce the questions.

1. The differential equation $[1 + (y')^2]^{1/3} = y''$ has the degree and order respectively
(a) 2, 1 (b) 1, 1
(c) 1, 3 (d) 3, 1

2. The differential equation $\frac{dy}{dx} = 2xy + xy^4 = 0$ is linear equation of
(a) Cauchy (b) Bessel
(c) Bernoulli's (d) None of these

3. The differential equation $\frac{dy}{dx} = \frac{x^2 + xy + 2y}{2x^2 + y^2}$ is
(a) Exact (b) Homogenous
(c) Cauchy (d) None of these

4. The differential equation $(1 - x^2)y'' - 2xy' + n(n + 1)y = 0$
(a) The Gauss equation
(b) The Legend equation
(c) The Bessel equation
(d) None of these

5. The differential equation $\frac{d^3y}{dx^3} + y^2 = x^4$ is
(a) Non-homogenous and linear
(b) Homogeneous and non-linear
(c) Non-homogeneous and non-linear
(d) None of these

6. The differential equation $\frac{d^2y}{dx^2} = 0$ has the primitive
(a) $Y = Ax^3 + Bx^2 + C$
(b) $Y = Ax^2 + Bx^2 + C$
(c) $Y + Ax^3 + Bx$
(d) None of these

7. The differential equation $(x - x^2)y'' + \}y - (\alpha - \beta + Z)xy - \alpha\beta y = 0$ is
(a) The legend equation
(b) The Bessel equation
(c) The Gauss equation
(d) None of these

8. The differential equation $\frac{\partial^2 y}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}$ is
(a) One dimensional wave equation
(b) One dimensional heat equation
(c) None of these

9. The linear partial differential equation $\frac{\partial^2 u}{\partial x^2} + c^2 \frac{\partial^2 u}{\partial y^2} = 0$ is
(a) 2 dimensional Poisson equation
(b) 2 dimensional Laplace equation
(c) Both (a) and (b)
(d) None of these

ANSWERS

1. b	2. a	3. a	4. b	5. c
6. b	7. c	8. c	9. b	

PAPER III

COMPULSORY QUESTION

Q.8. Write only the correct answer in the Answer Book. Don't reproduce the questions.

1. 0, 0.1 are the direction consines of
(a) x-axis (b) y-axis
(c) z-axis (d) None of these

2. $AB = BC \Rightarrow B = C$ when
(a) A is non singular
(b) $|A| = 0$ (c) A^{-1} exists
(d) None of these

3. The angle between the planes $x - y - 2z + 3 = 0$ and $2x + y - z = 5$ is
(a) 0 (b) $\frac{\pi}{2}$ radians
(c) $\frac{\pi}{3}$ radians (d) None of these

4. If AB = BA, when A and B square matrices, the multiplication is said to be
(a) Associative
(b) Reflexive
(c) Commutative
(d) None of these

5. The perpendicular distance of the point (3, 1, 2) from the plane $2x + y - z = 4$ is
(a) 2 (b) 4
(c) $\frac{1}{\sqrt{6}}$ (d) None of these

6. $\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 1$ represents
(a) Sphere
(b) Ellipsed
(c) Hyperbolized of one sheet
(d) Hyperbolized of two sheets

7. The radius of the sphere $x^2 + y^2 + z^2 - 4x + 2y - 6z = 1$ is
(a) 1 (b) 5
(c) 10 (d) None of these

8. The equation of surface of revolution obtained by revolving the curve $x = z^2$, $y = 0$ about the x-axis is
(a) $x^2 + y^2 + z^4$ (b) $x^2 + y^2 + z^2$
(c) $x^2 = z$ (d) None of these

9. $ax^2 + by^2 + 2hxy + 2gx = 2fy + c = 0$ represents a parabola when
(a) $h^2 < ab$ (b) $h^2 > ab$
(c) $h^2 = ab$ (d) None of these

10. u, p, b constitute a triple of orthogonal unit vectors which is
(a) Right handed (b) Left handed
(c) Orhtonomral (d) None of these

11. An equivalence relation satisfies the following three properties
(a) Reflexive, symmetric, transitive
(b) Reflexive, anti symmetric, transitive
(c) Not reflexive, symmetric, transitive
(d) None of these

12. If M and N are any two n × n square matrices, then det. (MN) equals
(a) det M + Let N (b) det M det N
(c) Matrix MN (d) None of these

13. If A is a square matrix, then
(a) det 3A = det A (b) det 3A = 3 det A
(c) det A t ≠ det A (d) det A = A
(e) None of these

14. Which of the following mapping is a Linear Transformation?
(a) $T(a_1, a_2, a_3) = (a_1, a_2)$
(b) $T(a, b, c) = a(a, 1)$
(c) $T(x, y, z) = (x + 1, y, z)$
(d) $T(x, y) = (x + 1, y + 1)$

15. where R is the set of all real numbers is a
(a) Field
(b) Commutative ring
(c) Ring with identity
(d) Division ring
(e) None of these

16. Which of the following are subspaces of R^2?
(a) $\{(a, a): a \in R\}$ (b) $\{(a, a^2): a \in R\}$
(c) $\{a, a + 1): a \in R\}$ (d) $\{(a^2, a): a \in R\}$

17. Matrix A is called involuntary if
(a) $A^2 - A$ (b) $A^2 = 1$
(c) $A^{k-1} = A$ (d) $A' = A$
(e) None of these

18. Which of the following statements for groups is wrong?
(a) $(g^{-1})^{-1} = g$, for ever g in G.
(b) The inverse of identify element e is a itself in G
(c) A group contains at least the identity element
(d) There is a concept of an empty group
(e) None of these

19. Given Ψ: G → G′ from G into G′ is a group homomoprhism. Then Ψ is called epimorphism if
(a) G′ = G
(b) Ψ is 1 – 1
(c) Ψ is not G′
(d) Ψ is 1 – 1 onto G′ both
(e) None of these

20. A cyclic group of order n is generated by
(a) n elements (b) Two elements
(c) One element (d) n – 1 elements
(e) None of these

ANSWERS

1. c	2. d	3. c	4. c	5. c
6. d	7. d	8. c	9. a	10. a
11. a	12. b	13. b	14. d	15. d
16. a	17. b	18. d	19. b	20. c

PAPER – IV

COMPULSORY QUESTIONS

Q.8. Write only the correct answer in the Answer Boo. Don't reproduce the questions.

1. The function $f(x) = \frac{x^2 - 9}{x - 3}$ is discontinuous at
(a) x = 0 (b) x = 3
(c) x = 1 (d) None of these

2. f(x) = sin x has a minimum value at
(a) x = 0 (b) $x = \frac{\pi}{2}$
(c) $x = \frac{3\pi}{2}$ (d) None of these

3. $\lim_{x \to \infty} \left(1 + \frac{1}{x}\right)^x$ is
(a) 0 (b) 1
(c) e (d) –e

4. Derivative of function f(x) = ln x at x = 0 is
(a) 1 (b) 0
(c) ∞ (d) None of these

12. Forces of 7.5 and 3 units, acting one particle, are in equilibrium. The angle between the last pair of force is
(a) 120° (b) 90°
(c) 60° (d) None of these

13. ABCD is square. Equal force $\vec{P}$ are acting along AB, CB and BC. The resultant is $2\vec{P}$ acting along
(a) DC (b) AB
(c) AC (d) None of these

14. If μ and λ be the coefficient and angle of friction, respectively the μ = ?
(a) sin λ (b) tan λ
(c) cos λ (d) None of these

15. If radial and transverse velocities of a particle be non-zero constants, then the path described by the particle is
(a) an ellipse (b) a spiral
(c) a circle (d) None of these

16. If m be the mass $\vec{r}$ the position vector and $\vec{v}$ be the velocity of a particle then its angular momentum is
(a) $m(\vec{r} \cdot \vec{v})$ (b) $m(\vec{r} + \vec{v})$
(c) $m(\vec{r} \times \vec{v})$ (d) None of these

17. The path described by a particle moving with zero velocity and acceleration is a
(a) Circle (b) Straight line
(c) Point (d) None of these

8. If three forces acting on a body are in equilibrium, then they
(a) Are parallel (b) Meet in a point
(c) Are-coplanar (d) None of these

9. The moment of inertia of a uniform solid sphere of mass m and radius r is
(a) $\frac{4}{3}mr^3$ (b) $\frac{2}{5}mr^2$
(c) $\frac{3}{4}mr^2$ (d) None of these

. An alternate to Newton's law of restitution is called
(a) Laplace's hypothesis
(b) Cauchy's hypothesis
(c) Poisson hypothesis
(d) None of these

ANSWERS

1. c	2. b	3. c	4. b	5. b
6. b	7. b	8. a	9. b	10. a
11. b	12. b	13. b	14. b	15. b
16. c	17. c	18. d	19. a	20. d

PAPER II

COMPULSORY QUESTION

Q.8. Write only the correct answer in the Answer Book. Don't reproduce the questions.

1. An invariant is a tensor of rank
(a) 0 (b) 1
(c) 2 (d) None of these

2. The tensor A is a ———— tensor
(a) Covariant (b) Contravariant
(c) Mixed (d) None of these

3. A^{ijk}_{pqrs} is a tensor of rank ————
(a) 3 (b) 4
(c) 7 (d) None of these

4. The production of two tensors of ranks 2 and 3 is a tensor of rank ————
(a) 2 (b) 3
(c) 5 (d) 6
(e) None of these

5. $\delta^p_q \delta^q_r = ?$
(a) δ^p_{qr} (b) δ^{pq}_q
(c) δ_r (d) None of these

6. The ORDER and DEGREE of differential equation $\frac{d^2y}{dx^2} = \left[y + \left(\frac{dy}{dx}\right)^2\right]^{3/4}$ are
(a) 2 and 3 (b) 4 and 2
(c) 2 and 4 (d) None of these

7. The differential equation for the primitive $y = A \cos x + B \sin ax$, A and B being arbitrary constants, is

(a) $(D^2 - a^2) y = 0$ (b) $(D^2 + a^2)y = 0$
(c) $(D^2 + a)y = 0$ (d) None of these

8. An INTEGRATING FACTOR for $\cos x \frac{dx}{dy} - y \sin x = 1$ is

(a) cos x (b) tan x
(c) sex x (d) None of these

9. The function $= At^2 + Br + C$ gives a differential equation of order ______

(a) 1 (b) 2
(c) 3 (d) None of these

10. "Infinitely many differential equations may have the same integrating factors". This statement is

(a) Never true (b) May be true
(c) Semi true (d) Always true

11. If m is the degree of a given differential equation, then

(a) m can be zero
(b) m is any non-negative integer
(c) m is any integer
(d) m is any natural number

12. Which one of the following differential equations is not-linear?

(a) $(x^2D^2 + D + 1) = 0$

(b) $\frac{d^2y}{dx^2} + x\frac{dy}{dx} + x^2y^2 = 0$

(c) $x^2\frac{d^2y}{dx^2} + x\frac{dx}{dy} + x^2y = 0$

(d) None of these

13. The number of arbitrary constants appearing in the particular solution of an nth order differential equation is

(a) n (b) n – 1
(c) n + 1 (d) None of these

14. The particular integral of the differential equation $(D^2 + a^2)y = \cos ax$ is

(a) $-\frac{x}{2a}\cos ax$ (b) $\frac{x}{2a}\sin ax$
(c) $-\frac{x}{2a}\sin ax$ (d) None of these

15. $\frac{\partial^2 y}{\partial t^2} = c^2\frac{\partial y^2}{\partial x^2}$ is

(a) Heat equation
(b) Wave equation
(c) Equation of a vibrating string
(d) None of these

16. Use number magnification method. Find the positive real root of the equation $1.5x - x^2 = 0$ for x near 1.3. Give answer correct to two decimal places

(a) 1.30 (b) 1.31
(c) 1.33 (d) None of these

ANSWERS

1. a	2. d	3. b	4. c	5. c
6. d	7. a	8. b	9. b	10. a
11. d	12. a	13. a	14. a	15. c
16. b				

PAPER II

COMPULSORY QUESTION

Q.8. Write only the correct answer in the Answer Book. Don't reproduce the questions.

1. The number of identify elements in a group is

(a) 0 (b) 1
(c) 2 (d) None of these

2. A field must contain at least

(a) One element (b) Two elements
(c) Three elements (d) None of these

3. A basis of vector space contains

(a) Only the zero vector
(b) No zero vectors
(c) Zero as well as non-zero vectors
(d) None of these

4. Every vector space is

(a) A group (b) A Ring
(c) A Field (d) None of these

5. Matrix A is nilpotent if
(a) $A^n \neq 0, \forall n$ (b) $A^n = 0$ for some n
(c) $A^n = 0, \forall n$ (d) None of these

6. A unit matrix of order n has the rank
(a) 0 (b) 1
(c) n (d) None of these

7. The matrix equation XA = B has unique solution if
(a) 0
(b) A is singular
(c) A is not invertible
(d) None of these

8. The determinant of a triangle matrix is the product of its entries on
(a) First row (b) Second row
(c) Main diagonal (d) None of these

9. In any conic the harmonic mean between the segments of focal chord is
(a) The geometric mean
(b) Zero
(c) Semi-latus-rectum
(d) None of these

10. $A = r\cos\theta$ is an asymptote of the curve
(a) $r = a\cos\theta$ (b) $r = a\sin\theta$
(c) $r = a\tan\theta$ (d) None of these

11. The radius of curvature of $y = r^2 - x^2$ for $x\,[-r, r]$ is
(a) $\frac{1}{r}$ (b) r
(c) 2r (d) None of these

12. The distance from the origin to the plane $x + 2y - z - 4 = 0$ is
(a) $\frac{1}{\sqrt{6}}$ (b) $\frac{\sqrt{6}}{4}$
(c) $\frac{4}{\sqrt{6}}$ (d) None of these

13. The rectangular coordinates of the point with spherical coordinates $(5, 5\pi, 5\pi)$ are
(a) (5, 0, 0) (b) (0, 5, 0)
(c) (0, 0, 5) (d) None of these

14. $a^2x^2 + b^2y^2 - c^2z^2 = 1$ is hyperbolic of
(a) 1 sheet (b) 2 sheets
(c) 3 sheets (d) None of these

15. The principle normal at point P on a curve is the intersection of normal plane at P and
(a) The curve
(b) Tangent plane
(c) Osculating plane
(d) None of these

16. A curve is not a straight line if its curvature is
(a) Zero (b) Non-zero
(c) One (d) None of these

17. The relations $t = kn$, $n = \bar{t}b$, $b = \bar{t}n$ are known as
(a) Gauss-Bonnet equations
(b) Secret-Frenet formulae
(c) Tissot equations
(d) None of these

18. A set of n + 1 vectors in n-dimensional vector space
(a) Must be linearly independent
(b) Must be linearly dependent
(c) Must be a basis
(d) None of these

19. Which of the following terms is not used in algebra?
(a) Homomorphism (b) Homeomorphism
(c) Epimorphism (d) None of these

20. No group of order 28 can have subgroup of order
(a) 7 (b) 11
(c) 14 (d) None of these

ANSWERS

1. b	2. b	3. b	4. c	5. [illegible]
6. c	7. d	8. d	9. c	10. [illegible]
11. d	12. d	13. d	14. b	15. [illegible]
16. b	17. b	18. b	19. b	20. [illegible]

PAPER IV
COMPULSORY QUESTION

Q.8. Write only the correct answer in the Answer Book. Don't reproduce the questions.

1. Which property of the real line makes it a complete metric space?
   (a) Reflexivity of its order
   (b) Existence of Supremes
   (c) Transitivity
   (d) None of these

2. A function of $f: R \to R$ is continuous at x_0 if
   (a) $\lim_{x \to x_0} f(x)$ exists
   (b) $\lim_{x \to x_0} f(x)$ does not exists
   (c) $\lim_{x \to x_0} f(x) = f(x_0)$
   (d) None of these

3. An other name for analytic function is
   (a) Homomorphism (b) Meromorphism
   (c) Holomorphic (d) None of these

4. Algebraically complete number system is
   (a) Rational number system
   (b) Irrational number system
   (c) Real number system
   (d) None of these

5. The famous Roll's Theorem implies
   (a) Zorn's Lemma
   (b) Continum hypothesis
   (c) Completeness
   (d) None of these

6. For a constant real function f
   (a) max f < min f
   (b) min f < max < f
   (c) max f ≤ min f
   (d) None of these

7. Improper integrals are those integrals which
   (a) Have zero integrand
   (b) Have non infinite intervals of integration
   (c) Are definite integrals
   (d) None of these

8. Every series of number is a
   (a) Number
   (b) Sequence of numbers
   (c) Vectors
   (d) None of these

9. Every Reimann integrable function is
   (a) Differentiable
   (b) Analytic
   (c) Riemann-steltye's integrable
   (d) None of these

ANSWERS

1. c	2. c	3. c	4. a	5. b
6. d	7. d	8. b	9. a	

PAPER I (2004)
COMPULSORY QUESTION

Q.8. Write only the correct answer in the Answer Book. Don't reproduce the questions.

1. Let G be a cyclic group of order 12. Then G has
   (a) 3 distinct subgroups
   (b) 4 distinct subgroups
   (c) 6 distinct subgroups
   (d) None of these

2. Let Q and Z be the additive groups of rational and integers respectively. Then
   (a) the group Q/Z is cyclic
   (b) Every element of Q/Z is of infinite order
   (c) Every element of Q/Z is of finite order
   (d) None of these

3. Suppose A, B are matrices such that the product AB exists and is zero matrix. Then
   (a) A must be zero matrix
   (b) B must be zero matrix
   (c) Neither A nor B need to be zero matrix
   (d) None of these

4. Let A be an n × n matrix, with rank A < n. Then
   (a) Determinant A may be positive
   (b) Determinant A must be positive

(c) Determinant A may be negative
(d) None of these

5. A square matrix A such that $A^2 = A$ is called
(a) Involuntary (b) Idempotent
(c) Nilpotent (d) None of these

6. Let V be that real vector space of all functions on R to R, and let A = {x, cos x}. Then
(a) a is linearly independent
(b) a spans V
(c) a is linearly dependent
(d) None of these

7. The additive group of integers has
(a) 6 quotient groups of order 6 each
(b) 2 quotient groups of order 3 each
(c) 1quotient groups of order 6
(d) None of these

8. The determinant of a triangular matrix is the product of its entries on
(a) Last row (b) Main diagonal
(c) First row (d) None of these

9. Every elementary matrix is
(a) Non singular (b) Singular
(c) Involuntary (d) None of these

10. The equation $x^2 + y^2 - z^2 = 0$ represents
(a) Quadric cone
(b) A hyperbolic cylinder
(c) A hyperbolic paraboloid
(d) None of these

11. Let A be matrix, then its
(a) Row rank may be greater than its column rank
(b) Row rank may be less than its column rank
(c) Row rank = column rank
(d) None of these

12. A system of m homogeneous linear equations AX = 0 in n variables has a non-trivial solution if and only if
(a) Rank A = n (b) Rank A < n
(c) Rank A > n (d) None of these

13. M_2: R denote all 2 × 2 real matrices for real numbers. Le F: $M_2 \to R$, f(A) = det A for $A = M_2$, then
(a) f is onto R
(b) f is one-to-one
(c) f is neither onto nor one-to-one
(d) None of these

14. If J_n denotes the ring of integers mod n. Then
(a) J_7 is a field (b) J_6 is a field
(c) J_8 is a field (d) None of these

15. The rectangular coordinates of a point with spherical coordinates $\left(3, \frac{\pi}{6}, \frac{\pi}{4}\right)$ are
(a) (3, 1, –2) (b) $\left(\frac{3\sqrt{6}}{4}, \frac{3\sqrt{2}}{4}, \frac{3\sqrt{2}}{4}\right)$
(c) $\left(\sqrt{3}, \frac{1}{2}\right)$ (d) None of these

16. The distance of the point (3, 2, 3) from the plane 2x + 3y – z = 5 is
(a) $\frac{5}{\sqrt{14}}$ (b) $\frac{3}{\sqrt{14}}$
(c) $\frac{4}{\sqrt{14}}$ (d) None of these

17. Monger's form of the equation of a surface is
(a) f(x, y, z) = 0 (b) $f\left[\frac{x}{y}, \frac{y}{z}\right] = 0$
(c) z = f(x, y) = 0 (d) None of these

18. The only space curve whose curvature and torsion are both constant is
(a) A parabola (b) A circular helix
(c) A circle (d) None of these

19. If torsion is zero at all points of a curve, the curve is
(a) A helix (b) A straight line
(c) All on one plane (d) None of these

20. Let G be a group of order 13. Then
(a) G is non cyclic (b) G is non Abilean
(c) G is commutative (d) None of these

ANSWERS

1. a	2. d	3. c	4. a	5. b
6. b	7. b	8. d	9. a	10. c
11. c	12. b	13. b	14. c	15. a
16. c	17. c	18. b	19. b	20. a

PAPER II (2004)

COMPULSORY QUESTION

Q.8. Write only the correct answer in the Answer Book. Don't reproduce the questions.

1. The set of all ——— number form a sequence
   (a) Real (b) Rational
   (c) Irrational (d) None of these

2. f(x) = x, x rational = 0, x irrational in [0, 1]
   (a) f is discontinuous at $x = \frac{1}{2}$
   (b) f is discontinuous at $x = 0$
   (c) f is continuous at $x = \frac{1}{3}$
   (d) None of these

ANSWERS

1. b 2. b

PAPER III (2004)

COMPULSORY QUESTION

Q.8. Write only the correct answer in the Answer Book. Don't reproduce the questions. Each part carries one mark.

1. If M, L, T denote the dimensions of mass, length, time, respectively, then the dimensions of potential energy are
   (a) MLT^{1} (b) $ML^{2}T^{-1}$
   (c) $ML^{2}T^{2}$ (d) $ML^{2}T^{2}$

2. A force is described by its
   (a) Point of application only
   (b) Direction only
   (c) Point of application and direction only
   (d) None of these

3. The coefficient of static friction μ depends upon
   (a) Force for friction, $\vec{F}$
   (b) Normal reaction, $\vec{N}$
   (c) Nature of the surface
   (d) Direction of $\vec{F} \times \vec{N}$

4. If the angle between the vectors $\vec{a}$ and $\vec{b}$ is $\frac{\pi}{4}$, then $\frac{\vec{a} \times \vec{a}}{\vec{a} \cdot \vec{b}}$ is equal to
   (a) 1 (b) $1\sqrt{2}$
   (c) $\sqrt{\frac{3}{2}}$ (d) $\frac{1}{2}$
   (e) None of these

5. If a particle executing harmonic motion describes a periodic path, $x(t) = c \cos \frac{2at}{3 +}$ then its period of oscillation is
   (a) $\frac{2\pi}{c}$ (b) $\frac{2\pi\omega}{c}$
   (c) $\frac{2a\pi}{3}$ (d) $\frac{3\pi}{a}$
   (e) None of these

6. The general motion of a rigid body is
   (a) A rotation
   (b) A translation
   (c) A vibration
   (d) A translation followed by a rotation
   (e) None of these

7. The power of a force is
   (a) The time rate of change of potential energy
   (b) The time rate of change of kinetic energy
   (c) The time rate of change of Linear momentum
   (d) The time rate of change of Angular momentum
   (e) None of these

8. The impulse is equal to the
   (a) Increment in linear momentum
   (b) Increment in angular momentum
   (c) Increment in linear velocity
   (d) Increment in angular velocity

(e) None of these

9. Tension in a string is
(a) An external force
(b) A non-conservative force
(c) A constraint force (d) A sliding force
(e) None of these

10. If φ is any scalar function, the $\vec{\nabla} \cdot (\vec{\nabla} \times \phi)$ is
(a) A vector (b) A scalar
(c) Zero (d) Undefined

11. The moment of inertia of uniform rod of mass m and length 2a about a perpendicular bisector is equal to
(a) $\frac{ma^3}{3}$ (b) $\frac{2ma^2}{3}$
(c) $\frac{ma^2}{3}$ (d) $\frac{4ma^2}{3}$
(e) None of these

12. If I, J, K are the unit vectors along x, y, z-axes, respectively, then $2i \times (5k \times 3j)$ is
(a) 0 (b) 1
(c) 30k (d) −30i
(e) None of these

13. For the simple harmonic motion of a particle, the force
(a) Is minimum
(b) Is periodic
(c) Is constant and towards a fixed point
(d) Is towards a fixed point and directly proportional to the acceleration
(e) None of these

14. As apse is point on a central orbit which lies from the centre at
(a) A maximum distance
(b) A minimum distance
(c) Both maximum and minimum distance
(d) A maximum or a minimum distance
(e) None of these

15. A force is said to be conservative if, and only if
(a) $(\vec{\nabla} \times \vec{\nabla}) \cdot \vec{F} = 0$ (b) $\vec{\nabla} \cdot \vec{F} = \vec{0}$
(c) $\vec{\nabla} \times \vec{F} = 0$ (d) $(\vec{\nabla} \times \vec{F}) \cdot \vec{\nabla} = \vec{0}$
(e) None of these

16. If three forces are represented in magnitude, direction and position by the sides of a triangle taken in order, then they
(a) Are equivalent to a single force
(b) Are in equilibrium
(c) Are equivalent to a couple
(d) Are equivalent to a force and a couple
(e) None of these

17. With Sun at one focus, each plane describes
(a) Circle (b) Ellipse
(c) Parabola (d) Conics
(e) None of these

18. Any three vectors only if
(a) $\vec{A} \cdot (\vec{B} \times \vec{C}) = \vec{0}$ (b) $\vec{A} \times (\vec{B} \cdot \vec{C}) = \vec{0}$
(c) $\vec{A} \cdot (\vec{B} \cdot \vec{C}) = \vec{0}$ (d) $\vec{A}\, m(\vec{B} \times \vec{C}) = \vec{0}$
(e) None of these

19. If the angular momentum
$\vec{h} = (at - 1)l + 5l - 3k$ of a particle is conserved, then a is equal to
(a) 1 (b) −1
(c) 0 (d) $\vec{2}$
(e) None of these

20. When two rough bodies are in contact with each other then in the case of limiting friction
(a) No friction force is acting between them
(b) Friction force is acting but neither body is on the point of sliding along the other
(c) One of the bodies is on the point of sliding along with other
(d) One body slides along the other
(e) None of these

ANSWERS

1. d	2. c	3. d	4. a	5. b
6. d	7. c	8. c	9. c	10. b
11. c	12. a	13. d	14. b	15. b
16. d	17. b	18. c	19. c	20. c

PAPER IV ([illegible])

COMPULSORY QUESTION

Q.8 Write only the correct answer in the Answer Book. Don't reproduce the questions

1. Any vector is a tensor of rank
   (a) -1 (b) 0
   (c) 1 (d) 3
   (e) None of these

2. A homogeneous differential equation always
   (a) Possesses a non-trivial solution
   (b) Possesses a trivial solution
   (c) No solution at all
   (d) Possess a singular solution
   (e) None of these

3. If the number 0.3062, 0.25026 and 2.51392 are rounded, then absolute error in the number $x = 0.3062 - 0.25026 + 2.51392$ is
   (a) 0.6×10^{-4} (b) 0.5×10^{-4}
   (c) 0.4×10^{-4} (d) 0.4×10^{-3}
   (e) None of these

4. The general solution of the equation $x^2p + y^2q = (x^4 - y^4)$ is
   (a) $(x^2 - y^3)/3$ (b) $(x^3 - y^3)/3$
   (c) $(x^4 - y^4)/4$ (d) $(x^3 + y^3)/3$

5. The process of contraction always reduces the rank of a mixed tensor by
   (a) 1 (b) 2
   (c) 3 (d) 4
   (e) None of these

6. A fourth order partial differential equation in $u(x, y)$ is linear if it is of
   (a) Degree one in $u(x, y)$ and its 4th order partial derivatives w.r.t. x
   (b) Degree one in $u(x, y)$ and its 4th order partial derivatives w.r.t. y
   (c) Degree one in $u(x, y)$ and its all partial derivatives upto order four
   (d) Degree one in $u(x, y)$ and its all 4th order partial derivatives
   (e) None of these

7. Gauss-Seidel is a
   (a) A direct method for system of linear equations
   (b) An iterative method for system of nonlinear equations
   (c) A direct method for system of nonlinear equations
   (d) An iterative method for system of linear equations
   (e) None of these

8. The second order partial differential equation $\frac{\partial^2 z}{\partial x^2} + 2\frac{\partial^2 z}{\partial x \partial y} + \frac{\partial^2 z}{\partial y^2} = 0$ is
   (a) a hyperbolic homogeneous
   (b) a hyperbolic non-homogeneous
   (c) a parabolic homogeneous
   (d) an elliptic homogeneous
   (e) None of these

9. If A_{ij} is skew symmetric tensor of rank 2 and η_i is tensor of rank one, then $A_{ij}\,\eta_i\,\eta_j$ is equal to
   (a) 1 (b) δ_{ij}
   (c) 0 (d) 4
   (e) None of these

10. The equations $y''' + 4y'' + y = \sin x$, $y(0) = 1$, $y'(0) = 1$, define
    (a) A third order initial value problem
    (b) An initial value problem
    (c) A boundary value problem
    (d) A mixed boundary value problem
    (e) None of these

11. The numbers 4.0643 and .37487 are rounded, then the relative error in product of these number $x = 4.0643 \times .37487$ is
    (a) $.2564 \times 10^{-4}$ (b) $.2564 \times 10^{-6}$
    (c) $.2564 \times 10^{-5}$ (d) $.2564 \times 10^{-3}$
    (e) None of these

12. The differential equation $(3x^2 - y^3)dx + (x^3 + 3xy^2)dy = 0$ has a singular point
    (a) $(0, 0)$ (b) $(1, \sqrt{3})$
    (c) $(-1, \sqrt{3})$ (d) No singular point
    (e) None of these

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13. If $A_{ij-\ldots l_m}$ and $B^{kk22\,k_n}$ are covariant and contra variant tensors of ran k m and. then $A_{ij2-\ldots l_m}$ and $B^{kk2\,k_n}$ is

(a) A covariant tensor of rank (m + n)
(b) A contra variant tensor of rank (m + n)
(c) A contra variant tensor of ran mn
(d) A mixed tensor of ran mn
(e) None of these

14. The differential equation $y^1 + y^2 - 4 = 0$ has singular solution

(a) 1 (b) −2
(c) 4 (d) −4
(e) None of these

15. The quantity $\delta^i_m \delta^j_m \delta^k_y \epsilon_{ijk} A\delta^p_{qr}$ is a tensor of rank

(a) Zero (b) One
(c) Three (d) Four
(e) None of these

16. The differential equation $(1 - x^2)y^{11} - 2xy^1 + n(n+1)y = 0$ is

(a) Bessel equation
(b) Legend equation
(c) Hyper geometric equation
(d) Poisson equation

17. The value of the tonsorial expression $\epsilon_{132}\epsilon_{123}\epsilon_{131} + \epsilon_{321}\epsilon_{112}\epsilon_{331}$ is equal to

(a) 1 (b) −1
(c) 0 (d) 6
(e) None of these

18. A primitive of an ordinary differential equation is

(a) Its green general solution
(b) Its particular solution
(c) Its complementary solution
(d) Its complete solution
(e) None of these

19. The error in Simpson's $\frac{1}{3}$ rd rule is of the order

(a) h^2 (b) h^3
(c) h^{12} (d) h^5
(e) None of these

20. The partial differential equation
$$\frac{\partial^2 y}{\partial x^2} = \frac{1}{c^2}\frac{\partial^2 y}{\partial x^2}$$

(a) Transverse vibrations of a string
(b) Longitudinal vibrations of a bar
(c) Sound wave in space
(d) Longitudinal sound waves
(e) None of these

ANSWERS

1. c	2. b	3. d	4. a	5. b
6. c	7. a	8. d	9. b	10. d
11. d	12. a	13. b	14. e	15. c
16. a	17. c	18. d	19. b	20. d

PAPER I (2005)

COMPULSORY QUESTIONS

Q.8. Write only the correct answer in the Answer Book. Do not reproduce the question. Each part carries one mark.

1. If the vectors $a\hat{i} + \hat{j} + \hat{k}$, $\hat{i} + b\hat{j} + \hat{k}$ and $\hat{i} + \hat{j} + c\hat{k}$ $(a \neq b \neq c \neq 1)$ are coplanar, then $\frac{1}{1-a} + \frac{1}{1-b} + \frac{1}{1-c} = \ldots$

(a) 1 (b) 0
(c) −4 (d) None of these

2. The points with position vectors $\vec{a}, \vec{b}, \vec{c}$ are collinear if ————

(a) $[\vec{a}\,.\,\vec{b}\,.\,\vec{c}] = 0$
(b) $(\vec{a} + \vec{b})\,\vec{c} = 0$
(c) $\vec{a} \times (\vec{b} \times \vec{c}) = 0$
(d) $(\vec{a} \times \vec{b}) + (\vec{b} \times \vec{c}) + (\vec{c} \times \vec{a}) = 0$

3. For non-zero vectors $\vec{a}$, $\vec{b}$ and $\vec{c}$ $|(\vec{a} \times \vec{b})\,\vec{c}| = |\vec{a}|\,|\vec{b}|\,|\vec{c}|$ holds if and only if

(a) $\vec{a} \cdot \vec{b} = 0 = \vec{b} \cdot \vec{c}$
(b) $\vec{b} \cdot \vec{c} = 0 = \vec{c} \cdot \vec{a}$
(c) $\vec{c} \cdot \vec{a} = 0 = \vec{a} \cdot \vec{b}$
(d) $\vec{a} \cdot \vec{b} = \vec{b} \cdot \vec{c} = \vec{c} \cdot \vec{a} = 0$
(e) None of these

4. Let $\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$, $\vec{b} = b_1\hat{i} + b_2\hat{j} + b_3\hat{k}$ and $\vec{c} = c_1\hat{i} + c_2\hat{j} + c_3\hat{k}$ be three non-zero vectors such that $\vec{c}$ is a unit vector perpendicular to both the vectors $\vec{a}$ and $\vec{b}$. If the angle between $\vec{a}$ and $\vec{b}$ is then $|\vec{a}\,(\vec{b} \times \vec{c})|^2 =$ ___
(a) 0
(b) $\frac{1}{2}\left(a_1^2 + a_2^2 + a_3^2\right)\left(b_1^2 + b_2^2 + b_3^2\right)$
(c) $\frac{3}{4}\left(a_1^2 + a_2^2 + a_3^2\right)\left(b_1^2 + a_2^2 + a_3^2\right)\left(c_1^2 + c_2^2 + c_3^2\right)$
(d) $\frac{1}{4}\left(a_1^2 + a_2^2 + a_3^2\right)\left(b_1^2 + b_2^2 + b_3^2\right)$
(e) None of these

5. If $\vec{a} = (1, 1, 1)$ $\vec{c} = (0, 1)$ are given vectors, then vector b satisfying the equations $\vec{a} \times \vec{b} = \vec{c}$ and $\vec{a} \cdot \vec{b} = 3$ is
(a) $\frac{2}{3}(5, 1, 1)$ (b) $\frac{1}{3}(5, 2, 2)$
(c) $\frac{5}{3}(1, 1, 1)$ (d) $\frac{1}{3}(5, 1, 2)$
(e) None of these

6. If $\vec{a}$ $\vec{b}$ and $\vec{c}$ are any three coplanar unit vectors, then ___
(a) $\vec{a} \cdot \left(\vec{b} \times \vec{c}\right)$ (b) $\left(\vec{a} \times \vec{b}\right) \cdot \vec{c} = 0$
(c) $\left(\vec{a} \times \vec{b}\right) \times \vec{c} = 1$ (d) $\left(\vec{a} \times \vec{b}\right) \times \vec{c} = 0$
(e) None of these

7. Let $\vec{a}$, $\vec{b}$ and $\vec{c}$ be three non-coplanar vectors and $\vec{p}$, $\vec{q}$, $\vec{r}$ be three vectors defined by the relations

$$\vec{p} = \frac{\vec{b} \times \vec{c}}{[\vec{a}\ \vec{b}\ \vec{c}]},\quad \vec{q} = \frac{\vec{c} \times \vec{a}}{[\vec{b}\ \vec{a}\ \vec{c}]},\quad \vec{r} = \frac{\vec{a} \times \vec{b}}{[\vec{a}\ \vec{b}\ \vec{c}]}$$

Then $(\vec{a} + \vec{b}) \cdot \vec{p} + (\vec{b} + \vec{c}) \cdot \vec{q} + (\vec{c} + \vec{a}) \cdot r$ = ___
(a) 0 (b) 1
(c) 2 (d) 3
(e) None of these

8. If $\begin{vmatrix} a & a^2 & 1+a^3 \\ b & b^2 & 1+b^3 \\ c & c^2 & 1+c^3 \end{vmatrix} = 0$ vectors $\vec{p} = (1, a, a^2)$, $\vec{q} = (1, b, b^2)$, $\vec{r} = (1, c, c^2)$ are coplanar, then $[\vec{p}\ \vec{q}\ \vec{r}] =$ ___
(a) -1 (b) -2
(c) 0 (d) -4
(e) None of these

9. The resultant of any number of couples acting in the same plane on a rigid body is ___
(a) A force
(b) A couple
(c) A force and a couple
(d) A force or a couple
(e) None of these

10. If μ and λ are the coefficient and the angle of friction, then ___
(a) $\tan\lambda = \frac{1}{\mu}$ (b) $\tan\mu = \frac{1}{\lambda}$
(c) $\tan^{-1}\lambda = \mu$ (d) $\tan^{-1}\mu = \lambda$

11. If three forces acting on a rigid body are in equilibrium then they must be
(a) Concurrent
(b) Parallel
(c) Non-coplanar
(d) Concurrent or parallel and coplanar
(e) None of these

12. If three forces acting on a particle are in equilibrium, then they are
(a) Not necessarily parallel
(b) Necessarily perpendicular
(c) necessarily parallel
(d) Not necessary coplanar
(e) None of these

13. The work done by a force $\vec{F} = 2\hat{i} - \hat{j} - \hat{k}$ through a displacement $\vec{r} = 3\hat{i} + 2\hat{j} - 2\hat{k}$ is ___
(a) 3 (b) 6
(c) 9 (d) 12
(e) None of these

14. A cricket ball is thrown with a velocity of 30 m/s, then the directions in which the ball may be thrown so as to give a range of 45 m($g = 10 m/s^2$) are ———
(a) 15° or 75° (b) 30° or 60°
(c) 36° or 54° (d) 45° or 54°
(e) None of these

15. If the radial and transverse velocities of a particle be non-zero constants, then the path described by the particle is ———
(a) An ellipse (b) A cardioid
(c) A spiral (d) A circle
(e) None of these

16. If a particle is thrown with a velocity $\sqrt{2gR}$ from the earth's surface, R being earth's radius, then the particle will ———
(a) Never come back
(b) Come back after a time g^{R}
(c) Come back after a time $2g^{R}$
(d) Come back after a time $\sqrt{gR}$
(e) None of these

17. The acceleration of a freely falling body under gravity ———
(a) Varies as the inverse of the distance traveled
(b) Varies as the square of the distance traveled
(c) Is uniform
(d) Is zero
(e) None of these

18. If a particle moves on a cycloid, then its motion is ———
(a) Linear (b) Simple harmonic
(c) Parabolic (d) Elliptic
(e) None of these

19. The science which is concerned with the relations between the forces action on rigid bodies and the resulting motion is called ———
(a) Kinematics
(b) Kinetics
(c) Statistics
(d) Quantum mechanics
(e) None of these

20. If the collision is inelastic, then the coefficient of restitution e satisfies the condition
(a) $e = 0$ (b) $e = 1$
(c) $0 \le e \le 1$ (d) $0 < e < 1$
(e) None of these

ANSWERS

1. b	2. a	3. e	4. e	5. e
6. b	7. b	8. c	9. b	10. c
11. a	12. b	13. c	14. d	15. d
16. a	17. c	18. d	19. b	20. b

PAPER II

COMPULSORY QUESTION

Q.8. Write the correct answer in the Answer Book. Do not reproduce the question. Each part carry one mark.

1. The solution of a differential equation subject to a condition satisfied at one particular pint only is called ———
(a) A boundary value problem
(b) A two-point boundary value problem
(c) An initial value problem
(d) A two-point initial value problem
(e) None of these

2. The order and degree of the differential equation $\left[\dfrac{1+\left(\dfrac{dy}{dx}\right)^2}{\dfrac{d^2y}{dx^2}}\right]^{3/2} = b$ are respectively ———
(a) 1 and 3 (b) 1 and 2
(c) 3 and 2 (d) 2 and 3
(e) None of these

3. The function obtained after solving a differential equation is called ——— Choose the odd one out.
(a) A solution (b) An integral
(c) A primitive (d) A root
(e) None of these

4. In the linear equation $F(D)y = f(x)$, the function $f(x)$ is called the ———— Choose the odd one out
(a) Input function
(b) Forcing function
(c) Excitation & Function
(d) Response function
(e) None of these

5. The singular of the differential equation $f(x, y, p) = 0$, where $p = dy, dx$, is obtained by eliminating p from ————
(a) $f(x, y, zp) = 0$ and $\frac{\partial f}{\partial p} = 0$
(b) $f(x, y, p) = 0$ and $\frac{\partial f}{\partial x} = 0$
(c) $f(x, y, p) = 0$ and $\frac{\partial f}{\partial y} = 0$
(d) $f(x, y, p) = 0$ and $\frac{\partial f(x, y, p)}{\partial p} = 0$
(e) None of these

6. Infinitely many differential equations have the same integrating factors. This statement is ————
(a) Never true
(b) May be true
(c) Semi-true
(d) Always true
(e) May not be true

7. If m is the degree of a given differential equation, then ————
(a) m can be zero
(b) m is any non-negative integer
(c) m is any integer
(d) m is any natural number
(e) None of these

8. A general solution of an nth order differential equation contains ————
(a) n − 1 arbitrary constants
(b) n arbitrary constants
(c) n + 1 arbitrary constants
(d) No constant
(e) None of these

9. An equation of the form $y = px + f(p)$, where $p = \frac{dy}{dx}$, is called ————
(a) Bernoulli's equation
(b) Euler's equation
(c) Clairaut's equation
(d) Bessel's equation
(e) None of these

10. A homogeneous differential equation of the form $a_0x^n \frac{d^n n}{dx^n} + a_1x^{n-1}\frac{d^{n-1}y}{dx^{n-1}} + \dots + n_y^n = f(x)$, where $a_0, a_1, \dots, a_n$ are constants and $f(x)$ is a function of x, is known as ————
(a) Bernoulli's equation
(b) Lagrange's equation
(c) Legendre's equation
(d) Cauchy-Euler equation
(d) None of these

11. If an index appears only once in a term of the tensor equation, then it is called a
(a) Bound index
(b) Free index
(c) Sliding index
(d) Directional index
(e) None of these

12. The summation convention was introduced by ————
(a) Ricci
(b) Levi Civita
(c) Einstein
(d) Christoffel
(e) None of these

13. A second rank tensor in an N, dimensional space has ————
(a) N! components
(b) N^2 components
(c) (n − 1)! components
(d) N components
(e) None of these

14. A second rank symmetric tensor in a four dimensional continuum will have only ———— independent components.
(a) 16
(b) 12
(c) 10
(d) 6
(e) None of these

ANSWERS

1. a	2. d	3. c	4. d	5. b
6. b	7. d	8. b	9. c	10. d
11. b	12. b	13. d	14. a	

16. If $\bar{r}$ is a position vector, then $\nabla \cdot \hat{r}$ is equal to
(a) 3 (b) 0
(c) Undefined (d) 1
(e) None of these

17. The work done by a force $\bar{F} = 2\hat{i} - \hat{j} - \hat{k}$ through the distance $\bar{r} = 3\hat{i} + 2\hat{j} - 5\hat{k}$ is
(a) 0 (b) 3
(c) 12 (d) 9
(e) None of these

18. An apse is a point on a central orbit which lies from the center at
(a) A maximum distance
(b) A minimum distance
(c) Both maximum and minimum
(d) A maximum or a minimum distance
(e) None of these

19. If M an m are masses of the sun and the planet respectively. Then they attract each other with a force
(a) $m\ddot{a}$ (b) $\frac{GMm}{r^2}$
(c) $\frac{GMm}{r}$ (d) $\frac{GMm}{r^3}$
(e) None of these

20. With sum at one focus, each planet describes
(a) Circle (b) Parabola
(c) Ellipse (d) Hyperbola
(e) None of these

ANSWERS

1. c	2. b	3. a	4. c	5. a
6. d	7. a	8. e	9. a	10. a
11. c	12. b	13. a	14. c	15. a
16. a	17. d	18. a	19. b	20. c

PAPER – II
COMPULSORY QUESTION

Q.8. Write only the correct answer in the Answer Book. Do not reproduce the question. Each part carries one mark.

1. The differential equation $y'' + y - e^{x} = 0$ is
(a) Linear
(b) Non-linear
(c) Non-homogeneous
(d) of Degree 2
(e) None of these

2. Solving $u'' + 6u' + 9u = 0$ with $u(0, 2)$ and $U'(0) = 0$ is
(a) Initial value problem
(b) Boundary value problem
(c) Initial/boundary value problem
(d) None of these

3. A general solution of 4th order differential equation contains
(a) 2 constants (b) 3 constants
(c) 4 constants (d) No constants
(e) None of these

4. The partial differential equation $x^2p + y^2q = z^2$, where $p = \frac{\partial z}{\partial x}, q = \frac{\partial z}{\partial y}$, is
(a) Linear
(b) Nonlinear
(c) Algebraic
(d) Not partial differential equation
(e) None of these

5. A vector is a tensor of rank
(a) Zero (b) One
(c) -1 (d) Three
(e) None of these

6. The partial differential equation $\frac{\partial^2 z}{\partial x^2} + 2\frac{\partial^2 z}{\partial x \partial y} + \frac{\partial^2 z}{\partial y^2} = 0$ is
(a) Hyperbolic
(b) Parabolic
(c) Elliptic
(d) Non-homogeneous
(e) None of these

. If W is a linearly independent set of vectors in vector space V, then

(a) dim V = O(W) (b) O(W) > dim V

(c) O(W) ≤ dim V (d) None of these

0. The matrix multiplication is

(a) Neither associative nor commutative

(b) Both associative and commutative

(c) Commutative but not associative

(d) None of these

1. If AB denote the product of two matrices, then

(a) rank (AB) exceeds both rank (A) an rank (B)

(b) rank (AB) exceeds at least one of rank (A) and rank (B)

(c) ran (AB) cannot exceed rank of the either matrix

(d) None of these

2. A homogenous system of linear equations has a non-trivial solution if

(a) The number of unknown's exceeds the number of equations

(b) The number of equations exceeds the number of unknown

(c) The number of unknown proceeds the number of equations

(d) None of these

3. The center of the circle $3x^2 + 3y^2 - 10x + 17y = 0$ is

(a) $(5, {}^{17}/_{6})$ (b) $({}^{5}/_{3}, -{}^{17}/_{6})$

(c) $(-{}^{5}/_{3}, {}^{17}/_{6})$ (d) None of these

. The locus of the intersection of perpendicular tangents to a conic is

(a) Parabola (b) Circle

(c) Hyperbola (d) None of these

. The conjugate lines through a focus of at 1 ellipse are at

(a) Acute angle (b) Right angle

(c) Obtuse angle (d) None of these

. The length of one arch of the cycloid $x = a(\theta = \sin\theta)$, $y = a(1 - \cos\theta)$ is

(a) 16a (b) 4a

(c) 8a (d) None of these

17. The envelope of the family of lines $x \cos c + y \sin c = p$ is

(a) $x^2 + y^2 = c$ (b) $x^2 - y^2 = 2p^2$

(c) $x^2 + y + = p^2$ (d) None of these

18. If we transform the equation $x^2 + y^2 - z^2 = 9$, we get

(a) $p^2 \cos 2\phi$ (b) $p^2 \cos 2\phi = 2$

(c) $p^2 \cos 9\phi$ (d) None of these

19. The surface $\bar{r} = (u, v, u^2 + v^2)$ is elliptic if

(a) $V < 0$ (b) $V \leq 0$

(c) $V > 0$ (d) None of these

20. A necessary and sufficient condition for a curve to be a straight line is that

(a) The curvature K is zero at all points

(b) K = 0 at some points

(c) K ≠ 0 at some points

(d) None of these

ANSWERS

1. a	2. a	3. b	4. a	5. a
6. b	7. a	8. c	9. c	10. a
11. a	12. b	13. b	14. d	15. a
16. c	17. b	18.	19. c	20. a

PAPER – IV

COMPULSORY QUESTION

Q.8. Write only the correct answer in the Answer Book. Do not reproduce the question.

1. In the extended real number system, the superman of the empty set is

(a) $+\infty$ (b) ϕ

(c) $-\infty$ (d) None of these

2. $e^{ix} = \cos x + i \sin x$, for all real x, is know as

(a) DeMoivere's theorem

(b) Euler's formula

(c) Cauchy's formula

(d) None of these

3. The set of all rational numbers is

(a) Nowhere denser in IR

(b) Open in IR

18. The winding number of a contour with respect to a point not on the contour is
(a) An imaginary number
(b) An irrational number
(c) An integer
(d) None of these

19. If function f is analytic in the interior D of a circle and Γ is any contour in D, then $\int_{\Gamma} f(z)\,dz$ is
(a) Positive
(b) Negative
(c) Zero
(d) None of these

20. A singularity of the complex function f is a point z_0 if every neighbour hood of z_0 contains at least one point at which f is analytic and
(a) f is analytic at z_0
(b) f is not analytic at z_0
(c) f may and may not be analytic at z_0
(d) None of these

ANSWERS

1. d	2. b	3. a	4. c	5. a
6. a	7. a	8. c	9. b	10. a
11. d	12. c	13. a	14. c	15. a
16. a	17. c	18. c	19. c	20. b

FEDERAL PUBLIC SERVICE COMMISSION COMPETITIVE EXAMINATION FOR RECRUITMENT TO POSTS IN BPS-17, UNDER THE FEDERAL GOVERNMENT, 2007, PURE MATHEMATICS

PAPER - I

COMPULSORY QUESTION

Q.8. Write only the correct answer in the Answer Book. Do not reproduce the question.

1. The additive group of integers has
(a) 3 quotient groups of order 2 each
(b) 5 quotient groups of order 5 each
(c) One quotient groups of order 5
(d) None of these

2. Let Q and Z be the additive groups of rationals and integers respectively. Then the group Q/Z
(a) Is cyclic
(b) Is a finite group
(c) Has no element of order 6
(d) None of these

3. Every field contains more than
(a) One element
(b) Three element
(c) Two element
(d) None of these

4. Suppose, A, B are matrices such that AB exists and is zero matrix. Then
(a) A must be zero matrix
(b) B must be zero matrix
(c) Neither A nor B needs be zero matrix
(d) None of these

5. The unit matrix of order n has rank
(a) Zero
(b) n
(c) 1
(d) None of these

6. Let V be the real vector space of all functions on R to R, and let A = {x^2, sin x}. Then
(a) A spans V
(b) A is linearly independent
(c) A is linearly dependent
(d) None of these

7. If the matrix equation AX = O, where A is an n × n matrix, a nontrivial solution, then
(a) Determinant A is zero
(b) Matrix A is non-singular
(c) Determinant A is non zero
(d) None of these

8. Let Jn denote the ring of integers mod n. Then
(a) J_8 is a field
(b) J_5 is a field
(c) None of these

ANSWERS

1. d	2. c	3. a	4. b	5. b
6. a	7. b	8. b		

14. If μ and λ are the coefficient and angle of friction, then

(a) $\tan \lambda \frac{1}{\mu}$ (b) $\tan \mu = \frac{-1}{\lambda}$

(c) $\tan^{-1} \lambda = \mu$ (d) $\tan^{-1} \mu = \lambda$

15. If M, L, T denote the dimensions of mass, length and time respective then the dimension of work is

(a) ML^2T^{-1} (b) MLT^{-2}

(c) ML^2T^2 (d) ML^2T^{-2}

16. Any force system in the fundamental plane may be reduced

(a) To a single force

(b) To a single couple

(c) Either to a single force or to a couple

(d) To two forces

(e) None of these

17. The transverse component of acceleration is

(a) $r\ddot{\theta}$ (b) $\ddot{r} - r(\dot{\theta})^2$

(c) $2\dot{r}\dot{\theta} + r\ddot{\theta}$ (d) $2\ddot{r} + r\ddot{\theta}$

(e) None of these

18. In CGS system the unit of angular momentum is

(a) gm cm/sec^3 (b) gm cm^2/sec

(c) gm cm^2/sec^2 (d) gm^2 cm/sec^2

(e) None of these

19. In usual notation, the maximum range of the projectile down the plane inclined at an angle β is

(a) $\frac{V_0^2}{g(1+\sin\beta)}$ (b) $\frac{V_0^2}{g(1-\sin\beta)}$

(c) $\frac{V_0^2}{g(1-\cos\beta)}$ (d) $\frac{2V_0^2}{g(1-\sin\beta)}$

(e) None of these

20. The work done by a force $\vec{P} = 2\hat{i} - \hat{j} + 3\hat{k}$ through $\vec{r} = 3\hat{i} + 2\hat{j} + 5\hat{k}$ is

(a) 15 (b) 19

(c) 21 (d) 10

(e) None of these

ANSWERS

1. b	2. b	3. b	4. c	5. a
6. b	7. a	8. d	9. b	10. a
11. a	12. c	13. c	14. c	15. b
16. c	17. d	18. b	19. a	20. b

FEDERAL PUBLIC SERVICE COMMISSION COMPETITIVE EXAMINATION FOR RECRUITMENT TO POSTS IN BPS-17, UNDER THE FEDERAL GOVERNMENT, 2007, PURE MATHEMATICS

PAPER – 3

COMPULSORY QUESTION

Q.8. Write the correct answer in the Answer Book. Do not reproduce the question. Each part carry one mark.

(i) If $\vec{a} = 2\hat{i} - 3\hat{j}$ and $\vec{b} = a + b$ is equal to

(a) $3\sqrt{5}$ (b) $2\sqrt{5}$

(c) $\sqrt{3}$ (d) 20

(e) None of these

(ii) Th differential equation $a_0x^4(y)^{(4)} + a_1x^3y^{(4)} + a_2x^4(y)^{(3)} + x_3y^2 + a_4x^{1/4}\,a_4x^{1/4}(y) = f(x)$, where q, a_1, a_2 and q_4 are all constants is the

(a) Euler's (b) Cauchy's

(c) Legendr's (d) Cauchy-Euler's

(e) None of these

(iii) For gradient operator and the position vector $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$, and the vector $\vec{\nabla} \times \vec{r} =$

(a) $2\vec{r}$ (b) $1/2\vec{r}$

(c) 0 (d) 3x

(e) None of these

(iv) For the symbols ϵ_{ijk} and δ_{jk} is the product of ϵ_{ijk} and g_{jk} is equal to θ

(a) 1 (b) 4

(c) 1/2 (d) −1

(e) None of these

Product of the tensors A_j^i and B_m^{kl} is a tensor rank

(a) 3
(b) 1
(c) 5
(d) 0
(e) None of these

A general solution of an *nth* order differential equation contains ______ arbitrary constants

(a) n - 1
(b) n + 2
(c) n + 3
(d) No constants
(e) None of these

The order and degree of the differential equation $\left[\frac{2+\left(\frac{dy}{dx}\right)^3}{\frac{d^3y}{dx^3}}\right] = a$ are

(a) 3 and 1
(b) 3 and 7
(c) 4 and 1
(d) 1 and 3
(e) None of these

The differential equation $\frac{d^3y}{dx^3} + \frac{\partial y}{\partial x} y^2\ 2x$ is =

(a) Nonlinear
(b) Linear
(c) Exact
(d) Ordinary
(e) None of these

The equation $\frac{dy}{}$

(xii) The equation of the form $c^2 \frac{\partial^2 u}{\partial x^2} = \frac{\partial u}{\partial y}$ is known as

(a) Wave
(b) Heat
(c) Lapalace's
(d) Poisson
(e) None of these

(xiii) The partial differential equation $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$ has exactly

(a) One solution
(b) Two solutions
(c) No solution
(d) Four
(e) None of these

(xiv) Every first order ordinary linear differential has as unique solution. This statement is

(a) Always true
(b) Never true
(c) May be true
(d) May not be true
(e) None of these

(xv) The dot product of two arbinary vectors is a tensor of tank

(a) One
(b) Two
(c) Three
(d) Four
(e) None of these

(xvi) The sum of the tensor δ_i^k . A_{klm}^{ij}, $\delta_j^i B$ and δ_j^m is

FEDERAL PUBLIC SERVICE COMMISSION COMPETITIVE EXAMINATION FOR RECRUITMENT TO POSTS IN BPS-17 UNDER THE FEDERAL GOVERNMENT, 2005

PAPER – I

PART – I (MCQ)

(COMPULSORY)

Q.1. Select the option/answer and fill in the appropriate ox on the Answer Sheet.

(i) The two non zero vectors $\bar{A}$ and $\bar{B}$ are parallel if

(a) $\bar{A} \times \bar{B} = 0$ (b) $|\bar{A} \times \bar{B}| = 1$
(c) $\bar{A} \cdot \bar{B} = 0$ (d) $|\bar{A}| = |\bar{B}|$

(ii) The displacement of a point moving in a straight line is $S = 8t^2 + 3t - 5$, S is measured in meters and t in seconds. The velocity when the displacement is zero, is

(a) 3m/Sec (b) 14m/Sec
(c) 16m/Sec (d) 12m/Sec
(e) None of these

(iii) The volume of the parallel O piped with sides $\bar{A} = 6\hat{i} - 2\hat{j}$, $\bar{B} = \hat{j} + 2\hat{k}$, $\bar{C} = \hat{i} + \hat{j} + \hat{k}$ is

(a) 5 cubic units (b) 10 cubic units
(c) 15 cubic units (d) 20 cubic units
(e) None of these

(iv) Three vectors $\bar{A}$, $\bar{B}$, $\bar{C}$ are coplanar, the value of their scalar triple product is

(a) 0 (b) 1
(c) 2 (d) 3
(e) None of these

(v) If θ is the angle between the vectors $\bar{A}$ and $\bar{B}$ such that $|\bar{A} \times \bar{B}| = |\bar{A} \cdot \bar{B}|$ then θ is

(a) 0° (b) 45°
(c) 120° (d) 180°
(e) None of these

(vi) Which two of the following vectors are perpendicular to each other
$\bar{P} = 3\hat{i} + 2\hat{j} + 4k$, $\bar{Q} = 12\hat{i} + 8\hat{j} + 16\hat{k}$,
$\bar{R} = 2\hat{i} = 11\hat{j} + 4\hat{k}$

(a) $\bar{P}$ and $\bar{Q}$ (b) $\bar{Q}$ an R
(c) $\bar{R}$ and $\bar{P}$ (d) None of these

(vii) A force $\bar{F}$ is conservative if

(a) $\nabla \cdot \bar{F} = 0$ (b) $\nabla \times \bar{F} = 0$
(c) $\nabla \cdot (\nabla \times \bar{F}) - 0$ (d) $\nabla \times (\nabla \times \bar{F}) = 0$

(viii) If M, L, F denote the dimensions of mass, length and force respectively, then the dimensions of time

(a) $LM^{1/2} F^{1/2}$ (b) $L^{1/2} M^{1/2} F^{1/2}$
(c) $L^{1/2} MF^{1/2}$ (d) $LMF^{1/2}$
(e) None of these

(ix) If F is the magnitude of friction, R that of normal reaction and μ the coefficient of friction then

(a) $\mu = FR$ (b) $\mu = \frac{F}{R}$
(c) $\mu = \frac{F}{\sqrt{R}}$ (d) $\mu = \frac{F}{R^2}$
(e) None of these

(x) The radial component of acceleration is

(a) $r\ddot{\theta}$ (b) $\ddot{r} - r(\dot{\theta})^2$
(c) $2\dot{r}\,\dot{\theta} + r\ddot{\theta}$ (d) $2\ddot{r} + r\ddot{\theta}$
(e) None of these

(xi) The time rate of change of angular momentum is

(a) Velocity (b) Acceleration
(c) Force (d) Kinetic energy
(e) None of these

(xii) If a body possesses central symmetry, then the center of symmetry is

(a) The center of gravity
(b) The center of mass
(c) Point of symmetry
(d) Axis of symmetry
(e) None of these

(xiii) If $\bar{A}$ is any differentiable vector function, then $\nabla(\nabla \cdot \bar{A})$ is

(a) Zero

(b) $\nabla(\nabla \times \bar{A}) + \nabla^2 \bar{A}$

(c) $\nabla^2 \bar{A}$

(d) $\nabla \times \bar{A}$

(e) None of these

(xiv) If $\bar{r}$ is a position vector, then div $\bar{r}$ is equal to

(a) 0 (b) 1

(c) 3 (d) Undefined

(e) None of these

(xv) In CGS system the unit of angular momentum is

(a) gm cm/sec^3 (b) gm cm^2/sec

(c) gm cm^2/sec^2 (d) gm^2 cm/sec^3

(e) None of these

(xvi) In usual notation, the time of flight for the motion of projectile is

(a) $\frac{2V_0 \sin \infty}{g}$ (b) $V_0^2 \sin \infty$

(c) $\frac{V_0(1 + \sin \infty)}{g}$ (d) $\frac{2V_0(\sin \infty)}{g}$

(e) None of these

(xvii) In M and m are the masses of the sun and the planet respectively then they attract each other with a force

(a) Ma (b) $\frac{GMm}{r^2}$

(c) $\frac{GMm}{r}$ (d) $\frac{GMm}{r^3}$

(e) None of these

(xviii) The work done by a Force $\dot{F} = 4\hat{i} - 6\hat{j} + 3\hat{k}$ through the distance $r = 3\hat{i} - 10\hat{j} + 3\hat{k}$ is

(a) 57 (b) 60

(c) 61 (d) 63

(e) None of these

(xix) If Q is a force and AB is the arm of couple then moment of couple is

(a) AB × Q (b) 2AB × Q

(c) Q × AB (d) AB × 2Q

(e) None of these

(xx) If three coplanar forces P_1, P_2 and P_3 acting at a point in equilibrium then

(a) $\frac{P_1}{\cos\alpha} = \frac{P_2}{\cos\beta} = \frac{P_3}{\cos\gamma}$

(b) $\frac{P_1}{\sin\alpha} = \frac{P_2}{\beta n} = \frac{P_3}{\sin\gamma}$

(c) $\frac{P_1}{\sin\alpha} = \frac{P_2}{\sin\beta} = \frac{P_3}{\sin\gamma}$

(d) $\frac{P_1}{\sin\alpha} = \frac{P_2}{\sin\beta} = \frac{P_3}{\cos\gamma}$

(e) None of these

ANSWERS

1. a	2. e	3. b	4. a	5. b
6. d	7. a	8. e	9. b	10. d
11. d	12. b	13. b	14. c	15. b
16. c	17. b	18. e	19. c	20. c

FEDERAL PUBLIC SERVICE COMMISSION COMPETITIVE EXAMINATION FOR RECRUITMENT TO POSTS IN BPS-17 UNDER THE FEDERAL GOVERNMENT, 2008

PAPER – III

PART – I (MCQ)

(COMPULSORY)

Q.1. Select the option/answer and fill in the appropriate box on the Answer Sheet.

(i) Every vector space have at least subspaces

(a) 1 (b) 2

(c) 3 (d) 4

(ii) If A symmetrical and non-singular matrix, then A^{-1} is

(a) Symmetric and Singular

(b) Singular

(c) Symmetric

(d) Skew symmetric

(iii) Which of the following set of vectors are bases of IR

(a) {1 0 0 1}, {0 1 0 0}, {1 1 1 1}, {0 1 1 1}

(b) {[1 – 1 0 2], [3 – 1 2 1], [1 0 0],
[1 – 1 1 0]}

(c) {[−2 4 6 4], [0 1 2 4], [−1, 2 3 2], [−3 2 5 6]}

(d) {[0 0 1 1], [−1 −1 1 2], [−1 1 0 0], [−2 −1 1]}

(iv) Similar matrices has ———— characteristics polynomial

(a) Unique (b) Different

(c) Same (d) None of these

(v) A group which is formed of the four permutation 1, (ab), (cd), (ab) 0 (cd) is

(a) Abelian but cyclic

(b) Abelian but non cyclic

(c) Non abelian but cyclic

(d) Non-abelian but non cyclic

(vi) An n-dimensional vector space has ———— distinct sub-space

(a) n + 1 (b) n

(c) n − 1 (d) Neon of these

(vii) Order of permutation $\begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 4 & 3 & 2 & 5 & 1 \end{pmatrix}$ is

(a) 4 (b) 5

(c) 6 (d) 7

(viii) Characteristics root of every orthogonal matrix f odd order is

(a) 0 (b) 1

(c) −1 (d) 1 or −1

(ix) Which statement is true

(a) The function det: $\underset{n\times n}{M}(F) \to F$ is linear transformation

(b) If $A \in \underset{n\times n}{M}(F)$ has rank n, then det (A) = 0

(c) If B is a matrix obtained from a square matrix A by multiplying a row of A by multiplying a row of A by a scalar, then det (A) = det(B)

(d) The determinant of an upper triangle matrix equals the product of its diagonal entries

(x) Let V and W be vector space and let T: V → V be linear. If V is finite dimensional, then

(a) Nullity (T) + rank (T) = dm (W)

(b) Nullity (T) + rank (T) = dim (V)

(c) Nullity (T) + rank (T) = dim (V ∪ W)

(d) Nullity (T) + rank (T) = dim (V ∪ W)

(xi) Over the field of real number, the vector space of complex number has dimension 2 and basis is

(a) {1, i} (b) {1, −i}

(c) {−1, i} (d) {−i, i}

(xii) If a plane intersects only one nape, the curve of intersection is

(a) Circle (b) Parabola

(c) Ellipse (d) Hyperbola

(xiii) The cylindrical coordinate of a point whose spherical coordinate are $\left(4, \frac{\pi}{3}, \frac{\pi}{2}\right)$

(a) $\left(4, \frac{\pi}{3}, 0\right)$ (b) $\left(4, 0, \frac{\pi}{3}\right)$

(c) $\left(4, \frac{\pi}{2}, 0\right)$ (d) $\left(4, 0, \frac{\pi}{2}\right)$

(xiv) Angle between the straight lines

L: $\frac{x-2}{1} = \frac{y-3}{-1} = \frac{z+1}{1}$,

M: $\frac{x-2}{2} = \frac{y-3}{-1} = \frac{z+1}{3}$

(a) 40° (b) 40.1°

(c) 40.2° (d) 40.3°

(xv) If the tangent at a point on a parabola meets the directrix is K, the angle KSP (S focus) is

(a) Acute angle (b) Right angle

(c) Straight line (d) Obtuse angle

(xvi) Lines: $r = a_1 tb_1$ & $a_2 + sb_2$ lies in a plane of

(a) $a_1 \times a_2 = 0$

(b) $b_1 \times b_2 = 0$

(c) $(a_1 - a_2) \times (b_1 \times b_2) = 0$

(d) $(a_1 + a_2) \times (b_1 + b_2) = 0$

(xvii) The set of vectors {(1, 2, 5), (0, 0, 0); (6, 7, 8)} is

(a) Linearly independent

(b) Linearly dependent

(c) Orthogonal

(d) None of these

(xviii) The ring A = z + iz of Gaussian integer is

(a) Ideal ring (b) One sided ring

(c) Two sided ring

(d) Principal ideal ring

(xi) The region of Argand diagram defined by $|Z-1|+|Z+1| \le 4$ represents
(a) Interior of the ellipse
(b) Interior & boundary of the ellipse
(c) Interior of the circle
(d) Interior and boundary of the circle

(xii) If $f_{xx}(x_0, y_0) \cdot f_{yy}(x_0, y_0) - \left[f_{xy}(x_0, y_0)\right]^2 < 0$, then has at (x_0, y_0)
(a) Relative maximum
(b) Relative minimum
(c) Saddle point
(d) None of these

(xiii) Radius of convergence of the power series $\sum_{n=0}^{\infty} \frac{(-1)^{n+1}(x+1)^{2n}}{(n+1)^2 5^n}$ is
(a) 5 (b) $\frac{1}{5}$
(c) $\sqrt{5}$ (d) $\frac{1}{\sqrt{5}}$

(xiv) $\Gamma\left(\frac{-3}{2}\right) =$ ________
(a) $\frac{4}{3}\sqrt{\pi}$ (b) $\frac{3}{4}\sqrt{\pi}$
(c) $\frac{4}{3}\pi$ (d) $\frac{3}{4}\pi$

(xv) The domain of the derivative of the function

$$f(x) = \begin{cases} \tan; & |x| \le 1 \\ \frac{1}{2}(|x|-1); & |x| > 1 \end{cases}$$

(a) $IR - \{0\}$ (b) $IR - \{1\}$
(c) $IR - \{-1\}$ (d) $IR - \{-1, 1\}$

(xvi) The set of all rational number is
(a) Nowhere dense in IR
(b) Open in IR
(c) Everywhere dense in IR
(d) Closed in IR

(xvii) The function $z \to$ IR e(z); $z \to \bar{z}$ and $z \to |z|$ are
(a) Continuous function
(b) Discontinuous function
(c) Complex function
(d) Rational function

(xviii) The number of asymptotes of $x^2y^2(x^2-y^2)^2 = (x^2+y^2)^3$ cannot be exceed by
(a) 2 (b) 4
(c) 6 (d) 8

(xix) For the curve $x^3 + y^3 - 3axy = 0$, the tangents at the origin are $x = 0$ & $y = 0$, then origin is
(a) Node
(b) Cusp
(c) Isolated point
(d) Either a node or isolated point

(xx) $\lim_{n\to\infty} \frac{(n-i)^3}{2n^3+n+2} =$
(a) ∞ (b) 0
(c) $\frac{1}{2}$ (d) 1

ANSWERS

1. a	2. d	3. d	4. c	5. d
6. c	7. b	8. c	9. a	10. d
11. d	12. c	13. d	14. d	15. b
16. a	17. c	18. a	19. b	20. b

11. A straight line through (4, −3) cuts the axes such that the intercepts are equal in magnitude. The equation of th line is
(a) $x - y + 1 = 0$ (b) $x + y + 1 = 0$
(c) $x + y = 7$ (d) $x - y = 7$

12. The line $lx = my + n = 0$, $mx + ny + 1 = 0$ and $nx + ly + m = 0$(l, m, n are not all equal) are concurrent if
(a) $l^2 + m^2 + n^2 = 1$ (b) $lm + mn + nl = 1$
(c) $lm + mn + nl = 0$ (d) $l + m + n = 0$

13. If the equation $hyx + gx + fy + c = 0$ $(h \neq 0)$ represents two straight lines, then
(a) $2fgh = c^2$ (b) $2fg = ch$
(c) $fgh = c^2$ (d) $fg = ch$

14. The distance of the pole from the line $4\cos\theta + 3\sin\theta = \frac{30}{4}$ is
(a) 2 (b) 4
(c) 6 (d) 8

15. The equation of the circle on the chord $x\cos\alpha + y\sin\alpha - p = 0$ of the circle $x^2 + y^2 - a^2 = 0$, $(0 < p < a)$ as diameter is
(a) $x^2 + y^2 - a^2 + 2p(x\cos\alpha + y\sin\alpha - p) = 0$
(b) $x^2 + y^2 - a^2 - 2p(x\cos\alpha + y\sin\alpha - p) = 0$
(c) $x^2 + y^2 + a^2 - 4p(x\cos\alpha + y\sin\alpha + p) = 0$
(d) $x^2 + y^2 - a^2 + 4p(x\cos\alpha + y\sin\alpha - p) = 0$

16. From a point (x_1, y_1), if perpendicular tangents are drawn to the circle $x^2 + y^2 = 5$, then which one of the following equations represents the locus of (x_1, y_1)?
(a) $x^2 + 2y^2 = 5$ (b) $2x^2 + y^2 = 10$
(c) $x^2 + y^2 = 10$ (d) $2x^2 + xy + 2y^2 = 5$

17. The equation of th common chord of the circles $x^2 + y^2 - 6x = 0$ and $x^2 + y^2 - 4y = 0$ is
(a) $3x + 2y + 1 = 0$ (b) $3x + 2y = 0$
(c) $3x + 2y = 0$ (d) $3x - 2y - 1 = 0$

18. Let $S_1 = x_2 + y_2 + 2g_1x + 2f_1y + C_1 = 0$, $S_2 = x_2 + y_2 + 2g_2x + 2f_2y + C_2 = 0$ by two circles. The slope of th radical axis of the two circles is
(a) $\frac{(f_1 - f_2)}{(g_1 - g_2)}$ (b) $\frac{(g_1 - g_2)}{(f_1 - f_2)}$
(c) $\frac{(f_1 - f_2)}{(g_1 - g_2)}$ (d) $\frac{(g_1 - g_2)}{(f_1 - f_2)}$

19. If λ, μ are parameters, the orthogonal coaxial system of the system of circles $x^2 + y^2 + 2\lambda x + c = 0$, is
(a) $x^2 + y^2 + 2\mu y - c = 0$
(b) $x^2 + y^2 + 2\mu y + c = 0$
(c) $x^2 + y^2 + 2\mu x - c = 0$
(d) $x^2 + y^2 + 2\mu x + c = 0$

20. The condition that the straight line $\frac{1}{r} = \frac{1}{a}\cos\theta = \frac{1}{b}\sin\theta$ may touch the circle $r = 2c\cos\theta$ is
(a) $\frac{2c}{a} = 1 - \frac{b^2}{c^2}$ (b) $\frac{2c}{a} = 1 - \frac{c^2}{b^2}$
(c) $\frac{2c}{a} = 1 + \frac{c^2}{b^2}$ (d) $\frac{2a}{c} = 1 - \frac{c^2}{b^2}$

21. The equation of the parabola whose focus is (−3, 0) and the directrix is $x + 5 = 0$, is
(a) $x^2 = 4(y - 4)$ (b) $x^2 = 4(y + 4)$
(c) $y^2 = 4(x - 4)$ (d) $y^2 = 4(x + 4)$

22. If the normals at(x_1, y_1), $r = 1, 2, 3, 4$ on the rectangular hyperbola $xy = c^2$ meet in the point (α, β), then
(a) $x_1 + x_2 + x_3 + x_4 = \alpha$
(b) $x_1 + x_2 + x_3 + x_4 = \beta$
(c) $x_1 + x_2 + x_3 + x_4 = \frac{1}{\alpha}$
(d) $x_1 + x_2 + x_3 + x_4 = \frac{1}{\beta}$

23. The lines $x = ay + b$, $z = cy + d$ and $x = a'y + b'$, $z = c'y + d'$ are perpendicular if
(a) $aa' + cc' = 1$ (b) $aa' + cc' = -1$
(c) $bb' + dd' = 1$ (d) $bb' = dd' = -1$

36. A particle executing a simple harmonic motion of amplitude 5 cm has a speed of 8 cm/sec when at a distance 3 cms from the centre of the path. The period of the motion of th particle will be

(a) $\frac{\pi}{2}$ (b) π sec

(c) 2π sec (d) 4π sec

37. A particle of mass m moves in a straight line under an attractive force m μ x towards a fixed point O on the line, x being the distance of the particle from O. If x = a at time t = 0, then the velocity of the particle at a distance x is given by

(a) $\mu(a-x)$ (b) $\mu(x-a)$

(c) $\sqrt{\mu(a^{1/2}-x^2)}$ (d) $-\sqrt{\mu(a^2-x^2)}$

38. In order to keep a body in air above the earth for 12 seconds, the body should be thrown vertically up with a velocity of

(a) $\sqrt{6g}$ m/sec (b) $\sqrt{12g}$ m/sec

(c) 6g m/sec (d) 12g m/sec

39. In a particle of mass 4 gms moves in a horizontal circle under the action of a force of magnitude 400 dynes towards the centre of the circle with a speed 20 cms/sec, then the radius of the circle will be

(a) 2 cm (b) 4 cm

(c) 8 cm (d) 10 cm

40. The escape velocity from the earth is about 11 km/second. The escape velocity from the planet having twice the radius and the same mean density as the earth is about

(a) 5.5 km/second (b) 11 km/second

(c) 16.5 km/second (d) 22 km/second

41. If I, N, R, Q denote respectively the sets of integers, natural numbers, real numbers and rational numbers, then consider the following statements.

I. $I \subset N \subset Q \subset R$

II. $I \subset Q$ and $N \subset R$

III. $I \subset R$, $N \subset Q$

Of these statements

(a) I, II, III are correct

(b) I alone is correct

(c) II alone is correct

(d) II and III are correct

42. If p is a real number such that $0 < p < 1$ 1 and x and y are real numbers with $x < y$, then

(a) $p^x < p^y$ (b) $p^x > p^y$

(c) $p^y < p^x > 1$ (d) $p^y > 1 > p^x$

43. The real part of $(\sin x + i\cos x)^5$ is

(a) $-\cos 5x$ (b) $-\sin 5x$

(c) $\sin 5x$ (d) $\cos 5x$

44. The modulus of $\frac{1+2i}{1-(1-i)^2}$ is equal to

(a) 1 (b) $\sqrt{5}$

(c) $\sqrt{3}$ (d) 5

45. If a, b are any two integers and $b \neq 0$, then there exists a unique pair of integers q, r such that $a = bq_1 + r$, where

(a) $0 \le r \le |b|$ (b) $0 \le |r| < |b|$

(c) $0 \le r < |b|$ (d) $-|b| \le r \le |b|$

46. For any positive integers n and m, the least positive number in the set {xn + ym: x and y are integers} is

(a) l.c.m (n, m) (b) h.c.f. (n, m)

(c) n + m (d) nm

47. If the gcd of a and b is denoted by (a, b) then consider the following statements

I. If (a, b) = d, $a = a_1 d$, $b = b_1 d$, then $(a_1, b_1) = 1$

II. If (a, c) = d > 1, a | b, c | b, then ac | b

III. If (a, b) = 1 a | c and b | c, then ab | c

Of these statements

(a) I and II are correct

(b) II and III are correct

(c) I and III are correct

(d) I, II and III are correct

48. If $f(x) = 2x + 4x^2$, $g(x) = 2 + 6x + 4x^2$ are polynomials in K[X], where K is the ring of integers modulo 8, then the degree of f(x). g(x), is

(a) 1 (b) 2

(c) 3 (d) 4

49. If $x3 + 5x^2 - 3x + 2$ is divided by x + 1, then the remainder will be

(a) 5 (b) 9

(c) 10 (d) 11

61. In the group S_3, of all permutations of 1, 2, 3, the inverse of $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{pmatrix}$ is
(a) $\begin{pmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \end{pmatrix}$ (b) $\begin{pmatrix} 1 & 2 & 3 \\ 1 & 3 & 2 \end{pmatrix}$
(c) $\begin{pmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \end{pmatrix}$ (d) $\begin{pmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{pmatrix}$

62. F is a field containing n elements. If n ∈ {122, 123, 124, 125}, then
(a) n = 122 (b) n = 123
(c) n = 124 (d) n = 125

63. Let I be the set of all integers and let * be a binary operation in I defined by $a * b = a + b + 10\ \forall\, a, b \in I$, then (I, *) is an abelian group. The identity element of this group is
(a) 0 (b) 10
(c) −10 (d) 1

64. Consider the group $(Z_7^*, \otimes)$ of non-zero residue classes, under multiplication modulo 7. If $[X] \in Z_7^*$ is such that $[2] \otimes [X] = [5]$, then which one of the following is correct?
(a) [X] = [3] (b) [X] = [4]
(c) [X] = [5] (d) [X] = [6]

65. Consider the following statements relating to matrix operations:
I. If A is a m × n matrix, then B has to be a n × n matrix for AB and BA to be defined
II. If (A + B) and AB are defined, then A and B must be square matrices of the same order
III. If AB and BA are both defined, then AB = Q implies BA = Q, where Q is the null matrix.
Of these statements
(a) I and II are correct
(b) I and III are correct
(c) II and III are correct
(d) I, II and III are correct

66. If $2\begin{bmatrix} x & y \\ z & p \end{bmatrix} + 9\begin{bmatrix} -2 & 2 \\ 1 & 0 \end{bmatrix} = 18I$, where I is the identity matrix, then the values of x and y are
(a) x = 0, y = −9 (b) x = 9, y = 0
(c) x = 18, y = 2 (d) x = 18, y = −9

67. If $\begin{vmatrix} x+a & b & c \\ c & x+b & a \\ a & b & x+c \end{vmatrix} = 0$, then one of the values of x is
(a) (a + b + c) (b) −(a + b + c)
(c) $(a^2 + b^2 + c^2)$ (d) $(a^3 + b^3 + c^3)$

68. Consider the following statements relating to any two matrices A and B such that AB is defined
I. $(AB)^{-1} = B^{-1}A^{-1}$
II. $(AB)^T = B^TA^T$
III. Rank (AB) ≤ min (Rank A, Rank B)
Here A^T is the transpose and A^{-1} denotes the inverse of A
Of these statements
(a) I, II and III are correct
(b) I and II are correct
(c) I and II are correct
(d) II and III are correct

69. If $A = \begin{pmatrix} -1 & 2 \\ 3 & -5 \end{pmatrix}$, then its inverse is
(a) $\begin{pmatrix} -5 & -2 \\ -3 & -1 \end{pmatrix}$ (b) $\begin{pmatrix} 5 & 2 \\ 3 & 1 \end{pmatrix}$
(c) $\begin{pmatrix} -5 & 2 \\ 3 & -1 \end{pmatrix}$ (d) $\begin{pmatrix} 5 & 3 \\ 2 & 1 \end{pmatrix}$

70. The system of equations, $x - y + 2z = 0$, $x + z = 0$, $x + y - z = 0$ has
(a) a unique solution
(b) finitely many solutions
(c) infinitely many solutions
(d) no solution

71. The set Q of all rational numbers is
(a) ordered and complete
(b) neither ordered nor complete
(c) ordered but not complete
(d) complete but not ordered

72. Which one of the following statements is not correct?
(a) every bounded and infinite sequence of real numbers has at least one limit point
(b) every convergent real number sequence is bounded

every monotonic ... [illegible]
(c) every ... is convergent
(d) every increasing sequence of positive numbers diverges or has a single limit point

73. f is a function from R to R and 'a' is a real number. If f is given to be continuous at x = a, then $\lim_{x \to a} f(x)$
(a) never exists
(b) may or may not exist
(c) always exists and is equal to f(a)
(d) always exists but may or may not be equal to f(a)

74. A continuous function f from a bounded closed interval [a, b] to R
(a) is always unbounded
(b) may be bounded or unbounded
(c) is always bounded and attains its bounds
(d) is always bounded but may or may not attain its bounds

75. $\frac{d}{dx} \arcsin \sqrt{\frac{x-\beta}{\alpha-\beta}}$, $\alpha > x > \beta$ is equal to
(a) $\frac{1}{\sqrt{(\alpha-x)(x-\beta)}}$
(b) $\frac{1}{2(\alpha-x)(x-\beta)}$
(c) $\frac{1}{(\alpha-x)(x-\beta)}$
(d) $\frac{1}{2\sqrt{(\alpha-x)(x-\beta)}}$

76. The function f defined on $A = (0, 1) \cup (2, 3)$ by f(x) = 1 if 0 < x < 1 and f(x) = 2 if 2 < x < 3 is
(a) not differentiable at some points of A
(b) differentiable on but f′ does not exist at some points of A
(c) derivatives of all orders exist and are equal to zero through f is not a constant function
(d) not a function that has the properties (a), (b), (c)

77. A cube is expanding in such a way that its edge is changing at a rate of 5 cm/sec. If its edge is 4 cm long, then the rate of change of its volume is
(a) 100 cm^3/sec
(b) 120 cm^3/sec
(c) 180 cm^3/sec
(d) ...

x = 3, is
(a) $\frac{(\ln 3)}{2x(\ln x)^2}$
(b) $\frac{3(\ln 3)}{x(\ln x)^2}$
(c) $\frac{-(\ln 3)}{2x(\ln x)^2}$
(d) $\frac{-3(\ln 3)}{x(\ln x)^2}$

79. If $f(x) = e^x$ and $g(x) = \ln x$, then $(g \circ f)'(x)$ is equal to
(a) 0
(b) 1
(c) e
(d) 1 + e

80. If $\frac{x}{y} + \frac{y}{x} = 2$, then $\frac{dy}{dx}$ is
(a) 1
(b) 2
(c) −1
(d) −2

81. If f(x) = ax + b, x ∈ [−1, 1], then the point c ∈ (−1, 1), where
$f'(c) = \frac{f(1) - f(-1)}{2}$
(a) does not exist
(b) can be any c ∈ (−1, 1)
(c) can be only $\frac{1}{2}$
(d) can be only $-\frac{1}{2}$

82. If $f(x) = (1-x)^{5/2}$ and
$f(x) = f(0) + xf'(0) + \frac{x^2}{2} f''(\theta x)$, then the value of θ as x → 1 is
(a) $\frac{1}{5}$
(b) $\frac{4}{5}$
(c) $\frac{9}{25}$
(d) $\frac{16}{25}$

83. The maximum and minimum values of $\left(x^4 - \frac{2}{3}x^3 - 2x^2 + 2x\right)$ in the interval [0, 3] are respectively.
(a) 51 and 0
(b) $\frac{23}{48}$ and 0
(c) $\frac{1}{3}$ and 0
(d) $\frac{23}{48}$ and $\frac{1}{3}$

84. The equation of the ...

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85. The length of the subtangent of the rectangular hyperbola $x^2 - y^2 = a^2$ at the point $(a, 2\sqrt{a})$ is

(a) $\sqrt{2}\,a$ (b) $\frac{a^{3/2}}{\sqrt{2}}$

(c) $2a$ (d) $\frac{1}{2a}$

86. The normal to the parabola $x^2 = 4ay$ at the point $(2a, a)$ on it cuts the parabola again at a point whose coordinates are

(a) $(-2a, a)$ (b) $(-4a, 4a)$

(c) $(-6a, 9a)$ (d) $(-8a, 16a)$

87. If z be a function of x and y, then consider the following statements

I. $\frac{\partial^2 z}{\partial x \partial y} = \frac{\partial^2 z}{\partial y \partial x}$ at the point $x = a, y = b$.

II. The partial derivatives $\frac{\partial^2 z}{\partial x \partial y}$ and $\frac{\partial^2 z}{\partial y \partial x}$ are both continuous in x and y at the point $x = a, y = b$

Of the above

(a) statement I implies and is implied by statement II

(b) it is possible that I is true but II is false, but if II is true, then I is necessarily true

(c) it is possible that I is false but II is true, but if I is true, then II is necessarily true

(d) it is possible that I is true but II is false and it is also possible that II is true but I is false

88. If $u = \sin^{-1}\left(\frac{x^{1/3} + y^{1/3}}{x^{1/2} + y^{1/2}}\right)^{1/2}$ then $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y}$ is

(a) $\frac{1}{12}\tan u$ (b) $-\frac{1}{12}\tan u$

(c) $\frac{12}{\tan u}$ (d) $\frac{-12}{\tan u}$

89. The point of inflexion on the curve $a^2y^2 = x^2(a^2 - x^2)$ is

(a) $(0, 0)$ (b) $\left(-\sqrt{\frac{3}{2}}\,a, 0\right)$

(c) $\left(-\sqrt{\frac{3}{2}}\,a, 1\right)$ (d) $(1, 1)$

90. If $y = x^3 + 2x^2 - 4x + 4$, then which one of the following statements is incorrect?

(a) y is increasing for $x < -2$

(b) y is decreasing for $-2 < x < -\frac{1}{2}$

(c) y is decreasing for $-\frac{1}{2} < x < 1$

(d) y is decreasing for $x > 1$

91. If f is a continuous function, then $\lim_{n \to \infty} \sum_{r=0}^{n-1} \left\{ f\left(\frac{r}{n}\right) \right\} \frac{1}{n}$ can be expressed as

(a) $\int_0^1 f\left(\frac{1}{x}\right) dx$ (b) $\int_0^1 f(x)\, dx$

(c) $\int_0^1 x\, f(x)\, dx$ (d) $\int_0^1 \frac{1}{x} f\left(\frac{1}{x}\right) dx$

92. If $f(x)$ is continuous in $[a, b]$ and $F(x)$ is any function such that $F'(x) = f(x)$, then

(a) $\int_a^b F(x)dx = f(b) - f(a)$

(b) $\int_a^b F(x)dx = f(a) - f(b)$

(c) $\int_a^b f(x)dx = F(b) - F(a)$

(d) $\int_a^b f(x)dx = F(a) - F(b)$

93. If $f(x) = f(2a - x)$, then $\int_0^{2a} f(x)dx$ is

(a) $2\int_0^a f(x)dx$ (b) $-2\int_0^a f(x)dx$

(c) $\int_0^a f(2a - x)dx$ (d) 0

94. $\int_0^{\pi/2} \frac{\sin^{3/2} x}{\sin^{3/2} x + \cos^{3/2} x}\, dx$ is equal to

(a) 0 (b) 1

(c) $\frac{\pi}{4}$ (d) $\frac{\pi}{2}$

95. The value of the definite integral

$\int_{-a}^{a} |x|\, dx$ is equal to

(a) a (b) a^2
(c) 0 (d) 2a

96. The length of the cycloid with parametric equation $x(t) = (t + \sin t)$, $y(t) = (1 - \cos t)$ between (0, 0) and (π, 2) is

(a) 4 units (b) 4π units
(c) 2 units (d) $\frac{\pi}{2}$ units

97. Volume generated by the revolution of the curve $r = a(1 - \cos\theta)$ is

(a) $\frac{32}{3}\pi a^3$ (b) $\frac{32}{5}\pi a^3$
(c) $\frac{8}{3}\pi a^3$ (d) $\frac{32}{9}\pi a^3$

98. The series $\sum_{n=0}^{n} (2x)^n$ converges

(a) for x with $-1 \le x \le 1$
(b) for x with $-\frac{1}{2} < x < \frac{1}{2}$
(c) for x with $-2 < x < 2$
(d) for x with $-\frac{1}{2} \le x \le \frac{1}{2}$

99. If p and are positive real numbers, then the series $\frac{2^p}{1^q} + \frac{3^p}{2^q} + \frac{4^p}{3^q} + \ldots$ ad inf is convergent for

(a) $p < q - 1$ (b) $p < q + 1$
(c) $p \ge q - 1$ (d) $p \ge q + 1$

100. Which one of the following series is not convergent?

(a) $\frac{1}{2\sqrt{2}} + \frac{1}{3\sqrt{3}} + \frac{1}{4\sqrt{4}} + \ldots$ ad inf
(b) $1\frac{1}{2} - 1\frac{1}{3} + 1\frac{1}{4} - 1\frac{1}{5} + \ldots$ ad inf
(c) $\frac{1}{2} - \frac{1}{3} + \frac{1}{4} - \frac{1}{5} + \ldots$ ad inf
(d) $x + x^2 + x^3 + x^4 \ldots$ where $|x| < 1$

ANSWERS

1.(a)

2.(a) Given homogenous equation can be reduced to the one with variables separable by substituting $y = vx$.

3.(c) $I.F. = e^{\int 1/x\, dx} = e^{\log x} = x$

$\therefore$ Solution given differential equation is

$y \cdot x = \int x^2 \cdot x\, dx + c = \frac{1}{4}x^4 = c$

when $x = 1$ $y = 1$

$\therefore \quad 1 = \frac{1}{4} + c \quad$ or $\quad c = \frac{3}{4}$

Hence, the solution is

$y \cdot x = \frac{1}{4}x^4 + \frac{3}{4}$ or $4xy = x^4 + 3$

4.(c)

5.(c) Given family of straight lines passing through the origin is

$$y = mx \qquad \text{(i)}$$

Differentiating w.r.t. x,

$$\frac{dy}{dx} = m$$

Substituting this value in (i), we get

$$y = \frac{dy}{dx} \cdot x \quad \text{or} \quad xdy - ydx = 0$$

6.(d)

7.(a) Equation of family of circles whose radius r and centre is (a, 0), which lies on the x-axis, is

$$(x - a)^2 + y^2 = r^2 \qquad \text{(i)}$$

Differentiating w.r.t. x,

$$2(x - a) + 2y\frac{dy}{dx} = 0 \quad \text{or} \quad x - a = -y\frac{dy}{dx}$$

Substituting this value in (i)

$$\left(-y\frac{dy}{dx}\right)^2 + y^2 = r^2$$

$$y^2\left[1 + \left(\frac{dy}{dx}\right)^2\right] = r^2$$

8.(a) Writing D for $\frac{d}{dx}$, given differential equation is

$$(D^4 - 6D^3 + 12D^2 - 8D)y = 0$$

or $\quad D(D-2)^3 y = 0$

Hence, the general solution is

$$y = C_1 e^{0 \cdot x} + (C_2 + C_3 x + C_4 x^2)e^{2x}$$

or $y = C_1 + (C_2 + C_3 x + C_4 x^2)e^{2x}$

9.(a) Given equation is

$$(4D^2 + 4D + 5)y = 0$$

or $\quad 4\left(D^2 + D + \frac{5}{4}\right)y = 0$

Roots of the equation

$D^2 + D + \frac{5}{4} = 0$ are $-\frac{1}{2} \pm i$

Hence, the two independent solution are

$$e^{-1/2x}\cos x \text{ and } e^{-1/2x}\sin x$$

10.(c) Given equation is

$(D^2 + D)y = x^2 + 2x + 4$ or

$D(D+1)y = x^2 + 2x + 4$

$\therefore$ P.I. $= \frac{1}{D(D+1)}(x^2 + 2x + 4)$

$= \frac{1}{D}(1+D)^{-1}(x^2 + 2x + 4)$

$= \frac{1}{D}\left(1 - D + \frac{D^2}{2} \ldots\right)(x^2 + 2x + 4)$

$= \frac{1}{D}(x^2 + 2x + 4 - 2x - 2 + 1)$

$= \frac{1}{D}(x2 + 3) = \frac{x^3}{3} + 3x$

11.(d) Suppose the equation of the line is

$\frac{x}{a} - \frac{y}{a} = 1$ or $x - y = a$

Since it passes through (4, −3)

$\therefore \quad 4 - (-3) = a$ or $a = 7$.

Hence, the equation of the line is $x - y = 7$.

12.(d) Since the lines are concurrent

$$\therefore \quad \begin{vmatrix} l & m & n \\ m & n & l \\ n & l & m \end{vmatrix} = 0$$

or $\quad$ ... $= 0$

or $\quad l^3 + m^3 + n^3 - 3mnl = 0$

or $\quad l + m + n = 0$

13.(d) Since $hxy + gx + fy + c = 0$ represents two straight lines

$$\therefore \quad 2 \cdot \frac{f}{2} \cdot \frac{g}{2} \cdot \frac{h}{2} - c\left(\frac{h}{2}\right)^2 = 0$$

or $\quad \frac{1}{4}fgh = \frac{1}{4}ch^2$ or $fg = ch$ $\quad (\because h \neq 0)$

14.(c) Given equation

$$4\cos\theta + 3\sin\theta = \frac{30}{r}$$

In Cartesian form can be written as

$$4x + 3y = 30$$

Hence, distance of pole (origin) from the line

$$\pm\frac{-30}{\sqrt{16+9}} = \frac{30}{5} = 6$$

15.(b) $\qquad$ 16.(c)

17.(b) $\qquad$ 18.(d)

19.(c) $\qquad$ 20.(b)

21.(d) Let P(x, y) be any point on the parabola

By definitions

$$\sqrt{(x+3)^2 + (y-0)^2} = \frac{x+5}{\sqrt{1}}$$

22.(a) Let $(x_r, y_r) = \left(ct_r, \frac{c}{t_r}\right)$, $r = 1, 2, 3, 4$

The equation of the normal at $\left(ct, \frac{c}{t}\right)$ to $xy = c^2$ is

$$ty = t^3 x + c - ct^4$$

Since it passes through (α, β)

$$\therefore \quad ct^4 - \alpha t^3 + \beta t - c = 0$$

The roots of this equation are t_1, t_2, t_3, t_4

$$\therefore \quad t_1 + t_2 + t_3 + t_4 = \frac{\alpha}{c}$$

Hence,

$x_1 + x_2 + x_3 + x_4 = c(t_1 + t_2 + t_3 + t_4)$

$= c \times \frac{a}{c} = a$

23.(b) Given lines are $\frac{x-b}{a} = \frac{y}{1} = \frac{z-d}{c}$ and

$\frac{x-b}{a} = \frac{y}{1} = \frac{z-d}{c}$

Hence, the lines are perpendicular if

$aa' + 1.1 + cc' = 0$

or $aa' + cc' = -1$

24.(c) A general equation of a cone which touches the three coordinate plane is $\sqrt{fx} + \sqrt{gy} + \sqrt{hx} = 0$.

25.(a) Let (α, β, γ) be any point on the cylinder.

Any generator through this has the equation

$\frac{x-\alpha}{0} = \frac{y-\beta}{0} = \frac{z-\gamma}{1}$ (parallel to z-axis)

This intersect the given curve at the point $(\alpha, \beta, r+\gamma)$, where r is given by

$\alpha^2 + \beta^2 + (r+\gamma)^2 = 1,\ \alpha + \beta + (r+\gamma) = 1$

Eliminating r,

$\alpha^2 + \beta^2 + [1 - \alpha - \beta]^2 = 1$

or $\alpha^2 + \beta^2 + 1 + \alpha^2 + \beta^2 - 2\alpha + 2\alpha\beta - 2\beta = 1$

or $2\alpha^2 + 2\beta^2 + 2\alpha\beta - 2\alpha - 2\beta = 0$

or $\alpha^2 + \beta^2 + \alpha\beta - \alpha = 0$

Hence, the equation of the cylinder is $x^2 + y^2 + xy - x - y = 0$

26.(b)

27.(a) Angle between $\vec{A} - 2\hat{i} - \hat{j} + 2\hat{h}$ and $\vec{B} = 6\hat{i} - 3\hat{j} + 6\hat{h}$

$\cos\theta = \frac{2 \times 6 + (-1)(-3) + 2(6)}{\sqrt{4+1+4}\sqrt{36+9+36}} = \frac{27}{3 \times 9} = 1$

$\Rightarrow \theta = 0°$

28.(c) We have $R^2 = P^2 + Q^2 + 2PQ\cos\theta$. For our case $P = Q = R$ which gives, $\cos\theta = -1/2 = \theta = 120°$.

29.(c) This comes form equalisation of clockwise and anti-clockwise moments.

30.(a)

31.(c) Vectors at right angles to each other are independent

32.(d) Net amount = (anti-clockwise moment – clockwise moment)

$= (5W \times 2) - (6W \times 3)$

$= 10W - 18W = -8W$

33.(c) Highest height achieved does not depend upon mass and is directly proportional to the square of velocity of projection.

34.(c) Here acceleration = Force/Mass = $W \times g/m$

Distance moved in time $T = 1/2\ (Wg/mn)T^2$

35.(a)

36.(b) We have $v = W\sqrt{a^2 - x^2}$, which gives $W = 8/\sqrt{25-9} = 2$. Therefore, the period $T = 2\pi/W = 2\pi/2 = \pi$

37.(c) It is case of S.H.M.

38.(c)

39.(b) We have $F = mv^2/r$. Thus, $r = mv^2/F = 4 \times (20)^2/400 = 4$ cm.

40.(d) Escape velocity

$= \sqrt{2GM/R} = \sqrt{\frac{2G}{R} \times \frac{4}{3}\pi R^3 \rho} = R\sqrt{\frac{8\pi}{3} G\rho}$

41.(d)

42.(b)

43.(c) $(\sin x - i\cos x)^5$

$= \left[\cos\left(\frac{\pi}{2} - x\right) + i\sin\left(\frac{\pi}{2} - x\right)\right]^5$

$= \cos 5\left(\frac{\pi}{2} - x\right) + i\sin 5\left(\frac{\pi}{2} - x\right)$

(by De Moivre's Theorem)

$= \sin 5x + i\cos 5x$

Hence, the real part is $\sin 5x$

44.(a) Given complex number

$= \frac{1+2i}{1-(1-i)^2} = \frac{1+2i}{(1-(1+i^2-2i))}$

$= \frac{1+2i}{1+2i} = 1 = 1 + 0i$

Hence, modulus of given number = 1.

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45.(c)

46.(a)

47.(c)

48.(d)

49.(b) When $f(x) = x^3 + 5x^2 - 3x + 2$ is divided by $x + 1$, the remainder will be
$f(-1) = (-1)^3 + 5(-1)^2 - 3(-1) + 2$
$= -1 + 5 + 3 + 2 = 9.$

50.(c) Since α, β, γ are roots of the equation
$2x^3 - 3x^2 + 6x + 1 = 0$

$\therefore \quad \alpha + \beta + \gamma = \frac{3}{2}, \alpha\beta + \beta\gamma + \gamma\alpha$

$= 3, \alpha\beta\gamma = -\frac{1}{2}$

Hence,
$\alpha^2 + \beta^2 + \gamma^2 = (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \beta\gamma + \gamma\alpha)$
$= \left(\frac{3}{2}\right)^2 - 2(3) = \frac{9}{4} - 6 = \frac{-15}{4}$

51.(c) Since $1, \alpha_1, \alpha_2, \ldots, \alpha_{n-1}$ are the roots of the equation $x^n - 1 = 0$,

$\therefore \quad (x - 1)(x - \alpha_1) \ldots (x - \alpha_{n-1}) = x^n - 1$

or $\quad (x - \alpha_1)(x - \alpha_2) \ldots (x - \alpha_{n-1})$

$= \frac{x^n - 1}{x - 1} = x^{n-1} + \ldots + x + 1$

54.(a) I. $\quad 0R1 \Leftrightarrow 3.0 + 4.1 = 5$ (false)

II. $\quad 1R\frac{1}{2} \Leftrightarrow 3.1 + 4\cdot\frac{1}{2} = 5$ (true)

III. $\quad \frac{2}{3}R\frac{3}{4} \Leftrightarrow 3\cdot\frac{2}{3} + 4\cdot\frac{3}{4} = 5$ (true)

$\therefore \quad \frac{3}{2}R\frac{1}{4} \Leftrightarrow 3\cdot\frac{3}{2} + 4\cdot\frac{1}{4} = 5$ (false)

Hence, only II and III are correct.

55.(b) Since $\alpha_0\beta = I$ and $\beta_0\alpha = I$, therefore, β is the inverse of α, also a 1 – 1 map.

56.(d)

57.(c) Reflexivity: $(x, x) \in R$ as $x - x = 0$ is divisible by 51

Symmetry: Let $(x, y) \in R$: $x - y$ is divisible by 51

$\therefore \quad y - x$ is also divisible by 51

$\therefore \quad (y, x) \in R$

Transitivity: Let $(x, y) \in R$, $(y, z) \in R$

$\therefore \quad x - y$ is divisible by 51, and $y - z$ is divisible by 51

$\therefore \quad (x - y) + (y - z) = x - z$ is also divisible by 51

Hence, R is a reflexive relation

67.(b) $\begin{vmatrix} x+a & b & c \\ c & x+b & a \\ a & b & x+c \end{vmatrix} = 0$

Adding elements of 2nd and 3rd columns to 1st column.

$$\begin{vmatrix} x+a+b+c & b & c \\ x+a+b+c & x+b & a \\ x+a+b+c & b & x+c \end{vmatrix} = 0$$

$$\text{or } (x+a+b+c)\begin{vmatrix} 1 & b & c \\ 1 & x+b & a \\ 1 & b & x+c \end{vmatrix} = 0$$

Thus, either $x+a+b+c=0 \Rightarrow x-(a+b+c)$

68.(a)

69.(b) Here $|A| = 5-6 = -1$

Also, Adj. $A = \begin{vmatrix} -5 & -2 \\ -3 & -1 \end{vmatrix}$

Hence, $A^{-1} = \frac{1}{|A|}$ Adj $A = \begin{bmatrix} 5 & 2 \\ 3 & 1 \end{bmatrix}$

70.(c) 71.(c)

72.(c) 73.(c)

74.(c)

75.(d) $\frac{d}{dx}[\sin^{-1}\sqrt{\frac{x-\beta}{\alpha-\beta}}$

$$= \frac{1}{\sqrt{1-\frac{x-\beta}{\alpha-\beta}}} \times \frac{1}{2\sqrt{\frac{x-\beta}{\alpha-\beta}}} \times \frac{1}{\alpha-\beta}$$

$$= \frac{1}{2}\frac{\sqrt{\alpha-\beta}}{\sqrt{\alpha-x}} \times \frac{\sqrt{\alpha-\beta}}{\sqrt{x-\beta}} \times \frac{1}{\alpha-\beta}$$

$$= \frac{1}{\sqrt{(\alpha-x)(x-\beta)}}$$

76.(c)

77.(d) Volume of cube, whose edge is l cm,

$V = l^3 \quad \therefore \quad \frac{dv}{dt} = 3l^2 = \frac{dl}{dt}$

But $\frac{dl}{dt} = 5$ cm / sec

When $l = 4$ cm

$\frac{dv}{dt} = 3 \times (4)^2 \times 5 = 240$ cm^3/sec

78.(c) Let $y = f(x) \log_{x^2} 3 = \frac{\log 3}{\log x^2} = \frac{\log 3}{2\log x}$

$$\therefore \frac{dy}{dx} = \frac{\log 3}{2} \times \frac{1}{(\log x)^2} \times \frac{1}{x}$$

$$= -\frac{\log 3}{2x(\log x)^2}$$

79.(b) Here $(gof)(x) = g[f(x)] = g(e^x) = \log e^x = x$

Hence, $(gof)'(x) = 1$

80.(a) $\frac{x}{y} + \frac{y}{x} = 2$ or $x^2 + y^2 = 2xy$

Differentiating w.r.t. to x,

$$2x + 2y\frac{dy}{dx} = 2y + 2x\frac{dy}{dx}$$

or $(y-x)\frac{dy}{dx} = y - x \quad \therefore \frac{dy}{dx} = 1$

81.(b) By Largrange' Mean Value theorem,

$\exists\, c \in (-1, 1)$ such that

$$f'(c) = \frac{f(1)-f(-1)}{2}$$

or $\quad a' = \frac{(a+b)-(-a+b)}{2} = a$

Hence, it holds for arg $c \in (-1, 1)$.

82.(c) $f(x) = (1-x)^{5/2}$

$\therefore \quad f'(x) = -\frac{5}{2}(1-x)^{3/2},$

$f''(x) = \frac{15}{4}(1-x)^{1/2}$

Thus, we have

$$(1-x)^{5/2} = 1 + x\left(-\frac{5}{2}\right) + \frac{x^2}{\lfloor 2}\frac{15}{4}(1-\theta x)^{1/2}$$

When $x \to 1$,

$$0 = 1 - \frac{5}{2} + \frac{1}{2} \times \frac{15}{4}(1-\theta)^{1/2}$$

or $\quad \frac{15}{8}(1-\theta)^{1/2} = \frac{3}{2}$

or $\quad (1-\theta)^{1/2} = \frac{4}{5}$

$\therefore \; 1-\theta = \frac{16}{25}$ or $\theta = 1 - \frac{16}{25} = \frac{9}{25}$

83.(d) Let $y = x^4 - \frac{2}{3}x^3 - 2x^2 + 2x$

$\therefore \quad \frac{dy}{dx} = 4x^3 - 2x^2 - 4x + 2, \frac{d^2y}{dx^2}$

$= 12x^2 - 4x - 4$

Now from max. or min. $\frac{dy}{dx} = 0$

This gives

$4x^3 - 2x^2 - 4x + 2 = 0$

or $\quad (x-1)(4x^2 + 2x - 2) = 0$

$\Rightarrow \quad x = 1, x = -1, x = \frac{1}{2}$

But $\quad -1 \notin [0, 2]$

At $\quad x = 1, \frac{d^2y}{dx^2} = 12 - 4 - 4 > 0$

$\therefore \quad y$ is min. at $x = 1$ and min. value $= \frac{1}{3}$

At $\quad x = \frac{1}{2}$

$\frac{d^2y}{dx^2} = 12\left(\frac{1}{2}\right)^2 - 4\left(\frac{1}{2}\right) - 4$

$= 3 - 2 - 4 < 0$

$\therefore \quad y$ is at $x = \frac{1}{2}$ and max. value $= \frac{23}{48}$

84.(b)

85.(a) Differentiating the given curve $x^2 - y^2 = a^2$ w.r. to x

$2x - 2y\frac{dy}{dx} = 0 \quad$ or $\quad \frac{dy}{dx} = \frac{x}{y}$

$\therefore \quad \left(\frac{dy}{dx}\right)_{(a,\sqrt{2}a)} = \frac{a}{\sqrt{2}a} = \frac{1}{\sqrt{2}}$

Hence, length of sub-tangent at $(a, \sqrt{2}a)$

$= y \cdot \frac{dx}{dy} = \sqrt{2}a \times \sqrt{2} = 2a$

86.(c) Differentiating

$x^2 = 4ay$ w.r. to x $\quad$ (i)

$2x = 4a\frac{dy}{dx} \quad$ or $\quad \frac{dy}{dx} = \frac{x}{2a}$

$\therefore \quad \left(\frac{dy}{dx}\right)_{(2a,a)} = \frac{2a}{2a} = 1$

$\therefore$ Equation of normal at $(2a, a)$ is

$y - a = -\frac{1}{1}(x - 2a) = -x + 2a$

or $\quad y = -x + 3a \quad$ (ii)

Substituting this value of y in (i)

$x^2 = 4a(-x + 3a) = -4ax + 12a^2$

or $\quad x^2 + 4ax - 12a^2 = 0$

or $\quad (x + 6a)(x - 2a) = 0$

$\therefore \quad x = -6a, x = 2a$

So that $y = 9a, y = a$

Hence, the required point is $(-6a, 9a)$.

87.(b)

88.(b) $u = \sin^{-1}\left(\frac{x^{1/3} + y^{1/3}}{x^{1/2} + y^{1/2}}\right)^{1/2}$

or $z = \sin u = x^{-1/12}\left(\frac{1 + y^{1/3}}{1 + y^{1/2}}\right)^{1/2}$

Thus, z is a homogenous function in x and y of order $-\frac{1}{12}$

$\therefore \quad$ By Euler's theorem

$x\frac{\partial z}{\partial x} + y\frac{\partial z}{\partial y} = -\frac{1}{12}z$

or $\quad x\cos u\frac{\partial u}{\partial x} + y\cos u\frac{\partial u}{\partial y}$

$= -\frac{1}{12}\sin u$

or $\quad x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = -\frac{1}{12}\tan u$

89.(b)

90.(b)

91.(b)

92.(c)

93.(a)

94.(c) Let $\quad I = \int_0^{\pi/2} \frac{\sin^{3/2}x\,dx}{\sin^{3/2}x + \cos^{3/2}x}$

Also $I = \int_0^{\pi/2} \frac{\sin^{3/2}\left(\frac{\pi}{2} - x\right)dx}{\sin^{3/2}\left(\frac{\pi}{2} - x\right) + \cos^{3/2}\left(\frac{\pi}{2} - x\right)}$

$$= \int_0^{\pi/2} \frac{\cos^{3/2} x}{\cos^{3/2} x + \sin^{3/2} x}\, dx$$

Adding

$$2I = \int_0^{\pi/2} \frac{\sin^{3/2} x + \cos^{3/2} x}{\sin^{3/2} x + \cos^{3/2} x} = \int_0^{\pi/2} dx$$

$$= \frac{\pi}{2} \quad \Rightarrow \quad I = \frac{\pi}{4}$$

95.(b) $\int_{-a}^{a} |x|\, dx = \int_{-a}^{0} -x\, dx + \int_0^a x\, dx$

$$= 2\int_0^a x \cdot dx = 2 \cdot \frac{x^2}{2}\Big]_{0}^{a} = a^2$$

96.(a) Required length of the cycloid

$$= \int ds = \int \frac{ds}{dt} dt = \int_0^{\pi} \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}\, dt$$

$$= \int_0^{\pi} \sqrt{(1 + \cos t)^2 + (\sin t)^2}\, dt$$

$$= \pi \int_0^{\pi} a^2(1 - \cos\theta)^2 \sin^2\theta \times a \sin\theta(2\cos\theta - 1)d\theta$$

$$= \pi a^3 \int_0^{\pi} (1-\cos\theta)^2 (1 - \cos^2\theta)(2\cos\theta - 1)\sin\theta\, d\theta$$

$$= -\pi a^3 \int_{1}^{-1} (1 - t)^2 (1 - t^2)(2t - 1)dt$$

(Putting $t = \cos\theta$)

$$= -\pi a^3 \int_{-1}^{1} (4t^2 + 5t^4 - 2t^6 - 2t^5 - 1)dt$$

$$= -\pi a^3 \left(2t^2 - \frac{4}{3}t^3 + t^5 - \frac{1}{2}t^6 - \frac{1}{3}t^6 - t\right)_{-1}^{1}$$

$$= -\pi a^3 \left(-\frac{8}{5}\right) = \frac{8}{5}\pi a^3$$

98.(b)

99.(a) Here

$$u_n = \frac{(n+1)^p}{n^q} = \left(\frac{n+1}{n}\right)^p \frac{1}{n^{q-p}}$$

1. A 2 × 2 matrix which commutes with every 2 × 2 matrix is of the form

(a) $\begin{pmatrix} a & 0 \\ 0 & 0 \end{pmatrix}$ (b) $\begin{pmatrix} a & 0 \\ 0 & a \end{pmatrix}$

(c) $\begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix}$ (d) $\begin{pmatrix} a & b \\ b & a \end{pmatrix}$

2. For any two 2 × 2 symmetric matrices A and B consider the following statements:

i. (A+B) is a symmetric matrix.
ii. AB is a symmetric matrix.
iii. $AB = (BA)^T$
iv. $A^TB = AB^T$

(where A^T denotes the transpose of A)
Of these statements:

(a) II, III and IV are correct
(b) I, III and IV are correct
(c) I and II are correct
(d) I, II, III and IV are correct

3. The determinant of

$$\begin{bmatrix} r & q & s & t \\ -q & r & -t & s \\ -s & t & r & -q \\ -t & -s & q & r \end{bmatrix}$$

is equal to

(a) $r^2 + q^2 + s^2 + t^2$ (b) $(r^2 + q^2 + s^2 + t^2)^2$
(c) $(r^2 + q^2 + s^2 + t^2)^4$ (d) 0

4. Let a 3 × 3 matrix A have determinant 5. If $B = 4A^2$, then the determinant of B is equal to

(a) 20 (b) 100
(c) 320 (d) 1600

5. Consider the following statements:

Assertion

(A): If M is an n × n matrix with rank n-1, then M can be made non-singular by changing one element.

Reason

(R): An n × n matrix with rank n-1 has a non-vanishing minor of order n-1.

Of these statements:

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not a correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

6. The system of equations

$2x + y = 5$
$x - 3y = -1$
$3x + 4y = k$

Is consistent, when k is

(a) 1 (b) 2
(c) 5 (d) 10

7. For the set of real numbers IR, consider the following statements:

i. Every non-empty subset of IR which is bounded below has an infimum.
ii. Every non-empty subset of IR which is bounded above has a supremum.
iii. Every bounded subset of IR has an infimum and a supremum.

Of these statements:

(a) I and II are independent statements
(b) III implies I but not conversely
(c) III implies II but not conversely
(d) I, II and III are all equivalent.

8. If $[x]$ denotes the greatest integer function, then which one of the following is not necessarily true for all x and y?

(a) $[x + n] = [x] + n$, for every integer n
(b) $[x + y] = [x] + [y]$
(c) $[2x] = [x] + \left[x + \frac{1}{2}\right]$
(d) $[3x] = [x] + \left[x + \frac{1}{3}\right] + \left[x + \frac{2}{3}\right]$

9. $\lim\limits_{x \to \infty} x \sin \frac{1}{x}$ equals

(a) 1 (b) 0
(c) −1 (d) $-\infty$

10. Let f be defined on IR by setting $f(x) = x$, if x is rational and $f(x) = 1 - x$ if x is irrational, then

(a) f is continuous on IR.

(b) f is continuous only at $x = \frac{1}{2}$

(c) f is continuous everywhere except at $x = \frac{1}{2}$

(d) f is discontinuous everywhere

11. If $f(x) = \begin{cases} ax + b, \text{ if } x \le \frac{\pi}{4} \\ \cos x, \text{ if } x > \frac{\pi}{4} \end{cases}$ then f(x) is differentiable at $x = \frac{\pi}{4}$ if

(a) $a = -\frac{1}{\sqrt{2}}, b = \frac{1+\frac{\pi}{4}}{\sqrt{2}}$ (b) $a = \frac{1}{\sqrt{2}}, b = \frac{1+\frac{\pi}{4}}{\sqrt{2}}$

(c) $a = b = \frac{1}{\sqrt{2}}$ (d) $a = b = -\frac{1}{\sqrt{2}}$

2. For functions f and g consider the following four statements:

I. None of fog and gof may be define.

II. Only one of fog and gof may be defined.

III. fog and gof may both be defined but may not be equal.

IV: fog and gof may be both defined and equal.

Of these statements, the number of correct statements is

(a) one (b) two
(c) three (d) four

. If $\sin y = x \sin(a + y)$, then $\frac{dy}{dx}$ equals.

(a) $\frac{\sin^2(a+y)}{\sin a}$ (b) $\frac{\sin^2(a+y)}{\sin y}$

(c) $\frac{\sin^2(a+y)}{\sin a \sin y}$ (d) $\frac{\sin(a+y)}{\sin^2 a}$

14. The continuity and the differentiability of f and its first few derivatives, as per Taylor's theorem under suitable conditions, there exists some $\theta \in (0,1)$ such that

$$f(a+h) = f(a) + hf'(a) + \frac{h^2}{2} f''(a) + \frac{h^3}{6} f'''(a + \theta h)$$

if $f(x) = \sin x$, $a = \frac{\pi}{3}$, then as h tends to 0, the value of θ tends to

(a) 0 (b) $\frac{1}{4}$

(c) $\frac{1}{2}$ (d) 1

15. The function f, defined by $f(x) = x^3 - 6x^2 + 12x - 4$ for all $x \in$ IR is

(a) decreasing for $x \ge 2$ and increasing for $x < 2$.

(b) decreasing for $x \le 2$ and increasing for $x > 2$.

(c) decreasing in every interval.

(d) increasing in every interval.

16. The function f defined by $f(x) = x^{1/x}$ has a maximum at

(a) 1 (b) e
(c) $\log_e 2$ (d) 2

17. Asymptotes of the curve $(y-x)^2(y-2x) - (y-x)(y-6x) + x - 5y + 3 = 0$ parallel to the line $y - x = 0$ are

(a) $y - x - 1 = 0$ and $y - x + 1 = 0$

(b) $y - x - 4 = 0$ and $y - x + 1 = 0$

(c) $y - x - 1 = 0$ and $y - x - 1 = 0$

(d) $y - x + 1 = 0$ and $y - x + 1 = 0$

18. The point at which the tangent to the curve $y = 2x^2 - x + 1$ is parallel to $y = 3x + 9$ is

(a) (3, 9) (b) (-2,1)
(c) (2,9) (d) (1,2)

19. The subtangent to the curve $y = \sin x$ at the point $\left(\frac{\pi}{3}, \frac{\sqrt{3}}{2}\right)$ has length

(a) $\frac{\pi}{3}$ (b) $\frac{\sqrt{3}}{2}$

(c) $\sqrt{3}$ (d) 1

20. The radius of curvature of the cycloid given by
$x = a(\theta + \sin\theta)$,
$y = a(1 - \cos\theta)$
at the point "θ" is
(a) $4a\sin\theta$ (b) $4a\cos\theta$
(c) $4a\cos(\theta/2)$ (d) $4a\sin(\theta/2)$

21. Envelope of the one-parameter family of straight lines $x\cos\alpha + y\sin\alpha = a$ where α is a parameter, is
(a) $xy = a^2$ (b) $y^2 = 4ax$
(c) $x^2 + y^2 = a^2$ (d) $x^2 - y^2 = a^2$

22. If $u = \tan{-1}\left(\frac{x+y}{\sqrt{x}+\sqrt{y}}\right)$, then $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y}$ equals
(a) $2\cos 2u$ (b) $\frac{1}{4}\sin 2u$
(c) $\frac{1}{4}\tan u$ (d) $2\tan 2u$

23. If $z = e^x\sin y$, $x = \log_e t$ and $y = t^2$, then $\frac{dz}{dt}$ is given by the expression
(a) $\frac{e^t}{t}(\sin y - 2t^2\cos y)$
(b) $\frac{e^t}{t}(\sin y + 2t^2\cos y)$
(c) $\frac{e^t}{t}(\cos y + 2t2\sin y)$
(d) $\frac{e^t}{t}(\cos y - 2t^2\sin y)$

24. For the curve $y^2 = (x-2)(x-5)^2$, the point $(5, 0)$ is a
(a) node (b) single cusp
(c) double cusp (d)conjugate point

5. For the curve, $y^2 = (x-2)^2(x-5)$
(a) (6, 4) is the only point of inflexion.
(b) (6,-4) is the only point of inflexion.
(c) there is no point of inflexion.
(d) (6,4) and (6,-4) both are points of inflexion.

26. Consider the curve
$y = x^2(x - 3a)$, $a > 0$
and the following statements about its graph:
I. It possesses no symmetries.
II. It meets the y-axis only at the origin.
III. y has a maximum at (0,0) and a minimum at $(2a, -4a^3)$.
Of these statements:
(a) II and III are correct.
(b) I and III are correct
(c) I and II are correct.
(d) I, II and III are correct.

27. $\lim_{n\to\infty}\left(\frac{n!}{n^n}\right)^{\frac{1}{n}}$ is equal to
(a) 0 (b) 1
(c) e (d) $1/e$

28. $\lim_{n\to\infty}\sum_{k=1}^{n}\frac{k}{k^2+n^2}$ is equal to
(a) 0 (b) $\frac{1}{2}\log 2$
(c) $\tan^{-1}2$ (d) ∞

29. $\int_0^{\pi}\frac{\sin 4x}{\sin x}\,dx$ is equal to
(a) 0 (b) π
(c) ∞ (d) 8π

30. The curve revolving around x-axis generating a sphere, is
(a) $y2 = 4ax$ (b) $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$
(c) $x^2 + y^2 = r^2$ (d) $x^2 - y^2 = r^2$

31. The area of the surface of the solid generated by the revolution of the line segment $y = 2x$ from $x = 0$ to $x = 2$, about the x-axis is equal to
(a) $\pi\sqrt{5}$ (b) $2\pi\sqrt{5}$
(c) $4\pi\sqrt{5}$ (d) $8\pi\sqrt{5}$

32. The area of the cardioids $r = a(1 - \cos\theta)$ is
(a) $3\pi^2 a$ (b) $6\pi a^2$
(c) πa^2 (d) $\frac{3}{2}\pi a^2$

33. The length of ... sec x from $x = 0$ to $x = \pi/3$ is

(a) $\log(2+\sqrt{3})$ (b) $\log(\sqrt{2}+3)$

(c) $\log(\sqrt{5}+1)$ (d) $\log(\sqrt{3}+1)$

34. The interval of convergence of the infinite series $x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \ldots\ldots$ Is

(a) $-1 < x < 1$ (b) $-1 < x \le 1$

(c) $-1 \le x < 1$ (d) $-1 \le x \le 1$

35. The series $\frac{1}{1.2} + \frac{2}{3.4} + \frac{3}{5.6} + \ldots\ldots \infty$ is

(a) divergent (b) convergent

(c) oscillating (d) None of the above

36. Consider the following statements:

i. Sequence (u_n) converges.

ii. $\sum_{n=0}^{\infty}$ converges. iii. $\sum_{n=0}^{\infty} |u_n|$ converges.

iv. $\sum_{n=0}^{\infty} |u_n|$ converges.

Of these statements:

(a) I implies II (b) III implies I

(c) I implies III (d) II implies III

39. The singular solution of the equation $y = px + f(p)$, $p = \frac{dy}{dx}$ is obtained on eliminating p between original equation and the equation

(a) $x - f'(p) = 0$ (b) $x + f'(p) = 0$

(c) $y - f'(p) = 0$ (d) $y + f'(p) = 0$

40. The singular solution of the differential equation $y = px + p^3$, $\left(P = \frac{dy}{dx}\right)$ is

(a) $4y^3 + 27x^4 = 0$ (b) $4x^4 + 27y^3 = 0$

(c) $4y^2 + 27x^3 = 0$ (d) $4x^3 + 27y^2 = 0$

41. The differential equation $ydx - 2xdy = 0$ represents

(a) a family of straight lines

(b) a family of parabolas

(c) a family of circles

(d) a family of catenaries

42. The orthogonal trajectories of the family of parabolas $y = ax^2$ are given by the solution of the differential equation

(a) $\frac{dy}{dx} = \frac{2y}{x}$ (b) $\frac{dy}{dx} = -\frac{2y}{x}$

(c) $\frac{dy}{dx} = \frac{x}{2y}$ (d) $\frac{dy}{dx} = -\frac{x}{2y}$

45. A particular integral of the differential equation $(D_2 + 4)\,y = x$, is

(a) $x e^{-2x}$ (b) $x \cos 2x$
(c) $x \sin 2x$ (d) $x/4$

46. The solution of the differential equation $y'' + (3i - 1)\,y' - 3iy = 0$ is

(a) $y = C_1e^{x} + C_2e^{3ix}$
(b) $y = C_1e^{-x} + C_2e^{3ix}$
(c) $y = C_1e^{x} + C_2e^{-3ix}$
(d) $y = C_1e^{-x} + C_2e^{-3ix}$

47. The angle between the lines $x + y = 0$ and $x - y = 0$ is

(a) 45° (b) 90°
(c) 135° (d) 180°

48. The coordinates of the vertices of a triangle ABC are (0,0), (1,-1) and (3,2) respectively. The length of the perpendicular drawn from A (0,0) to the side BC is

(a) $\frac{5}{\sqrt{2}}$ (b) $\frac{5}{\sqrt{13}}$
(c) $\frac{7}{\sqrt{2}}$ (d) $\frac{7}{\sqrt{13}}$

49. If the equation $2xy + 3x + 4y + c = 0$ represents a pair of straight lines, then the value of c is equal to

(a) 6 (b) 0
(c) −2 (d) 8/3

50. Consider the following statements:

Assertion

(A) The equation of the line through two points $(2, \pi)$ and $\left(1, \frac{\pi}{2}\right)$ is $\frac{2}{r} = 2 \sin \theta - \cos \theta$.

Reason (R):

The equation above is written in polar coordinates.

Of these statements:

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but is not a correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

51. The equation of the circle with radius [illegible] and diameters $x + y = 6$ and $x + 2y = 4$ is

(a) $x^2 + y^2 - 16x + 4y = 32$
(b) $x^2 + y^2 - 16x - 4y = 32$
(c) $x^2 + y^2 + 16x + 4y = 32$
(d) $x^2 + y^2 + 16x - 4y = 32$

52. Consider a circle $x^2 + y^2 = r^2$. If the line $ax + by + c = 0$ is a normal to the circle, then

(a) $c = 0$ (b) $c = r$
(c) $c = -r$ (d) $c = 1$

53. The line $y = 2x + c$ is a tangent to the parabola $y^2 = 16x$, if c equals

(a) 0 (b) −1
(c) −2 (d) 2

54. If the length of the latus rectum of an ellipse is one-third of its major axis, then the eccentricity of the ellipse is

(a) $\frac{2}{3}$ (b) $\frac{\sqrt{3}}{2}$
(c) $\frac{2\sqrt{2}}{3}$ (d) $\frac{\sqrt{2}}{3}$

55. The length of the latus rectum of the hyperbola $\frac{x^2}{9} - \frac{y^2}{16} = 1$ is

(a) $\frac{4}{3}$ (b) $\frac{3}{4}$
(c) $\frac{32}{3}$ (d) $\frac{3}{32}$

56. Match List I with list II and select the correct answer using the codes given below the lists:

List I
(Equation of conics)
I. $3y = 4 + 5(x-2)^2$
II. $x^2 + 3y^2 = 5$
III. $xy = 8$
IV. $2x^2 - y^2 = 3$

List II
(The description)
(a) A circle
(b) an ellipse with the two axes unequal
(c) a parabola
(d) a rectangular hyperbola with coordinate axis as asymptotes
(e) a non-rectangular hyperbola with co-ordinate axes as axes

Codes:
(a) I-A, II-B, III-E, IV-D
(b) I-C, II-A, III-D, IV-E
(c) I-D, II-E, III-C, IV-A
(d) I-C, II-B, III-D, IV-E

57. The value of λ for which the lines $3x + 2y + z - 5 = 0 = x + y - 2z - 3$ and $2x - y - \lambda z = 0 = 7x = 10y - 8z - 8z$ are perpendicular to each other is
(a) -1 (b) -2
(c) 2 (d) 1

58. Consider the following statements:
Assertion (A):
The plane $y + z + 1 = 0$ is parallel to x-axis.
Reason (R):
Normal to the plane is parallel to the x-axis
Of these statements:
(a) Both A and R true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not a correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

59. The shortest distance of the point (10,11,8) from the sphere $x^2 + y^2 + z^2 - 2x + 2y = 2$ is
(a) 16 (b) 14
(c) 15 (d) 19

60. Which of the following are generators of the cone $5yz - 8zx - 3xy = 0$?
I. $\frac{x}{1} = \frac{y}{2} = \frac{z}{3}$ II. $x = y = z$
III. $\frac{x}{5} = \frac{y}{-4} = \frac{z}{1}$ IV. $\frac{x}{1} = \frac{y}{1} = \frac{z}{-1}$
Select the correct answer using the codes given below:
Codes:
(a) I, III and IV (b) I, II and III
(c) II, III and IV (d) I, II and IV

61. The equation $x^2 + 4y^2 = 5 - 2y$ represents the surface of
(a) a circular cylinder
(b) a parabolic cylinder
(c) an elliptic cylinder
(d) a cone

62. When we study the motion of the planet earth is treated as
(a) a particle (b) a rigid body
(c) an oblate spheroid
(d) a hard mass of finite size

63. If a train passes an electric pole in 8 seconds and a bridge of the same length as the train in 16 seconds. Then the train is moving with
(a) uniform velocity
(b) retardation
(c) uniform acceleration
(d) None of the above

64. If $\bar{A} = 3\hat{i} - 2\hat{j} + k, \bar{B} = 2\hat{i} - 4\hat{j} - 3\hat{k}$ and $2\bar{A} - 3\bar{B} - 5\bar{C}$ is
(a) 45° (b) 60°
(c) 90° (d) 180°

65. If $\bar{A} = 2\hat{i} + 3\hat{j} - 4k$ and $\bar{B} = 5\hat{i} + 2\hat{j} + 4k$ the the angle between $\bar{A}$ and $\bar{B}$ is
(a) 45° (b) 60°
(c) 90° (d) 180°

66. Forces 13.10.5 kg. Wt. Act along the sides BC, CA and AB of an equilateral triangle ABC. The direction of their resultant has slope.
(a) $\frac{5}{11}\sqrt{3}$ (b) $\frac{5}{7}\sqrt{3}$
(c) $\frac{3}{11}\sqrt{5}$ (d) $\frac{11}{3}\sqrt{5}$

67. If a particle starting with a velocity u an subjected to a uniform acceleration a trav for n sec, then the distance described by in nth second is
(a) $u + \frac{a}{2}(2n-1)$ (b) $u + \frac{a}{2}(2-n)$
(c) $u + \frac{n}{2}(2a-1)$ (d) $u + \frac{n}{2}(2a+1)$

68. A stone is thrown vertically upwards w a velocity 98 meters per seconds. velocity will be zero after ($g = 9.8 m/sec^2$)
(a) 15 seconds (b) 20 seconds
(c) 10 seconds (d) 5 seconds

69. A body of mass $\frac{1}{2}$ kg is subjected to a force $(\hat{i} + 2\hat{j} + \hat{k})$ N. If the body is initially at rest, then its velocity after 4 seconds will be
(a) $2\sqrt{6}$ m/sec (b) $4\sqrt{6}$ m/sec
(c) $8\sqrt{6}$ m/sec (d) $16\sqrt{6}$ m/sec

70. Newton's third law says
(a) to every action there is an unequal and opposite reaction
(b) to every action there is an equal and opposite reaction
(c) to every action there is an equal and same reaction
(d) None of the above

71. A particle moving in S.H.M. has the speeds 8 cm/sec and 6 cm/sec at distances 3 cm and 4 cm respectively from the center of its path. The period of its motion is
(a) $\frac{\pi}{2}$ sec (b) 2π sec
(c) π sec (d) $\frac{\pi}{3}$ sec

72. The amplitude of a particle executing S.H.M. is 20 cm and time period is 1 sec. The maximum velocity is
(a) 10π cm/sec (b) 20π cm/sec
(c) 30π cm/sec (d) 40π cm/sec

73. If the greatest height to which a man can throw a stone is h, then the greatest distance to which he can throw it will be
(a) $h/2$ (b) h
(c) $2h$ (d) $4h$

74. The angle of projection of a body for attaining maximum range is
(a) $> 45°$ (b) $< 45°$
(c) $45°$ (d) $90°$

75. Angular velocity of the minutes hand of a watch is
(a) $\frac{\pi}{1800}$ radian/sec (b) $\frac{\pi}{180}$ radian/sec
(c) $\frac{\pi}{360}$ radian/sec (d) None of the above

76. A particle moves in a plane with an acceleration F which is always directed towards a fixed point O in the plane. It is found that the path is given by $r = a(1 + 2\cos\theta)$. The apsidal distances are equal to
(a) $2a$ and $a/2$ (b) $3a$ and a
(c) $4a$ and $2a$ (d) $5a$ and $3a$

77. For any two $r, s \in \mathbb{R}^+$ with $r < s$, there always exists an $n \in \mathbb{Z}^+$, $n > 1$ such that (where $\mathbb{R}^+$ an $\mathbb{Z}^+$ stand for the sets of positive reals and integers respectively)
(a) $nr > s$ (b) $nr = s$
(c) $nr < s$ (d) $nr \neq s$

78. Consider the following statements:
Assertion (A)
There exists a bijection between the set N of natural numbers and the set Z of integers.
Reason (R)
N is a proper subset of Z.
Of these statements
(a) Both A and R are true, and R is the correct explanation of A
(b) Both A and R are true. But R is not the correct explanation of A
(c) A is true, but R is false
(d) A is false, but R is true

79. If a,b,c,d are four positive real numbers and further if abcd = 1, then the least value of $(1 + a)(1 + b)(1 + c)(1 + d)$ is
(a) 16 (b) 14
(c) 12 (d) 10

80. If ω is an imaginary cube root of unity, $x = a + b$, $y = a\omega + b\omega^2$, $z = a\omega^2 + b\omega$, then xyz equals
(a) $a + b$ (b) $a^2 + b^2$
(c) $a^3 + b^3$ (d) $a^4 + b^4$

81. The correct polar form of the complex number $1 - i$ is
(a) $\sqrt{2}\, e^{\frac{i\pi}{4}}$ (b) $e^{-\frac{i\pi}{4}}$
(c) $\sqrt{2}\, e^{-\frac{i\pi}{4}}$ (d) $e^{\frac{i\pi}{4}}$

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82. The HCF of two polynomials $f(x)$ and $g(x)$ is $(x-1)$ and their LCM is $(x-1)^3(x-2)(x-3)$. If $f(x)$ is $(x-1)^3$, then $g(x)$ is
(a) $(x-2)(x-3)$
(b) $(x-1)^2(x-2)$
(c) $(x-1)(x-2)(x-3)^2$
(d) $(x-1)(x-2)(x-3)$

83. Consider the following pairs of numbers:
I. 45689, 45690
II. 689975, 689976
III. 875498, 784588
IV. 1000,1001
The pairs consisting of relatively prime numbers are
(a) I, II and III (b) I, II and IV
(c) II, III and IV (d) I and II

84. The numbers of candidates appearing for Hindi, English and Mathematics are 60, 84 and 108 respectively. if the same number of candidates have to be seated in every room and all the candidates in a room should be appearing for the same subject, then the minimum number of rooms required to seat all the candidates is
(a) 12 (b) 21
(c) 24 (d) 63

85. The polynomial equation $x^3 + 3x^2 + 4x - 7 = 0$ has over the complexfield
(a) 3 roots (b) only 2 roots
(c) only 1 root (d) no root

86. If the polynomial $8x^4 - 2x^3 - bx - 3$ is divisible by $2x^2 + x - 3$, then the values of a and b are respectively.
(a) 10 and 31 (b) 13 and 10
(c) 8 and 5 (d) –10 and –13

87. Consider the following statements:
I. $x^2 - 9$ is a factor of $x^3 + 5x^2 = 9x - 45$.
II. $f\left(\frac{b}{a}\right)$ is the remainder when the polynomial $f(x)$ is divided by $ax + b$.
III. $f\left(-\frac{a}{b}\right)$ is the remainder when the polynomial $f(x)$ is divided by $ax + b$.
IV. 0 is the remainder when the polynomial $f(x)$ is divided by a linear factor $x - a$.
Of these statements
(a) I and II are correct.
(b) I alone is correct.
(c) III alone is correct.
(d) III and IV are correct.

88. If kx^4 and lx^4 leave the remainders p and $\frac{1}{p}$ when divided by $ax + b$ and $bx + a$ respectively, then the value of kl is

89. The simplified form of $\frac{x^{4a/b} + (xy)^{2a/b} + y^{4a/b}}{x^{6a/b} - y^{6a/b}}$ is
(a) $x^{2a/b} + y^{2a/b}$ (b) $x^{2a/b} + y^{2a/b}$
(c) $\frac{1}{x^{2a/b} - y^{2a/b}}$ (d) $\frac{1}{x^{2a/b} + y^{2a/b}}$

90. If α, β, γ and δ are the roots of the equation 36x4 – 216x3 + 401x2 – 231x + 40 = 0, then $\alpha^2 + \beta^2 + \gamma^2 + \delta^2$ equals
(a) 36 (b) $\frac{247}{18}$
(c) $\frac{293}{18}$ (d) $\frac{1049}{18}$

91. If $3 + \sqrt{-5}$ is a root of the equation $x^3 - 9x^2 + 32x - 42 = 0$, then the real root of the equation is
(a) 3 (b) –3
(c) $3 - \sqrt{5}$ (d) $3 + \sqrt{5}$

92. If 1 is a twice repeated root of the equation $ax^3 + bx^2 + bx + d = 0$, then
(a) $a = . = b$ (b) $-a = d = -b$
(c) $a = d = -b$ (d) $a = d = d$

93. If the polynomial equation $x^3 + px^2 + qx + r = 0$ has the roots α, β, γ, then the roots of the polynomial equation $x^3 - qx^2 + prx - r^2 = 0$ are
(a) $\frac{\alpha}{\beta}, \frac{\beta}{\gamma}, \frac{\gamma}{\alpha}$ (b) $\alpha + \beta, \beta + \gamma, \gamma + \alpha$
(c) $\alpha\beta, \beta\gamma, \gamma\alpha$ (d) $\alpha + \frac{\beta}{\gamma}, \beta + \frac{\gamma}{\alpha}, \gamma + \frac{\alpha}{\beta}$

94. In a survey 60% of those surveyed owned a car and 80% of those surveyed owned a TV. If 55% owned both a car and a TV, then percentage of those surveyed who owned a car or a TV but not both will be
(a) 30% (b) 5%
(c) 25% (d) 15%

95. If A = {1,2,3,4,5}, then the number of subsets of A which contain 2 but not 4 is
(a) 2 (b) 4
(c) 6 (d) 8

96. If T be the set of the triangles in a plane and R means "has twice the area of", then R is
(a) not reflexive
(b) reflexive but neither symmetric nor transitive
(c) reflexive and symmetric
(d) reflexive and transitive

97. The mapping $f: R \to R$ given by $f(x) = x^2$ is
(a) one-to-one and onto
(b) one-to-one but not onto
(c) neither one-to-one nor onto
(d) onto but not one-to-one

98. Let I be the set of integers. If R be a relation defined on I by setting $(a, b) \in R$ iff $a - b$ is an even integer, $a, b \in I$, then
(a) R is reflexive but not symmetric
(b) R is transitive but neither reflexive nor symmetric
(c) R is reflexive and symmetric but not transitive

If Q^+, Q^-, Q^* respectively denote the sets of positive, negative and non-zero rationals, then which one of the following pairs is an abelian group?
(a) $(Q^+, \oplus)$ (b) $(Q^-, \oplus)$
(c) $Q, \odot$ (d) $Q^*, \oplus$

ANSWERS

Q 1 (b) The matrix $\begin{pmatrix} a & 0 \\ 0 & a \end{pmatrix}$ commutes with every 2 × 2 matrix, because

$$\begin{pmatrix} a & 0 \\ 0 & a \end{pmatrix}\begin{pmatrix} x & y \\ z & t \end{pmatrix} = \begin{pmatrix} ax & ay \\ az & at \end{pmatrix}$$

and $$\begin{pmatrix} x & y \\ z & t \end{pmatrix}\begin{pmatrix} a & 0 \\ 0 & a \end{pmatrix} = \begin{pmatrix} ax & ay \\ az & at \end{pmatrix}$$

Q 2 (b) Since A, B are symmetric matrices,
$A^T = A, B^T = B$
Now $(A + B)^T = B^T + A^T = B + A = A + B$
$\Rightarrow$ A + B is symmetric
$(AB)^T = B^T A^T = BA$
$\Rightarrow$ AB is not symmetric
Next, $(BA)^T = A^T B^T = AB$
Finally, $A^T B = AB$ as $AB^T = AB$

Q 3 (c)

Q 4 (b) $|B| = |4A^T| = 4|A|^2 = 4 \times 5^2 = 100$

Q 5 (c)

Q.13. (a) Differentiating w.r. to x.

$$\cos y\frac{dy}{dx}=\sin(a+y)+x\cos(a+y)\frac{dy}{dx}$$

$$\frac{dy}{dx}=\frac{\sin(a+y)}{\cos y-x\cos(a+y)}$$

Or $$=\frac{\sin(a+y)}{\cos y-\frac{\sin y}{\sin(a+y)}\cos(a+y)}$$

$$\frac{\sin^2(a+y)}{\cos(a+y)\cos y-\sin y\cos(a+y)}$$

$$=\frac{\sin^2(a+y)}{\sin(a+y-y)}=\frac{\sin^2(a+y)}{\sin a}$$

Q.14. (c)

Q.15. (d) Here $f'(x)=3x^2-12x+12$
$=3(x^2-4x+4)=3(x-2)^2$

Since $f'(x)>0\ \forall\ x$, therefore, $f(x)$ is increasing in every interval.

Q.16. (b) $\log f(x)=\frac{1}{x}\log x$

Differentiating,

$$\frac{1}{f(x)}f'(x)=-\frac{1}{x^2}\log x+\frac{1}{x}.\frac{1}{x}=\frac{1-\log x}{x^2}$$

For f(x) to be maximum, $f'(x)=0$

$\Rightarrow\quad 1-\log x=0$ or $\log x=1=1=\log e$

$\therefore\quad x=e$

Q.17. (c)

Q.18. (d) Slope of tangent to given curve at (x_1,y_1).

$$\frac{dy}{dx}=4x_1-1$$

Since, the tangent is parallel to $y=3x+9$,

$4x_1-1=3$ or $x_1=1$ Also, then
$y_1=2(1)^2-1+1=2$

Hence, the point of contact is (1, 2).

Q.19. (c) Here $\frac{dy}{dx}=\cos x$

$$\therefore\quad\left(\frac{dy}{dx}\right)_{\left(\frac{\pi}{3},\frac{\sqrt{3}}{2}\right)}=\cos\frac{\pi}{3}=\frac{1}{2}$$

Hence, length of subtangent =

$$y\frac{dx}{y}=\frac{\sqrt{3}}{2}\times\left(\frac{2}{1}\right)=\sqrt{3}$$

Q.20. (c) $\frac{dy}{d\theta}=a(\sin\theta),\frac{d^2y}{d\theta^2}=a\cos\theta$

Q.21. (c)

Q.22. $u=\tan^{-1}\left(\frac{x+y}{\sqrt{x}+\sqrt{y}}\right)=\tan^{-1}\left(\sqrt{x}\frac{1+\frac{y}{x}}{1+\sqrt{\frac{y}{x}}}\right)$ or

$$\tan u=x^{\frac{1}{2}}\left[\frac{1+\frac{y}{x}}{1+\sqrt{\frac{y}{x}}}\right]$$

$\therefore\quad z=\tan u$ is a homogeneous function of order ½.

Thus, by Euler's theorem,

$$x\frac{\partial z}{\partial x}+y\frac{\partial z}{\partial y}=\frac{1}{2}z$$

or $$x\sec^2u\frac{\partial u}{\partial x}+y\sec^2u\frac{\partial u}{\partial y}=\frac{1}{2}\tan u$$

or $$x\frac{\partial u}{\partial y}+y\frac{\partial u}{\partial y}=\frac{1}{2}\frac{\tan u}{\sec^2u}=\frac{1}{4}\sin 2u$$

Q.23. (b) $\frac{dz}{dt}=\frac{dz}{dt}\frac{du}{dt}+\frac{\partial z}{\partial y}\frac{dy}{dt}$

$$=e^x\sin y.\frac{1}{t}+e^x\cos y.2t$$

$$=\frac{e^x}{t}\left[\sin y_1+2\cos y\right)$$

Q.24. (d) Shifting the origin to the point (5, 0) by using the transformation $X=x-5$, $Y=y$, the equation becomes $Y^2=X^2(X-7)$.

$\therefore$ Equation of tangent at new origin is $Y2=-7x2$ or $Y=\pm\sqrt{7iX}$

Since the two tangent are imaginary, therefore, new origin or (5, 0) is a conjugate point.

Q.25. (c) Q.26. (b)

Q.27. (d) Q.28. (b)

$$\lim_{n\to\infty}\sum_{k=1}^{n}\frac{k}{k^2+n^2}=\lim_{n\to\infty}\sum_{k=1}^{n}\frac{1}{n}\left[\frac{k/n}{(k/n)^2+1}\right]$$

$$=\int_0^1\frac{x}{x^2+1}\,dx=\left[\frac{1}{2}\log(x^2+1)\right]_0^1=\frac{1}{2}\log 2$$

Q.29. (a)

$$=\int_0^{\pi}\frac{\sin 4x}{\sin x}\,dx=\int_0^{\pi}\frac{2\sin 2x\cos 2x}{\sin x}$$

$$=2\int_0^{\pi}\frac{2\sin x\cos x\,(2\cos^2 x-1)}{\sin x}\,dx$$

$$=4\int_0^{\pi}(2\cos^3 x-\cos x)\,dx=4\left[0-(\sin x)_0^{\pi}\right]=0$$

Q.30. (c)

Q.31. (d) Required surface area =

$$2\pi\int_0^2 2x\sqrt{1+(2)^2}\,dx=4\sqrt{5}\pi\int_0^2 x\,dx=8\sqrt{5}\,\pi$$

Q.32. (d)

Q.33. (a) Required length of curve = $\int_0^{\pi/3}\frac{ds}{dx}\,dx$

$$=\int_0^{\pi/3}\sqrt{1+\left(\frac{1}{\sec x}\cdot\sec x\tan x\right)^2}$$

$$dx=\int_0^{\pi/3}\sqrt{1+\tan^2 x}\,dx=\int_0^{\pi/3}\sec x\,dx$$

$$=\left[\log(\sec x+\tan x)\right]_0^{\pi/3}$$

$$=\log(2+\sqrt{3})-\log 1=\log(2+\sqrt{3})$$

Q.34. (a)

Q.35. (b) Here

$$u_n=\frac{n}{(2n-1)\cdot 2n},\; u_{n+1}=\frac{n+1}{(2n+1)(2n+2)}$$

$$\therefore\quad \underset{n\to\infty}{Lt}\frac{u_n}{u_{n+1}}=\underset{n\to\infty}{Lt}\frac{n}{(2n-1)n}\times\frac{(2n+1)(2n+2)}{n+1}$$

$$=\underset{n\to\infty}{Lt}\left[\frac{1}{\left(1+\frac{1}{n}\right)}\times\frac{\left(2+\frac{1}{n}\right)\left(2+\frac{2}{n}\right)}{\left(2-\frac{1}{n}\right)}\right]=2>1.$$

Hence, Σu_n is covergent.

Q.36. (b) The necessary condition for a series of positive terms to converge is that $\lim\limits_{n\to\infty}u_n=0$

Q.37. (b) $(x+2y^3)\frac{dy}{dx}=y$

Or $\quad y\frac{dx}{dy}-x=2y^3$ or $\frac{dx}{dy}-\frac{1}{y}\cdot x=2y^2$

$$\therefore\quad I.F.=e^{\int-\frac{1}{y}}=e^{-\log y}=e^{\log y^{-1}}=y^{-1}=\frac{1}{y}$$

Hence, solution is

$$x\cdot\frac{1}{y}=\int 2y^2\cdot\frac{1}{y}\,dy+c=y^2+c\quad\text{or } x=cy+y^3$$

Q.38. (a) Q.39. (b)

Q.40. (b) $y=px+p^3\ \ldots(i)$

Here $f'(p)=3p^2$

Eliminating p between (i) and $x=3p^2=0$ we get

$$y=px-p\cdot\frac{x}{3}=\frac{2}{3}px$$

$$\text{or } y^2=\frac{4}{9}x^2\left(-\frac{x}{3}\right)\text{ or } 27y^2+4x^3=0$$

Q.41. (b) $y\,dx-2x\,dy=0$ or $\frac{dx}{x}-\frac{2}{y}\,dy=0$

Integrating, $\log x-2\log y=\log c$

or $\quad\frac{x}{y^2}=c$ or $y^2=ax$, which represents a family of hyperbolas.

Q.42. (d) Q.43. (a) Q.44. (b)

Q.45. (d) $P.I.=\frac{1}{D^2+4}x$

$$=\frac{1}{4\left(1+\frac{D^2}{4}\right)}x=\frac{1}{4}\left(1+\frac{D^2}{4}\right)^{-1}x$$

$$=\frac{1}{4}\left(1-\frac{D^2}{4}+\ldots\right)x=\frac{1}{4}(x-0)=\frac{1}{4}x$$

Q.46. (c) Q.47. (b)

Q.48. (b) Equation of the line BC is

$$\frac{y-2}{x-3}=\frac{2+1}{3-1}=\frac{3}{2}$$

or $3x-2y-5=0$. Hence, length of perpendicular from A(0,0) on the side BC

$$=\pm\frac{3\times 0-2\times 0-5}{\sqrt{3^2+(-2)^2}}=\pm\frac{-5}{\sqrt{9+4}}=\frac{5}{\sqrt{13}}$$

Q.49. (d) Since the equation $2xy+3x+4y+c=0$, represents a pair of straight lines

$$\therefore\quad \begin{vmatrix} 0 & 3/2 & 1 \\ 3/2 & 0 & 2 \\ 1 & 2 & c \end{vmatrix}=0$$

or $-3/2\left(\frac{3}{2}c-2\right)+1(3-0)=0$ or $c=8/3$

Q.50. (d)

Q.51. (a) Point of intersection of the diameters $x+y$, $x+2y=4$ is $(8,-2)$

Hence, equation of circle with center at $(8,-2)$ and radius 10 is

$(x-8)^2+(y+2)^2=100$

or $x^2+y^2-16x+4y=32$

Q.52. (a)

Q.53. (d) The line $y=2x+c$ will touch the parabola $y^2=16x=4(4x)$ $c=\frac{4}{2}=2$

Q.54. (c) Given that $\frac{2b^2}{a}=\frac{2}{3}b$ or $b=\frac{1}{3}a$

Hence, eccentricity,

$$e=\sqrt{\frac{a^2-b^2}{a^2}}=\sqrt{\frac{a^2-\frac{1}{9}a^2}{a^2}}=\sqrt{\frac{8}{9}}=\frac{2\sqrt{2}}{3}$$

Q.55. (c) Latus rectum $=\frac{2b^2}{a}=2\times\frac{16}{3}=\frac{32}{3}$

Q.56. (d)

Q.57. (d) Let a,b,c and a_1, b_1, c_1 be the direction ratios of the two given lines.

Then $3a+2b+c=0$, $a+b-2c=0$

and $2a_1-b_1-\lambda c_1=0$, $7a_1+10b_1-8c_1=0$

$\Rightarrow\quad \frac{a}{-5}=\frac{b}{7}=\frac{c}{1};\frac{a_1}{10\lambda+8}=\frac{b_1}{-7\lambda+16}=\frac{c_1}{27}$. Since two lines are perpendicular.

If $aa_1+bb_1+cc_1=0$

or $-5(10\lambda+8)+7(-7\lambda+16)+1\times 27=0$

$\Rightarrow\quad \lambda=1$

Q.58. (c)

Q.59. (c)

Q.60. (a) Since 1, 2, 3; 5, -4, 1 and 1, 1, -1 satisfy the given cone, therefore, I, III and IV are generators of the given cone.

Q.61. (a) Q.62. (b) Q.63. (a)

Q.64. (d) Here

$2\vec{A}-3\vec{B}-5\vec{C}=2(3\hat{i}-2\hat{j}-\hat{k})-3(2\hat{i}-4\hat{j}-\hat{k})-5(-\hat{i}+2\hat{j}+2\hat{k})$

$=5\hat{i}-2\hat{j}-\hat{k}$. Hence, its magnitude $=$

$\left|5\hat{i}-2\hat{j}-\hat{k}\right|=\sqrt{25+4+1}=\sqrt{30}$

Q.65. (c) If θ be the angle between $\vec{A}$ and $\vec{B}$, then

$$\cos\theta=\frac{\vec{A}.\vec{B}}{|\vec{A}||\vec{B}|}=\frac{2\times 5+3\times 2-4\times 4}{\sqrt{2^2+3^2+(-4)^2}\sqrt{5^2+2^2+4^2}}=0$$

$\therefore\quad \theta=90^\circ$

Q.66. (a) Q.67. (a)

Q.68. (a) Q.69. (c)

Q.70. (b) Q.71. (c)

Q.72. (d) Q.73. (c)

Q.74. (c) Q.75. (a)

Q.76. (c) Q. 76. (c)

Q.77. (a) Q.78. (b)

Q.79. (a) Q.80. (d)

Q.81. (c) Let $1-i=r(\cos\theta+i\sin\theta)$

or $r\cos\theta=1$, $r\sin\theta=-1$

$\therefore\quad \tan\theta=-1$ or $\theta=-\pi/4$, $r=\sqrt{2}$

Thus,

$1-i=\sqrt{2}\,[\cos(a+\pi/4)+i\sin(-\pi/4)]$

$=\sqrt{2}e^{-\frac{\pi}{4}i}$

Q.82. (d) Using the result,

$f(x)\,g(x)=$ L.C.M. $\times$ H.C.F.

$(x-1)^3\times g(x)$

$=(x-1)\times(x-1)^3(x-2)(x-3)$

or $g(x)=(x-1)(x-2)(x-3)$

Q.83. (d) Q.84. (c)

Q.85. (b) Given equation $x^3 + 3.1x2 + 3.\frac{4}{3}x - 7 =$ 0 is a cubic equation. Whose solution by Garden method is found. Transforming given equation by using

$z = 1\,x + 1$, it becomes

$(z-1)2 + 3(z-1)^2 + 4(z-1) - 7 = 0$

or $z^3 + z - 9 = 0$. Here $H = \frac{1}{3}$, $G = -9$

Since,

$G^2 + 4H^3 = (-9)^2 + 4\left(\frac{1}{3}\right)^3 = 81 + \frac{4}{9} > 0$

Therefore, given equation has one real and two complex roots.

Q.86. (b) Since $2x^2 + x - 3 = (x-1)(2x+3)$ is a factor of

$f(x) = 8x^4 \quad -ax^2 + bx - 3$

$f(1) = 0$ and $f\left(-\frac{3}{2}\right) = 0$

$\Rightarrow \quad 8 - 2 - a + b - 3 = 0$ and

$\left(-\frac{3}{2}\right)^4 - 2\left(-\frac{3}{2}\right)^3 - a\left(-\frac{3}{2}\right)^2 + b\left(-\frac{3}{2}\right) - 3 = 0$

or $\quad a - b - 3 = 0$

and $3a + 2b - \quad = 0$ ...... (ii)

Solving (i) and (ii), $a = 13$, $b = 10$

Q.87. (b)

Q.88. (c) By hypothesis,

$k\left(-\frac{b}{a}\right)^4 = p.\,l\left(-\frac{b}{a}\right)^4 = \frac{1}{p}$

Multiplying,

$Kl = l = p \times \frac{1}{p} = 1$

$$\frac{x^{\frac{4a}{b}} + (xy)^{\frac{2a}{b}} + y^{\frac{4a}{b}}}{x^{\frac{6a}{b}} - y^{\frac{6a}{b}}}$$

Q.89. (c) $= \dfrac{x^{\frac{4a}{b}} + x^{\frac{2a}{b}}y^{\frac{2a}{b}} + y^{\frac{4a}{b}}}{\left(x^{\frac{2a}{b}} - y^{\frac{2a}{b}}\right)\left(x^{\frac{4a}{b}} + x^{\frac{2a}{b}} + y^{\frac{4a}{b}}\right)}$

$= \dfrac{1}{x^{\frac{2a}{b}} - y^{\frac{2a}{b}}}$

Q.90. (b) Since $\alpha, \beta, \gamma, \delta$ are the roots of the equation

$36x^4 - 216x^3 + 401x^2 - 231x + 40 = 0$

$\therefore \quad \alpha + \beta + \gamma + \delta = \frac{216}{36} = 6$

$\alpha\beta + \beta\gamma + \gamma\delta + \delta\alpha + \alpha\gamma + \beta\delta = \frac{401}{36}$

Hence, $\alpha^2 + \beta^2 + \gamma^2 + \delta^2$

$= (\alpha + \beta + \gamma + \delta)^2 - 2\,((\alpha\beta + \beta\gamma + \gamma\delta + \delta\alpha + \alpha\gamma + \beta\delta)$

$= 6^2 - 2 \times \frac{401}{36} = \frac{247}{18}$

Q.91. (a) Since $3 + \sqrt{5}i$ is a root, $\therefore 3 - \sqrt{5}i$ is also root. Thus,

$\left[x - \left(3 + \sqrt{5}i\right)\right]\left[x - \left(3 - \sqrt{5}i\right)\right] = x^2 - 6x + 14$

is a factor of a given equation. Thus,

$x^3 - 9x^2 + 32x - 42 = (x^2 - 6x + 14)(x - 3)$

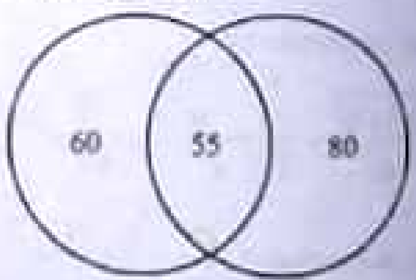
Hence, the real root is 3.

Q.92. (c) Q.93. (c)

Q.94. (a) Here $n(C) = 60$, $n(T) = 80$, $n(C \cap T) = 55$

$n(C \cup T) = n(C) + n(T) - n(C \cap T)$

$= 60 + 80 - 55 = 85$



Hence, percentage of people who owned a car or TV but not both

$= n(C \cup T) - n(C \cap T) = 85 - 55 = 30\%$

Q.95. (d) Q.96. (a)

Q.97. (d) Q.98. (d)

Q.99. (b) Q.100. (d)

MODEL TEST PAPER 3

1. A unit vector perpendicular to the two vector $\hat{i}+2\hat{j}-\hat{k}$ and $2\hat{i}+3\hat{j}+\hat{k}$ is

   (a) $5\hat{i}-3\hat{j}-\hat{k}$ (b) $\frac{1}{\sqrt{35}}(5\hat{i}-3\hat{j}-\hat{k})$

   (c) $\hat{i}-\hat{j}-2\hat{k}$ (d) $\frac{1}{\sqrt{6}}(\hat{i}-\hat{j}-2\hat{k})$

2. The points A, B, C whose position vectors are $\vec{a}=3\hat{i}-4\hat{j}-4\hat{k}$, $\vec{b}=2\hat{i}-\hat{j}-\hat{k}$, $\hat{k}=\vec{c}-\hat{i}-3\hat{j}-5\hat{k}$ $\vec{c}=\hat{i}-3\hat{j}-5\hat{k}$ are the vertices of

   (a) an isosceles triangle.
   (b) an acute-angled triangle.
   (c) an obtuse-angled triangle.
   (d) a right-angled triangle.

3. The distance between the line $\vec{r}=(\hat{i}+\hat{j})+\lambda(2\hat{i}+\hat{j}+4\hat{k})$ and the plane $\vec{r}.(-2\hat{i}+\hat{k})=5$ is

   (a) 7 (b) $\sqrt{5}$

   (c) $7/\sqrt{5}$ (d) $7\sqrt{5}$

4. The solution of the vector equation $\vec{r}\times\vec{a}=\vec{a}\times\vec{b}$ is (where k is any real number)

   (a) $\vec{r}=k\vec{a}-\vec{a}$ (b) $\vec{r}=k\vec{a}+\vec{a}$

   (c) $\vec{r}=k\vec{b}+\vec{a}$ (d) $\vec{r}=k\vec{a}+\vec{b}$

5. The vector equation of the line passing through the point with position vector $(2\hat{i}-3\hat{j}-5\hat{k})$ and perpendicular to the plane $\vec{r}.(6\hat{i}-3\hat{j}-5\hat{k})+2=0$ is (λ is a scalar in each case)

   (a) $\vec{r}=(-6\hat{i}+3\hat{j}+5\hat{k})+\lambda(2\hat{i}-3\hat{j}-5\hat{k})$
   (b) $\vec{r}=(2\hat{i}+3\hat{j}+5\hat{k})+\lambda(6\hat{i}-3\hat{j}-5\hat{k})$
   (c) $\vec{r}=(6\hat{i}+3\hat{j}+5\hat{k})+\lambda(2\hat{i}-3\hat{j}-5\hat{k})$
   (d) $\vec{r}=(-2\hat{i}+3\hat{j}+5\hat{k})+\lambda(6\hat{i}-3\hat{j}-5\hat{k})$

6. If three forces acting upon a rigid body be represented in magnitude, direction and line of action by the sides of a triangle taken in order, then

   (a) they will be in equilibrium
   (b) they will reduce to a single force
   (c) they will reduce to a couple
   (d) they will reduce to a single force and a couple

7. A necessary and sufficient condition that a body acted upon by coplanar forces be in equilibrium is that the sum of the moments of all the forces about

   (a) a point is zero
   (b) any two points is zero
   (c) any three non-collinear points is zero
   (d) an axis is zero

8. If a particle in equilibrium is subjected to four forces, viz.,

   $F_1=2\hat{i}-5\hat{j}+6\hat{k}$, $F_2=\hat{i}-3\hat{j}-7\hat{k}$

   $F_3=2\hat{i}-2\hat{j}+3\hat{k}$ and F_4,

   then F_4 is equal to

   (a) $-5\hat{i}+4\hat{j}+4\hat{k}$ (b) $5\hat{i}+4\hat{j}-4\hat{k}$

   (c) $3\hat{i}+2\hat{j}-\hat{k}$ (d) $3\hat{i}+\hat{j}+10\hat{k}$

9. A steam boat is moving with velocity v_1 when steam is shut off. If the retardation at any subsequent time is equal to the magnitude of the velocity at that time, then the velocity in time t after the steam is shut off is

   (a) v_1e^{t} (b) v_1e^{-t}

   (c) $2v_1e^{t}$ (d) $2v_1e^{t}$

10. The least force that will move a weight W along a rough horizontal plane, where λ is the angle of friction, is

    (a) W tan λ (b) W cos λ

    (c) W sin λ (d) W cot λ

11. The magnitude of a force which is acting on a body of mass 1kg for 5 seconds to produce velocity of 1 m/sec in it, is
(a) 1,000 dynes (b) 2,000 dynes
(c) 10,000 dynes (d) 20,000 dynes

12. If a body of mass m kg is carried by a lift moving with upward acceleration f, then the pressure on the plane of the lift is
(a) $mf - mg$ (b) $mg - mf$
(c) $mg + mf$ (d) $(mg)(mf)$

13. If the equation of motion of a particle executing a simple harmonic motion is $\frac{d^2x}{dt^2} + 16x = 0$, then its frequency will be
(a) $\frac{\pi}{2}$ (b) $\frac{\pi}{8}$
(c) $\frac{2}{\pi}$ (d) $\frac{8}{\pi}$

14. If the particle is projected from a horizontal plane with a velocity u at an angle α, then the time of flight of the particle will be
(a) $\frac{u \sin \alpha}{g}$ (b) $\frac{u \cos \alpha}{g}$
(c) $\frac{2u \cos \alpha}{g}$ (d) $\frac{2u \cos \alpha}{g}$

15. The escape velocity of a projectile from the earth is approximately
(a) 101 km/sec (b) 7 km/sec
(c) 11.2 km/sec (d) 112 km/sec

16. The hexadecimal number $(A\ 2F.\ D)_{16}$ when converted into the decimal system is equal to
(a) 2608.6125 (b) 2607.8125
(c) 2507.8125 (d) 2607.0125

17. In a flow chart, a rectangle is a
(a) start/stop box. (b) decision box.
(c) computation box. (d) input/output box.

18. An algorithm is
(a) a table of logarithms
(b) a collection of results.
(c) a chart of formulae.
(d) a step-by-step procedure for finding the solution of a problem.

Directions. Q.19 – 21: These questions consist of two statements, one labeled the Assertion (A) and the other labeled the Reason (R). You are to examine these two statements carefully and decide if the Assertion A and the Reason R are individually true an if so, whether the reason is a correct explanation of the Assertion. Select your answer to these items using the codes given below and mark your answer sheet accordingly.

Codes:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A an R are true an R is not a correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

19. Assertion
(A): If x and y are odd positive integers, then there exists no natural numbers such that $x^2 + y^2 = n^2$.
Reason (R):
If n is a natural number, then either $n^2 = 4m$ for some integer $m \geq 1$ or $n^2 = 4l + 1$ for some integer $l > 0$.
(a) (b)
(c) (d)

20. *Assertion (A):* Given any arbitrarily large numbers L and M, there exists a number N such that $\int_L^N x^{-1} dx > M$
Reason (R):
For any $A > 1$, $\int_L^{AL} x^{-1}\ dx = \log A$ and for a suitable A, log A can exceed M.
(a) (b)
(c) (d)

21. Assertion
(A): $y = e^{2x}$ and $y = xe^{2x}$ are two solution of differential equation $\frac{d^2y}{dx^2} - 4\frac{dy}{dx} + 4y = 0$. The general solution of the equation is $y = (a + bx)\, e^{2x}$, where a and b are arbitrary constants.

Second

If u and v are two solutions of the second order homogeneous linear differential equation, then $au + bv$ is the general solution of the equations where a and b are arbitrary constants.

(a)
(b)
(c)
(d)

If $x = \sqrt[3]{6}$, $y = \sqrt[4]{11}$, $z = \sqrt[6]{17}$, then

(a) $x < y < z$
(b) $x > y > z$
(c) $x < z < y$
(d) $x > z > y$

The fraction $\frac{1}{3}$ is

(a) equal to 0.3333333

(b) less than 0.3333333 by $\frac{1}{3.10^7}$

(c) greater than 0.3333333 by $\frac{1}{3.10^8}$

(d) greater than 0.3333333 by $\frac{1}{3.10^7}$

Let z_1 and z_2 be two non-zero complex numbers, then $|z_1 + z_2|^2 = |z_1|^2 + |z_2|^2$, if

(a) $z_1 - z_2$ is purely imaginary
(b) $z_1 - z_2$ is real
(c) $z_1 + z_2$ is purely imaginary
(d) $z_1 + z_2$ is real

If P is a point in the Argand diagram representing the complex number $4\left(\cos\frac{4\pi}{3} + i\sin\frac{4\pi}{3}\right)$ and OP is rotated through an angle $\frac{2\pi}{3}$ in the anti-clockwise direction, then P in the new position represents

(a) $4(\cos \pi/2 + i \sin \pi/2)$
(b) 4
(c) $4(\cos \pi/3 + i \sin \pi/3)$
(d) $3 + 2i$

220 cannot be the sum of the first n cubes for a suitable n, because 220 is

(a) not an odd number
(b) not a square
(c) not a cube
(d) divisible by 10

27. If $f(x)$ is a polynomial in x and a, b are unequal, then the remainder in the division of $f(x)$ by $(x-a)(x-b)$ is

(a) $\frac{(x-a)f(a)-(x-b)f(b)}{a-b}$

(b) $\frac{(x-a)f(a)-(x-a)f(b)}{a-b}$

(c) $\frac{(x-b)f(b)-(x-b)f(a)}{a-b}$

(d) None of the above

28. If $x + a$ is a factor of $x^4 - a^2x^2 + 3x - a$, then the value of a is

(a) 0
(b) 1
(c) 2
(d) 3

29. A quadratic equation with rational coefficients and with one root as $2 - \sqrt{3}$ is

(a) $x^2 - 4x + 1 = 0$
(b) $x^2 + 4x + 1 = 0$
(c) $x^2 - 4x - 1 = 0$
(d) $x^2 + 4x - 1 = 0$

30. If α, β are the roots of the equation $ax^2 + bx + c = 0$, then the value of $(\alpha - \beta)^2$ is

(a) $\frac{b^2 + 4ac}{a^2}$
(b) $\frac{b^2 - 4ac}{a^2}$
(c) $\frac{-b^2 - 4ac}{2}$
(d) $\frac{-b^2 - 4ac}{2}$

31. A repeated root of the equation $x^3 + 3x^2 + 3x + 1 = 0$ is

(a) 0
(b) -1
(c) 1
(d) 2

32. The values of $i^{\frac{1}{6}}$ are

(a) $\sin\frac{n\pi}{12} + i\cos\frac{n\pi}{12}$, $n = 1,5,9,13,17,21$

(b) $\cos\frac{n\pi}{12} + i\sin\frac{n\pi}{12}$, $n = 1,5,9,13,17,21$

(c) $\sin\frac{n\pi}{6} + i\cos\frac{n\pi}{6}$, $n = 1,5,9,13,17,21$

(d) $\cos\frac{n\pi}{6} + i\sin\frac{n\pi}{6}$, $n = 1,5,9,13,17,21$

42. If T is a linear transformation from R^2 to R^2 for which $T(1,0) = (a, b)$, $T(0, 1) = (c, d)$, then $T(x, y)$, $x, y \in R$ is

(a) $(ax = by, cx + dy)$

(b) $(ax + dy, bx + cy)$

(c) $(ax + cy, bx + dy)$

(d) None of the above

43. If $A = \begin{pmatrix} 1 & 1 & 2 \\ 2 & 1 & 0 \end{pmatrix}$ and $B = \begin{pmatrix} 1 & 2 \\ 2 & 0 \\ -1 & 1 \end{pmatrix}$, the $(AB)^T$

(AB^T denotes the transpose of AB) is

(a) $\begin{pmatrix} 1 & 4 \\ 4 & 4 \end{pmatrix}$ (b) $\begin{pmatrix} 1 & 4 \\ 1 & 4 \end{pmatrix}$

(c) $\begin{pmatrix} 1 & 4 \\ 4 & 1 \end{pmatrix}$ (d) $\begin{pmatrix} 1 & 1 \\ 4 & 4 \end{pmatrix}$

44. If $A = \begin{pmatrix} 0 & c \\ \beta & 0 \end{pmatrix}$, then $A^3 + A = 0$ whenever

(a) $\alpha\beta = 0$ (b) $\alpha\beta = 1$

(c) $\alpha\beta \neq 0$ (d) $\alpha\beta = -1$

45. If the value of the determinant

$$A = \begin{vmatrix} a_1 & a_2 & a_3 & a_4 \\ b_1 & b_2 & b_3 & b_4 \\ c_1 & c_2 & c_3 & c_4 \\ d_1 & d_2 & d_3 & d_4 \end{vmatrix}$$ is equal to K, then the

value of the determinant

$$B = \begin{vmatrix} a_2 & a_1 + 3a_2 & a_3 & a_4 \\ b_2 & b_1 + 3b_3 & b_3 & b_3 \\ c_2 & c_1 + 3c_3 & c_3 & c_4 \\ d_2 & d_1 + 3d_3 & d_3 & d_4 \end{vmatrix}$$ is equal to

(a) 3K (b) –K

(c) K (d) –3K

46. If $A = \begin{pmatrix} 1 & 0 & -1 \\ -2 & -1 & 0 \\ -1 & 0 & 0 \end{pmatrix}$, then inverse of matrix A will be

(a) $\begin{pmatrix} -1 & 0 & 1 \\ 2 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix}$ (b) $\begin{pmatrix} 1 & -2 & -1 \\ 0 & -1 & 0 \\ -1 & 0 & 0 \end{pmatrix}$

(c) $\begin{pmatrix} 0 & 0 & 1 \\ 2 & 2 & 0 \\ 1 & 0 & 1 \end{pmatrix}$ (d) $\begin{pmatrix} 0 & 0 & -1 \\ 0 & -1 & 2 \\ -1 & 0 & -1 \end{pmatrix}$

47. Consider the equation AX = B, where $A = \begin{pmatrix} -1 & 2 \\ 2 & -1 \end{pmatrix}$, $b = \begin{pmatrix} 3 \\ 1 \end{pmatrix}$ then

(a) the equation has no solution

(b) there exists a non zero unique solution

(c) there exists has infinitely many solutions

(d) the equation has infinitely many solutions

48. The limit of a convergent sequence of rational numbers

(a) need not exist at all

(b) exists and is always rational

(c) exists and is always irrational

(d) exists but it may be rational or irrational.

49. If $f(x) = \begin{cases} x^2 & 0 \le x \le \frac{1}{2} \\ x & \frac{1}{2} \le x \le 1 \end{cases}$ then $\underset{x \to \frac{1}{2}}{\text{Lt}} f(x)$

(a) has the value $\frac{1}{2}$ (b) has the value 1

(c) has the value 2 (d) does not exist

50. If $\underset{x \to \frac{1}{2}}{\text{Lt}} \dfrac{\sin 2x + a \sin x}{x^3} = b$, where b is finite then the values of a and b respectively will be

(a) –2, -1 (b) 2, 1

(c) –2, 1 (d) 2, -1

51. The function f defined by

$$f(x) = \begin{cases} \dfrac{x^2}{a} - a & (0 < x < a) \\ a - \dfrac{a^3}{x^2} & (x \ge a) \end{cases}$$

(a) is not continuous on $(0, \infty)$

(b) is not differentiable on $(0, \infty)$

(c) is differentiable on $(0, \infty)$

(d) is differentiable on $(0, \infty)$ except at $x = a$.

52. If $y = \sin x$, then for any positive integer $\dfrac{d^n y}{dx^n}$ is given by

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66. Let $f(x, y) = x^2 \sin \frac{1}{x} + y^2 \sin \frac{1}{y}$ when $(x, y) \neq (0, 0)$

$f(x, 0) = x^2 \sin \frac{1}{x}$ when $x \neq 0$

$f(0, y) = y^2 \sin \frac{1}{y}$ when $y \neq 0$

$f(0, 0) = 0$, then at (0, 0)

(a) fx is continuous but no fy.
(b) fy is continuous but not fx.
(c) fx and fy are both continuous.
(d) neither fx nor fy is continuous.

67. The double point on the curve $(x-2)^2 = y(y-1)^2$ is

(a) (1, 2) (b) (3, 4)
(c) (4, 3) (d) (2, 1)

68. The value of $\int_0^{\pi/2} \frac{\sqrt{\cos x}}{\sqrt{\cos x} + \sqrt{\sin x}} dx$ is

(a) $\pi/2$ (b) $\pi/4$
(c) $\pi/8$ (d) $\pi/6$

69. The value of $\int \frac{dx}{(e^x - 1)}$ (c is constant of integration) is

(a) $\log(e^{-x} - 1) + c$ (b) $\log|(e^{-x} - 1)| + c$
(c) $\log(e^{-x} - e^x) + c$ (d) $\log(e^x - 1)$

70. The length of the arc of the parabola $y^2 = 4kx$ measured from the vertex to one extremity of the latus rectum is

(a) $k\{\sqrt{2} + \log(1 + \sqrt{2})\}$

(b) $\frac{1}{k}\{\sqrt{2} + \log(1 + \sqrt{2})\}$

(c) $k\{\sqrt{2} - \log(1 + \sqrt{2})\}$

(d) $\frac{1}{k}\{\sqrt{2} - \log(1 + \sqrt{2})\}$

71. The area of the cardioid $r = a(1 + \cos\theta)$ is equal to

(a) $4\pi a^2$ (b) $8\pi a$

(c) $\frac{3\pi a^2}{2}$ (d) $2\pi a^2$

72. The volume of the solid generate by revolving the curve $x = a\cos t$, $y = b\sin t$ about the x-axis is

(a) $4\pi ab^2$ (b) $\frac{4\pi ab^2}{3}$

(c) $4\pi ab^2$ (d) $\frac{4\pi a^2 b}{3}$

73. The arc of the sine curve $y = \sin x$ from $x = 0$ to $x = \pi$ is revolved about the x-axis. The area of the surface of the solid generated is

(a) $2\pi\left[\sqrt{2} + \log\left(\sqrt{2} + 1\right)\right]$

(b) $\frac{2\pi^2}{3}\left[\sqrt{2} + \log\left(\sqrt{2} + 1\right)\right]$

(c) $\frac{\pi}{3}\left[\sqrt{2} + \log\left(\sqrt{2} + 1\right)\right]$

(d) $\frac{\pi^2}{3}\left[\sqrt{2} + \log\left(\sqrt{2} + 1\right)\right]$

74. The series whose nth term is $t_n = \sqrt{n^2 + 1} - n$

(a) converges to the sum 0
(b) converges to the sum $\frac{1}{2}$
(c) converges to the sum 1
(d) diverges

75. The series $1 + \frac{2}{6} + \frac{2.5}{6.12} + \frac{2.5.8}{6.12.18} + \ldots$ is

(a) divergent
(b) convergent
(c) oscillates finitely
(d) oscillates infinitely

76. Match List I with List II and select answer using the codes given below the lists

List I

I. $\sum_1^\infty (-1)^{n-1} \frac{n^2}{n+1}$ II. $\sum_1^\infty \frac{n^n}{n}$

III. $\sum_1^\infty \frac{(-1)^n \log n}{n^2}$ IV. $\sum_2^\infty \frac{(-1)^n}{\log n}$

List II

(A) Diverges (B) Converges
(C) Converges conditionally
(D) Converges absolutely

Codes:
(a) I-B, II-D, III-A, IV-C
(b) I-D, II-C, III-B, IV-A
(c) I-D, II-A, III-B,
(d) I-C, II-A, III-D, IV-B

77. The number of arbitrary constants in the complete primitive of the differential equation $\phi\left(x, y, \frac{dy}{dx}, \frac{d^2y}{dx^2}\right)=0$ is
(a) 1 (b) 2
(c) 3 (d) 4

78. The differential equation of the system of circles touching the x-axis at origin is
(a) $(x^2-y^2)\frac{dy}{dx}-2xy=0$
(b) $(x^2-y^2)\frac{dy}{dx}+2xy=0$
(c) $(x^2+y^2)\frac{dy}{dx}-2xy=0$
(d) $(x^2+y^2)\frac{dy}{dx}+2xy=0$

79. The singular solution of the equation $p^3-4xyp+8y^2=0\left(p=\frac{dy}{dx}\right)$ is
(a) $27y=4x$ (b) $27y=4x^3$
(c) $27y=4x^2$ (d) $27y^2=4x$

80. The solutions of the differential equation $2y(y'+2)-xy'^2=0$ are the functions
I. $y=0$ II. $y=-4x$
Select the correct answer using the codes given below:
Codes:
(a) None of I and II is a singular solution
(b) Both I an II are singular solutions
(c) I is a singular solution but II is not
(d) II is a singular solution but I is not

81. The orthogonal trajectories of the system of curves $r^n \sin n\theta=K^n$ are
(a) $r^n \cos n\theta=a$ (b) $r\cos\theta=a$
(c) $r^2 \cos n\theta=a$ (d) $r^n \tan n\theta=a$

82. The equation whose solution family is self orthogonal is
(a) $p-\frac{1}{p}=x^2, p=\frac{dy}{dx}$
(b) $(px+y)(x+yp)-4p=0, p=\frac{dy}{dx}$
(c) $(px-y)(x+yp)-4p=0, p=\frac{dy}{dx}$
(d) $(px+y)(x-yp)-4p=0, p=\frac{dy}{dx}$

83. Which of the following statements associated with a first order non-linear differential equation $f\left(x, y, \frac{dy}{dx}\right)=0$ are correct?
I. Its general solution must contain only one arbitrary constant
II. Its singular solution can be obtained by substituting particular value of the arbitrary constant in its general solution
III. Its singular solution is an envelope of the general solution which also satisfies the equation
Select the correct answer using the codes given below:
Codes:
(a) I, II and III (b) I and II
(c) I and III (d) II and III

84. A particular integral of $\frac{d^2y}{dx^2}-(a+b)\frac{dy}{dx}+aby=Q(x)$ is
(a) $e^{ax}\int\left\{e^{(b-a)x}\left(\int Qe^{-bx}dx\right)\right\}dx$
(b) $e^{ax}\int\left\{e^{(b-a)x}\left(\int Qe^{-ax}dx\right)\right\}dx$
(c) $e^{ax}\int\left\{e^{(b-a)x}\left(\int Qe^{bx}dx\right)\right\}dx$
(d) $e^{ax}\int\left\{e^{(b-a)x}\left(\int Qe^{ax}dx\right)\right\}dx$

85. The solution of the differential equation $x\left[y\frac{d^2y}{dx^2}+\left(\frac{dy}{dx}\right)^2\right]=y\frac{dy}{dx}$ is
(a) $ax+by=c$ (b) $ax^2+by=0$
(c) $ax^2+by^2=1$ (d) $ax+by^2=0$

86. A variable line passes through the fixed point (a, b). The locus of the middle point of the segment intercepted between the axes is given by

(a) $\frac{a}{x} + \frac{b}{y} = 1$ (b) $\frac{a}{x} + \frac{b}{y} = 2$

(c) $\frac{x}{a} + \frac{y}{b} = 1$ (d) $\frac{x}{a} + \frac{y}{b} = 2$

87. The joint equation of the pair of lines through the origin which are perpendicular to the lines represented by $ax^2 + 2hxy + by^2 = 0$ is

(a) $bx^2 + 2hxy + ay^2 = 0$

(b) $ax^2 - 2hxy + by^2 = 0$

(c) $bx^2 - 2hxy + ay^2 = 0$

(d) $bx^2 - 2hxy - ay^2 = 0$

88. If $u = ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ represents two straight lines, then the third pair of straight lines through the four points in which the lines $u = 0$ meet the axis is

(a) $u + 4(fg - ch)xy = 0$

(b) $cu + 4(fg - ch)xy = 0$

(c) $cy + (fg - ch)xy = 0$

(d) $u + (fg - ch)xy = 0$

89. The polar equation $r = \frac{2}{4\cos\theta + 5\sin\theta}$ represents

(a) a straight line (b) a parabola

(c) a hyperbola (d) an ellipse

90. If the normal drawn at a point $(at_1^2, 2at_1)$ of the parabola $y^2 = 4ax$ meets it again in the point $(at_2^2, 2at_2)$ then

(a) $t_2 = -t_1 - \frac{2}{t_1}$ (b) $t_2 = \frac{t_1}{2}$

(c) $t_2 = t_1 + \frac{1}{t_1}$ (d) $t_1 t_2 = -1$

91. The two circles $x^2 + y^2 - 4x + 2y + 4 = 0$ and $x^2 + y^2 - 10x - 6y + 18 = 0$

(a) touch each other internally

(b) touch each other externally

(c) intersect each other

(d) neither intersect nor touch each other

92. The circles $r = 2a\cos(\theta - \alpha)$ and $r = 2b\cos(\theta - \beta)$ intersect at an angle

(a) $\frac{\alpha - \beta}{2}$ (b) $\alpha - \beta$

(c) $\frac{\alpha + \beta}{2}$ (d) $\alpha + \beta$

93. The standard equation of the ellipse, the length of whose major axis is 8 and the distance between whose directrices is 16 is given by

(a) $\frac{x^2}{5} + y^2 = 1$ (b) $\frac{x^2}{16} + \frac{y^2}{12} = 1$

(c) $\frac{x^2}{12} + \frac{y^2}{16} = 1$ (d) $\frac{x^2}{4} + \frac{y^2}{16} = 1$

94. If a circle cuts the rectangular hyperbola $xy = c^2$ in points (x_r, y_r) $(r = 1, 2, 3, 4)$, then

(a) $x_1 x_2 x_3 x_4 = -c^2$ (b) $x_1 x_2 x_3 x_4 = c^4$

(c) $x_1 x_2 x_3 x_4 = -c^4$ (d) $x_1 x_2 x_3 x_4 = c^8$

95. The polar coordinates of the center of the circle $r = 6\cos(\theta - \alpha)$ are

(a) $(0, 0)$ (b) $(0, \alpha)$

(c) $(6, \alpha)$ (d) $(3, \alpha)$

96. If a straight line makes angles $60°$, $45°$ and $\theta°$ respectively which the axes of co-ordinates OX, OY and OZ, then θ is equal to

(a) $15°$ (b) $30°$

(c) $45°$ (d) $60°$

97. If the plane $\frac{x}{2} + \frac{y}{3} + \frac{z}{6} = 1$ cuts the axes of co-ordinates at points A, B, C, then the area of the triangle ABC is

(a) 18 sq. units (b) 36 sq. units

(c) $3\sqrt{14}$ sq. units (d) $2\sqrt{14}$ sq. units

98. The diameter of the circle $x^2 + y^2 + z^2 = 9$, $x + y + z = 3$, is

(a) $2\sqrt{3}$ (b) $\sqrt{3}$

(c) $\sqrt{6}$ (d) $2\sqrt{6}$

The equation to the cone which passes through the three coordinates axes as well as the lines $\frac{x}{1}=\frac{y}{-2}=\frac{z}{3}$ and $\frac{x}{3}=\frac{y}{-1}=\frac{z}{1}$ is

(a) $yz - 2zx + 3xy = 0$
(b) $3yz - zx + xy = 0$
(c) $yz + zx + xy = 0$
(d) $3yz + 16zx + 12xy = 0$

50. The equation of the cylinder, generated by lines parallel to the fixed line $\frac{x}{l}=\frac{y}{m}=\frac{z}{n}$ and having the ellipse $z = 0$, $ax^2 + by^2 = 1$ as the guiding curve, is

(a) $a(nx + lz)^2 + b(ny + mz)^2 = n^2$
(b) $a(nx - lz)^2 + b(ny - mz)^2 = n^2$
(c) $a(nx + lz)^2 - b(ny - mz)^2 = n^2$
(d) $a(nx - lz)^2 - b(ny - mz)^2 = n^2$

ANSWERS

Q.1. (b) A vector perpendicular to $\hat{i}+2\hat{j}-\hat{k}$ and $2\hat{i}+3\hat{j}+\hat{k}$ is

$$(\hat{i}+2\hat{j}-\hat{k})\times(2\hat{i}+3\hat{j}+\hat{k})$$

$$=5\hat{i}-3\hat{j}-\hat{k}$$

Hence, a unit vector perpendicular to given vector

$$=(2+6\lambda)\hat{i}+(-3-3\lambda)\hat{j}+(-5-5\lambda)\hat{k}$$

$$=(2\hat{i}-3\hat{j}-5\hat{k})+\lambda(6\hat{i}-3\hat{j}-5\hat{k})$$

Q.6. (a) If follows from the *Triangle of forces.*

Q.7. (c)

Q.8. (a) Since four forces keep the particle in equilibrium.

$$\vec{F}_1+\vec{F}_2+\vec{F}_3+\vec{F}_4=\vec{0}$$

$$\vec{F}_4=-(\vec{F}_1+\vec{F}_2+\vec{F}_3)=-(5\hat{i}-4\hat{j}-4\hat{k})$$

$$= -5\hat{i}+4\hat{j}+4\hat{k}$$

Q.9. (b)
Q.10. (c)
Q.11. (d)
Q.12. (b)
Q.13. (c)
Q.14. (c)

Q.15. (c) Escape velocity $=\sqrt{2gr}$

$$=\sqrt{2\times\frac{9.87}{1000}\times 4000}=11.2 \text{ km/sec}$$

Q.16. (b)
Q.17. (c)
Q.18. (d)
Q.19. (b)
Q.20. (a)
Q.21. (d)

Sum of roots $= (2-\sqrt{3})(2+\sqrt{3}) = 4$

Product of roots $= (2-\sqrt{3})(2+\sqrt{3})$

$= 4-3 = 1$

Hence, the required root is $x^2 - 4x + 1 = 0$

Q.30. (b): Since α, β are roots of $ax^2 + bx + c = 0$

$\therefore \quad \alpha+\beta = -\frac{b}{a}, \alpha\beta = \frac{c}{a}$

Hence,

$(\alpha-\beta)^2 = (\alpha+\beta)^2 - 4\alpha\beta = \left(-\frac{b}{a}\right)^2 - 4\left(\frac{c}{a}\right)$

$= \frac{b^2}{a^2} - \frac{4c}{a} = \frac{b^2-4ca}{a^2}$

Q.31. (b): Given equation is $(x+1)^3 = 0$

$\therefore$ The root -1 is repeated thrice.

$i = \cos \frac{\pi}{2} + i \sin \frac{\pi}{2} = \cos\left(2k\pi + \frac{\pi}{2}\right) + i \sin\left(2k\pi + \frac{\pi}{2}\right) \therefore$

$i^{\frac{1}{6}} = \left[\cos\left(2k\pi + \frac{\pi}{2}\right) + i \sin\left(2k\pi + \frac{\pi}{2}\right)\right]^{\frac{1}{6}}$

$\frac{2k\pi + \frac{\pi}{2}}{} + i \sin \frac{2k\pi + \frac{\pi}{2}}{}, k = 0, 1, 2, 3, 4, 5$

$AB = \begin{pmatrix} 1 & 1 & 2 \\ 2 & 1 & 0 \end{pmatrix}\begin{pmatrix} 1 & 2 \\ 2 & 0 \\ -1 & 1 \end{pmatrix} = \begin{pmatrix} 1+2-2 & 2+2 \\ 2+2 & 4 \end{pmatrix} = \begin{pmatrix} 1 & 4 \\ 4 & 4 \end{pmatrix}$

$\therefore (AB)^T = \begin{pmatrix} 1 & 4 \\ 4 & 4 \end{pmatrix}$

Q.44. (d): A^2

$= \begin{pmatrix} \alpha\beta & 0 \\ 0 & \alpha\beta \end{pmatrix}\begin{pmatrix} 0 & \alpha \\ \beta & 0 \end{pmatrix} = \begin{pmatrix} 0 & \alpha^2\beta \\ \alpha\beta^2 & 0 \end{pmatrix}$

$A^3 \begin{pmatrix} \alpha\beta & 0 \\ 0 & \alpha\beta \end{pmatrix}\begin{pmatrix} 0 & \alpha \\ \beta & 0 \end{pmatrix} = \begin{pmatrix} 0 & \alpha^2\beta \\ \alpha\beta^2 & 0 \end{pmatrix}$

Thus, $A^3 + A = 0$ given

$\begin{pmatrix} 0 & \alpha^2\beta \\ \alpha\beta^2 & 0 \end{pmatrix}\begin{pmatrix} 0 & \alpha \\ \beta & 0 \end{pmatrix}$

$= \begin{pmatrix} 0 & \alpha^2\beta + \alpha \\ \alpha\beta^2 + \beta & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$

or $\quad \alpha^2\beta + \alpha = 0, \ \alpha\beta^2 + \beta = 0$

or $\quad \alpha(\alpha\beta + 1) = 0,$

$\beta(\alpha\beta + 1) = 0,$

$\therefore \quad \alpha = 0, \alpha\beta + 1 = 0,$

Q.46. (d): $|A| = \begin{vmatrix} 1 & 0 & -1 \\ 2 & -1 & 0 \\ -1 & 0 & 0 \end{vmatrix} = -1(0-1) = 1$

Also, $\text{Adj.}A = \begin{bmatrix} 0 & 0 & -1 \\ 0 & -1 & 2 \\ -1 & 0 & -1 \end{bmatrix}$

Hence $A^{-1} = \frac{1}{|A|}\text{Adj.}A = 1\begin{bmatrix} 0 & 0 & -1 \\ 0 & -1 & 2 \\ -1 & 0 & -1 \end{bmatrix}$

Q.47. (c) Q.48. (d)

Q.49. (d): L.H.L. =

$$\underset{x \to \frac{1}{2}-0}{\text{Lt}} f(x) = \underset{h \to 0}{\text{Lt}} f\left(\frac{1}{2}-h\right) = \underset{h \to 0}{\text{Lt}}\left(\frac{1}{2}-h\right)^2$$

$$= \left(\frac{1}{2}\right)^2 = \frac{1}{4}$$

$$\text{R.H.L.} = \underset{x \to \frac{1}{2}+0}{\text{Lt}} f(x) = \underset{h \to 0}{\text{Lt}} f\left(\frac{1}{2}-h\right) = \frac{1}{2}$$

Since, L.H.L. $\neq$ R.H.L.

$\therefore$ $\underset{x \to \frac{1}{2}}{\text{Lt}} f(x)$ does not exist.

Q.50. (a): $\underset{x \to 0}{\text{Lt}} \frac{\sin 2x + a \sin x}{x^3} = b \left(\frac{0}{0} \text{ form}\right)$

By L' Hospital's rule,

$$\underset{x \to 0}{\text{Lt}} \frac{2\cos 2x + a\cos x}{3x^2} = b$$

For this to be of $\frac{0}{0}$ form, $a = -2$

Then,

$$\underset{x \to 0}{\text{Lt}} \frac{2\cos 2x - 2\cos x}{3x^2} = b \qquad \left(\frac{0}{0} \text{ form}\right)$$

$$\therefore \quad \underset{x \to 0}{\text{Lt}}\ 2 \cdot \frac{-2\sin 2x + \sin x}{6x} = b$$

$$\text{and} \quad \underset{x \to 0}{\text{Lt}} \frac{-4\cos 2x + \cos x}{3} = b$$

$$\frac{-4+1}{3} = b \quad \text{or} \quad b = -1$$

Q.51. (d) Q.52. (c)

Q.53. (c): Put $x = \tan\theta$

$\therefore$ $y = \tan^{-1}\frac{2\tan\theta}{1-\tan^2\theta}$, $z = \tan^{-1}\frac{1-\tan\theta}{1+\tan\theta}$

or $y = \tan^{-1}(\tan 2\theta)$,

$z = \tan^{-1}\left[\tan\left(\frac{\pi}{4}+\theta\right)\right]$

or $y = 2\theta$, $z = \frac{\pi}{4}+\theta$

$\therefore$ $\frac{dy}{d\theta} = 2, \frac{dz}{d\theta} = 1$

Hence, $\frac{dy}{dz} = \frac{dy}{d\theta} \Big/ \frac{dz}{d\theta} = \frac{2}{1} = 2.$

Q.54. (c): $y = \tan^{-1}(\sec x + \tan x)$

$$= \tan^{-1}\left(\frac{1}{\cos x} + \frac{\sin x}{\cos x}\right) = \tan^{-1}\frac{1+\sin x}{\cos x}$$

$$= \tan^{-1}\frac{\left(\cos\frac{x}{2}+\sin\frac{x}{2}\right)^2}{\left(\cos^2\frac{x}{2}-\sin^2\frac{x}{2}\right)} = \tan^{-1}\frac{\cos\frac{x}{2}+\sin\frac{x}{2}}{\cos\frac{x}{2}-\sin\frac{x}{2}}$$

$$= \tan^{-1}\left(\frac{1+\tan\frac{x}{2}}{1-\tan\frac{x}{2}}\right) = \tan^{-1}\left[\tan\left(\frac{\pi}{4}+\frac{x}{2}\right)\right]$$

$$= \frac{\pi}{4}+\frac{x}{2}$$

Hence, $\frac{dy}{dx} = \frac{1}{2}$

Q.55. (a) Q.56. (d)

Q.57. (d): $\log(1+\sin x) = \sin x - \frac{(\sin x)^2}{2} + \frac{(\sin x)^3}{3}$

$$= \left(x - \frac{x^3}{|\underline{3}} + \frac{x^5}{|\underline{5}}\right) - \frac{1}{2}\left(x - \frac{x^3}{|\underline{3}} + \frac{x^5}{|\underline{5}}\right)$$

$$+ \frac{1}{3}\left(x - \frac{x^3}{|\underline{3}} + \frac{x^5}{|\underline{5}}\right)^3$$

$$= x - \frac{x^3}{3} - \frac{1}{2}x^2 + \frac{1}{3}x^3$$

$$= x - \frac{1}{2}x^2 + \frac{x^3}{6}$$

Q.58 (b)

Q.59 (b) Let p be the area of a sector of angle θ

$$\therefore \quad p = \frac{\pi r^2}{180}$$

By hypothesis,

$$\therefore \quad l = 2r + p = 2r + \frac{\pi r\theta}{180}$$

or $\quad \theta = (l - 2r)\dfrac{180}{r\pi}$

Also,

Area of sector, $A = \dfrac{\pi r^2\theta}{360} = \dfrac{\pi r^2}{360}(l - 2r)\dfrac{180}{r\pi}$

$$A = \frac{1}{2}(lr - 2r^2)$$

$$\frac{dA}{dr} = \frac{1}{2}(l - 4r), \frac{d^2A}{dr^2} = -1$$

But $\dfrac{dA}{dr} = 0$ gives, $r = \dfrac{l}{4}$

Hence, A is maximum when $r = \dfrac{l}{4}$

Maximum area =

$$\frac{1}{2}\left(l\frac{l}{4} - 2\frac{l^2}{16}\right) = \frac{1}{2}\left(\frac{l^2}{4} - \frac{l^2}{8}\right) = \frac{l^2}{16}$$

Q.60 (a) Q. 61 (d)

Q.62 (b):

$$\frac{dy}{dx} = \frac{dy/d\theta}{dx/d\theta} = \frac{a\sin\theta}{a(1-\cos\theta)}$$

$$= \frac{2\sin\frac{\theta}{2}\cos\frac{\theta}{2}}{2\cos^2\frac{\theta}{2}} = \tan\frac{\theta}{2}$$

Hence, ratio of sub-tangent to sub-normal

$$= \frac{y\dfrac{dx}{dy}}{y\dfrac{dy}{dx}} = \frac{1}{(dy/dx)^2} = \frac{1}{\tan^2\frac{\theta}{2}} = \cot^2\frac{\theta}{2}$$

Q.63 (a) Q.64 (b)

Q.65 (c) Q.66 (d)

Q.67 (d): Let $f = (x-2)^2 - y(y-1)^2$

$$\therefore f_x = 2(x-2), f_y = -(y-1)^2 - 2y(y-1)$$

$$\therefore \quad f_x = 0, f_y = 0 \text{ give}$$

$$x = 2, y = 1, y = \frac{1}{3}$$

So the double points can be $(2, 1)$, $\left(2, \frac{1}{3}\right)$

But $\left(2, \frac{1}{3}\right)$ does not satisfy $f = 0$

Hence, the double point is $(2, 1)$

Q.68. (b): Let $I = \displaystyle\int_0^{\pi/2} \frac{\sqrt{\cos x}}{\sqrt{\cos x} + \sqrt{\sin x}}\,dx$

Also, I

$$= \int_0^{\pi/2} \frac{\sqrt{\cos\left(\frac{\pi}{2} - x\right)}}{\sqrt{\cos\left(\frac{\pi}{2} - x\right)} + \sqrt{\sin\left(\frac{\pi}{2} - x\right)}} \quad \text{(ii)}$$

$$dx = \int_0^{\pi/2} \frac{\sqrt{\sin x}}{\sqrt{\sin x} + \sqrt{\cos x}}\,d$$

Adding (i) and (ii)

$$2I = \int \frac{\sqrt{\cos x} + \sqrt{\sin x}}{\sqrt{\cos x} + \sqrt{\sin x}}\,dx = \int_0^{\pi/2} dx = \frac{\pi}{2}$$

$$\therefore \quad I = \frac{\pi}{4}$$

Q.69 (b)

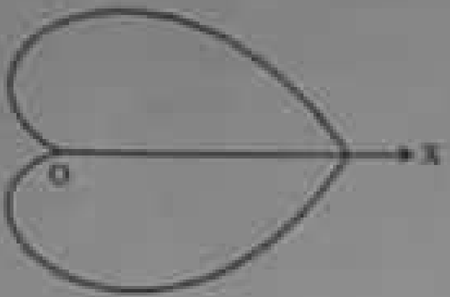
Q.70 (a)

Q.71 (c): Required area of cardioide

$$= 2\int_0^{\pi} \frac{1}{2} r^2 \, d\theta$$

$$= \int_0^{\pi} a^2 (1 + \cos\theta)^2 \, d\theta$$

$$= a^2 \int_0^{\pi} 4\cos^4 \frac{\theta}{2} \, d\theta$$



$$= 8a^2 \int_0^{\pi/2} \cos^4 t \, dt = 8a^2 \times \frac{3}{4} \times \frac{1}{2} \times \frac{\pi}{2} = \frac{3}{2} \pi a^2$$

Q.72 (b)

Q.73 (a)

Q.74 (d): Here $t_n = \sqrt{n^2 + 1} - n$

$$= \frac{\left(\sqrt{n^2+1} - n\right)\left(\sqrt{n^2+1} + n\right)}{\sqrt{n^2+1} + n} = \frac{1}{\sqrt{n^2+1} + n}$$

$$= \frac{1}{n\left[1 + \sqrt{1 + \frac{1}{n^2}}\right]}$$

Consider the series Σu_n, where $u_n = \frac{1}{n}$

Also, $\underset{n \to \infty}{Lt} \frac{t_n}{u_n} = \underset{n \to \infty}{Lt} \frac{1}{1 + \sqrt{1 + \frac{1}{n^2}}}$

$$= \frac{1}{1 + \sqrt{1 + 0}} = \frac{1}{2} \neq 0, \infty$$

Both the series Σt_n and Σu_n converge or diverge together. But as Σu_n is divergent, therefore, Σt_n is also divergent.

Q.75 (a)

Q.76 (c)

Q.77 (b)

Q.78 (a): Let the equation of circle through origin be

$x^2 + y^2 + 2gx + 2fy = 0$

Since it touches y-axis

$\therefore \quad g^2 - c = 0$

So the circle is

$x^2 + y^2 + 2fy = 0$

Differentiating w.r. to x.

$2x + 2y\frac{dy}{dx} + 2f\frac{dy}{dx} = 0$

or $x + (y + f)\frac{dy}{dx}$

Eliminating f.

$x + \left(y - \frac{x^2 + y^2}{2y}\right)\frac{dx}{dy} = 0$

or $2xy + (y2 - x2)\frac{dx}{dy} = 0$

Q.79 (d)

Q.80 (c)

Q.81 (a)

Q.82 (d)

Q.83(a)

Q.84 (b)

Q.85 (c)

Q.86 (d): Suppose the equation of line is

$\frac{x}{A} + \frac{y}{B} = 1$

$\therefore$ It meets co-ordination axes at P(A, O), Q(O, B)

Since, (a, b) is the mid-point of PQ

$\therefore \quad a = \frac{A}{2}, b = \frac{B}{2}$

or, $\quad A = 2a, B = 2b$

Substituting these values in (1),

$\frac{x}{2a} + \frac{y}{2b} = 1$ or $\frac{x}{a} + \frac{y}{b} = 2$

Q.87 (c)

Q.88.(a) Since given equations represent a pair of lines.

$abc + 2fgh - af2 - bg2 - ch2 = 0$

Joint equation of axes is $xy = 0$

$\therefore$ Equation of the through the intersection of given pair an co-ordinate axis is

$ax^2 + 2hxy + by2 + 2gx + 2fy + c + 2\lambda xy = 0$

or $ax^2 + 2hy(h + \lambda) + by^2 + 2gx + 2fy + c = 0$

This will represent a pair of lines if

$Abc + 2fg(h + \lambda) - af^2 - bg^2 - c(h + \lambda)^2 = 0$

or $\lambda = \frac{2}{c}(fg - ch)$ [(Using (i)]

Hence, the required equation is

or $u + \frac{4}{c}(fg - ch)\, xy = 0$

Q.89.(a) Q.90.(a)

Q.91.(b) Q.92.(b)

Q.93.(b) Q.94.(b)

Q.95.(c)

Q.96.(d): Direction cosines of the line are

$l = \cos 60^\circ = \frac{1}{2}, m = \cos 45^\circ = \frac{1}{\sqrt{2}}, n = \cos\theta$

Using the formula, $l^2 + m^2 + n^2 + 1$, we get

$\frac{1}{4} + \frac{1}{2} + \cos 2\theta = 1$ or $\cos 2\theta = \frac{1}{4}$

$\therefore \cos\theta = \frac{1}{2}$ $\therefore \theta = 60^\circ$

Q.97.(c) Q.98.(d)

Q.99.(d): Let the equation of the cone which passes through three co-ordinate axes be

$fyz + gzx + hxy = 0$

Thus, $-6f + 3g - 2h = 0$ (i)

and $-f + 3g - 3h = 0$ (ii)

From (i) and (ii)

$$\frac{f}{-9+3} = \frac{g}{2-18} = \frac{h}{-18+3}$$

or $\frac{f}{-3} = \frac{g}{-16} = \frac{h}{-15}$ or $\frac{f}{3} = \frac{g}{16} = \frac{h}{15}$

Hence, the require equation of cone is

$3yz + 16zx + 15xy = 0.$

Q.100.(b)

1. If A is a 2×2 non-singular square matrix, then Adj (adj A) is
(a) A^2 (b) A
(c) A^{-1} (d) None of the above

2. If $A=\begin{pmatrix}0&1&0&0\\0&2&0&1\\0&3&2&0\\1&4&1&0\end{pmatrix}$, then det (A^{-1}) is equal to
(a) 2 (b) $\frac{1}{2}$
(c) $-\frac{1}{2}$ (d) -2

3. If ω is one of the imaginary cube roots of unity, then the value of $\begin{vmatrix}1&\omega&\omega^2\\\omega&\omega^2&1\\\omega^2&1&\omega\end{vmatrix}$ is
(a) ω^3 (b) ω^2
(c) 0 (d) 1

4. If $A=\begin{bmatrix}1&1&1\\1&2&-3\\2&-1&3\end{bmatrix}$ and B = adj A, then the value of B_{23} (the element in B belonging to second row and 3rd column) is
(a) 4 (b) 5
(c) -7 (d) -4

5. The system of equations
$x-y+3z=4$
$x+z=2$
$x+y-z=0$ has
(a) a unique solution
(b) finitely many solutions
(c) infinitely many solutions
(d) No solution

6. If $f(x)=2x+1$, then $f\left(\frac{1}{x}\right)$ is
(a) $\frac{1}{2x+1}$ (b) $\frac{2x+1}{x}$
(c) $\frac{1}{x}$ (d) $\frac{2}{x}+1$

7. Let $f(x)=\begin{cases}x & \text{when } 0\le x<\frac{1}{2}\\ 1 & \text{when } x=\frac{1}{2}\\ 1-x & \text{when } 1/2<x<1\end{cases}$

then $f(x)$ is
(a) continuous at $x=\frac{1}{2}$
(b) not defined at $x=\frac{1}{2}$
(c) discontinuous at $x=\frac{1}{2}$
(d) continuous for all x, $0\le x<1$

8. Consider the following conditions:
I. $f(x)$ is well defined at $x=a$.
II. $\lim_{x\to a} f(x)$ must exist.
III. $f(x)$ is continuous.
IV. $f(x)\neq 0$ at $x=a$.
Which of these conditions are necessary for a function $f(x)$ to be derivable at a point $x=a$ of its domain?
(a) I and II (b) III and IV
(c) I, II and III (d) I, II and IV

9. If $x^y \cdot y^x=72$, then the value of $\frac{dy}{dx}$ at (2, 3) is

(a) $-\frac{9+\log 729}{4+\log 64}$ (b) $\frac{9+\log 729}{4+\log 64}$

(c) $-\frac{3+\log 9}{2+\log 8}$ (d) $\frac{3+\log 9}{2+\log 8}$

10. Match List I with List II and select the correct answer using the codes given below the lists.

List I **(Function)**

I. $e^{\sqrt{x}}$ II. $\frac{e^x}{x}$

III. 2^{x^2} IV. $\frac{1}{\sqrt{x}}$

List II
(Derivative)

(A) $2x^2 \log 2.2x$ (B) $-\frac{1}{2}x^{-3/2}$

(C) $\frac{(xe^x - e^x)}{x^2}$ (D) $\frac{e^{\sqrt{x}}}{2\sqrt{x}}$

Codes:
(a) I-D, II-B, III-C, IV-A
(b) I-C, II-D, III-A, IV-B
(c) I-A, II-C, III-B, IV-D
(d) I-D, II-C, III-A, IV-B

11. If $f(x) = \log x (\log x)$, then $f'(e)$ is equal to

(a) $\frac{1}{e}$ (b) e

(c) 1 (d) 0

12. $\frac{d}{dx}(\sin^2 \sqrt{x})$ is equal to

(a) $\frac{\sin^2 \sqrt{x}}{2\sqrt{x}}$ (b) $\sin 2\sqrt{x}$

(c) $\frac{\cos 2\sqrt{x}}{2\sqrt{x}}$ (d) $\cos 2\sqrt{x}$

13. The derivative of $\tan^{-1}\left[\frac{\sin x + \cos x}{\cos x - \sin x}\right]$ with respect to x is

(a) 0 (b) $\frac{1}{2}$

(c) 1 (d) 2

14. Consider the following statements:
I. Rolle's theorem ensures that there is a point on the curve, the tangent at which is parallel to the x-axis.
II. Lagrange's mean value theorem ensures that there is a point on the curve, the tangent at which is parallel to the y-axis.
III. Cauchy's mean value theorem can be deduced from Lagrange's mean value theorem.
IV. Rolle's mean value theorem can be deduced from Lagrange's mean value theorem.

Which of the above statements is/are correct?

(a) I and IV (b) II and IV

(c) I alone (d) I, III and IV

15. The value of $\sqrt{3}\sin x + \cos x$ will be greatest when x is equal to

(a) $\frac{\pi}{2}$ (b) $\frac{\pi}{4}$

(c) $\frac{\pi}{3}$ (d) $\frac{\pi}{6}$

16. If $z = \tan^{-1}\left(\frac{y}{x}\right)$, then $\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial y^2}$ is

(a) 0 (b) $\tan\left(\frac{y}{x}\right)$

(c) xy (d) x^2y^2

17. The derivative of $\sin^{-1} x$ w.r.t $\cos^{-1}\sqrt{1-x^2}$ is

(a) $\frac{1}{\sqrt{1-x^2}}$ (b) $\sin^{-1} x$

(c) $\cos^{-1} x$ (d) 1

18. If $x^5 + 4x^3 + 2x - 7 = a(x-1)^5 + b(x-1)^4 + c(x-1)^3 + d(x-1)^2 + e(x-1) + f$, then the values of a, b, c, d, e and f will be respectively

(a) 1, 5, 12, 20, 18 and 0
(b) 1, 5, 14, 22, 19 and 0
(c) 1, 24, 12, 20, 18 and 1
(d) 1, 14, 22, 5, 19 and 0

19. The particular integral of $(D^2 + 1)\, y = xe^{2x}$ is

(a) $\frac{e^{x}}{25}(5x-4)$ (b) $\frac{e^{2x}}{25}(4x-5)$

(c) $\frac{e^{x}}{25}(4x-5)$ (d) $\frac{e^{2x}}{25}(5x-4)$

20. The curve $y = x^3 - 3x^2 - 9x + 9$ has a point of inflexion at

(a) $x = -1$ (b) $x = 1$

(c) $x = -2$ (d) $x = 2$

21. Match List I with List II and select the correct answer using the codes given below the Lists:

List I
(Curves)
I. Cubical parabola
II. Catenary III. Astroid
IV. Semi-cubical parabola

List II
(Equations)
(A) $c \cosh (x/c)$
(B) $y^2 = ax^3$ (C) $y = ax^3$
(D) $x^{2/3} + y^{2/3} = a^{2/3}$

Codes:
(a) I-C, II-D, III-A, IV-B
(b) I-C, II-A, III-D, IV-B
(c) I-B, II-C, III-D, IV-A
(d) I-D, II-A, III-B, IV-C

22. If $[x]$ denotes the greatest integer not greater than x, then $\int_0^2 [x]\, dx$ is equal to

(a) 0 (b) 1

(c) 2 (d) 4

23. $\int \frac{dx}{(x+2)\sqrt{(x+1)}}$ equals

(a) $\tan^{-1}(x+2)^{1/2} + c$
(b) $\tan^{-1}(x+1)^{1/2} + c$
(c) $2\tan^{-1}(x+2)^{1/2} + c$
(d) $2\tan^{-1}(x+1)^{1/2} + c$

24. The value of

$$\lim_{n\to\infty}\left\{\frac{1}{n+1}+\frac{1}{n+2}+\frac{1}{n+3}+\dots\dots+\frac{1}{3n}\right\}$$

is equal to

(a) $\log_e 2$ (b) $\log_e 3$

(c) $\log_e \frac{1}{2}$ (d) $\log_e \frac{1}{3}$

25. $\int_0^{\pi} x f(\sin x)\, dx$ is equal to

(a) $\pi \int_0^{\pi} f(\sin x)\, dx$ (b) $\frac{\pi}{2} \int_0^{\pi} f(\sin x)\, dx$

(c) $\pi \int_0^{\pi/2} f(\sin x)\, dx$ (d) $\frac{\pi}{4} \int_0^{\pi} f(\sin x)\, dx$

26. The area bound by the curve $x = a\cos^3 t$, $y = a\sin^3 t$ equals

(a) $\frac{1}{2}\pi a^2$ (b) $\frac{1}{3}\pi a^2$

(c) $\frac{3}{4}\pi a^2$ (d) $\frac{3}{8}\pi a^2$

27. The perimeter of the cardioid $r = a)\,(1 + \cos\theta)$ is

(a) $6a$ (b) $4a$

(c) $8a$ (d) $12a$

28. If $u_n = \sqrt{n+1} - \sqrt{n}$, $v_n = \sqrt{n^4+1} - n^2$, then

(a) $\sum_{n=1}^{\infty} u_n$ converges but $\sum_{n=1}^{\infty} v_n$ diverges.

(b) $\sum_{n=1}^{\infty} u_n$ diverges but $\sum_{n=1}^{\infty} v_n$ converges

(c) $\sum_{n=1}^{\infty} u_n$ and $\sum_{n=1}^{\infty} v_n$ both converge.

(d) $\sum_{n=1}^{\infty} u_n$ and $\sum_{n=1}^{\infty} v_n$ both diverge.

29. Which one of the following tests does not give absolute convergence of a series?

(a) Root test (b) Comparison test

(c) Ratio test (d) Leibnitz test

30. If $\sum u_n$ is a series of positive terms, then

(a) convergence of $\sum (-1)^n u_n$ implies the convergence of $\sum u_n$

(b) ... convergence of $\sum(-1)^n u_n$

(c) convergence of $\sum(-1)^n u_n$ implies the divergence of $\sum u_n$

(d) divergence of $\sum u_n$ implies the divergence of $\sum(-1)^n u_n$

35. The integrating factor of a homogeneous equation $a(x, y)\,dy + b(x, y)\,dx = 0$, is

(a) $\frac{1}{ax+by}$, $ax-by \neq 0$

(b) $\frac{1}{bx+ay}$, $ax-by \neq 0$

(c) $\frac{1}{ax+by}$, $ax-by \neq 0$

(d) $\frac{1}{bx+ay}$, $bx-ay \neq 0$

32. The singular solution of the differential equation $(xp-y)2 = p^2 - 1$ is

$\left(\frac{dy}{dx}\right)^2 + x\frac{dy}{dx} - y = 0$?

(a) $x^2 + 4y = 0$ (b) $y = x + 1$

(c) $y + x = 1$ (d) $y2 - 4x = 0$

34. The solution of the differential equation

(b) $-x = y = z$

(c) $x = -y - z$

(d) $x = y = -z$

37. Match *List I* with *List II* an select the correct answer using the codes given below the lists

List I

(Differential equation)

I. $\frac{dy}{dx} = -\frac{x}{y}$ II. $\frac{dy}{dx} = \frac{y}{x}$

III. $\frac{dy}{dx} = -1$ IV. $\frac{dy}{dx} = -\frac{y}{x}$

List II

(Geometrical meaning of the solution)

(A) Family of curves having slope at any point (x, y) equal to negative of the slope of line joining origin and the point (x, y)

(B) Family of curves orthogonal to the line joining origin and the point (x, y)

(C) Family of straight lines through origin

(D) Family of straight lines having slope – 1.

Codes:

(a) I-A, II-B, III-D, IV-C

(b) I-B, II-A, III-C, IV-D

(c) I-B, II-A, III-D, IV-C

(d) I-A, II-B, III-C, IV-D

53. The equation of the cone, whose vertex is the origin and the base the circle, $x = a$, $y^2 + z^2 = b^2$, is given by
(a) $a^2(y^2 + z^2) = b^2x^2$ (b) $b^2(y^2 + z^2) = a^2x^2$
(c) $a^2(y^2 + x^2) = b^2z^2$ (d) $b^2(y^2 + x^2) = a^2z^2$

54. The equation of the right circular cylinder with axis, $x = 2y = -z$ and radius 4 is
(a) $5x^2 + 8x^2 + 8z^2 - 4xy + 2yz + 4xz = 0$
(b) $5x2 + 8x2 + 8z2 - 4xy + 2yz - 4xz = 0$
(c) $5x^2 + 8x^2 + 5z^2 - 4xy - 4yz - 8zx - 144 = 0$
(d) $5x^2 + 8xy^2 + 5z^2 + 4yz + 8zx - 4xy - 144 = 0$

55. The general solution of the equation $(2D + 1)2\, y = 0$, $D = \frac{d}{dx}$ is (A, B are arbitrary constants)
(a) $y = (A + Bx)e^{x}$ (b) $y = (A + Bx)e^{-x/2}$
(c) $y = (A + B)e^{-x/2}$ (d) $y = Ae^{-x/2}$

Directions: The following four (4) items consists of two statements, one labeled the Assertion A and the other labeled the Reason R. You are to examine these two statements carefully and decide if the Assertion A and the Reason R are individual true and if so, whether the reason is a correct explanation of the Assertion. Select your answer to these items using the codes given below and mark your Answer Sheet accordingly.

Codes:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true and R is not a correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

56. Assertion (A): The differential equation $\frac{d^2y}{dx^2} + x\frac{dy}{dx} + (x^2 + 5)y = e^x$ is a linear equation.
Reason (R): Every first degree equation is a linear equation.
(a) (b)
(c) (d)

57. Assertion (A): The function $f(x) = \frac{1}{4}x^4 - \frac{1}{3}x^3 - \frac{1}{2}x^2 + x$ has an extreme value at $x = 1$
Reason (R): $f'(1) = 0$

58. Assertion (A): Moment of a force F which is represente in length and direction by the line AB about a point C is equal to twice the area of the triangle ABC.
Reason (R): Area of a triangle is the product of base and altitude.
(a) (b)
(c) (d)

59. Assertion (A): Three equal forces acting clockwise along the sides of an equilateral triangle are in equilibrium.
Reason (R): Three forces acting at a point are in equilibrium if these can be represented in magnitude and direction by the sides of a triangle.
(a) (b)
(c) (d)

60. The centroid of a rigid body is a
(a) point which can be made to lie on or outside the body by changing the coordinate system
(b) fixed point in space regardless of the orientation of the boy.
(c) point on which the boy can be balanced by a force when it is subjected to external forces
(d) unique point fixed with respect to the body.

61. f two bodies A and B at rest having masses in the ratio 2:1 are subjected to forces in the ratio 1:2 for equal intervals of time, then
(a) their moments will be equal
(b) their velocities will be equal
(c) their energies will be equal
(d) None of the above

62. onsider the following statements:
If a particle moves under the action of a force $f(r)$ towards a fixed point, r being the distance of the particles from the fixed point, then the

I. linear momentum of the particle will be constant.

II. angular momentum of the particle about the center of force will be constant.

III. orbital path will necessarily be a plane curve.

IV. path will always be closed.

Which of the above statements are correct.

(a) I and II (b) II and III

(c) III and IV (d) II and IV

63. Keeping the sun in focus, the planets describe

(a) parabolas (b) ellipses

(c) circles d) hyperbolas

64. Consider the following statements:

A particle is executing simple harmonic motion in a straight line. If the period of oscillation is T, and amplitude of oscillation is A, then

I. speed is greatest when the acceleration is zero.

II. speed is least when the acceleration is least

III. A depends on T, but does not depend on initial condition

IV. A is independent of T but depends on initial conditions.

Which of the above statements are correct?

(a) I and II (b) II and III

(c) III and IV (d) I and IV

65. If the velocity at the maximum height of a projectile is half of its initial velocity 'u', then its horizontal range will be

(a) $\frac{3u^2}{2g}$ (b) $\frac{2g}{3u^2}$

(c) $\frac{\sqrt{3}u^2}{2g}$ (d) $\frac{2g}{\sqrt{3}u^2}$

66. If a body is projected at such an angle that the horizontal range is 4 times the greatest height, then the angle of projection is

(a) 45° (b) $\tan^{-1} 2$

(c) $\cot^{-1} 2$ (d) $\tan^{-1} \frac{1}{2}$

67. Match List I with List II and select the correct answer using the codes given below the lists.

List I

(Expressions are referred to plane polar)

I. $\dot{r}$ II. $\frac{1}{2}r^2\dot{\theta}$

III. $r\ddot{\theta}+2\dot{r}\dot{\theta}$

List II

(List of dynamical quantities)

(A) Angular momentum

(B) A component of acceleration

(C) A component of velocity

(D) A real velocity

Codes:

(a) I-C, II-A, III-D (b) I-C, II-D, III-B

(c) I-A, II-C, III-B (d) I-A, II-D, III-C

68. Match List I with List II and select the correct answer using the codes given below the lists:

List I

I. The velocity required to allow escape of a body from the gravitational attraction of earth.

II. Resultant of a force and a couple

III. The velocity necessary for an artificial satellite to revolve round the earth in a circular orbit.

IV. Work done by a force in moving a particle from one position to another.

List II

(A) 8 kilometers per sec

(B) Couple

(C) 16 kilometers per sec

(D) Vector quantity

(E) A force

(F) Scalar quantity

(G) 11 kilometers per sec

codes:

(a) I-A, II-B, III-D, IV-C

(b) I-G, II-B, III-C, IV-F

(c) I-G, II-E, III-A, IV-F

(d) I-C, II-E, III-G, IV-

69. If a particle moves in a circular path of radius r with a constant speed, then its
(a) velocity is perpendicular to the radius vector and acceleration is radially inwards.
(b) velocity is perpendicular to the radius vector and acceleration is radially outwards.
(c) velocity is along the radius vector and acceleration is radially outwards.
(d) velocity is along the radius vector and acceleration is radially inwards.

70. A particle starting from rest moves with constant acceleration. If it travels a distance S_1 in the first two seconds, S_2 in the next two seconds and S_3 in the next two seconds, then
(a) $S_1 : S_2 : S_3 = 1 : 1 : 2$
(b) $S_1 : S_2 : S_3 = 1 : 2 : 3$
(c) $S_1 : S_2 : S_3 = 1 : 3 : 5$
(d) $S_1 : S_2 : S_3 = 1 : 5 : 9$

71. The radix of the binary number system is
(a) 16 (b) 2
(c) 10 (d) 8

72. The hexadecimal number FF is equivalent to the decimal number
(a) 4080 (b) 625
(c) 552 (d) 255

73. The octal sums 670 + 255, 267 + 531 and 671 + 265 are respectively
(a) 1020, 1145 and 1156
(b) 1020, 1156 and 1145
(c) 1145, 1020 and 1156
(d) 1145, 1156 and 1020

74. A combination of bits used to represent is character is known as a
(a) byte (b) binary digit
(c) word (d) code

75. In a flow chart, a parallelogram box is used as
(a) a terminal box
(b) a processing box
(c) an input or output box
(d) a connector box

76. The value of $\left(-1+i\sqrt{3}\right)^{48}$, $i=\sqrt{-1}$, is
(a) 1 (b) 2
(c) 2^{24} (d) 2^{48}

77. If $2\cos\theta = x + \frac{1}{x}$ and $2\cos\phi = y + \frac{1}{y}$, then the value of $\cos((\theta+\phi))$ will be
(a) $\frac{x}{y}+\frac{y}{x}$ (b) $2\left[\frac{x}{y}+\frac{y}{x}\right]$
(c) $\frac{1}{2}\left[xy+\frac{1}{xy}\right]$ (d) $xy+\frac{1}{xy}$

78. The expression
$\left(27+\sqrt{756}\right)^{\frac{1}{3}}+\left(27-\sqrt{756}\right)^{\frac{1}{3}}$ is equal to
(a) 3 (b) 9
(c) 18 (d) 27

79. $\sqrt{2}$ and $\sqrt{3}$ are algebraic over Z.A polynomial satisfied by $\sqrt{2}+\sqrt{3}$ over Z is
(a) $x^4-5x^2+6=0$
(b) $x^4-3x^2+5x-6=0$
(c) $x^4-2x^2+3x+6=0$
(d) $x^4-10x^2+1=0$

80. If the HCF of two quadratic expressions is $(x + 2)$ an the LCM is $x^3 + 2x^2 - x - 2$, then the expressions are
(a) $x^2 - 3x + 2$ and $x^0 + x - 2$
(b) $x^2 + 3x + 2$ and $x^2 + x + 2$
(c) $x^2 + 3x + 2$ and $x^2 - x - 2$
(d) $x^2 + 3x + 2$ and $x^2 + x - 2$

81. The polynomial $2x^3 - 5x^2 + mx + n$ i divisible by $x^2 - 4$ is
(a) $(m, n) = (2, 3)$ (b) $(m, n) = (8, -20)$
(c) $(m, n) = (-8, 20)$ (d) $(m, n) = (4, 7)$

82. If α, β are the roots of the equation $x^2 + \alpha x + \beta = 0$, then (α, β) is equal to
(a) (0, 0) (b) (1, 0)
(c) (1, 1)
(d) $(\alpha, 0)$ for any real $\alpha \neq 0$

83. If α, β, γ are the roots of $x^3 + qx + r = 0$, then $\sum_{\alpha\beta\gamma} \frac{\alpha}{\beta+\gamma}$ is
(a) 0 (b) $q + r$
(c) q/r (d) -3

84. If α, β, γ are the roots of $x^3 - px^2 + qx - r = 0$ then the roots of the equation $r(x + q)^3 - (px + r)^3 = 0$ are
(a) $\alpha^2, \beta^2, \gamma^2$ (b) $\frac{1}{\alpha}, \frac{1}{\beta}, \frac{1}{\gamma}$
(c) $\frac{1}{\alpha^2}, \frac{1}{\beta^2}, \frac{1}{\gamma^2}$ (d) $\alpha^3, \beta^3, \gamma^3$

85. The set {2, 4, 6, 8} is a group with binary operation $\otimes_{10}$, multiplication modulo 10. The identity element of this group is
(a) 2 (b) 4
(c) 6 (d) 8

86. Let (S, +, •) be the system of non-negative integers. If we define a new binary operation (*) in S as $x * y = x + y + xy, \forall x, y \in S$, then with respect to this new operation (*), which one of the following statements is true?
(a) (S, *) contains an identity element 0
(b) (S, *) contains an identity element 1
(c) (S, *) does not contain an identity element.
(d) (S, *) contains an identity element –1

87. Let Z denote the set of integers. Let p be a prime number and Let $Z_2 = \{0, 1\}$
Let $f : Z \to Z$ and $g : Z \to Z$ be define as follows
$f(n) = pn$ for any $n \in Z$
$g(n) = 1$ if n is a perfect square = 0 otherwise
If we consider the composite function *gof* then
(a) *gof* is onto but *f* is not onto
(b) *gof* is one-to-one but *g* is not one-to-one
(c) *gof* is invertible but *g* is not invertible
(d) *f* and *g* are both one-to-one

88. The number of disjoint cycles of length > 1 in the permutation
$\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 \\ 5 & 2 & 7 & 6 & 3 & 4 & 1 \end{pmatrix}$ is
(a) 2 (b) 3
(c) 4 (d) 5

89. Let $f : G \to H$ be a group homomorphism from a group G into a group H, with kernel K. If the orders of G, H an K are 75, 45 an 15 respectively, then the order of the image $f(G)$ is
(a) 3 (b) 5
(c) 15 (d) 45

90. Consider the following statements with regard to S_3, the symmetric group of order 3
I. There exist two elements x and y in S_3 such that $(xy)^2 \neq x^2y^2$.
II. There are four elements satisfying $x^2 = e$ and three elements satisfying $y^3 = e$.
III. There is one cyclic sub-group of order 3 and three cyclic sub-groups of order 2.
Which of the above statements are correct?
(a) I and III (b) I an II
(c) II and III (d) I, II and III

91. Which one of the following rings is an integral domain?
(a) $Z_{[illegible]}$ (b) $Z_{[illegible]}$
(c) $Z_{[illegible]}$ (d) $Z_{[illegible]}$

92. Let U be an arbitrary non-empty set and S denote the set of all subsets of U. For X, Y ∈ S, if we define X + Y = {x : x ∈ X or x ∈ Y but x ∉ X ∩ Y} and X ∘ Y = X ∩ Y, then
(a) (S, +, ∘) is a commutative ring without unity.
(b) (S, +, ∘) is a commutative ring with unity and X ∘ X = X for all X ∈ S.
(c) for X, Y, Z ∈ S, X ∘ (Y + Z) ≠ X ∘ Y + X ∘ Z and so (S, +, ∘) is not a ring
(d) S can be made a ring if U and the null set φ are added to S.

93. Let $R = \{a + \sqrt{2}b : a, b \in Q\}$. R is a field with respect to usual addition and multiplication. If an element

If $a+\sqrt{2}b \neq 0$, then its multiplicative inverse is given by

(a) $\frac{1}{\left(a+\sqrt{b}\right)^2}$ (b) $\frac{1}{a-\sqrt{2b}}$

(c) $\frac{a}{a^2-2b^2}+\sqrt{2}\,\frac{-b}{a2-2b^2}$

(d) $\frac{a}{a^2-2b^2}+\sqrt{2}\,\frac{b}{a2-2b^2}$

94. If $W_1 = (0, a_2, a_3)$ and $W2 = (b_1, 0, b_3)$ are two subspaces of F^3, where F^n is a vector space of dimension n. the0n the subspace $W_1 \cap W_2$ is isomorphic to the vector space

(a) F^1 (b) F^2

(c) F^3 (d) $2F^4$

95. If the set of all triples (x_1, x_2, x_3) of real numbers R form a vector space V_3, then a subspace denoted by a vertical plane $y = x$ can be obtained by a linear combination of the sets

(a) (1, 1, 0) and (0, 0, 1)

(b) (1, 1, 0) and (1, 0, 0)

(c) (1, 0, 0,) and (0, 1, 0)

(d) (1, 0, 1) and (0, 0, 1)

96. In R^3, consider the following three statements about the subset
E = {(1, 0, 0), (0, 1, 0), (0, 0, 1), (1, 1, 1), (1, 1, 0)}

I. E is a linearly dependent set.

II. Any three vectors of E are linearly independent

III. Any four vectors of E are linearly dependent

Which of the above statements are correct?

(a) I, II and III (b) I and II

(c) I and III (d) II and III

97. If $T : R^2 \to R$ is a linear map for which $T(1, 1) = 3$ and $T(0, 1) = -2$, then $T(a, b)$ is equal to

(a) $5a - 2b$ (b) $2a - 5b$

(c) $2a + 5b$ (d) $5a + 2b$

98. The rank of the linear transformation $T : R^3 \to R^3$ defined by $T(x, y, z) = (y, 0, z)$ is

(a) 3 (b) 2

(c) 1 (d) 0

99. If $P_n(R)$ is the set of polynomials of order n over R, then match List I with List II an select the correct answer using the codes given below the lists

List I

(Linear transformations)

I. $T : R^3 \to R^2$
$T(x, y, z) = (x + y, y + z)$

II. $T : R^3 \to P_2(R)$
$T(a, b, c) = (a + b) + (b + c)x$

III. $T : P_2(R) \to P_3(R) : (C)$
$T(p(x)) = xp(x)$

IV. $T : R^2 \to R^3 : T(x, y) = (x, x + y, y)$

Codes:

(a) I-A, II-D, III-B, IV-C

(b) I-B, II-C, III-D, IV-A

(c) I-C, II-D, III-B, IV-A

(d) I-C, II-A, III-D, IV-B

List II

(Matrices)

(A) $\begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 0 & 1 \end{pmatrix}$ (B) $\begin{pmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$

(C) $\begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{pmatrix}$

(D) $\begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}$

100. If $A = \begin{bmatrix} 2 & 6 \\ 3 & 9 \end{bmatrix}$ $B = \begin{bmatrix} 3 & x \\ y & 2 \end{bmatrix}$, then in order that $AB = O$, the values of x and y will be respectively

(a) –6 and -1 (b) 6 and 1

(c) –6 and –3 (d) –5 and –14

ANSWERS

Q.1.(b)

Q.2.(b) Here $A = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 2 & 0 & 1 \\ 0 & 3 & 2 & 0 \\ 1 & 4 & 1 & 0 \end{pmatrix}$

Writing A = IA, we get

$$\begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 2 & 0 & 1 \\ 0 & 3 & 2 & 0 \\ 1 & 4 & 1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} A$$ (using elementary row transformations)

$R_2/2$ and $R_3/2$

$$\begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & \frac{1}{2} \\ 0 & \frac{3}{2} & 1 & 0 \\ 1 & 4 & 1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & \frac{1}{2} & 0 & 0 \\ 0 & 0 & \frac{1}{2} & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} A$$

$R_1 \leftrightarrow R_4$

$$\begin{pmatrix} 1 & 4 & 1 & 0 \\ 0 & 1 & 0 & \frac{1}{2} \\ 0 & \frac{3}{2} & 1 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & \frac{1}{2} & 0 & 0 \\ 0 & 0 & \frac{1}{2} & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix} A$$

$R_1 \to 4R_2$

$$\begin{pmatrix} 1 & 0 & 1 & -2 \\ 0 & 1 & 0 & \frac{1}{2} \\ 0 & \frac{3}{2} & 1 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & -2 & 0 & 1 \\ 0 & \frac{1}{2} & 0 & 0 \\ 0 & 0 & \frac{1}{2} & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix} A$$

$R_4 - R_2$

$$\begin{pmatrix} 1 & 0 & 1 & -2 \\ 0 & 1 & 0 & \frac{1}{2} \\ 0 & \frac{3}{2} & 1 & 0 \\ 0 & 0 & 0 & -\frac{1}{2} \end{pmatrix} = \begin{pmatrix} 0 & -2 & 0 & 1 \\ 0 & \frac{1}{2} & 0 & 0 \\ 0 & 0 & \frac{1}{2} & 0 \\ 1 & -\frac{1}{2} & 0 & 0 \end{pmatrix} A$$

$R_2 + R_4$

$$\begin{pmatrix} 1 & 0 & 1 & -2 \\ 0 & 1 & 0 & 0 \\ 0 & \frac{3}{2} & 1 & 0 \\ 0 & 0 & 0 & -\frac{1}{2} \end{pmatrix} = \begin{pmatrix} 0 & -2 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & \frac{1}{2} & 0 \\ 1 & -\frac{1}{2} & 0 & 0 \end{pmatrix} A$$

$R_4(-2)$

$$\begin{pmatrix} 1 & 0 & 1 & -2 \\ 0 & 1 & 0 & 0 \\ 0 & \frac{3}{2} & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 0 & -2 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & \frac{1}{2} & 0 \\ -2 & 1 & 0 & 0 \end{pmatrix} A$$

$R_3 - \frac{3}{2}R_2$

$$\begin{pmatrix} 1 & 0 & 1 & -2 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 0 & -2 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ -\frac{3}{2} & 0 & \frac{1}{2} & 0 \\ -2 & 1 & 0 & 0 \end{pmatrix} A$$

$R_1 - R_3$

$$\begin{pmatrix} 1 & 0 & 0 & -2 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} \frac{3}{2} & -2 & -\frac{1}{2} & 1 \\ 1 & 0 & 0 & 0 \\ -\frac{3}{2} & 0 & \frac{1}{2} & 0 \\ -2 & 1 & 0 & 0 \end{pmatrix} A$$

$R_1 + 2R_4$

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} -5/2 & 0 & -1/2 & 1 \\ 1 & 0 & 0 & 0 \\ -3/2 & 0 & 1/2 & 0 \\ -2 & 1 & 0 & 0 \end{pmatrix} A$$

$$\therefore \quad A^{-1} = \begin{pmatrix} -\frac{5}{2} & 0 & -\frac{1}{2} & 1 \\ 1 & 0 & 0 & 0 \\ -\frac{3}{2} & 0 & \frac{1}{2} & 0 \\ -2 & 1 & 0 & 0 \end{pmatrix}$$

$$\therefore \det(A^{-1}) = \begin{vmatrix} -\frac{5}{2} & 0 & -\frac{1}{2} & 1 \\ 1 & 0 & 0 & 0 \\ -\frac{3}{2} & 0 & \frac{1}{2} & 0 \\ -2 & 1 & 0 & 0 \end{vmatrix} = -\begin{vmatrix} 1 & 0 & 0 \\ -\frac{3}{2} & 0 & \frac{1}{2} \\ -2 & 1 & 0 \end{vmatrix}$$

$$= -\left\{1\left(0-\frac{1}{2}\right)\right\} = -\left(-\frac{1}{2}\right) = \frac{1}{2}$$

Q.3 (c): $$\begin{vmatrix} 1 & \omega & \omega^2 \\ \omega & \omega^2 & 1 \\ \omega^2 & 1 & \omega \end{vmatrix} = \begin{vmatrix} 1+\omega+\omega^2 & \omega & \omega^2 \\ \omega+\omega^2+1 & \omega^2 & 1 \\ \omega^2+1+\omega & 1 & \omega \end{vmatrix}$$

(Adding the second and third column to first column)

$$= \begin{pmatrix} 0 & \omega & \omega^2 \\ 0 & \omega^2 & 1 \\ 0 & 1 & \omega \end{pmatrix}$$

$= 0 \quad [\because \ 1+\omega+\omega^2 = 0]$

Q.4.(a): $A = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & -3 \\ 2 & -1 & 3 \end{pmatrix}$ and B = adj A

$$\therefore B23 = A32 = - \begin{vmatrix} 1 & 1 \\ 1 & -3 \end{vmatrix} = -(-3-1) = 4$$

Q.5.(c) The equation are

$x - y + 3z = -4$

$x + z = 2$

$x + y - z = 0$

$$\text{Here } D = \begin{vmatrix} 1 & -1 & 3 \\ 1 & 0 & 1 \\ 1 & 1 & -1 \end{vmatrix}$$

$= 1(0-1) - (-1)(-1-1) + 3(1-0)$

$= -1 - 2 + 3 = 0$

$$\text{But } D_x = \begin{vmatrix} 4 & -1 & 3 \\ 2 & 0 & 1 \\ 0 & 1 & -1 \end{vmatrix}$$

$= 4(-1) - 2(1-3) = -4 + 4 = 0$

$$D_y = \begin{vmatrix} 1 & 4 & 3 \\ 1 & 2 & 1 \\ 1 & 0 & -1 \end{vmatrix} = \begin{vmatrix} 1 & 4 & 3 \\ 0 & -2 & -2 \\ 0 & -4 & -4 \end{vmatrix} = 1(8-8) = 0$$

$$D_z = \begin{vmatrix} 1 & -1 & 4 \\ 1 & 0 & 2 \\ 1 & 1 & 0 \end{vmatrix} = 1(0-2)$$

$+1(0-2) + 1(0-2) + 4(1-0) = 0$

Since all of them are zero, therefore given system has infinitely many solutions.

Q.6.(d): Given $f(x) = 2x + 1$

$$\therefore \quad f\left(\frac{1}{x}\right) = \frac{2}{x} + 1$$

Q.7.(c): L.H.L =

$$\lim_{x\to\frac{1}{2}} - f(x) = \lim_{h\to 0}\left(\frac{1}{2} - h\right)\left[\because x = \frac{1}{2} h\right]$$

$$= \frac{1}{2} - 0 = \frac{1}{2}$$

$$\text{R.H.L. } \lim_{x\to\frac{1}{2}} - f(x) = \lim_{h\to 0}\left[1 - \left(\frac{1}{2} + h\right)\right]$$

$$= 1 - \frac{1}{2} = \frac{1}{2}$$

$$\therefore \quad \text{L.H.L} = \text{R.H.L} = \frac{1}{2}$$

But $f\left(\frac{1}{2}\right)=1$

Hence, the given function is discontinuous at $x=\frac{1}{2}$

Q.8 (a): For a function to be derivable, the necessary condition is that it is continuous i.e. $f(x)$ is define at $x=a$ and $\lim_{x\to a} f(x)$ exists.

Q.9 (a): We have $x^y . y^x = 72$

Taking log of both sides

$y\log x + x\log y = \log 72$

Differentiating w.r.t. x

$$\frac{y}{x} + \log x.\frac{dy}{dx} + \log y + \frac{x}{y}.\frac{dy}{dx} = 0$$

or $$\frac{dy}{dx}\left(\frac{x}{y}+\log x\right) = -\frac{y}{x} - \log y$$

∴ $$\left.\frac{dy}{dx}\right|_{(2,3)} = \frac{-\frac{3}{2}-\log 3}{\frac{2}{3}+\log 2}$$

$$= -\frac{3+2\log 3}{2+3\log 2}\cdot\frac{3}{2}$$

$$= -\frac{9+6\log 3}{4+6\log 2} = -\frac{9+\log 3^6}{4+\log 2^6}$$

$$= -\frac{9+\log 729}{4+\log 64}$$

Q.10 (d)

Q.11 (a): $f(x) = \log_x(\log x) = \dfrac{\log(\log x)}{\log x}$

$$= \frac{\log x.\frac{1}{\log x}.\frac{1}{x} - \log(\log x).\frac{1}{x}}{(\log x)^2}$$

$$= \frac{1-\log(\log x)}{x(\log x)^2} = \frac{1-\log(\log e)}{e(\log e)^2} = \frac{1-0}{e} = \frac{1}{e}$$

Q.12 (a): $\cos\sqrt{x}.\frac{1}{2\sqrt{x}} = \frac{\sin 2\sqrt{x}}{2\sqrt{x}}$

Q.13 (c): Let $y=\tan^{-1}\left(\frac{\cos x+\sin x}{\cos x-\sin x}\right)$

or $y=\tan^{-1}\left(\frac{1+\tan x}{1-\tan x}\right)$

or $y=\tan^{-1}\left[\tan\left(\frac{\pi}{4}+x\right)\right]$

or $y=\frac{\pi}{4}+x$

Differentiating w.r.t. x, $\frac{dy}{dx}=1$.

Q.14 (a)

Q.15 (c): $f(x)=\sqrt{3}\sin x+\cos x$

∴ $f'(x)=\sqrt{3}\cos x-\sin x$

At stationary points, $f'(x)=0$

∴ $\sqrt{3}\cos x - \sin x = 0$

or $\tan x=\sqrt{3}$ or $x=\frac{\pi}{3}$

Now $f''(x) = -\sqrt{3}\sin x-\cos x$

For $x=\frac{\pi}{3}$, $f''(x)=-\sqrt{3}\sin\frac{\pi}{3}-\cos\frac{\pi}{3} = -\sqrt{3}.\frac{\sqrt{3}}{2}-\frac{1}{2}$

$$=-\frac{3}{2}-\frac{1}{2}=-\frac{4}{2}=-2<0$$

Thus f(x) will be greatest at $x=\frac{\pi}{3}$

Q.16 (a): $z=\tan^{-1}\left(\frac{y}{x}\right)$

∴ $$\frac{\partial z}{\partial x} = \frac{1}{1+\frac{y^2}{x^2}}\cdot\left(\frac{-y}{x^2}\right) = \frac{-y}{x^2+y^2}$$

$$\frac{\partial^2 z}{\partial x^2}+\frac{-y}{\left(x^2+y^2\right)^2}\times 2x=\frac{2xy}{(x^2+y^2)^2}$$

Again, $\frac{\partial z}{\partial y}=\frac{1}{1+\frac{y^2}{x^2}}\cdot\frac{1}{x}=\frac{x}{x^2+y^2}$

Also, $\frac{\partial^2 z}{\partial y^2}+\frac{\partial^2 z}{\partial y^2}=0$

Q.17.(d): Let $y=\sin^{-1}x$ and $u=\cos^{-1}\sqrt{1-x^2}$

We have to find $\frac{dy}{du}$

Putting $x=\sin\theta$ in (i) and (ii), we get

$y=\sin^{-1}(\sin\theta)=\theta$

$\therefore\ \frac{dy}{d\theta}=1$

Again $u=\cos^{-1}$

$\sqrt{1-\sin^2\theta}=\cos^{-1}\sqrt{\cos^2\theta}$

$=\cos^{-1}(\cos\theta)=\theta$

$\therefore\quad \frac{du}{d\theta}=1$

Hence, $\frac{dy}{du}=\frac{dy}{d\theta}\div\frac{du}{d\theta}=1\div1=1$

[From (iii) and (iv)]

Q.18.(b): $x^5+4x^3+2x-7=a(x-1)^5+b(x-1)^4$
$+c(x-1)^3+d(x-1)^2+e(x-1)+f$

$=a(x^5-5x^4+11x^3-10x^2+5x-1)$
$+b(x^4-4x^3+6x^2-4x+1)$
$+c(x^3-3x^2+3x-1)$

$=d(x^2-2x+1)+e(x-1)+f$

Comparing the coefficient of x^5, x^4, x^3, x^2, x, constant term from both sides, we get

$a=1$ (i)

$-5a+b=0$ (ii)

$10a-4b+c=4$ (iii)

$-10a+6b-3c+\ =0$ (iv)

$5a-4b+3c-2d+e=2$ (v)

$-a+b-c+d-e+f=0$ (vi)

Solving (i) to (vi),

$a=1,\ b=5,\ c=14,\ d=22,\ e=19,\ f=0$

Q.19.(d): Particular integral $=\frac{1}{D^2+1}(xe^{2x})$

$=e^{2x}\left(\frac{1}{(D+2)^2+1}\right)x$

$=e^{2x}\frac{1}{D^2+4D+5}x$

$=e^{2x}\frac{1}{5}\left(1+\frac{4}{5}+\frac{1}{5}D^2\right)^{-1}x$

$=e^{2x}\frac{1}{5}\left[1-\left(\frac{4}{5}D+\frac{1}{5}D^2\right)+\ldots\right]x$

$=e^{2x}\frac{1}{5}\left[x-\frac{4}{5}.1+\frac{1}{5}.0\right]=\frac{e^{2x}}{25}(5x-4)$

Q.20.(b): $y=x^3-3x^2-9x+9$

$\therefore\quad \frac{dy}{dn}=3x^2-6x-9$

$\frac{d^2y}{dx^2}=6x-6=6(x-1)$

Now $\frac{d^2y}{dx^2}=0\Rightarrow 6(x-1)=0$ or $x=1$

Also, as $\frac{d^2y}{dx^2}$ changes sign as x passes through 1 and is zero at $x=1$.

Therefore, $x=1$ is a point of inflexion.

Q.21.(b)

Q.22.(b): $\int_0^2[x]dx=\int_0^1 0\,dx+\int_1^2 1\,dx$

$=0+|x|_1^2=(2-1)=1$

Q.23.(d): Put $x+1=t^2\ \therefore\ dx=2t\,dt$

$\therefore\ \int\frac{dx}{(x+2)\sqrt{x+1}}=\int\frac{2t\,dt}{(t^2+1)t}=\int 2\frac{dt}{1+t^2}$

$=2\tan^{-1}t=2\tan^{-1}\left(\sqrt{x+1}\right)+c$

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Q.24.(b): $\lim_{n\to\infty}\left(\frac{1}{n+1}+\frac{1}{n+2}+\dots\dots+\frac{1}{3n}\right)$

$$=\lim_{n\to\infty}\sum_{r=1}^{2n}\frac{1}{n+r}=\lim_{n\to\infty}\frac{1}{n}\sum_{r=1}^{2n}\frac{1}{\left(1+\frac{r}{n}\right)}$$

$$=\int_0^2\frac{1}{1+x}\,dx=\left[\log_e(1+x)\right]_0^2$$

$$=\log_e 3-\log 1=\log_e 3.$$

Q.25.(b): Let $I=\int_0^\pi x f(\sin x)\,dx$

By property to definite integral,

$$I=\int_0^\pi(\pi-x)\,f(\sin \pi-x)\,dx$$

or $\quad I=\int_0^\pi(\pi-x)\,f(\sin x)\,dx$

Adding (i) and (ii)

$$2I=\int_0^\pi \pi\, f(\sin x)\,dx$$

or $\quad I=\frac{\pi}{2}\int_0^\pi f(\sin x)\,dx$

Q.26.(d)

Q.27.(c): Here $r=a\,(1+\cos\theta)$

$\therefore \quad \frac{dr}{d\theta}=a\sin\theta$

$\therefore \quad \frac{dr}{d\theta}=\sqrt{r^2+\left(\frac{dr}{d\theta}\right)^2}$

$$=\sqrt{a^2(1+\cos\theta)^2+a^2\sin^2\theta}$$

$$=a\sqrt{2+2\cos\theta}$$

$$=a\sqrt{2.2\cos\frac{2\theta}{2}}=a\cos\frac{\theta}{2}$$

Hence, reqd. perimeter = 2 $2\int_0^\pi\frac{ds}{d\theta}\,d\theta$

$$=2\int_0^\pi 2a\cos\frac{\theta}{2}\,d\theta=4a.\sin\frac{\theta}{2}.2\Big|_0^\pi$$

$$8a=[1-0]=8a$$

Q.28.(b): $u_n=\left(\sqrt{n+1}-\sqrt{n}\right)\times\frac{\sqrt{n+1}+\sqrt{n}}{\sqrt{n+1}+\sqrt{n}}$

$$=\frac{(n+1)}{\sqrt{n+1}+\sqrt{n}}=\frac{1}{\sqrt{n+1}+\sqrt{n}}$$

$$=\frac{1}{\sqrt{n}\left[1+\sqrt{1+\frac{1}{n}}\right]}$$

Comparing it with $\sum w_n$, where $w_n=\frac{1}{n^{1/2}}$

We infer that $\sum u_n$ is divergent

Again, $u_n=$

$$\left(\sqrt{n^4+1}-n^2\right)\times\frac{\sqrt{n^4+1}+n^2}{\sqrt{n^4+1}+n^2}$$

$$=\frac{(n^4+1)-n^4}{n^2\left[1+\sqrt{1+\frac{2}{n^4}}\right]}=\frac{1}{n^2\left[1+\sqrt{1+\frac{1}{n^4}}\right]}$$

Comparing it with $\sum w_n'$ where $\sum w_n'=\frac{1}{n^2}$

We infer that wv_n is convergent.

Q.29.(d) Q.30.(b) Q.31.(d)

Q.32.(b): Given differential equation is $(xp-y)^2=p^2-1$

or $\quad y=px-\sqrt{p^2-1}$

and singular equation is obtained by eliminating c between (i) and

$$0=x-\frac{1}{2\sqrt{c^2-1}}\times 2c \quad \text{or} \quad c^2-1=\frac{c^2}{x^2}$$

or $\quad c=\frac{x}{\sqrt{x^2-1}}$

Substituting this value of c = (i)

$$y-cx-\frac{c}{x}=c\left(x-\frac{1}{x}\right)$$

$$=\frac{x}{\sqrt{x^2-1}}\left(\frac{x^2-1}{x}\right)=\sqrt{x^2-1}$$

$\therefore \; y^2 - x^2 = 1$ or $x^2 - y^2 = 1$

Q.33.(d): It can be seen, as discussed in Q.32. that while (b) and (c) are general solutions. (a) is a singular solution of the given clairut's equation.

Q.34.(d): Integrating $\frac{d^2s}{dt^2} = g, \frac{ds}{dt} = gt + c$

When $t = 0, \frac{ds}{dt} = u$ so that $c = u$

$\therefore$ Equation (*i*) becomes

$$\frac{ds}{dt} = gt + u$$

Again integrating,

$$s = \frac{gt^2}{2} + ut + d$$

When $t = 0, s = 0$, so that $d = 0$

$$\therefore \quad s = ut + \frac{1}{2}gt^2$$

Q.35.(a) Q.36.(c) Q.37.(c)

Q.38.(c): $z = \log x$ or $x = e^z$

$$\therefore \frac{dy}{dx} = \frac{dy}{dz} \cdot \frac{dz}{dx} = \frac{1}{x}\frac{dy}{dz} = \frac{1}{x}Dy$$

$$\therefore \frac{d^2y}{dx^2} = \frac{-1}{x^2}\frac{dy}{dz} + \frac{1}{x} \cdot \frac{d^2y}{dz^2} \cdot \frac{1}{x}$$

$$= \frac{1}{x^2}\left(\frac{d^2y}{dz^2} - \frac{dy}{dz}\right) = \frac{1}{x^2}$$

$$(D^2y - Dy) = \frac{1}{x^2}D(D-1)y$$

Hence, given equation becomes

$$\frac{1}{x}D(D-1)y + 2\frac{1}{x}Dy = 6x$$

or $(D^2 + D)y = 6x^2 = 6e^{2z}$

or $D(D + 1)y = 6e^{2z}$

Q.39.(a): Resultant of vectors

$(2\hat{i} + 4\hat{j} - 5\hat{k})$ and

$(\hat{i} + 2\hat{j} - 3\hat{k})$ is $3\hat{i} + 6\hat{j} - 2\hat{k}$

$(2\hat{i} + 4\hat{j} - 5\hat{k})$ and

$(\hat{i} + 2\hat{j} - 3\hat{k})$ is $3\hat{i} + 6\hat{j} - 2\hat{k}$

Hence, unit vector parallel to this vector

$$\frac{3\hat{i} + 6\hat{j} - 2\hat{k}}{\sqrt{9 + 36 + 4}} = \frac{3\hat{i} + 6\hat{j} - 2\hat{k}}{7}$$

Q.40.(c): Since $\bar{F}_1, \bar{F}_2, \bar{F}_3$ are in equilibrium.

Then $\bar{F}_1, \bar{F}_2, \bar{F}_3 = \bar{0}$

or $(x_1\hat{i} + x_2\hat{j} - x_3\hat{k}) + (y_1\hat{i} + y_2\hat{j} - y_3\hat{k}) + (z_1\hat{i} + z_2\hat{j} - z_3\hat{k}) = \bar{0}$

or $(x_1 + y_1 + z_1)\hat{i} + (x_2 + y_2 + z_2)\hat{j} + (x_3 + y_3 + z_3)\hat{k} = \bar{0}$

$x_1 = y_1 + z_1 = 0$

$x_2 + y_2 + z_2 = 0$

$x_3 + y_3 + z_3 = 0$

Q.41.(b): Equations of pair of bisectors of

$x^2 - 2pxy - y^2 = 0$ are $\frac{x^2 - y^2}{1-(-1)} = \frac{xy}{b}$

or $\quad x^2 - y^2 + \frac{2}{p}xy = 0$

Comparing (*i*) with $x^2 - 2qxy - y^2 = 0$

We get $\frac{2}{p} = -2q$ or $pq = -1$

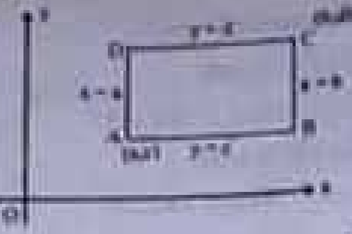
Q.42.(b): Applying two-point form equation of the diagonal AC is

$$\frac{y-c}{x-a} = \frac{d-c}{b-a}$$

or $(d-c)(x-a) - (b-a)(y-c) = 0$

or $(d-c)x + (a-b)y - da + ca + bc - ac = 0$

or $(d-c)x + (a-b)y + bc - ad = 0$



Q.43.(b): Here $a = \lambda$, $h = -4$, $b = 12$, $g = \frac{5}{2}$,

$f = -8$, $c = -3$

$\therefore$ Given equation represents a pair of straight lines if

$abc + 2fgh - af^2 - bg^2 - ch^2 = 0$

$$\lambda \times 12 \times (-3) + 2 \times (-8)\left(\frac{5}{2}\right)(-5) - \lambda(-8)^2$$

or

$$-12\left(\frac{5}{2}\right)^2 - (-3)(-5)^2 = 0$$

or $\quad -36\lambda + 200 - 64\lambda - 75 + 75 = 0$

Q.44.(a):

Q.45.(d): Let the equation of circle be

$x^2 + y^2 - 2hx - 2ky + c = 0$

Its intersection with x-axis, i.e., $y = 0$ is

$x^2 + 2hx + c = 0$

Since it touches x-axis,

$\therefore \quad (-2h)^2 - 4c = 0 \quad$ or $\; c = h^2$

Hence, equation (i) becomes

$x^2 + y^2 - 2hx - 2ky + h^2 = 0$

Q.46.(c):

Q.47.(c): Since $y = mx + c$ is normal to the circle, therefore it must pass through center (a, b) of the circle, i.e.,

$b = ma + c$

Q.48.(b): Radical axis of the circles

$S_1 \equiv x^2 + y^2 - x = 0$

and $S_2 \equiv x^2 + y^2 - \frac{3}{2}y - \frac{5}{2} = 0$

$S_1 - S_2 = 0$

i.e., $\quad -x + \frac{3}{2}y + \frac{5}{2} = 0$

or $\quad 2x - 3y - 5 = 0$

Q.49.(a): Since $2g_1g_2 + 2f_1f_2 = C_1 + C_2$

$(\text{for } 2 \times (-a) \times 0 + 2 \times 0(b) = \ldots)$

Therefore given circles are orthogonal

Q.50.(b): Equations of lines BS and B'S are respectively

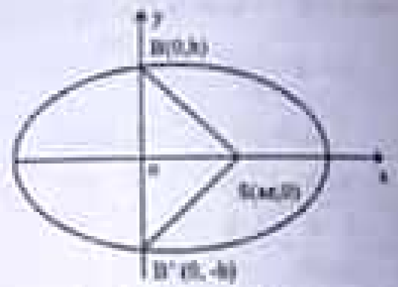
$$\frac{y-b}{x-0} = \frac{b-0}{0-ae} \text{ and } \frac{y+b}{x-0} = \frac{0+b}{ae-0}$$

or $\quad y = \frac{-b}{ae}x + b$ and $y = \frac{b}{ae}x - b$

Since these lines are at right angles,

$$\therefore \quad \left(\frac{-b}{ae}\right)\left(\frac{b}{ae}\right) = -1$$

or $\quad \frac{b^2}{a^2} = e^2$



or $\quad 1 - e^2 = e^2$ or $2e^2 = 1$

$\therefore \quad e = \frac{1}{\sqrt{2}}$

Q.51(d): Suppose equation of normal ... $y^2 = 4ax$ is $y = mx - 2am - am^3$

Also, equation of normal is $2x + y - 9 = 0$

So that $m = -2$ ...(i) and $-2am - am^3 = 9$ or $2am + am^3 = -9$

Using (i), $2a \times (-2) + a(-2)^3 = -9$ or $-4a -$ $8a = -9$ or $a = \frac{3}{4}$

Q.52.(b): $\begin{vmatrix} x-1 & y-4 & z+2 \\ 6 & 2 & 2 \\ 1 & 2 & -6 \end{vmatrix} = 0$

$\Rightarrow (x-1)(-12-4) - (y-4)(-36-2)$
$+ (z+2)(12-2)$

$$\Rightarrow \quad -16x + 38y + 10z - 116 = 0$$

$$\Rightarrow \quad -8x + 19y + 5z - 108 = 0$$

Hence, the line passing through the point (1, 4, -2) and parallel to the given planes is

$$\frac{x-1}{-8} = \frac{y-4}{19} = \frac{z+2}{5}$$

Q.53.(a): Equation of base circle is

$x = a,\ y^2 + z^2 = b^2$

or $\quad x = a$ (i)

and $\quad x^2 + y^2 + z^2 = a^2 + b^2 \qquad$ (ii)

Equation of any line through the vertex O (0, 0, 0) is

$$\frac{x}{l} = \frac{y}{m} = \frac{z}{n} = r \text{ (say)} \qquad \text{(iii)}$$

Any point on this line is (lr, mr, nr)

For the line (iii) to be generator of the line, it must intersect the base unit (i, ii)

i.e., $\quad lr = a \quad$ or $\quad r = a/l$

and $\quad l^2r^2 + m^2r^2 + n^2r^2 = a^2 + b^2$

Eliminating r between (iv) and (v)

$$\frac{a^2}{l^2}(l^2 + m^2 + n^2) = a^2 + b^2$$

or $\quad a^2(l^2 + m^2 + n^2) = l^2(a^2 + b^2)$

Hence, required equation of the one is (obtained by eliminating l, m, n, between (iii) and (iv))

$$a^2\left(\frac{x^2}{x^2} + \frac{x^2}{x^2} + \frac{2^2}{x^2}\right) = \frac{x^2}{x^2}(a^2 + b^2)$$

or $\quad a2\,(y^2 + z^2) = b^2x^2$

Q.54.(d): Equation of right circular cylinder with axis $\frac{x-\gamma}{l} = \frac{y-\beta}{m} = \frac{z-\gamma}{n}$ and radius a is

$$(x-a)^2 = (y-\beta) + (z-\gamma)^2$$

$$\frac{[l(x-a) + m(y-\beta) + n(z-\gamma)]^2}{l^2 + m^2 + n^2} = a^2$$

$\therefore$ Here $a = 4$, and $\frac{x-0}{2} = \frac{y-0}{1} = \frac{z-0}{-2}$

as the axis is $x = 2y = -z$

$$\therefore \quad (x-0)^2 + (y-0)^2 + (z-0)^2 - \frac{[2(x-0) + 1(y-0) - 2(z-0)]^2}{(2)^2 + (1)^2 + (-2)^2} = (4)^2$$

$$\Rightarrow \quad x^2 + y^2 + z^2 - \frac{(2x + y - 2z)^2}{4+1+4} = 16$$

$$\Rightarrow \quad 9x^2 + 9y^2 + 9z^2 - (4x^2 + y^2 + 4z^2 + 4xy - 4yz - 8xz) = 144$$

$$\Rightarrow \quad 9x^2 + 9y^2 + 9z^2 - 4x^2 + y^2 + 4z^2 + 4xy - 4yz - 8xz = 144$$

$$\Rightarrow \quad 5x^2 + 8y^2 + 5z^2 - 4x^2 + 4y^2 + 8xz - 144 = 0$$

Q.55.(b): $(2D + 1)^2 y = 0$, then roots of $(2D + 1)^2 = 0 \Rightarrow 2D + 1 = 0 \Rightarrow D = -\frac{1}{2}$

Hence, CF $= (A + Bx)e^{-\frac{x}{2}}$

$$\therefore \quad y = (A + Bx)e^{-\frac{x}{2}}$$

Q.56.(c): **Q.57.(d)**

Q.58.(c): Moment of a force $\vec{F}$ which is represented in length and direction by line AB about a point C is the multiplication of the force and perpendicular distance from AB to the point C.

Again area of the triangle ABC is $\frac{1}{2}$ (AB × perpendicular distance between AB and C)

$$\Rightarrow \quad \frac{1}{2} \text{ (Base × altitude)}$$

Hence A is true.

Area of a triangle is $\frac{1}{2}$ (product of base and altitude)

Hence R is false.

Q.59 (a): As three forces acting at a point represented by magnitude and direction of three sides of a triangle then it will be equilibrium. Hence A is true.

Three forces acting at a point are in equilibrium if these can be represented as a triangle. Hence B is true, the answer is (a).

Q.60 (d) Q.61 (d)

Q.62 (b) Q.63 (b)

Q.64 (d)

Q.65 (c): Since $u\cos\alpha = \frac{u}{2}$

or $\cos\alpha = \frac{1}{2}$ ∴ $\alpha = 60^\circ$

Hence, max, range

$= \frac{u^2\sin\alpha}{g} = \frac{u^2}{g}\frac{\sqrt{3}}{2} = \frac{\sqrt{3}u^2}{2g}$

Q.66 (a) Q.67 (b)

Q.68 (d) Q.69 (a)

Q.70 (c) Q.71 (b)

$x^3 = 27+\sqrt{756}+27-\sqrt{756}$

∴ $+3(27+\sqrt{756})^{1/3}+\left(27-\sqrt{756}\right)^{1/3}$

$+\left(27-\sqrt{756}\right)^{1/3}$

$\left[\left(27+\sqrt{756}\right)^{1/3}+\left(27-\sqrt{756}\right)^{1/3}\right] = 54 - 9x$

or $x^3 + 9x - 54 = 0$

or $(x-3)(x^2+3x+18) = 0$ ∴ $x = 3$

Q.79 (d) Q.80 (d)

Q.81 (c): Since $f(x) = 2x^3 - 5x^2 + mx + n$ is

Divisible by $x^2 - 4 = (x-2)(x+2)$

∴ i.e. $f(2) = 0$ and $f(-2) = 0$

or $2.2^3 - 5.2^2 + m.2 + n = 0$

and $2(-2)^3 - 5(-2)^2 + m(-2) + n = 0$

or $16 - 20 + 2m + n = 0$ and

$-16 - 20 - 2m + n = 0$

$2m + n - 4 = 0$ ...(i)

and $2m - n + 36 = 0$ ...(ii)

On solving (i) an (ii), we get

$m = -8$, $n = 20$.

or $\alpha^2\beta^2\gamma^2 = r^2$

It can also be verified that the required roots are $\alpha^2, \beta^2, \gamma^2$, because

$$\alpha^2 + \beta^2 + \gamma^2 = (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \beta\gamma + \gamma\alpha)$$

$= p^2 - 2q$

and $\alpha^2\beta^2 + \beta^2\gamma^2 + \gamma^2\beta^2 (\alpha\beta + \beta\gamma + \gamma\alpha)^2$

$- 2(\alpha\beta^2\gamma + \beta\gamma^2\alpha + \gamma\alpha^2\beta)$

$= q^2 - 2\alpha\beta\gamma\,(\alpha + \beta + \gamma)$

$= q^2 - 2rp$

Q.85 (c)

are $\begin{pmatrix} 4 & 6 \\ 6 & 4 \end{pmatrix}, \begin{pmatrix} 1 & 5 & 3 & 7 \\ 5 & 3 & 7 & 1 \end{pmatrix}$

Q.89 (b) Q.90 (d)

Q.91 (a) Q.92 (b)

Q.93 (c) Q.94 (a)

Q.95 (a): As $y = x$, $\therefore (x_1, x_2, x_3) = (x_1, x_1, x_3)$

$= (x_1, x_1, 0) + (0, 0, x_3)$

$= x_1 (1, 1, 0) + x_3 (0, 0, 1)$

Hence, (x_1, x_2, x_3) is a linear combination **(1, 1, 0) and (0, 0, 1)**

Q.96 (c) Q.97 (a)

Q.98 (b) Q.99 (a)

Q.100 (a): AB = O gives

$\begin{bmatrix} 6 + 6y & 2x + 12 \end{bmatrix} \quad \begin{bmatrix} 0 & 0 \end{bmatrix}$

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1. If $x = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix}$ then $|x'x|$ is equal to
(a) 2
(c) 6
(b) 4
(d) 8

2. If $A + B = \begin{bmatrix} 0 & 0 \\ 2 & 1 \end{bmatrix}$ and $A - B = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$ then A is equal to
(a) $\begin{bmatrix} 0 & 2 \\ 2 & 2 \end{bmatrix}$
(b) $\begin{bmatrix} 0 & 0 \\ 1 & 1 \end{bmatrix}$
(c) $\begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix}$
(d) $\begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$

3. Co-factors of the elements of the second of the row of the determinant $\begin{bmatrix} 1 & 2 & 3 \\ -4 & 3 & 6 \\ 2 & -7 & 9 \end{bmatrix}$ are
(a) −39, 3, 11
(b) 6, 5, 4
(c) 3, 11, −39
(d) 13, 1, 3

4. $\begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ a^2 & b^2 & c^2 \end{vmatrix}$ is equal to
(a) $(a-b)(c-b)(c-a)$
(b) $(a-b)(a-c)(a-b)$
(c) $a^2 + b^2 + c^2$
(d) $a + b + c$

5. If $A = \begin{bmatrix} 3 & 8 \\ 2 & 1 \end{bmatrix}$, then A^{-1} is
(a) $\frac{1}{13}\begin{bmatrix} 3 & 2 \\ 8 & 1 \end{bmatrix}$
(b) $\frac{1}{-13}\begin{bmatrix} 1 & -8 \\ -2 & 3 \end{bmatrix}$
(c) $\frac{1}{-13}\begin{bmatrix} 3 & 2 \\ 8 & 1 \end{bmatrix}$
(d) $\frac{1}{13}\begin{bmatrix} 1 & -8 \\ -2 & 8 \end{bmatrix}$

6. The value of α for which the system of equation $x + y = 0$, $y + 2z = 0$, $\alpha x + z = 0$ has more than one solution, is
(a) −1
(b) 0
(c) $\frac{1}{2}$
(d) 1

7. If [x] denotes the greatest integer less than or equal to the real number x, then the range of the function $f(x) = 1 + x - x[x - 1]$ is
(a) (4, 5)
(b) (4, ∞)
(c) the set of Z of all integers
(d) the set R of all real numbers

8. Which of the following statement is/are correct?
I. The sum and difference of any two irrational numbers need not be irrational
II. Product of any two irrational numbers is irrational
III. For any two distinct irrational numbers a and b, the number a/b is irrational

Select the correct answer using the codes given below:

Codes:
(a) I, II and III
(b) I and II
(c) II and III
(d) I alone

9. $\lim_{n\to\infty}\left\{\left(1+\frac{1}{n}\right)^n + \left(1+\frac{1}{n}\right)^{-n}\right\}$
(a) exists and is equal to 0
(b) does not exist
(c) exists and is equal to $e + \frac{1}{e}$
(d) exists and is equal to e

10. Consider the following statements: If
$$f(x) = \begin{cases} x & 0 \le x < 1 \\ 3 - x & 1 \le x \le 2 \end{cases}$$
then
I. $\lim_{x\to 1^-} f(x) = 1$
II. $\lim_{x\to 1^+} f(x) = 2$
III. $\lim_{x\to 1} f(x) = 2$

Of these statements

(a) I and III are correct

(b) I, II and III are correct

(c) I and II are correct

(d) II alone is correct

11. A function which is continuous nowhere on $\mathbb{R}$ but is bounded on R is f defined by

(a) $f(x) = x$, x rational;
$f(x) = -x$, x irrational

(b) $f(x) = n$, $n \le x < n + 1$, $n \in Z$

(c) $f(x) = -1$, x rational; $f(x) = 1$, x irrational

(d) $f(x) = 1$, $x \neq n$, $n \in Z$, $f(x) = 0$ otherwise

12. Let D = {f | f is differentiable on (0, 1)}, C = {f | f is continuous on (0, 1)} then

(a) $C \cap D = \phi$ (b) $C \subset D$

(c) $C = D$ (d) $D \subset C$

13. Let f, g, h, k be differentiable in (a, b). If F is defined as $F(x) = \begin{vmatrix} f(x) & g(x) \\ h(x) & k(x) \end{vmatrix}$ for all $x \in (a, b)$, then $F'(x)$ is given by

(a) $\begin{vmatrix} f'(x) & g(x) \\ h(x) & k(x) \end{vmatrix} + \begin{vmatrix} f(x) & g'(x) \\ h'(x) & k(x) \end{vmatrix}$

(b) $\begin{vmatrix} f'(x) & g'(x) \\ h(x) & k(x) \end{vmatrix} + \begin{vmatrix} f(x) & g(x) \\ h'(x) & k'(x) \end{vmatrix}$

(c) $\begin{vmatrix} f(x) & g'(x) \\ h(x) & k'(x) \end{vmatrix} + \begin{vmatrix} f'(x) & g(x) \\ h(x) & k'(x) \end{vmatrix}$

(d) $\begin{vmatrix} f(x) & g(x) \\ h'(x) & k'(x) \end{vmatrix} + \begin{vmatrix} f'(x) & g(x) \\ h(x) & k(x) \end{vmatrix}$

30. $\int_0 \cos h\frac{y}{a}$ is equal to

(a) $a \sin h\frac{y}{a}$ (b) $-\frac{1}{a}\sin h\frac{x}{a}$

(c) $a \sin h\frac{x}{a}$ (d) $\frac{1}{a}\sin h\frac{x}{a}$

31. For definite integrals, the formula of integration by parts is

(a) $\int_a^b u\,dv = uv]_a^b + \int_a^b v\,du$

(b) $\int_a^b u\,dv = uv]_a^b - \int_a^b v\,du$

(c) $\int_a^b u\,dv = uv]_a^b + \int_a^b u\,dv$

(d) $\int_a^b u\,dv = uv]_a^b - \int_b^a u\,dv$

32. The correct value of the volume of the prolate shpheroid formed by the revolution of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{a^2} = 1$ about x-axis is

(a) $\frac{1}{3}\pi a^2 b$ (b) $\frac{2}{3}\pi a^2 b^2$

(c) $\frac{4}{\sqrt{3}}\pi ab^2$ (d) $\frac{4}{3}\pi ab^2$

33. The length of the arc of the curve $6xy = x^4 + 3$ from $x = 1$ to $x = 2$ is

(a) $\frac{13}{12}$ units (b) $\frac{17}{12}$

(c) $\frac{19}{12}$ units (d) none of the above

34. The volume of the ellipsoid $\frac{x^2}{a^2} + \frac{y^2}{a^2} + \frac{z^2}{c^2} = 1$

(a) $\frac{4}{5}\pi$ abc cubic units

(b) $\frac{4}{3}\pi$ abc cubic units

(c) 4π abc cubic units

(d) none of the above

35. For a positive term series Σa_n, the ratio test states that

(a) the series converges if $\lim\limits_{n \to \infty} \frac{a_{n+1}}{a_n} > 1$

(b) the series converges if $\lim\limits_{n \to \infty} \frac{a_{n+1}}{a_n} < 1$

(c) the series diverges if $\lim\limits_{n \to \infty} \frac{a_{n+1}}{a_n} = 1$

(d) none of these

36. The series $x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \ldots$ is convergent for

(a) all real values of x

(b) $|x| < 1$ only

(c) $|x| \le 1$ (d) $-1 < x \le 1$

37. The differential equation of the system of circles touching the y-axis at the origin, is

(a) $x^2 + y^2 - 2xy\frac{dy}{dx} = 0$

(b) $x^2 + y^2 + 2xy\frac{dy}{dx}$

(c) $x^2 - y^2 + 2xy\frac{dy}{dx} = 0$

(d) $x^2 - y^2 - 2xy\frac{dy}{dx} = 0$

38. The differential equation $M(x, y)dx + N(x, y)dy = 0$ is an exact equation if

(a) $\frac{\partial M}{\partial y} + \frac{\partial N}{\partial x} = 0$ (b) $\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} = 0$

(c) $\frac{\partial N}{\partial y} + \frac{\partial M}{\partial x} = 0$ (d) $\frac{\partial N}{\partial y} - \frac{\partial M}{\partial x} = 0$

39. The equation $(4x + 3y + 1)dx + (3x + 2y + 1)dy = 0$ represents a family of

(a) circles (b) parabolas

(c) ellipses (d) hyperbolas

40. The general solution of the differential equation $(x^2 + y^2)dx - 2xydy = 0$ is

(a) $x^2 - cx - y^2 = 0$, where c is an arbitrary constant

(b) $(x - y)^2 = cx$, where c is an arbitrary constant

(c) $x + y + 2xy = c$, where is an arbitrary constant

(d) $y = x^2 - 2x + c$, where c is an arbitrary constant

41. The general solution of the differential equation $y = x\left(\frac{dy}{dx}\right) + \left(\frac{dy}{dx}\right)^2$ is

(a) $y = cx - c^2$, c is an arbitrary constant

(b) $y = cx + c$, c is an arbitrary constant

(c) $y = cx - c$, c is an arbitrary constant

(d) $y = cx + c^2$, c is an arbitrary constant

42. The singular solution of the differential equation $(xp - y)^2 = p^2 - 1$ is

(a) $x^2 + y^2 = 1$ (b) $x^2 - y^2 = 1$

(c) $x^2 + 2y^2 = 1$ (d) $2x^2 + y^2 = 1$

43. The orthogonal trajectories of the parabolas $y^2 = 4a(x + a)$, a being the parameter, are the curves given by

(a) $y^2 = 4b(x + b)$, b being a parameter

(b) $y^2 = 4b(x - b)$, b being a parameter

(c) $y^2 = 4bx$, b being a parameter

(d) $x^2 = 4by$, b being a parameter

44. The general solution of the differential equation $\frac{d^4y}{dx^4} + 2\frac{d^2y}{dx^2} + y = 0$ is (c_1, c_2, c_3 and c_4 are arbitrary constant)

(a) $y = (c_1 + c_2x)\sin x + (c_3 + c_4x)\cos x$

(b) $y = c_1 \sin x + c_2 \cos x + x \sin x \cos x$

(c) $y = c_1 \sin x + c_2 \cos x + c_3 \tan x + c_4 \cot x$

(d) $y = c_1 \sin x + c_2 \cos x + c_3 x + c_4$

45. The general solution of the differential equations $(D^2 + D - 2)y = e^x$ is given by

(a) $y = c_1 e^x + c_2 e^{-2x} + 1/3\, x e^x$

(b) $y = c_1 e^x + c_2 e^{2x}$

(c) $y = c_1 e^x + c_2 e^{-2x} + 1/6\, x^2 e^x$

(d) $y = 1/3\, x e^x + (c_1 + c_2 x) e^{-2x}$

46. A particular integral of the differential equation $\frac{d^2y}{dx^2} + a^2y = \sin ax$ is

(a) $\frac{1}{2a}\cos ax$ (b) $-\frac{1}{2a}x\cos ax$

(c) $\frac{1}{2a}x\cos 2ax$ (d) $-\frac{1}{2a}x\cos 2x$

47. P(1, 2) and Q(7, 0) are two vertices of a triangle PQR. If the slope of side PR is twice the slope of the side QR, then the locus of the third vertex R is

(a) $x^2 + y^2 - 5y + 2x - 14 = 0$

(b) $2x + 5y - 14 = 0$

(c) $x = 4, y = 1$

(d) $xy + 2x + 5y - 14 = 0$

48. If the three lines $x + y = 1$, $x - y = 1$ and $2x + 3y = k$ are concurrent, then the value of k is

(a) 1 (b) −1

(c) 0 (d) 2

49. The straight lines ... equation $(x^2 + y^2)\sin^2\alpha$ $= (x\cos\alpha - y)\sin\alpha)^2$ are inclined to each other at an angle

(a) 2α (b) $\frac{\pi}{2} + \alpha$

(c) α (d) $\frac{\pi}{2}$

50. If $\lambda x^2 - 10xy + 12y^2 + 5x - 16y - 3 = 0$ represents a pair of straight lines, then the value of λ is

(a) 1 (b) 2

(c) 3 (d) 4

51. The condition for the two circles $x^2 + y^2 + 2k_1x + k^2 = 0$ and $x^2 + y^2 = 2k_2y + k^2 = 0$ to touch each other externally is

(a) $k_1^2 + k_2^2 = k^2$ (b) $k_1^2 - k_2^2 = k^2$

(c) $k^2(k_1^2 - k_2^2) = k_1^2 k_2^2$

(d) $k^2(k_1^2 + k_2^2) = k_1^2 k_2^2$

52. The locus of the point of intersection of two normals to the parabola $y^2 = 4ax$ which are at right angles to one another

(a) $y^2 = a(x - 2a)$ (b) $y^2 = a(x + 2a)$

(c) $y^2 = a(x - 3a)$ (d) $y^2 = a(x + 3a)$

53. If the locus of the point of intersection of perpendicular tangents to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is a circle with centre at (0, 0), then the radius of the circle would be

(a) $a + b$ (b) ab

(c) b/a (d) $\sqrt{a^2 + b^2}$

54. The centre of the hyperbola $2x^2 + y^2 - 3xy - 5x + 4y + 6 = 0$ is the point

(a) (1, 2) (b) (2, 1)

(c) (1, 1) (d) (2, 2)

55. Th...

56. The parametric equations $x = \frac{a}{2}\left(t + \frac{1}{t}\right)$, $y = \frac{b}{2}\left(t - \frac{1}{t}\right)$ represents

(a) an ellipse

(b) a hyperbola

(c) a parabola

(d) a circle with centre at origin

57. A straight line passes through the point (2, −1, −1). It is parallel to the plane $4x + y$ $- z + 2 = 0$ and is perpendicular to the line $\frac{x}{1} = \frac{y}{-2} = \frac{z-5}{1}$. The equation of the straight line are

(a) $\frac{x-2}{4} = \frac{y+1}{1} = \frac{z+1}{1}$

(b) $\frac{x+2}{4} = \frac{y-1}{1} = \frac{z-1}{3}$

(c) $\frac{x-2}{-1} = \frac{y+1}{1} = \frac{z+1}{3}$

(d) $\frac{x+2}{-1} = \frac{y-1}{1} = \frac{z-1}{3}$

58. Perpendicular is drawn from the point (0, 3, 4) to the plane $2x - 2y + z = 10$. The coordinates of the foot of the perpendicular are

(a) $\left(-\frac{8}{3}, \frac{1}{3}, \frac{16}{3}\right)$ (b) $\left(\frac{8}{3}, \frac{1}{3}, \frac{16}{3}\right)$

(c) $\left(\frac{8}{3}, \frac{-1}{3}, \frac{16}{3}\right)$ (d) $\left(-\frac{8}{3}, \frac{1}{3}, \frac{-16}{3}\right)$

59. A sphere $x^2 + y^2 + z^2 = 9$ is cut by the plane $x + y + z = 3$. The radius of the circle so formed is

(a) $\sqrt{6}$ (b) $\sqrt{3}$

(c) 3 (d) 6

60. The equation $fyz + gzx + hxy = 0$ represents a

(a) a pair of planes (b) sphere

(c) cylinder (d) cone

61. The equation of a cylinder, whose generating lines have the directions cosines (l, m, n) and which passes through the ...

62. If three vector A, B, C are such that $\vec{A} + \vec{B} = \vec{C}$, then

(a) $\vec{A} \cdot \vec{C} = 0$ (b) $\vec{B} \cdot \vec{C} = 0$

(c) $\vec{A} \cdot \vec{C} > \vec{B} \cdot \vec{C}$ (d) $\vec{A} \cdot \vec{C} = \vec{B} \cdot \vec{C}$

63. The resultant of two forces P and Q is R. If one of the forces is reversed in direction the resultant becomes R, then

(a) $R^2 = P^2 + Q^2 + 2PQ \cos\alpha$

(b) $R^2 = P2 - Q^2 - 2PQ \cos\alpha$

(c) $R^2 + R^2 = 2(P^2 + Q^2)$

(d) $R^2 + R^2 = 2(P^2 - Q^2)$

64. Two weights of 10 gms and 2 gms hangs from the ends of a uniform lever one meter long and weighting 4 gms. The point in the lever about which it will balance is from the weight of 10 gms at a distance of

(a) 5 cm (b) 25 cm

(c) 45 cm (d) 65 cm

65. The angle between two forces F and 2F acting at a point when the resultant is perpendicular to F is

(a) 60° (b) 135°

(c) 120° (d) 150°

66. If a body is in equilibrium under the action of three coplanar forces, then

(a) they must act in a straight line

(b) they must meet in a point

(c) their horizontal and vertical components must be equal

(d) none of these

67. If a particle moves in one dimensions under the potential energy $V = V_0(a^{-ax} + bx)$, where V_0, a, b are positive constants, then the nature of the equilibrium position is

(a) unstable (b) stable

(c) neutral (d) undecided

68. Which one of the following statements is correct?

(a) Newton's three laws of motion are independent

(b) first law of motion is contained in the second law as a special case

(c) second law of motion can be deduced from the third law

(d) third law of motion can be deduced from the second law

69. Newton's second law of motion is given by

(a) velocity = $\frac{\text{mass}}{\text{force}}$

(b) acceleration = $\frac{\text{moving mass}}{\text{moving force}}$

(c) acceleration = $\frac{\text{moving force}}{\text{moving moved}}$

(d) none of the above

70. The pedal equation of the path of a central orbit is

(a) $F = \frac{h^2}{p^3}\frac{dr}{dp}$ (b) $F = \frac{h^2}{p^3}\frac{dp}{dr}$

(c) $F = \frac{h}{p^3}\frac{dp}{dr}$ (d) $F = \frac{h^2}{p^2}\frac{d^2p}{dr^2}$

71. For the motion of a particle in a plane in a central orbit, the angular velocity of the particle varies

(a) inversely as the distance

(b) inversely as the square of the distance

(c) directly as the distance

(d) directly as the square of the distance

72. A particle executing simple harmonic motion has acceleration 8 cm/sec^2 when it is at a distance 2 cm form the centre. The time period will be

(a) $\frac{2}{\pi}$ seconds (b) $\frac{1}{\pi}$ seconds

(c) $\frac{\pi}{2}$ seconds (d) π seconds

73. A particle is projected from a point on the x-axis in the vertical x-y plane. If the trajectory above the x-axis is given by $x^2 + 4y - 8 = 0$, then the velocity of projection is

(a) $\sqrt{3g}$ (b) $\sqrt{4g}$

(c) $\sqrt{5g}$ (d) $\sqrt{6g}$

74. A particle moving along a circular path with constant speed

(a) is not accelerated

(b) has a constant velocity

(c) has radial acceleration away from the centre

(d) has radial acceleration towards the centre

75. If a particle moves along a circle of radius 'a' so that $r = a$, transverse velocity is equal to

(a) $a\frac{d\theta}{dt}$ (b) $r\frac{d\theta}{dt}$

(c) $\frac{dr}{dt}\cdot\theta$ (d) $\frac{da}{dt}\cdot\theta$

76. The escape velocity for a body projected vertically upwards is 11.2 km/sec. If the body is projected in a direction making an angle of 60° with the vertical, then the escape velocity will be

(a) 11.2 km/sec (b) $5.6\sqrt{3}$ km/sec

(c) 5.6 km/sec (d) none of these

77. Which of the following statements are correct for the integers p, m and n?

I. If $p < m$, then $m < p$

II. If $p \neq m$, then either $m < p$ or $p < m$

III. $mn \leq mp$ if and only if $n \leq p$

IV. $mn < mp$ if and only if $n < p$, provided $m \neq 0$

Select the correct answer using the code s given below.

(a) II and IV (b) I and II

(c) I, II and IV (d) I, II and III

78. Consider the following numbers $\pi, \frac{22}{7}, \frac{223}{71}$.

The correct sequence in increasing order would be

(a) $\pi, \frac{22}{7}, \frac{223}{71}$ (b) $\pi, \frac{223}{71}, \frac{22}{7}$

(c) $\frac{223}{71}, \pi, \frac{22}{7}$ (d) $\frac{223}{71}, \frac{22}{7}, \pi$

79. If n is a positive integer, then $\sqrt{n+1} + \sqrt{n-1}$ is

(a) rational for only one value of n

(b) rational for more than one value of n

(c) rational for no value of n

(d) rational for at least one value of n

80. If w be an imaginary cube root of unit, then $(1 - w - w^2) + (1 + w - w^2)$ is equal to

(a) 64 (b) 32

(c) 16 (d) 8

81. The conjugate of $(1 + i)^2$ is given by

(a) $(1 - i)^2$ (b) $(1 + i)^{-1}$

(c) $-2i$ (d) $2i$

82. If m and x be positive integers and a and be integers such that $a \equiv b \pmod{m}$, then consider the following statement

I. $a + x \equiv b + x \pmod{m}$

II. $a - x \equiv b - x \pmod{m}$

III. $ax \equiv bx \pmod{m}$

IV. $a^x \equiv b^x \pmod{m}$

Of these statements

(a) only one of the statements is correct

(b) only two of the statements are correct

(c) only three of the statements are correct

(d) all the statements are true

83. The equation, in the set of integers, $2x \equiv 3$ (modulo 20) has

(a) a unique solution

(b) no solution

(c) infinite number of solutions

(d) only 2 solutions

84. Consider the following statements:

I. If d is g.c.d. of integers h and k, then $d = mh + nk$, where m and n are uniquely determined

II. If h and k are primes and $m \mid hk$, then $m \mid h$ or $m \mid k$.

Of these statements:

(a) I is true, but II is false

(b) II is true, but I is false

(c) both I and II are true

(d) both I and II are false

85. If the quotient on dividing $x^4 + 2x^3 + 2x^2 - 3x + 1$ by $x + 2$ is $x^3 + 2ax^2 + 2bx + c$, then the value of a, are respectively

(a) 1, 2, −7 (b) 0, 2, 7

(c) 0, −2, 7 (d) 0, 1, −7

86. If $a(x) = x^3 + 2x + 3$, $b(x) = 2x^2 + 3x$ and $c(x) = 2x + 2$ be three members of the ring $I_4[x]$ over the ring I of integers modulo 4 then consider the following statements:
I. deg $[a(x) + b(x)] = 3$
II. deg $[c(x)^2] = 0$
III. deg $[a(x), b(x)] = 4$
Of these statements
(a) I and II are correct
(b) I and III are correct
(c) II and III are correct
(d) I, II and III are correct

87. The G.C.D. and L.C.M. of f(x) an g(x) are respectively $x^2 + x - 2$ and $x^4 + 2x^3 - 3x^2 - 7x + 6$. If $f(x) = x^3 + 4x^2 + x - 6$, then g(x) is
(a) $x^3 - 2x^2 + 2$ (b) $x^3 - 2x^2 - 2$
(c) $x^3 - 2x + 2$ (d) $x^3 - 2x - 2$

88. If α, β, γ are the roots of the equation $x^3 + px^2 + qx + r = 0$, then the value of $(\alpha + \beta)(\beta + \gamma)(\gamma + \alpha)$ is
(a) $r + pq$ (b) $r^2 - pq$
(c) $r^2 + pq$ (d) $r - pq$

89. If $x + y + z = 6$ and $xy + yz + zx = 10$, then the value of $x^3 + y^3 + z^3 - 3xyz$ is
(a) 56 (b) 30
(c) 26 (d) 16

90. If the roots of the equation $x^4 - 6x^3 - 38x^2 - 3x + 17 = 0$ are greater by k than the roots of the equation $x^4 - 22x^3 + 130x^2 - 243x + 61 = 0$, then value of k would be
(a) 4 (b) −4
(c) 6 (d) −6

91. Given that $x^2 + 11$ is a factor, the number of the real roots of the equation $2x^4 - 11x^3 + 17x^2 - 11x + 15 = 0$ is
(a) 0 (b) 1
(c) 2 (d) 4

92. The equation $x^4 + x^2 - 16x + 20 = 0$ as a repeated root. The root is
(a) 2 repeated twice
(b) 2 repeated thrice
(c) −3 repeated twice
(d) −3 repeated thrice

93. Let U = {1, 2, 3, ..., 8} be a universal set and A = {1, 2, 3, 4} and B = {2, 4, 5, 7} be subsets of U, then $A' \cup B'$ is equal to
(a) {1, 2, 3, 5, 6, 7, 8}
(b) {1, 3, 4, 5, 6, 7, 8}
(c) {1, 2, 4, 5, 6, 7, 8}
(d) {1, 3, 5, 6, 7, 8}

94. If A = {1, 2, 5, 6} and B = {1, 2, 3}, then $(A \times B) \cap (B \times A)$ is
(a) {(1, 1) (1, 2) (2, 1) (2, 2)}
(b) {(1, 1) (2, 2) (5, 1) (1, 6)}
(c) {(1, 1) (2, 1) (6, 1) (1, 2)}
(d) {(2, 3) (3, 1) (3, 2) (5, 3)}

95. On the set R of a real numbers we define $x \sim y$ and only if $xy \geq 0$, then the relation ~ is
(a) reflexive but not symmetric
(b) symmetric but not transitive
(c) transitive but not reflexive
(d) an equivalence relation

96. If $f(x + 1) - 2f(x) + f(x - 1) = 2$ for all x, then
(a) $f(x) = -x$ (b) $f(x) = x$
(c) $f(x) = x^3$ (d) $f(x) = x^2$

97. Consider the following statements:
I. Two equivalent classes are either identical or they have a various intersection
II. The quotient set of a set S relative to an equivalence relation is a subset of S
III. The partition of a set S into disjoint subset is define an equivalence relation on S
Of these statements
(a) I and II are correct
(b) I and III are correct
(c) II and III are correct
(d) I, II and III are correct

98. On the set of integers define a relation R by setting $(a, b) \in R$, if and only if a^2 and b^2 are not prime to each other. This relation is not an equivalence relation because it fails to be

(a) reflexive (b) symmetric
(c) antisymmetric (d) transitive

99. Consider $A = \{3n : n \in Z\}$, $B = \{4n : n \in Z\}$. The subgroups of the additive group Z is
(a) $\{A \cup B, +\}$ (b) $\{A \cap B, +\}$
(c) $\{A \cup B, +\}$ (d) $\{A \cap B, \cdot\}$

100. If Q and Z are the sets of rational numbers and integers respectively, then which one of the following triples is a field?
(a) $(Q, +, \times)$ (b) $(Q, -, \times)$
(c) $(Z, +, \times)$ (d) $(Z, -, \times)$

ANSWERS

1.(d) $x'x = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix} = \begin{bmatrix} 3 & 6 \\ 6 & 14 \end{bmatrix}$

Hence, $|x'x| = \begin{vmatrix} 3 & 6 \\ 6 & 14 \end{vmatrix}$

$= 3 \times 14 - 6 \times 6 = 42 - 36 = 0$

2.(c) Adding the given matrix equations, we get
$(A + B) + (A - B)$

$= \begin{bmatrix} 0 & 0 \\ 2 & 1 \end{bmatrix} + \begin{bmatrix} 0 & 2 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 2 \\ 2 & 2 \end{bmatrix}$

$2A = \begin{bmatrix} 0 & 2 \\ 2 & 2 \end{bmatrix}$

$\therefore A = \frac{1}{2}\begin{bmatrix} 0 & 2 \\ 2 & 2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix}$

3.(a) Cofactors of the elements of second row of the determinant
$\begin{vmatrix} 1 & 2 & 3 \\ -4 & 3 & 6 \\ 2 & -7 & 9 \end{vmatrix}$ are $(-1)^{2+1}(18 + 21)$, $(-1)^{2+2}$
$(9 - 6)$, $(-1)^{2+3}(-7 - 4)$.

4.(b) $\begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ a^2 & b^2 & c^2 \end{vmatrix}$

$= \begin{vmatrix} 1 & 0 & 0 \\ a & b-a & c-a \\ a^2 & b^2-a^2 & c^2-a^2 \end{vmatrix}$

(Operating $C_2 \to C_2 - C_1, \to C_3, C_1$)

$= (b - a)(c - a)\begin{vmatrix} 1 & 0 & 0 \\ a & 1 & 1 \\ a^2 & b+a & c+a \end{vmatrix}$

$= (b - a)(c - a)[(c + a) - (b + a)]$

$= (a - b)(c - a)(c - b)$
$= (a - b)(a - c)(c - b)$

5.(b) Here $|A| = \begin{vmatrix} 3 & 8 \\ 2 & 1 \end{vmatrix} = 3 - 16 = -13$

Since, $|A| \neq 0$

$\therefore A^{-1}$ exists

Also, Adj. $A = \begin{bmatrix} 1 & -8 \\ -2 & 3 \end{bmatrix}$

Hence, $A^{-1} = \frac{1}{|A|}\text{Adj. } A = -\frac{1}{13}\begin{bmatrix} 1 & -8 \\ -2 & 3 \end{bmatrix}$

6.(a) 7.(d)

8.(c)

9.(c) $\lim_{n\to\infty}\left\{\left(1+\frac{1}{n}\right)^n + \left(1+\frac{1}{n}\right)^{-n}\right\}$

$= \lim_{n\to\infty}\left(1+\frac{1}{n}\right)^n + \lim_{n\to\infty}\left\{\left(1+\frac{1}{n}\right)^n\right\}^{-1}$

$= e + e^{-1} = e + \frac{1}{e}$

10.(c) I. $\lim_{x\to1-} f(x) = \lim_{h\to0} f(1 - h)$,

where $x = 1 - h$, $h > 0$

$= \lim_{h\to0}(1 - h) = 1 - 0 = 1$

II. $\lim_{x\to1+} f(x) = \lim_{h\to0} f(1 + h)$,

where $x = 1 + h$, $h > 0$

III. Since

$= \lim_{x\to1-} f(x) = 1 \neq 2 = \lim_{x\to1+} f(x)$

11.(c)

12.(d) Since every differentiable function is continuous but converse is not true,
$\therefore$ $D \subset C$.

13.(b) Here $F(x) = \begin{vmatrix} f(x) & g(x) \\ h(x) & k(x) \end{vmatrix}$

$= f(x)\,k(x) - g(x)\,h(x)$

$F'(x) = f(x)\,k'(x) + f'(x)\,k(x) - g(x)\,h'(x)$
$- g'(x)\,hx$

$= [f'(x)\,k(x) - g'(x)\,h(x)]$
$+ [f(x)\,k(x)\,g(x)\,h'(x)]$

$= \begin{vmatrix} f'(x) & g'(x) \\ h(x) & k(x) \end{vmatrix} + \begin{vmatrix} f(x) & g(x) \\ h'(x) & k'(x) \end{vmatrix}$

31.(b)

32.(d) Reqd. volume $= 2\pi\int_0^a y^2\,dx$

$$= 2\pi\int_0^a \frac{b^2}{a^2}(a^2 - x^2)\,dx = \frac{2\pi b^2}{a^2}\left(a x^2 - \frac{x^3}{3}\right)_0^a$$

$$= \frac{2\pi b^2}{a^2}\,a^3 - \frac{a^3}{3} = \frac{2\pi b^2}{a^2}\cdot\frac{2a^3}{3} = \frac{4}{3}\pi a b^2$$

33.(b) 34.(b)

35.(a)

36.(d) Here $|u_n| = \frac{|x|^n}{n}$, $|u_{n+1}| = \frac{|x^{n+1}|}{n+1}$

$$\therefore \left|\frac{u_n}{u_{n+1}}\right| = \left|\frac{x_n}{x^{n+1}}\right| \cdot \frac{n+1}{n}\left|\frac{1}{x}\right|\left(1+\frac{1}{n}\right)$$

$$\therefore \lim_{n\to\infty}\left(\frac{u_n}{u_{n+1}}\right) = (1+0)\frac{1}{|x|} = \frac{1}{|x|}$$

$\therefore$ By ratio test, $\Sigma|u_n|$ is convergent if

$\frac{1}{|x|} > 1$ or $|x| < 1$.

The series becomes $1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \ldots$

which by Leibnitz test is convergent.

40.(a) Since the given equation is of the homogeneous form

$$\frac{dy}{dx} = \frac{x^2 + y^2}{2xy}$$

Put $y = ux$

$$\therefore \quad u + x\frac{du}{dx} = \frac{x^2 + u^2x^2}{2x\cdot ux} = \frac{1+u^2}{2u}$$

or $\quad x\frac{du}{dx} = \frac{1+u^2}{2u} - u = \frac{1-u^2}{2u}$

or $\quad \frac{2u}{1-u^2}\,du = \frac{dx}{x}$

$\therefore$ Its solution is $\log(1-u^2) + \log x = \log c$

or $\quad (1-u^2)x = c \quad$ or $\quad \left(1-\frac{y^2}{x^2}\right)x = c$

or $\quad (x^2 - y^2) = cx$

41.(d) 42.(b)

43.(c)

44.(a) Given equation is $(D^4 + 2D^2 + 1)y = 0$ or $(D^2+1)^2 y = 0$

48.(c)

49.(a) $(x^2+y^2)\sin^2\alpha = (x\cos\alpha - y\sin\alpha)^2$

$= x^2\cos^2\alpha + y^2\sin^2\alpha$

$- 2xy\cos\alpha\sin\alpha\ x^2\cos 2\alpha\ + xy\sin 2\alpha$

$= 0.$

If θ be the angle between the tow lines, then

$$\tan\theta = \frac{2\sqrt{\frac{1}{4}\sin^2 2\alpha + 0}}{\cos^2\alpha + 0} = \frac{\sin^2\alpha}{\cos^2\alpha} = \tan 2\alpha$$

50.(b) 51.(d)

52.(c) 53.(d)

54.(b) 55.(a)

56.(b)

57.(c) Suppose a, b, c be the direction ratios of the required line. Since this line is parallel to the given plane and perpendicular to the given line.

$\therefore\ 4a + b + c = 0$ (i)

and $a - 2b + c = 0$ (ii)

Solving (i) and (ii), $b = -a$, $c = -3a$

Thus, the required lien through (2, −1, −1) are

$$\frac{x-2}{a} = \frac{y-(-1)}{-a} = \frac{z-(-1)}{-3a}$$

or $$\frac{x-2}{-1} = \frac{y+1}{1} = \frac{z+1}{3}$$

59.(a) 60.(d)

61.(b) 62.(c)

63.(c) 64.(b)

65.(c) 66.(d)

67.(a) 68.(a)

69.(c) 70.(b)

71.(b) 72.(d)

73.(d) 74.(d)

75.(a) 76.(c)

77.(b)

78.(c) Here $\pi = 3.14159$, $\frac{22}{7} = 3.1428571$

$\frac{223}{71} = 3.140845$

79.(c)

80.(a) Since w is an imaginary cube root of unity.

$\therefore\ w^2 + w + 1 = 0,\ w^3 = 1$

Thus $(1 - w - w^2) + (1 + w - w^2)^6$

$= (1+1)^6 + (-w^2 - w^2)^6$

$= 0 + (-2w^2)^6 = 64 \cdot w^{12} = 64\,(w^3)4 = 64$

81.(c) $(1+i)^2 = 1 + i^2 + 2i = 1 - 1 + 2i = 2i$

$\therefore$ Its conjugate is $-2i$.

82.(d) 83.(b)

84.(c) 85.(c)

86.(d) 87.(c)

$= -[p^3 + p^2(-p) + p(q) + (-r)]$

$= -[p^3 - p^3 + pq - r] = r - pq$

89.(b)

90.(b)

91.(c)

92.(a)

93.(d) Here

$A^c \cup B^c = \{5, 6, 7,\} \cup \{1, 3, 6, 8\}$

$= \{1, 3, 5, 6, 7, 8\}$

94.(a) Here $A \times B = \{(1,1), (1, 2), (1, 3), (2, 1), (2, 2), (2, 3), (5, 1), (5, 2), (5, 3), (6, 1), 6, 2), (6, 3)$

$B \times A = \{(1, 1), (1, 2), (1, 5), (1, 6), (2, 1), (2, 2), (2, 6), (3, 1), (3, 2), (3, 5), (3, 6)\}$

$= \{(1, 1), (1, 2), (2, 1), (2, 2)\}$

95.(d)

96.(d)

97.(d)

98.(d)

99.(b)

100.(a) The set of rationals with respect to addition and multiplication forms an ordered field.